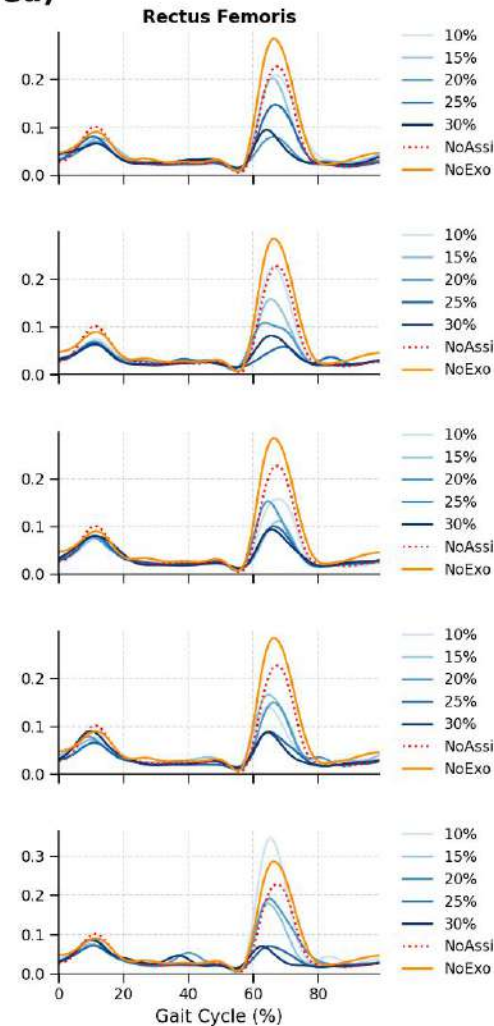
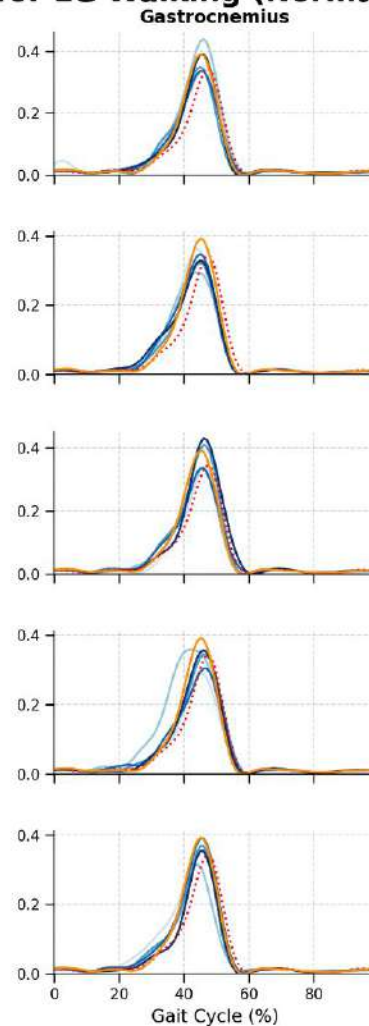
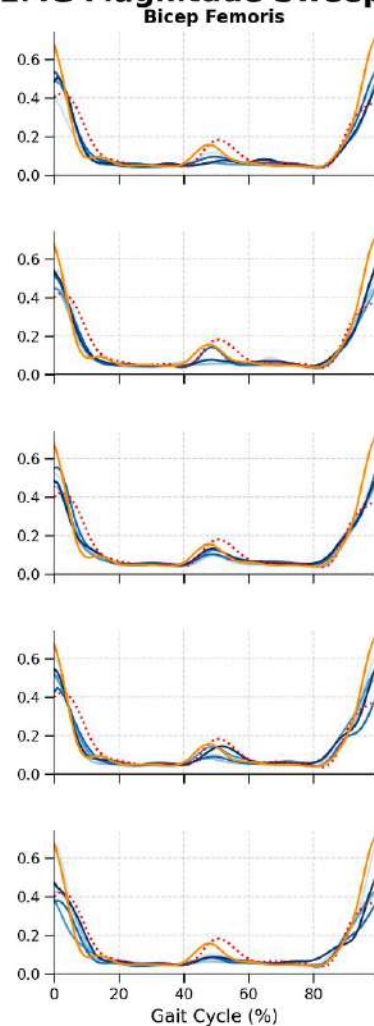
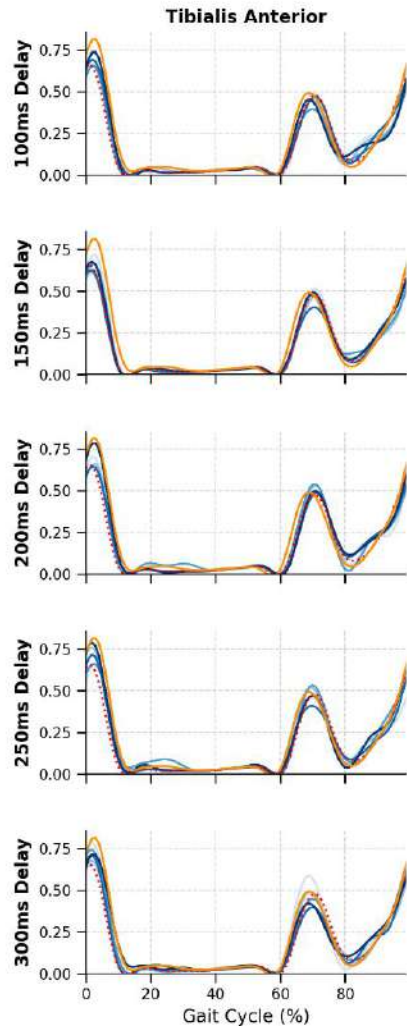


Magnitude & Delay Sweeps with Heat Maps

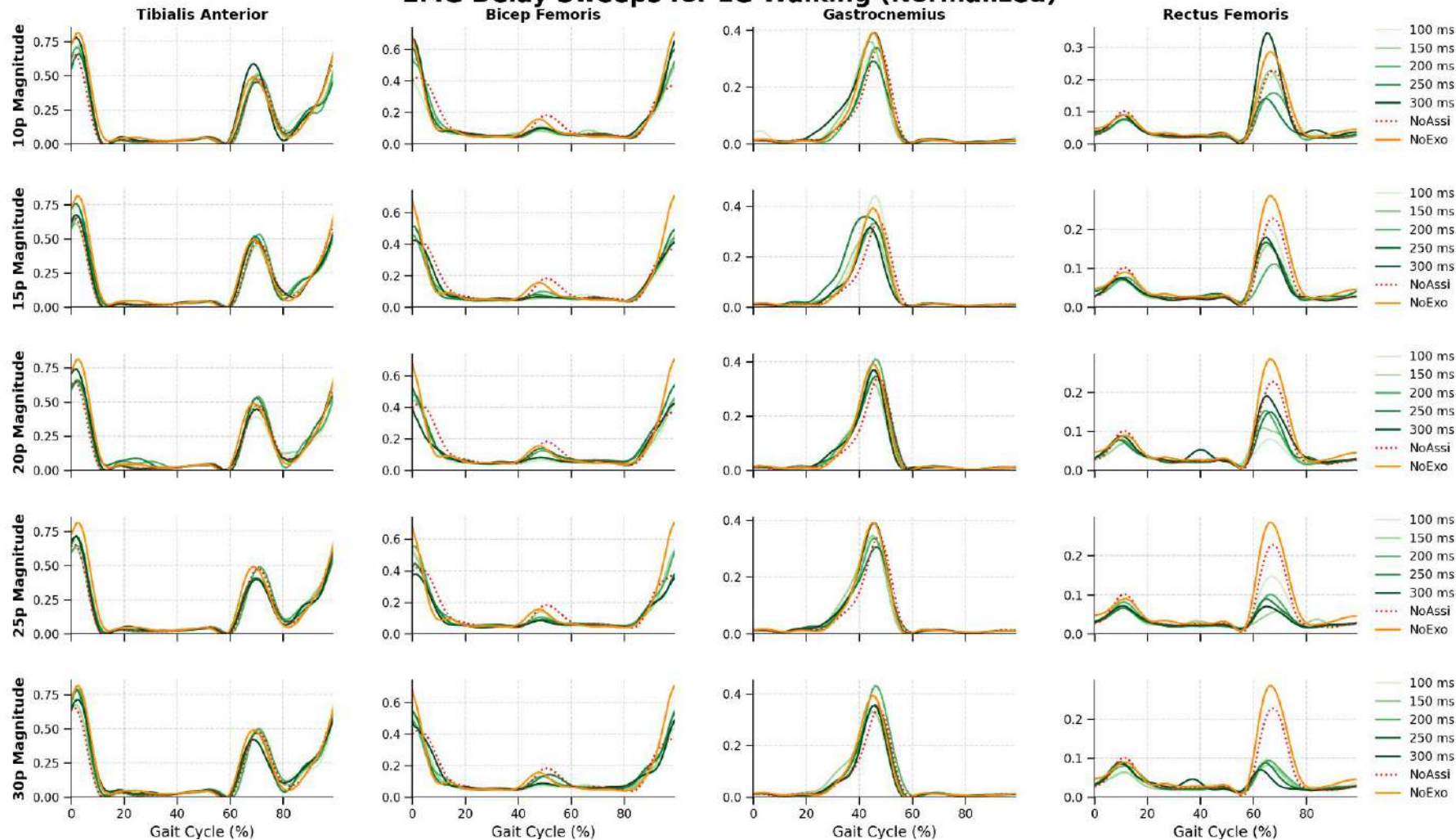
Subject: AB01_Jimin

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

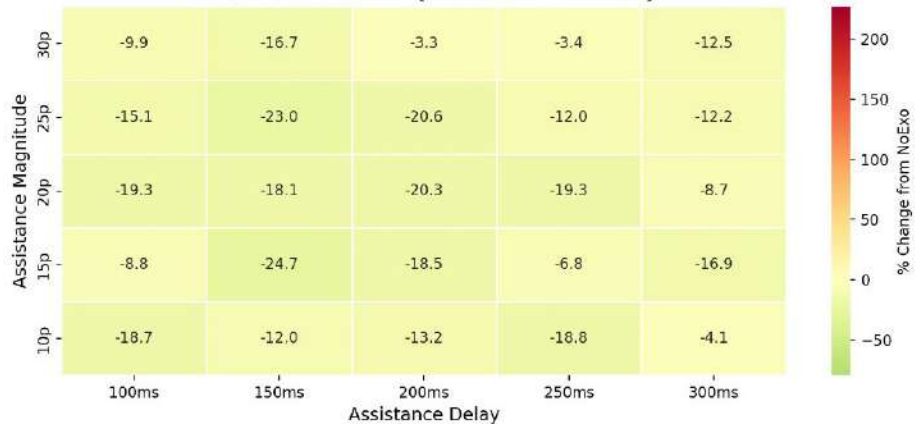


EMG Delay Sweeps for LG Walking (Normalized)

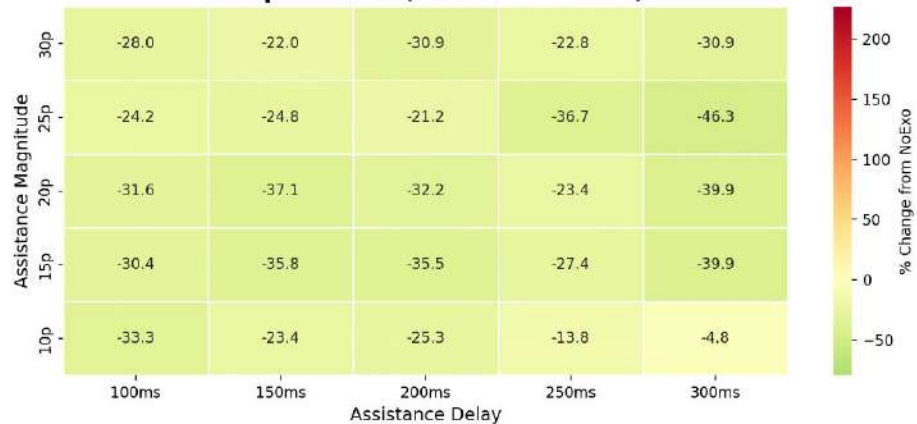


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

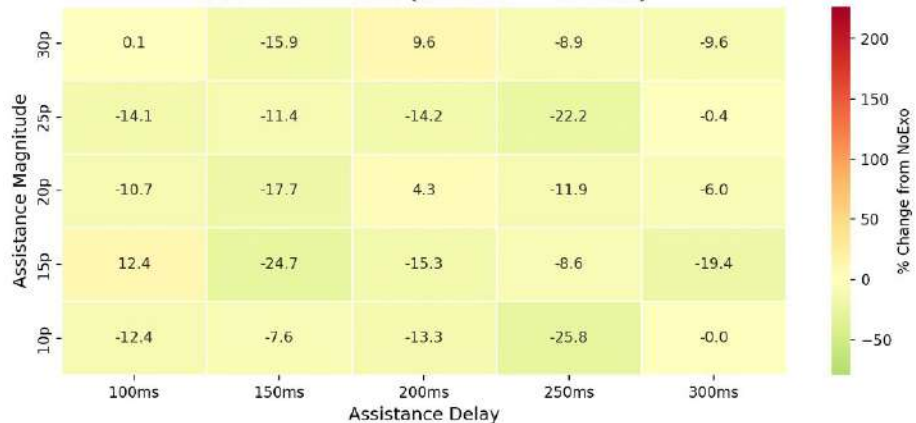
Tibialis Anterior (NoAssi = -19.5%)



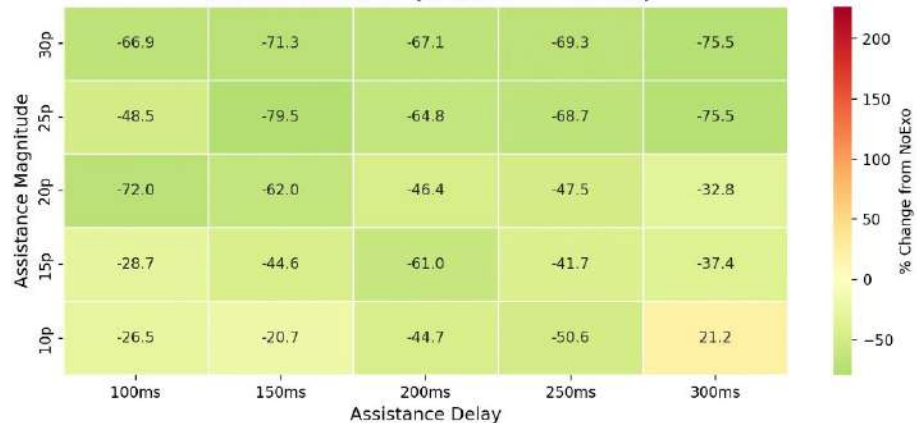
Bicep Femoris (NoAssi = -40.5%)



Gastrocnemius (NoAssi = -12.6%)

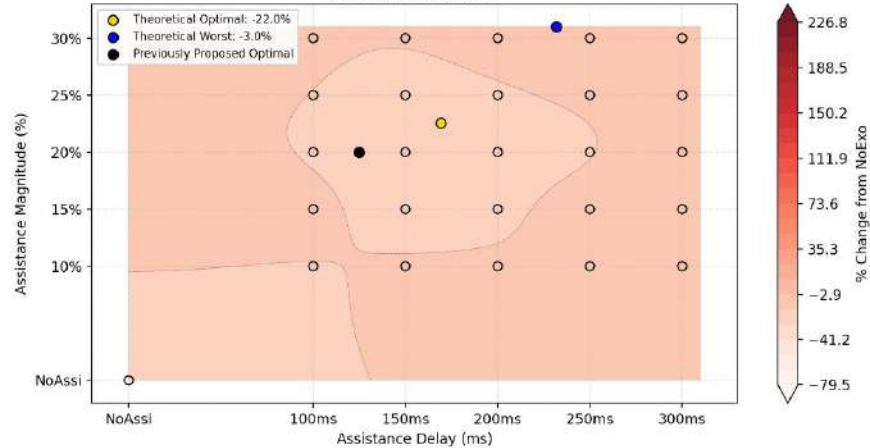


Rectus Femoris (NoAssi = -20.2%)

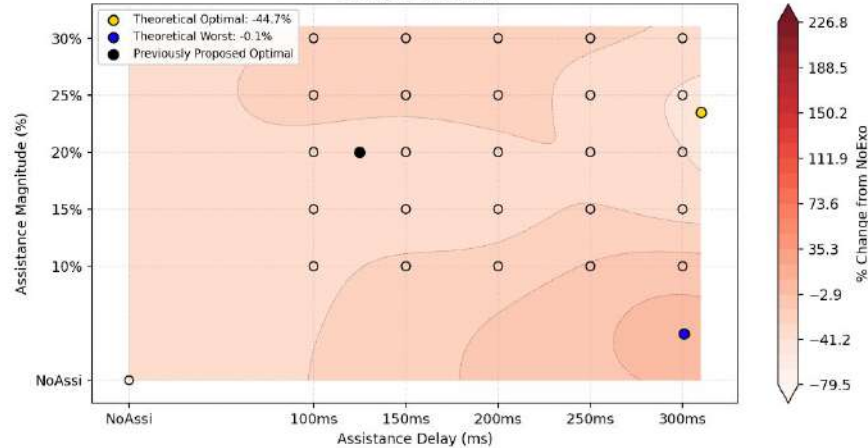


Peak EMG Activation Contour Map - LG Walking

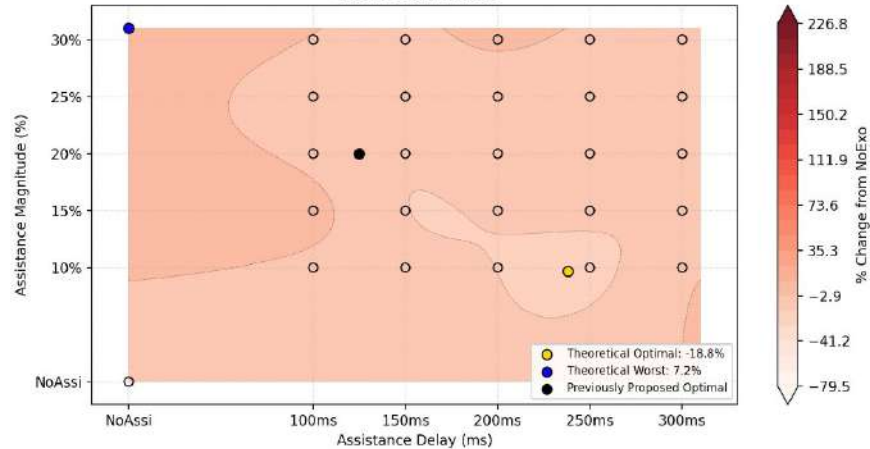
Tibialis Anterior



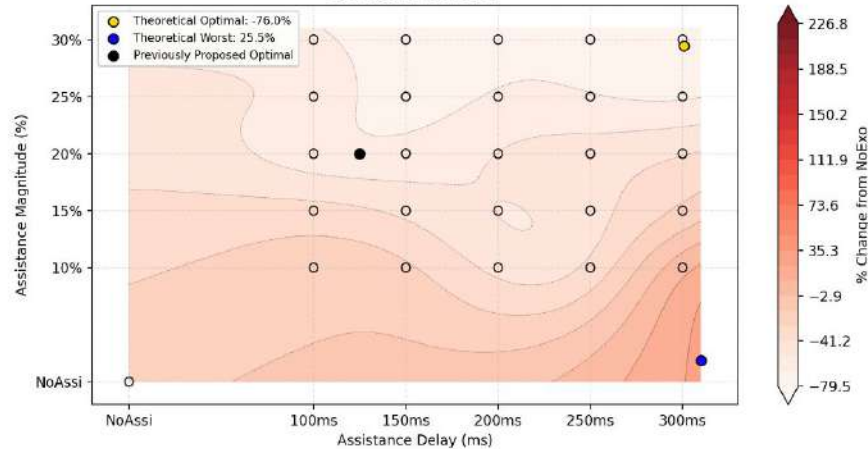
Bicep Femoris



Gastrocnemius

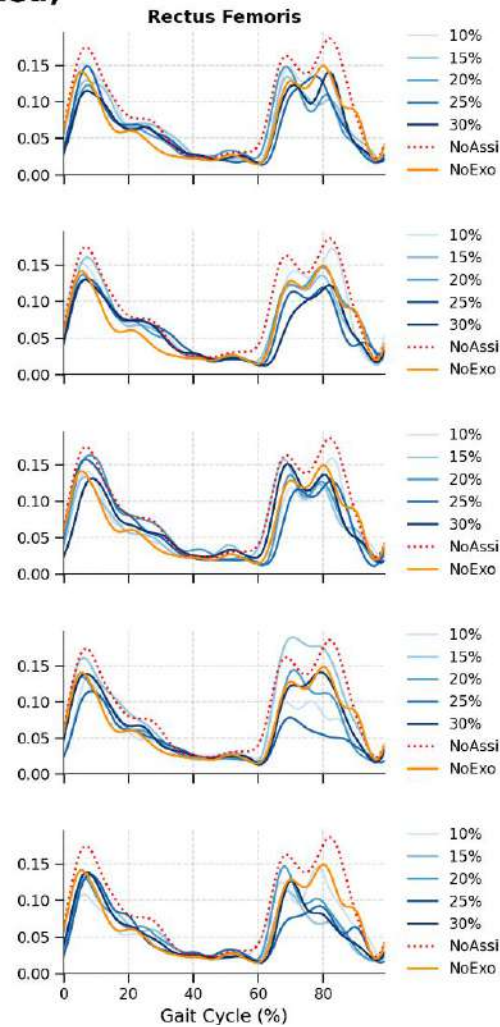
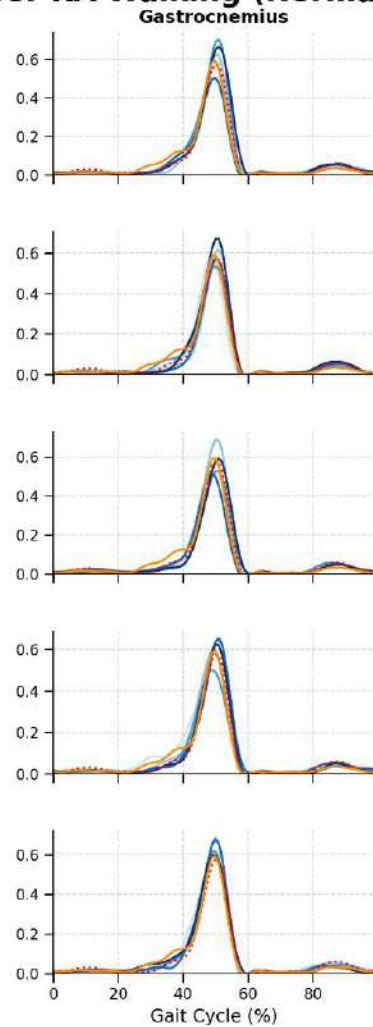
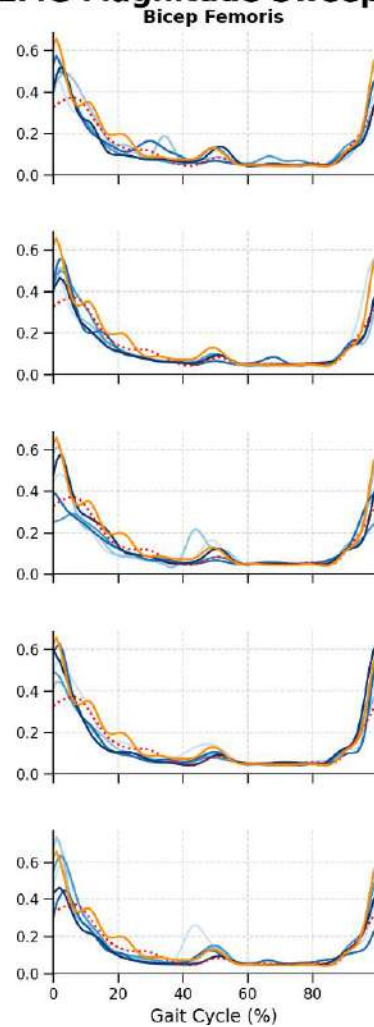
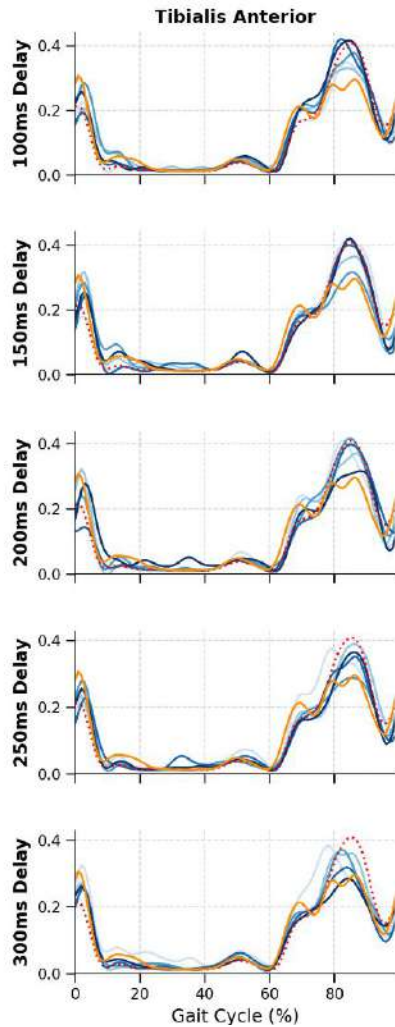


Rectus Femoris

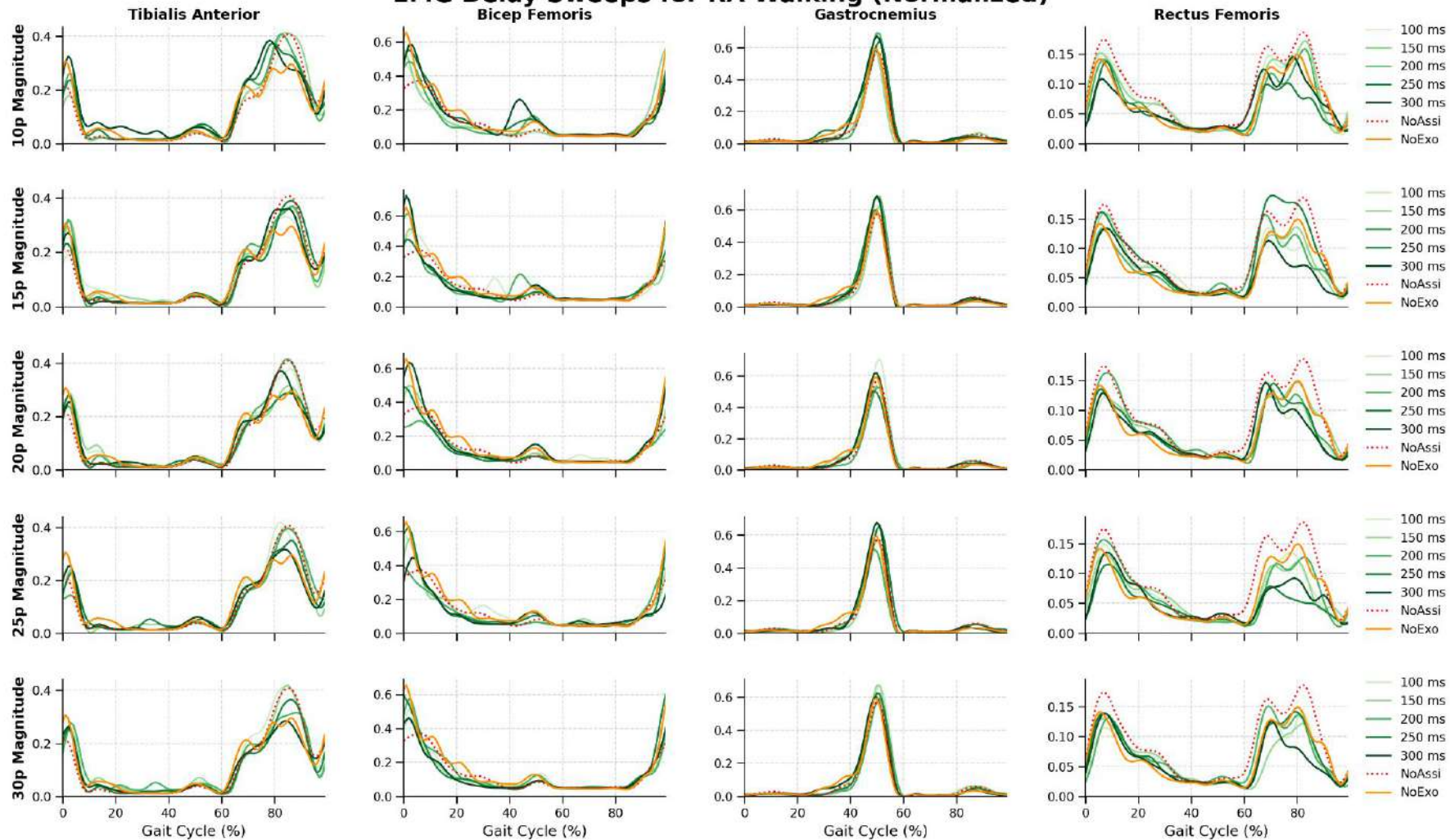


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

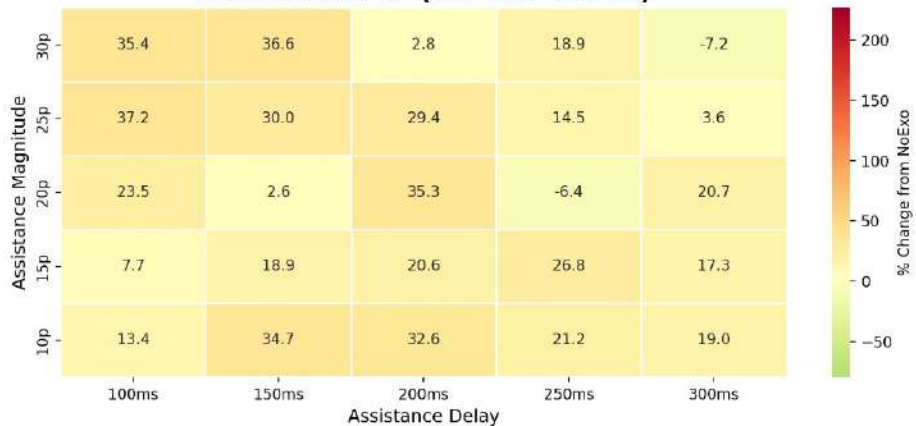


EMG Delay Sweeps for RA Walking (Normalized)

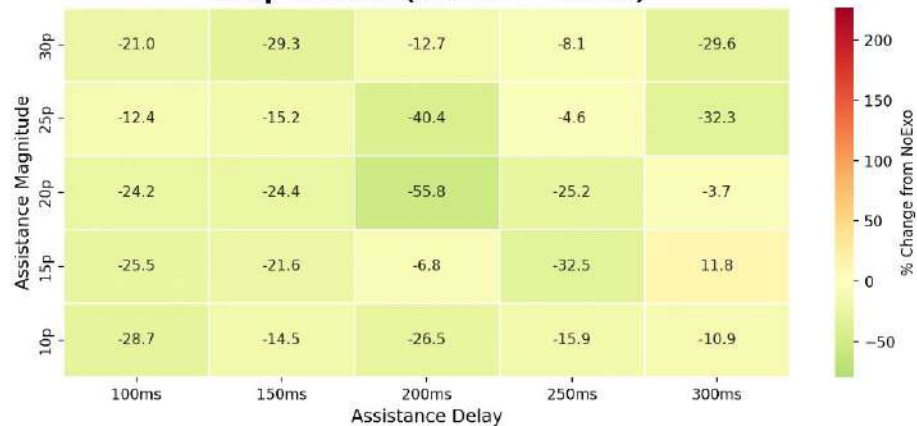


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

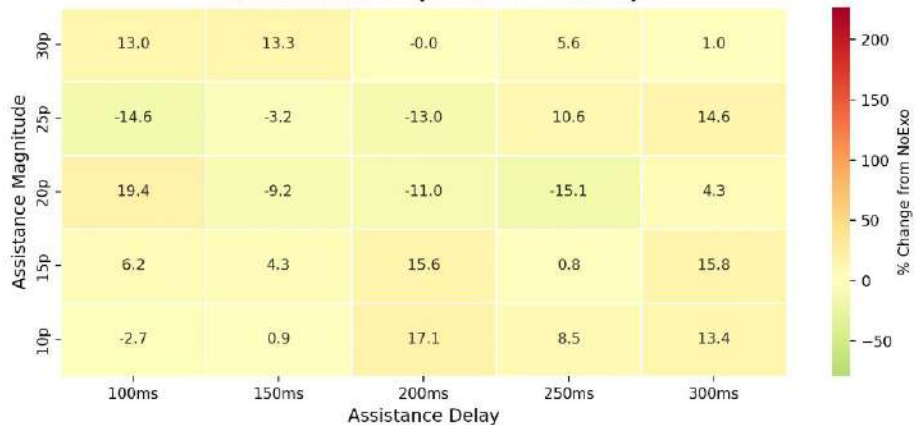
Tibialis Anterior (NoAssi = 32.8%)



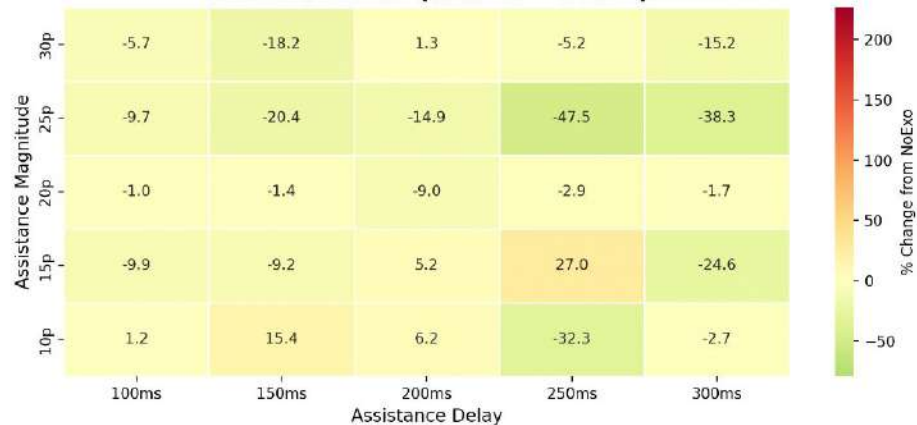
Bicep Femoris (NoAssi = -43.1%)



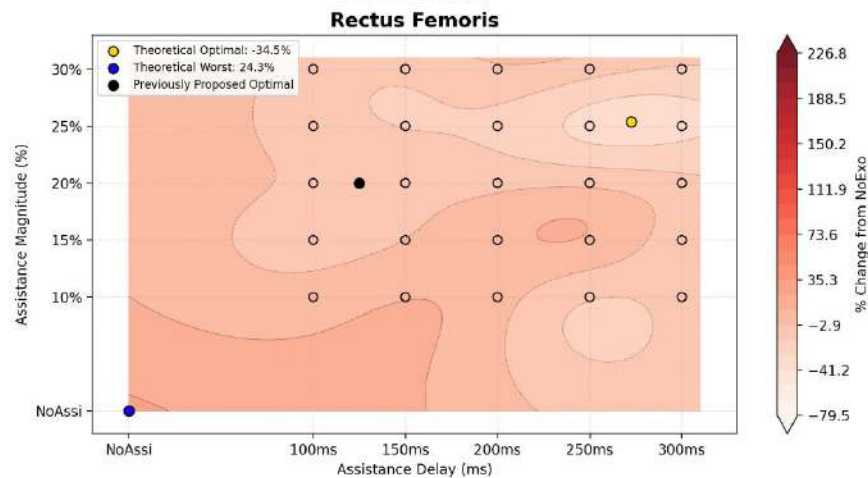
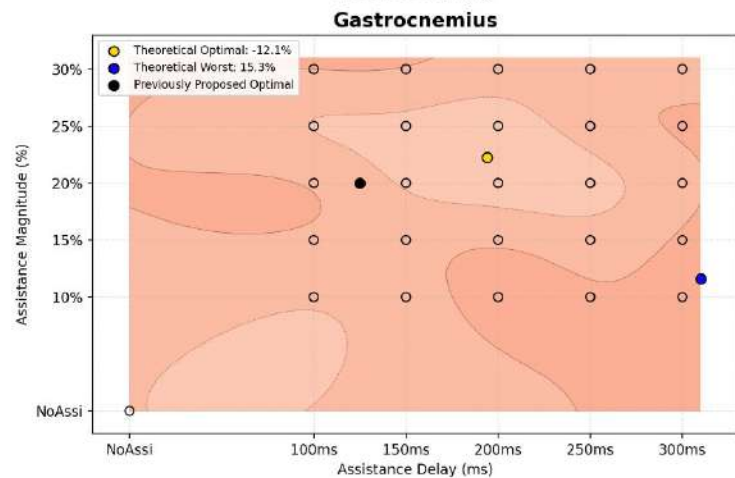
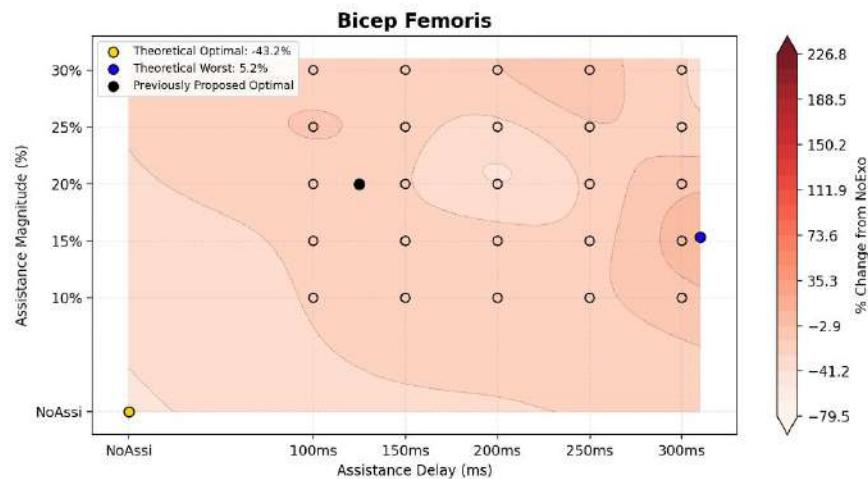
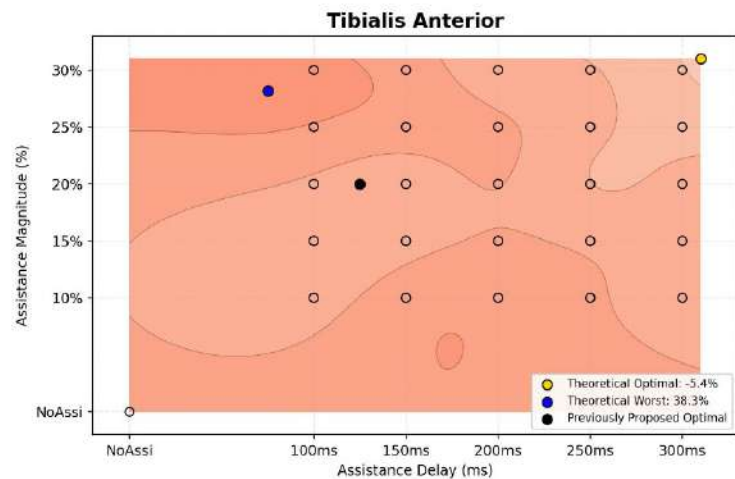
Gastrocnemius (NoAssi = -2.7%)



Rectus Femoris (NoAssi = 24.8%)

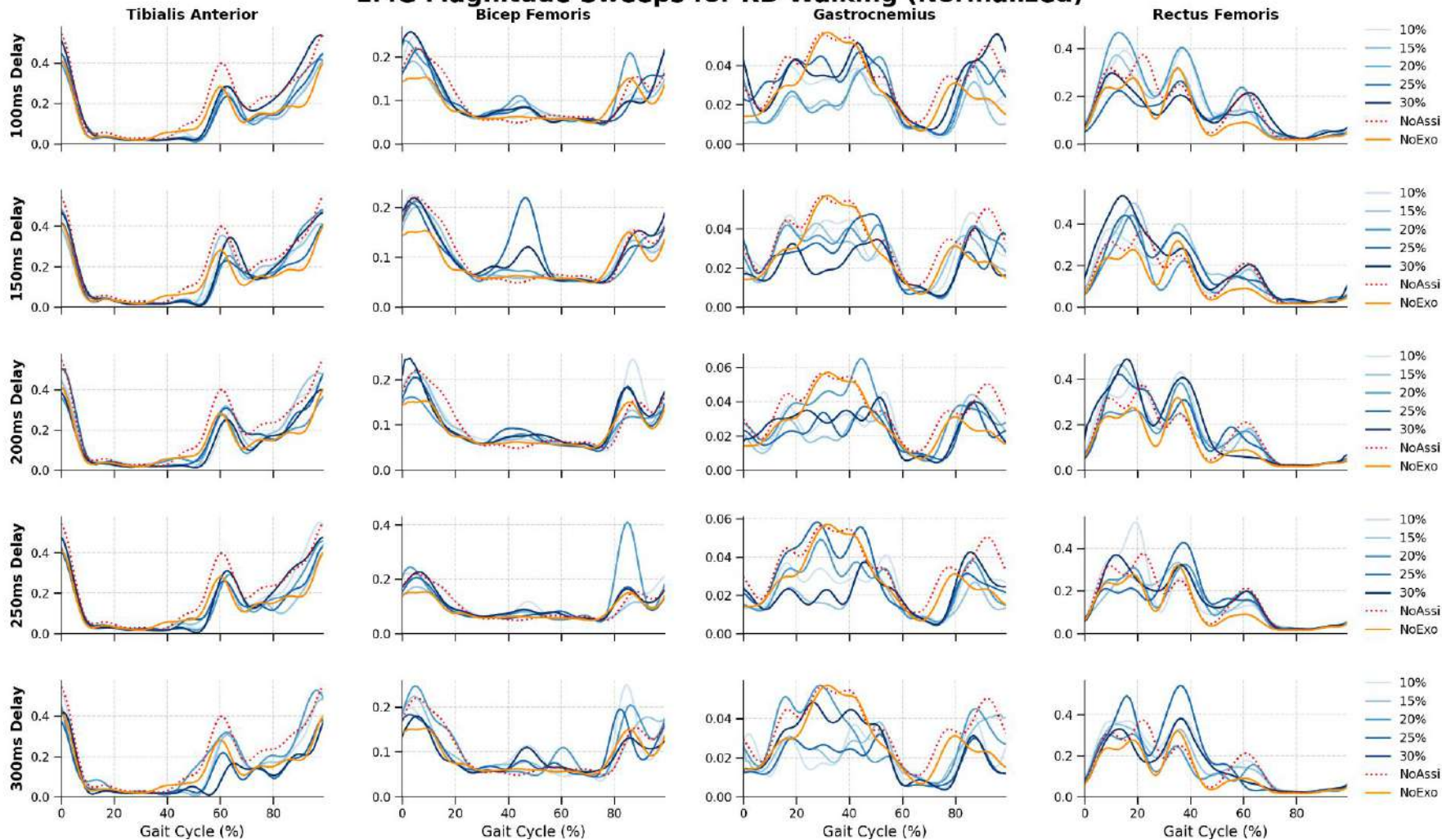


Peak EMG Activation Contour Map - RA Walking

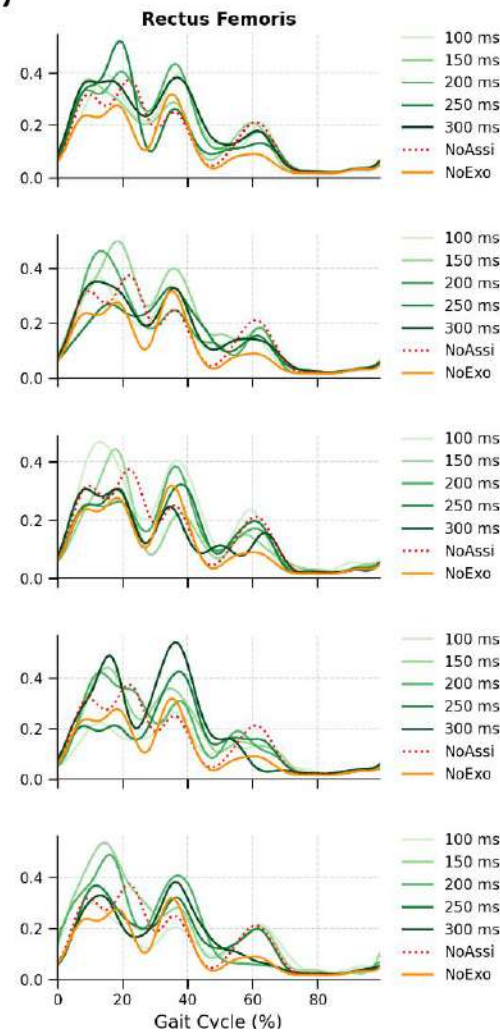
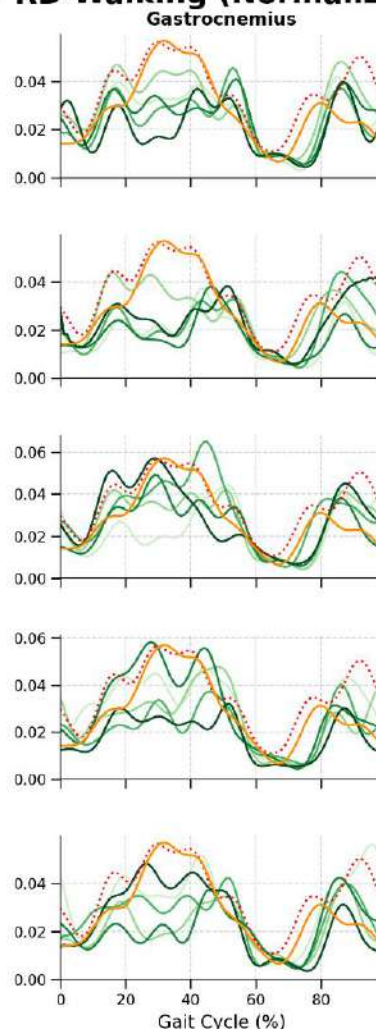
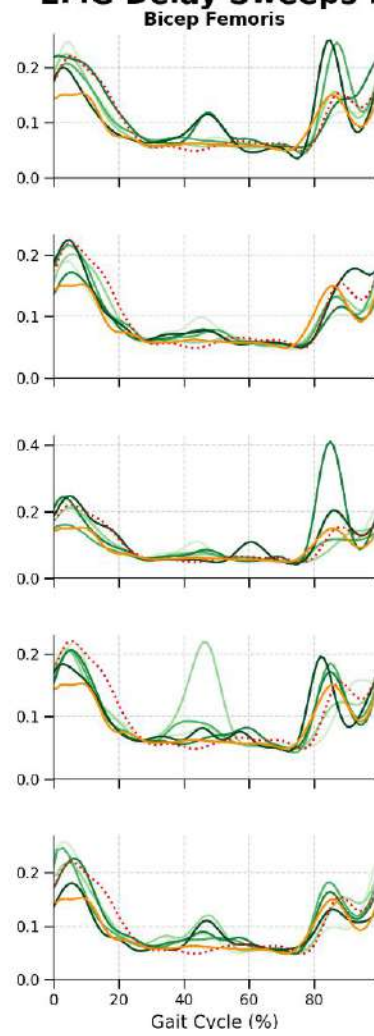
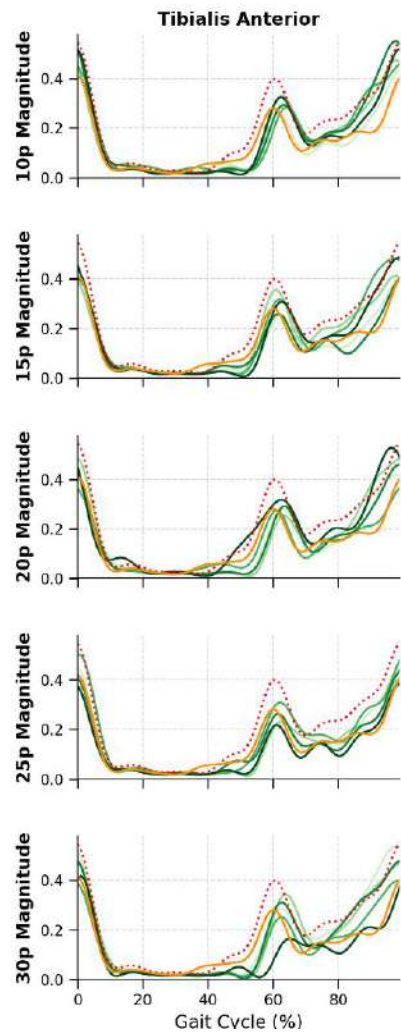


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

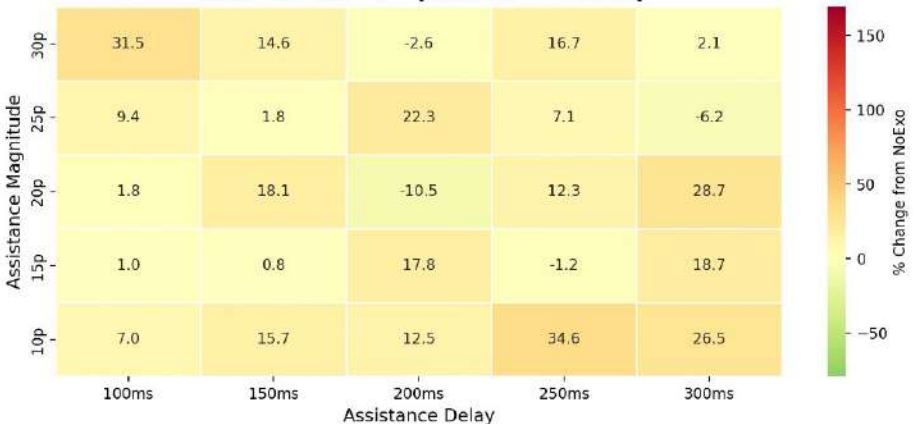


EMG Delay Sweeps for RD Walking (Normalized)



Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

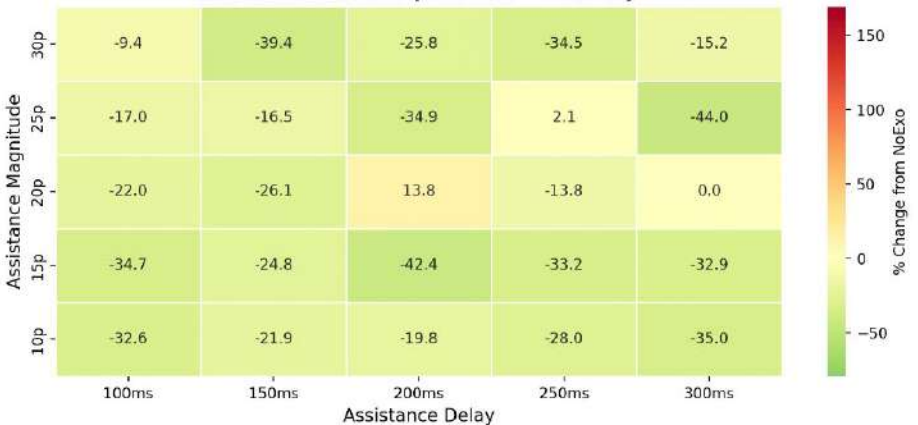
Tibialis Anterior (NoAssi = 34.4%)



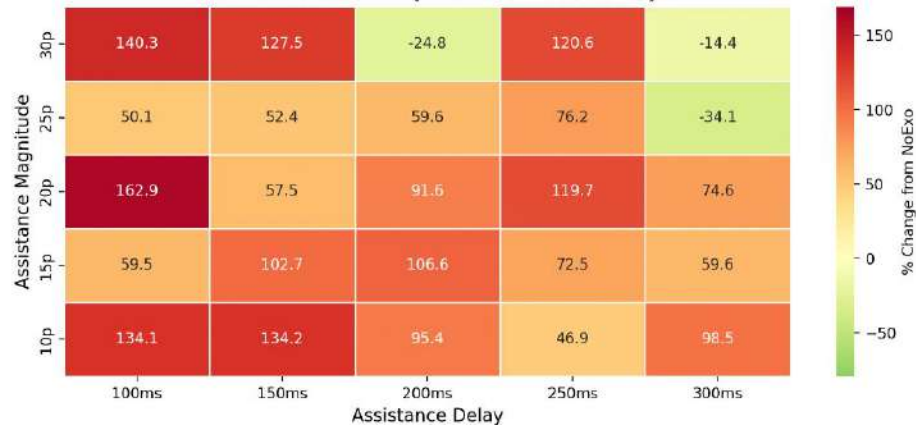
Bicep Femoris (NoAssi = 44.1%)



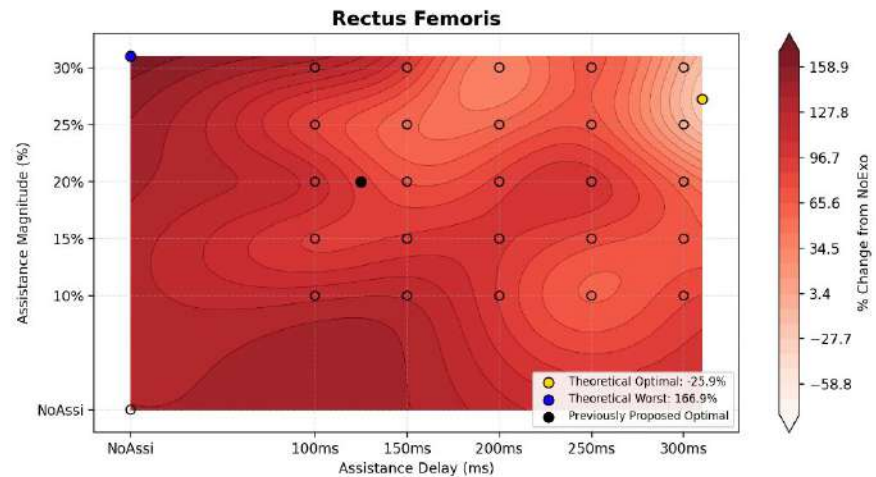
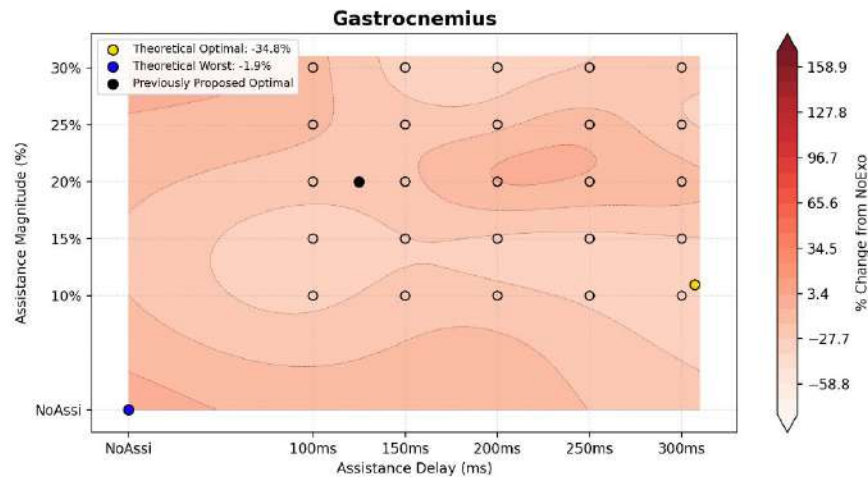
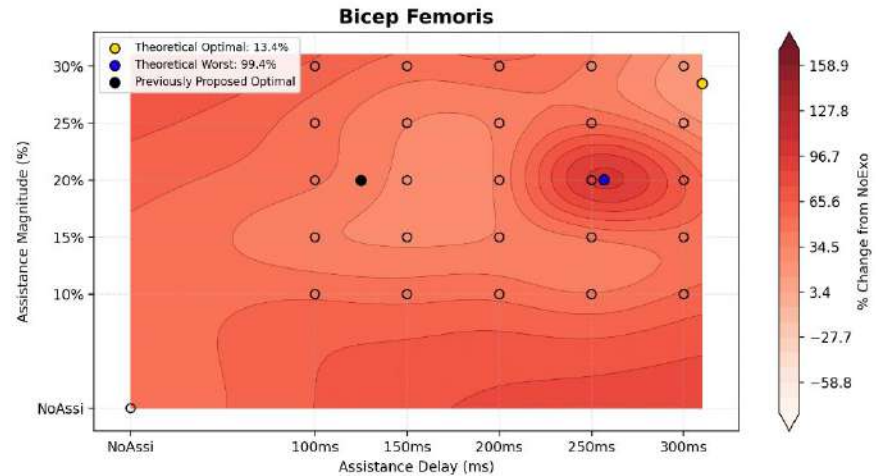
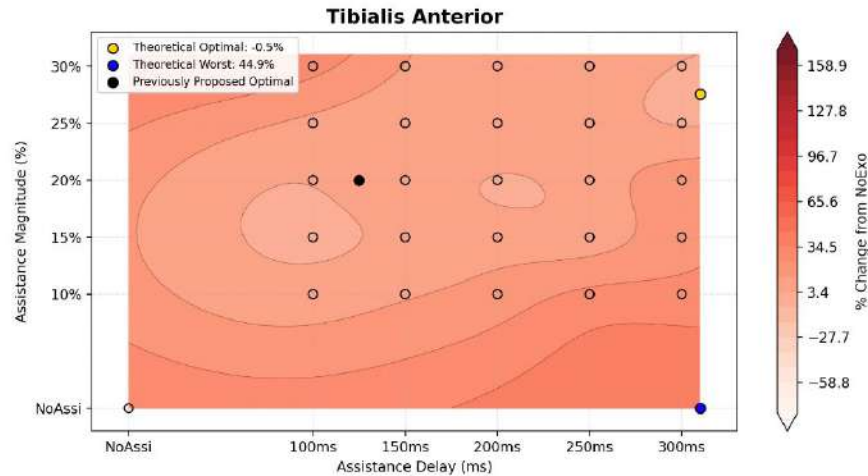
Gastrocnemius (NoAssi = -0.9%)



Rectus Femoris (NoAssi = 136.9%)



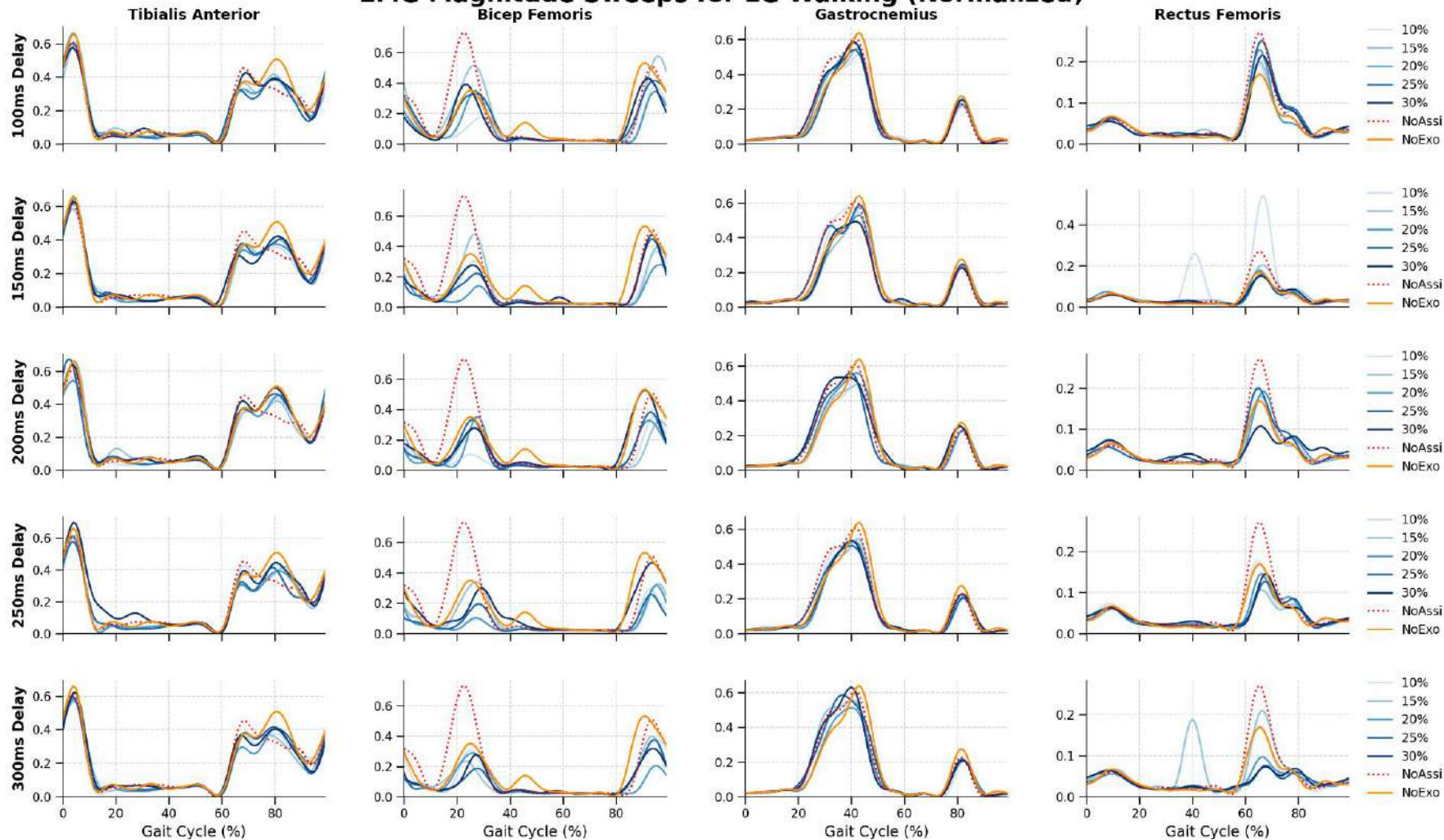
Peak EMG Activation Contour Map - RD Walking



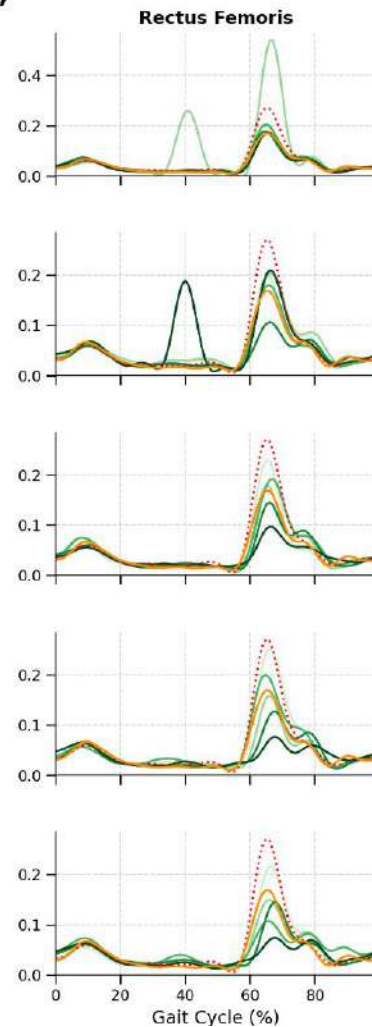
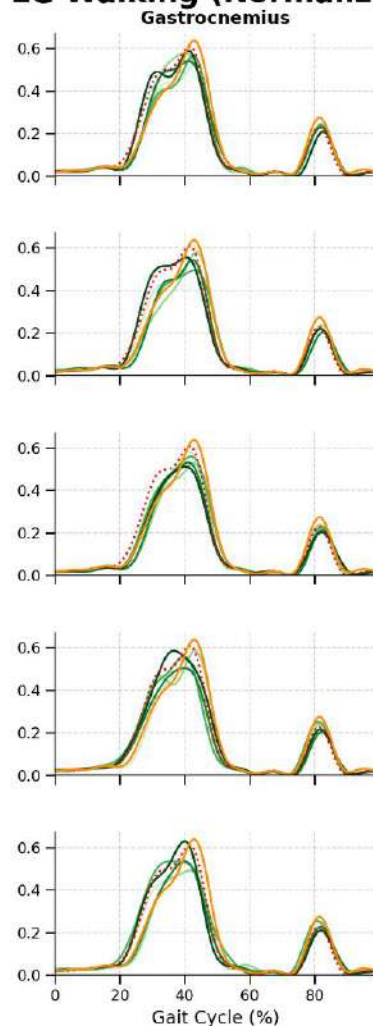
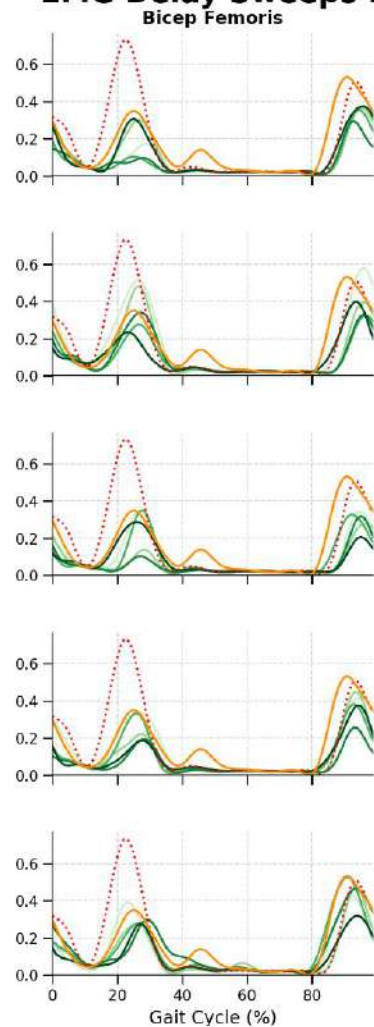
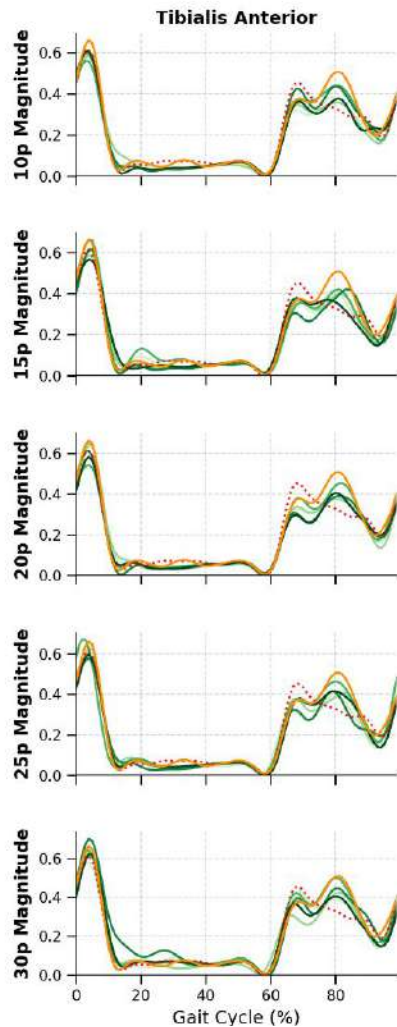
Subject: AB02_Rajiv

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

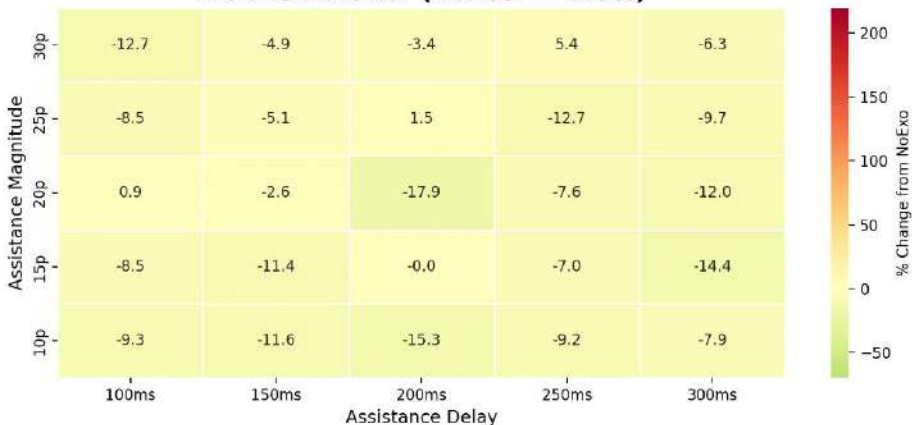


EMG Delay Sweeps for LG Walking (Normalized)

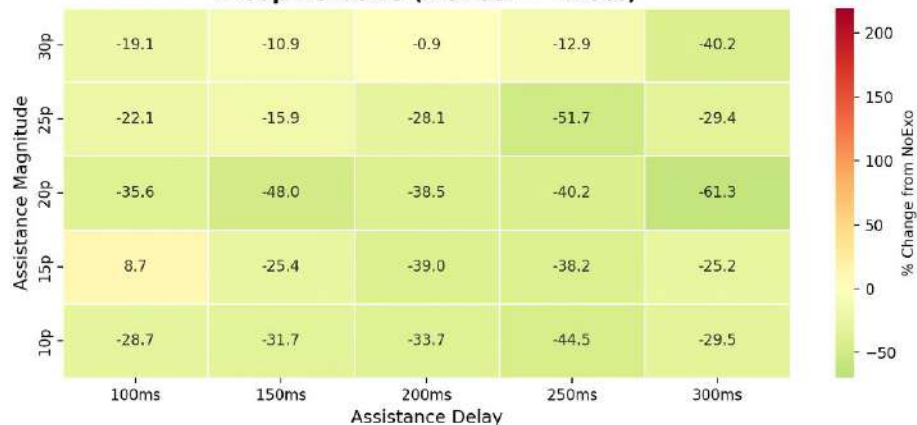


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

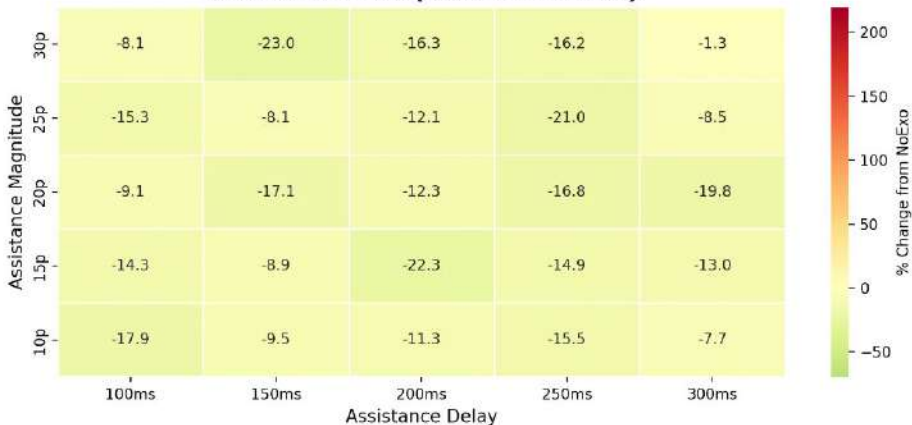
Tibialis Anterior (NoAssi = -7.5%)



Bicep Femoris (NoAssi = -4.8%)



Gastrocnemius (NoAssi = -4.7%)

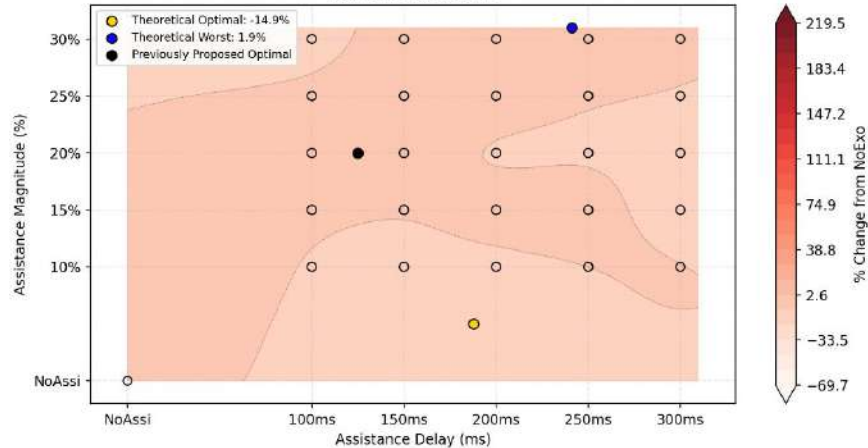


Rectus Femoris (NoAssi = 60.3%)

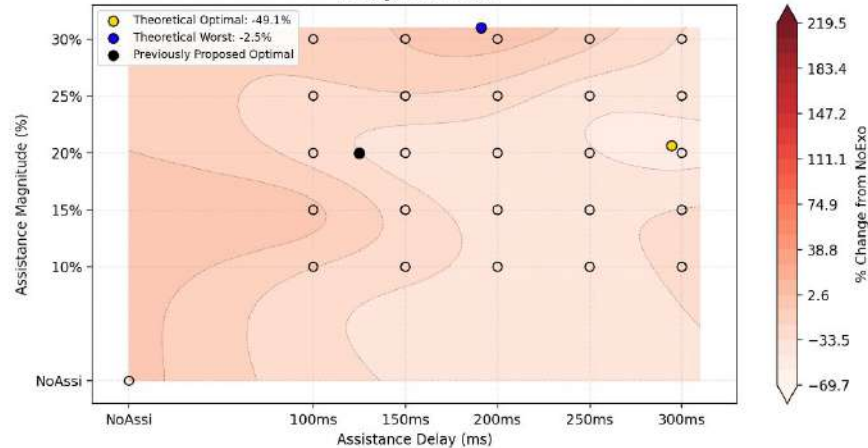


Peak EMG Activation Contour Map - LG Walking

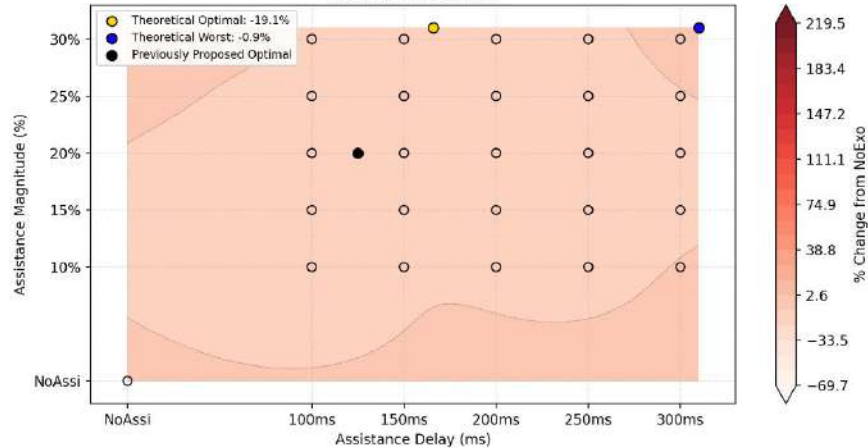
Tibialis Anterior



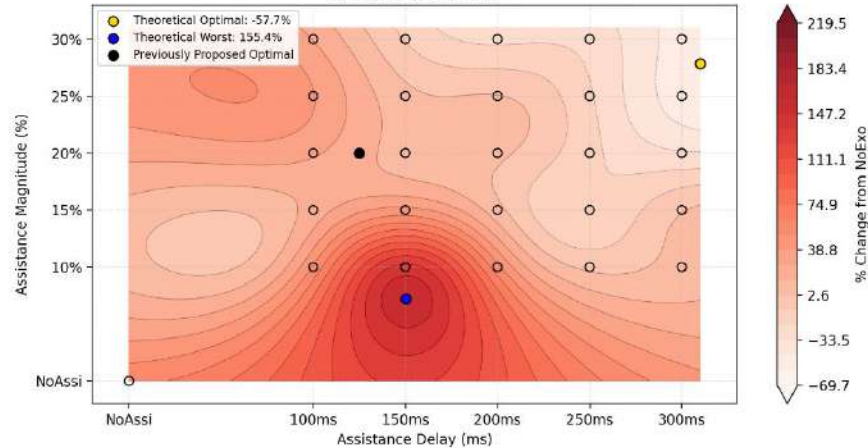
Bicep Femoris



Gastrocnemius

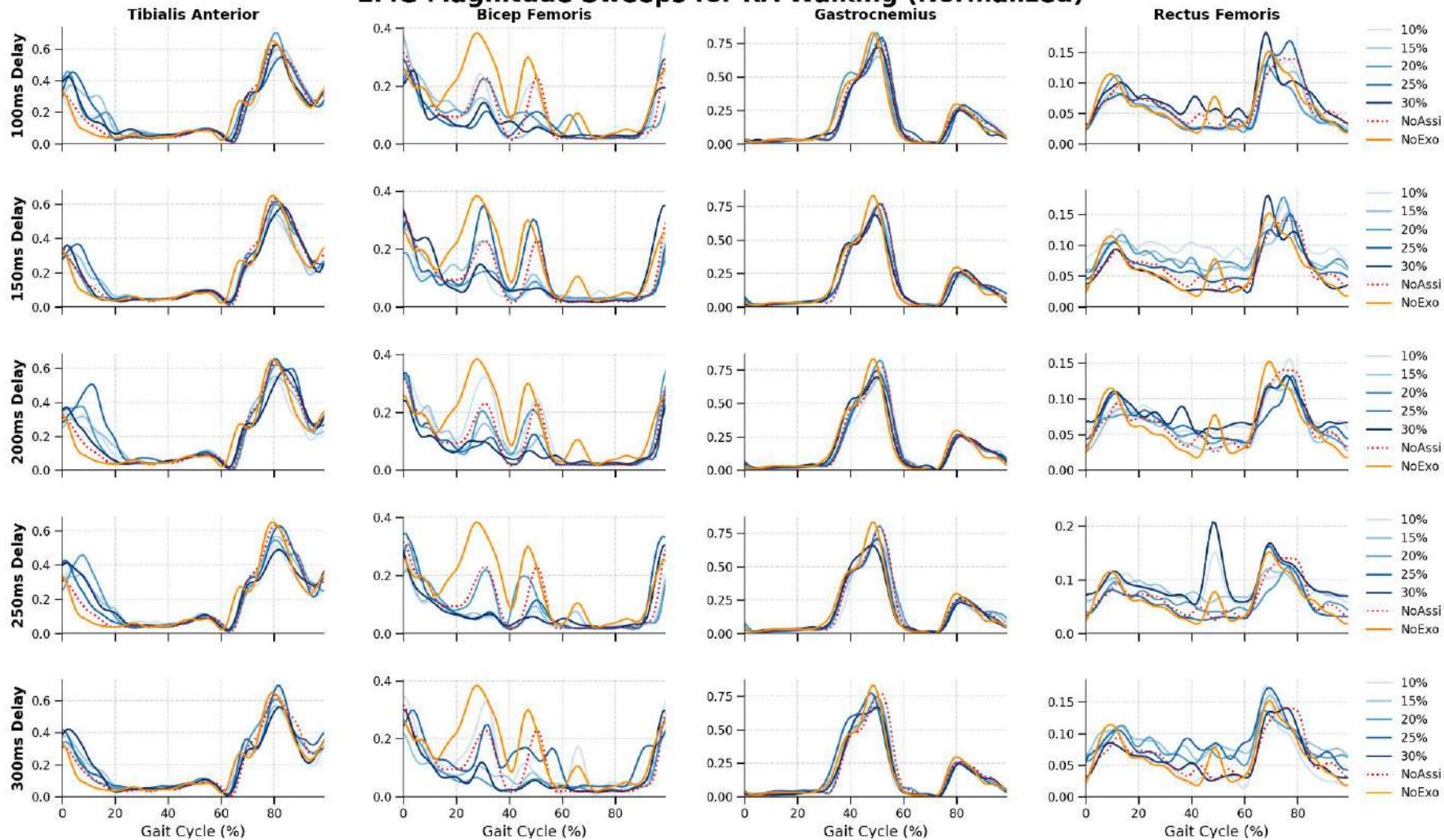


Rectus Femoris

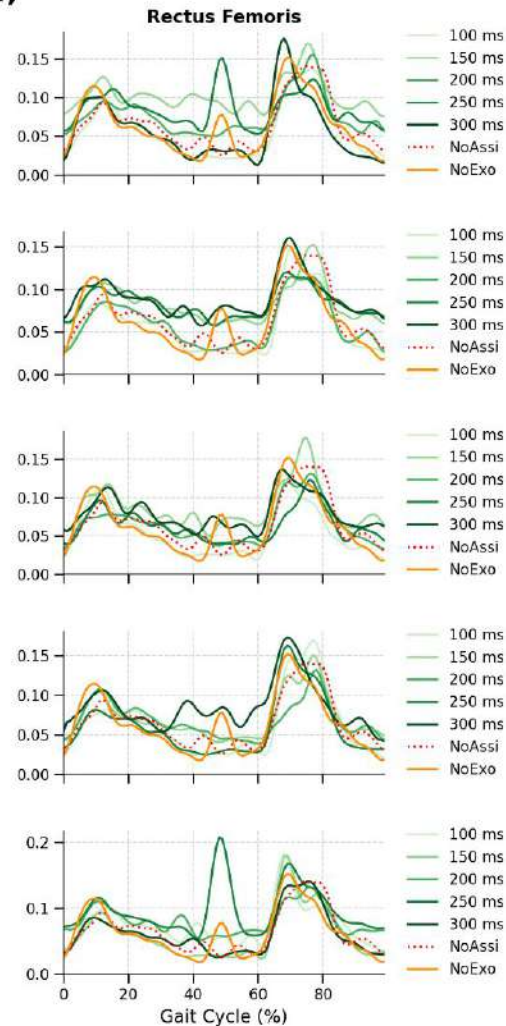
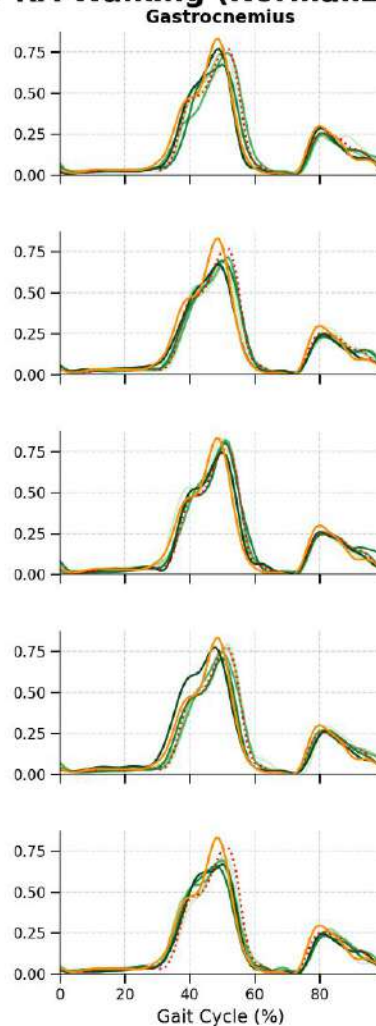
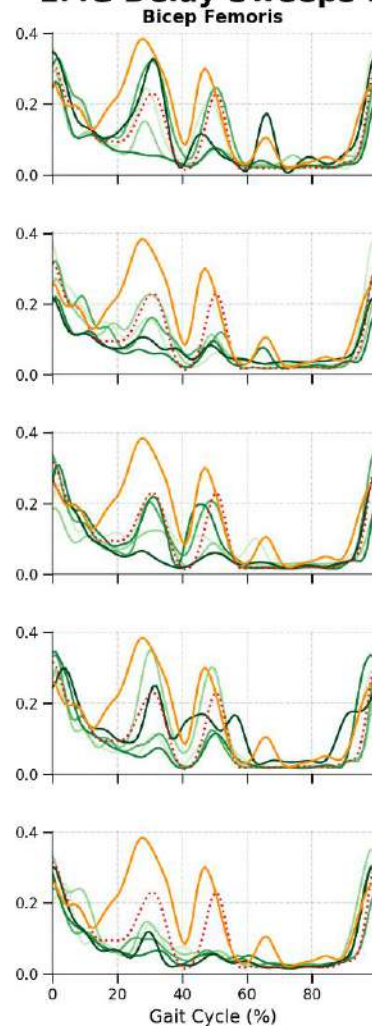
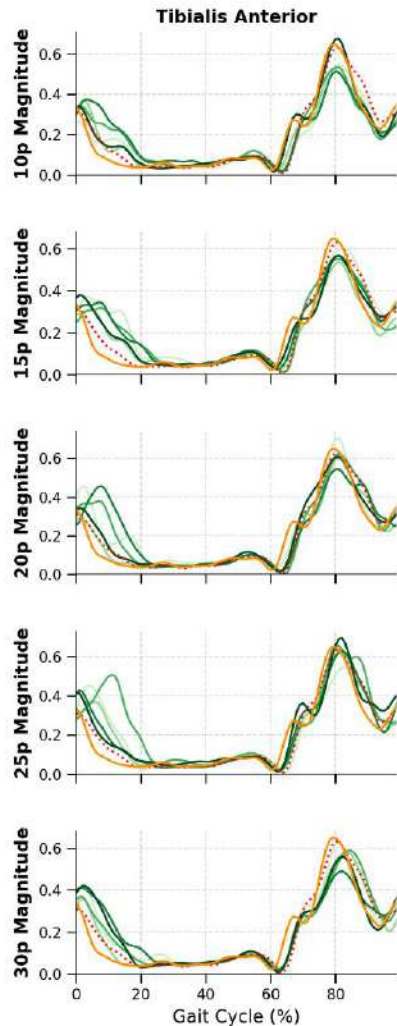


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

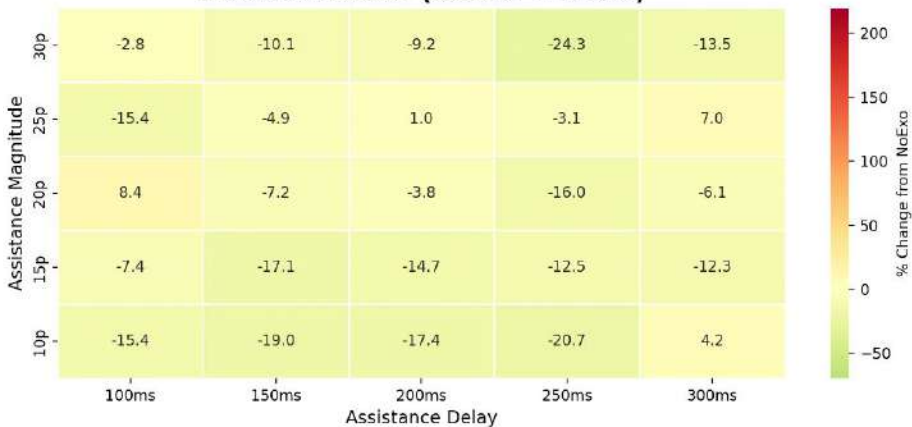


EMG Delay Sweeps for RA Walking (Normalized)

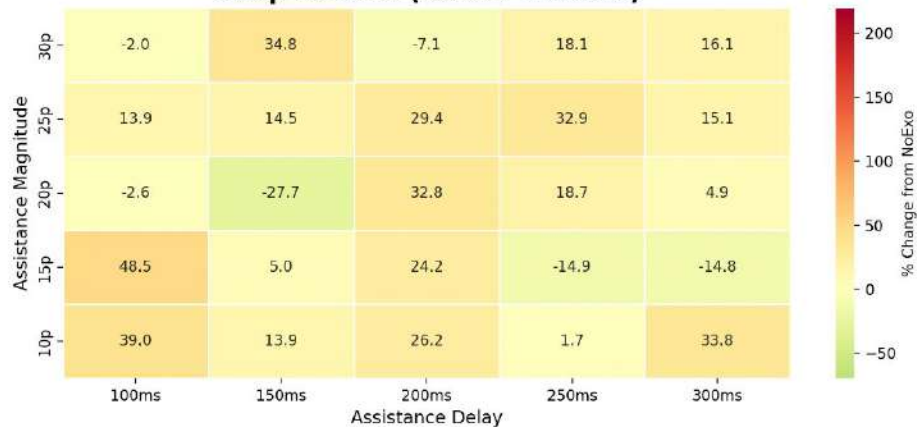


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

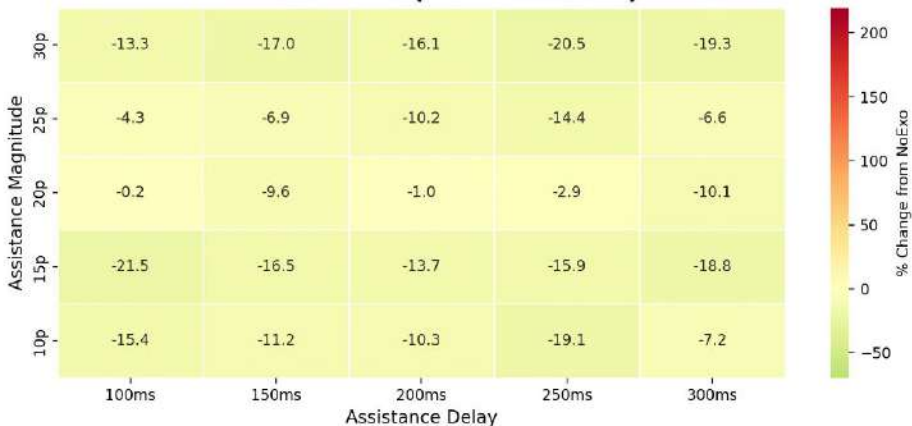
Tibialis Anterior (NoAssi = -2.6%)



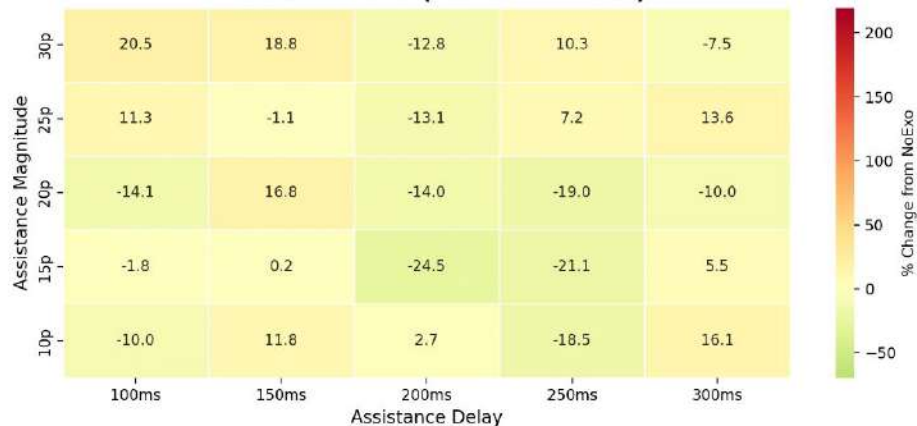
Bicep Femoris (NoAssi = 22.1%)



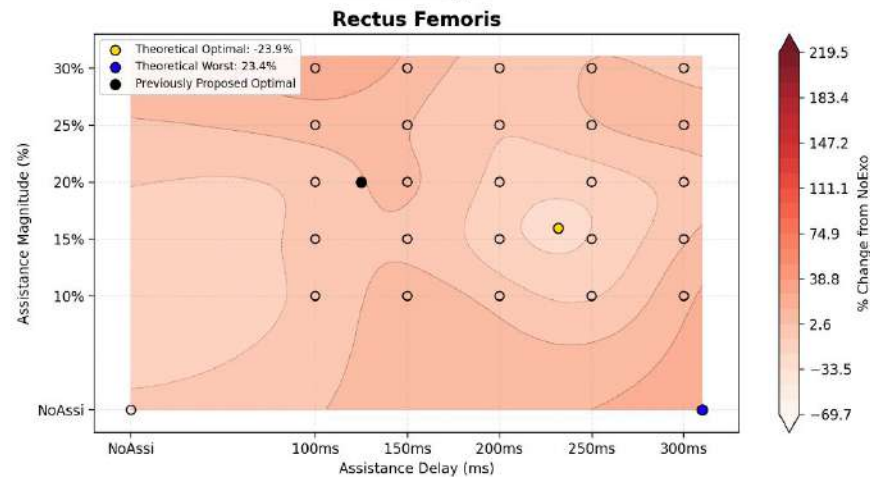
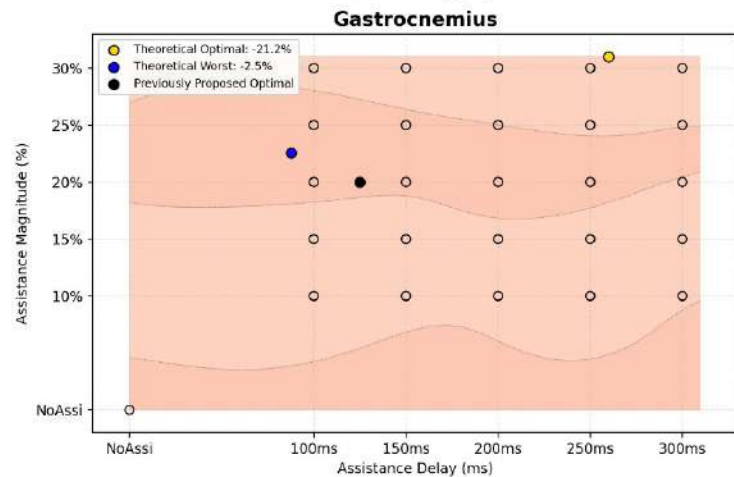
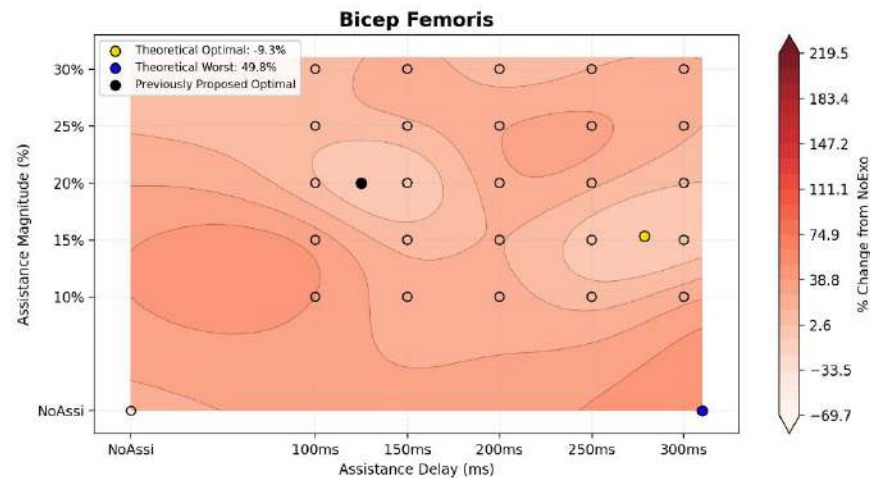
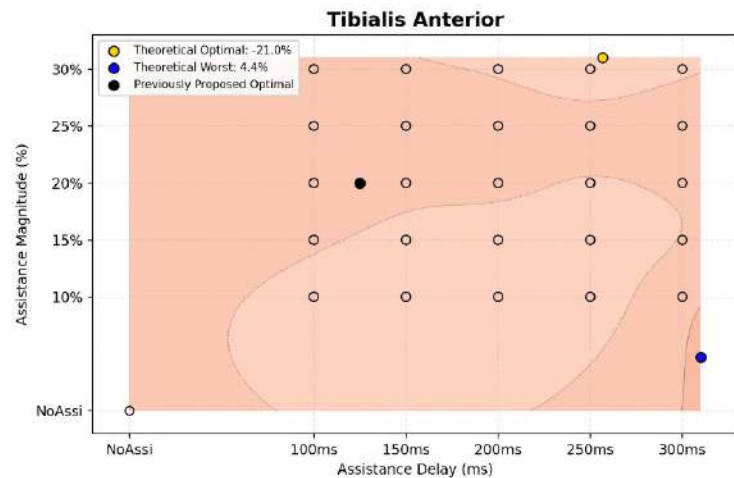
Gastrocnemius (NoAssi = -6.9%)



Rectus Femoris (NoAssi = -7.7%)

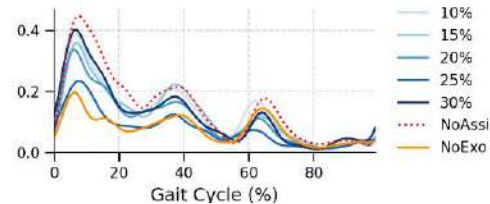
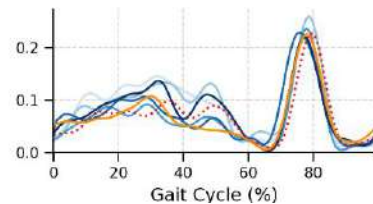
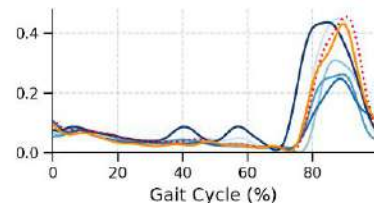
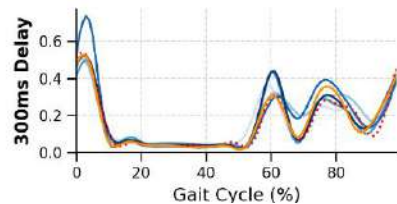
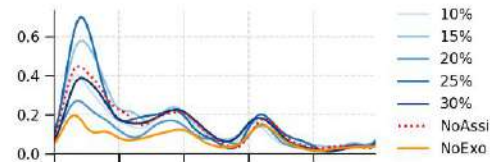
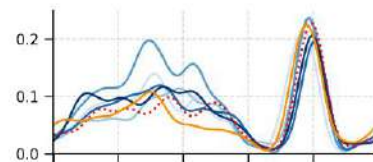
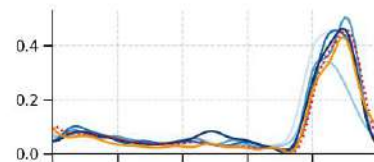
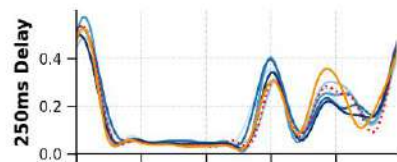
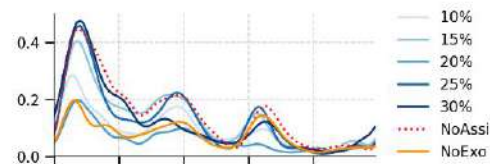
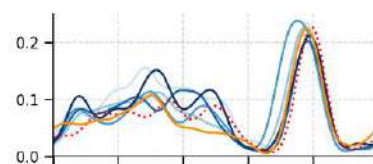
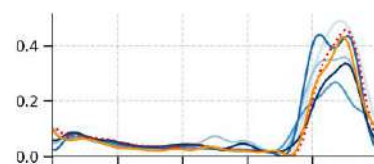
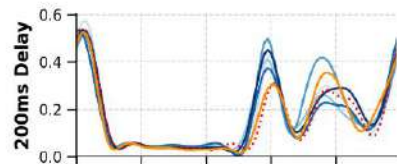
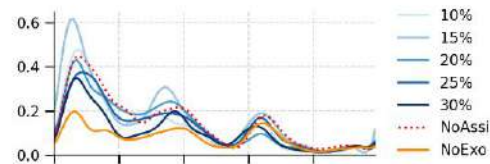
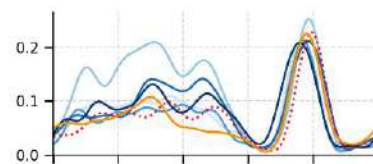
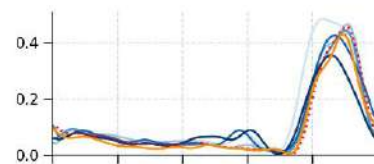
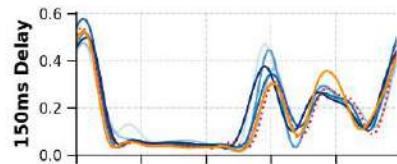
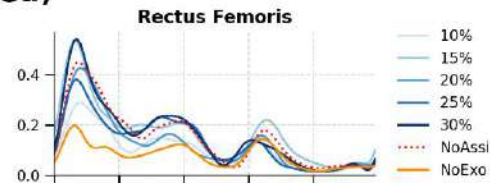
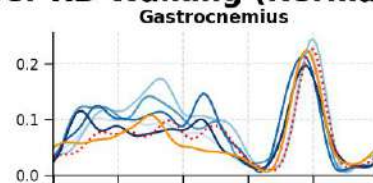
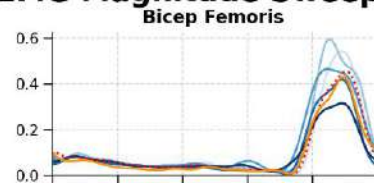
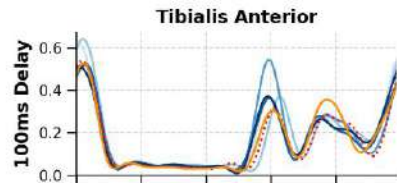


Peak EMG Activation Contour Map - RA Walking

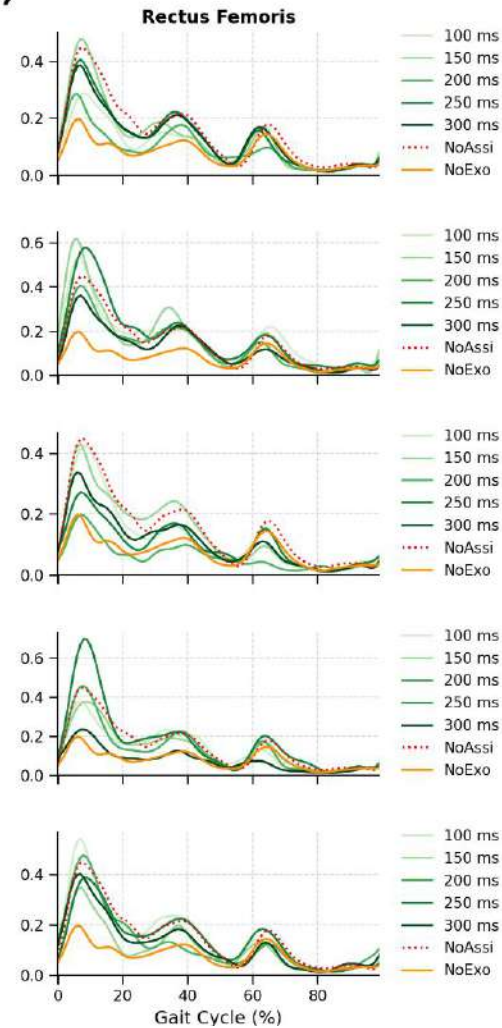
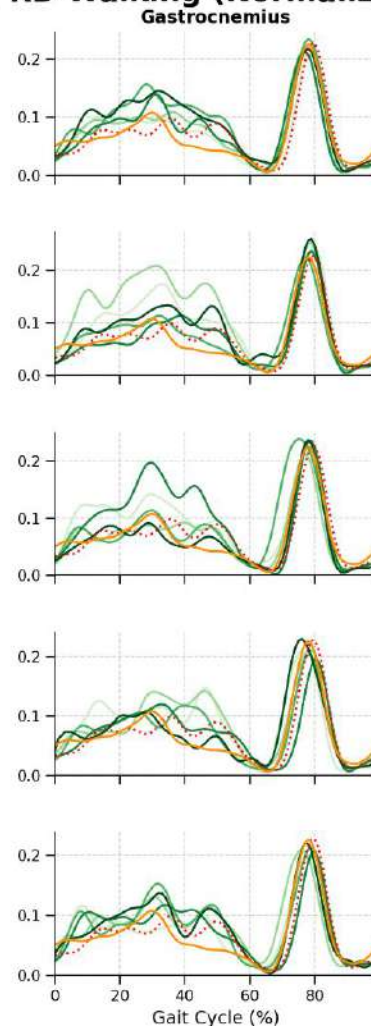
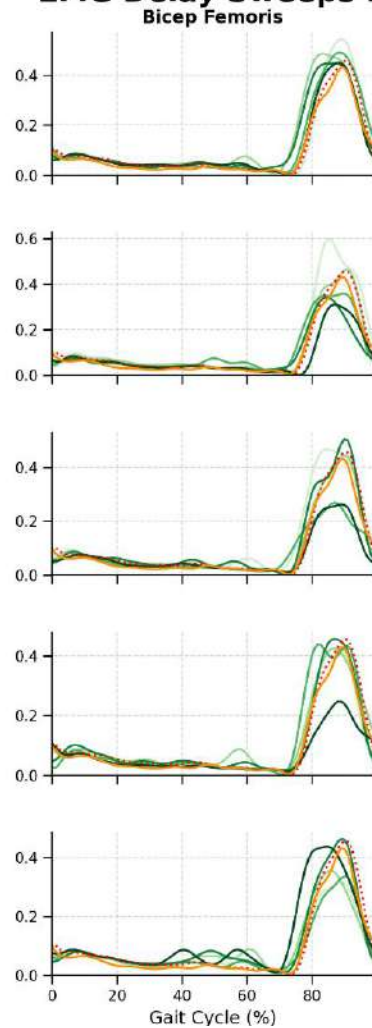
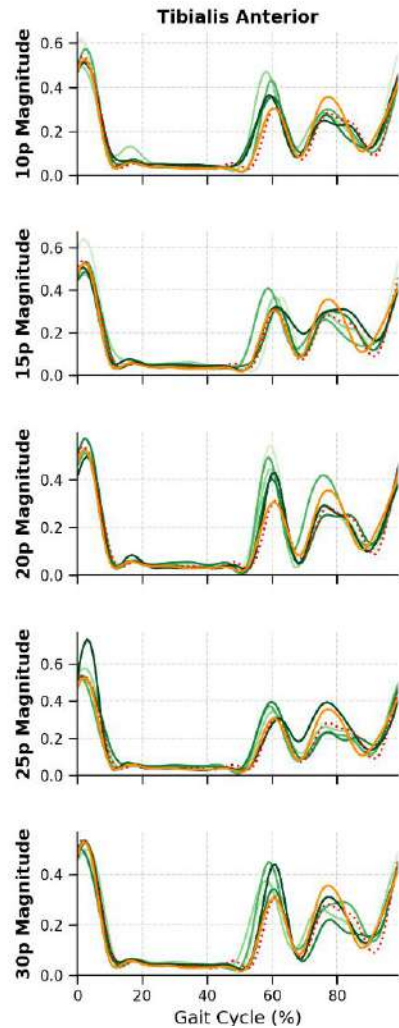


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

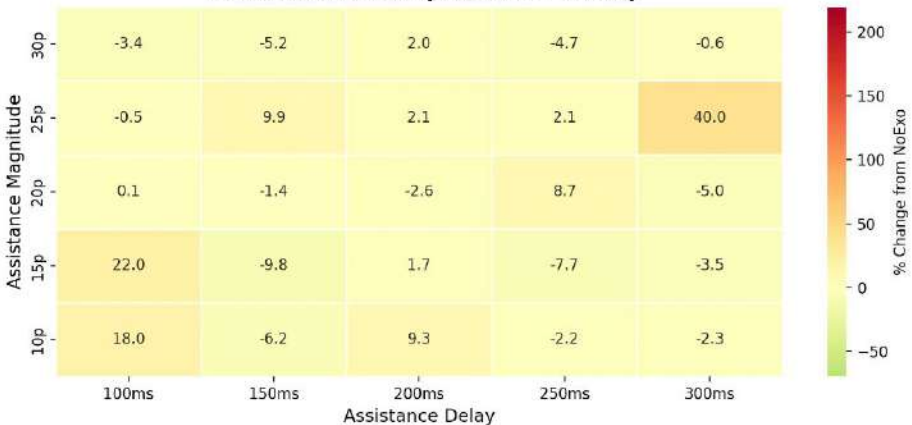


EMG Delay Sweeps for RD Walking (Normalized)

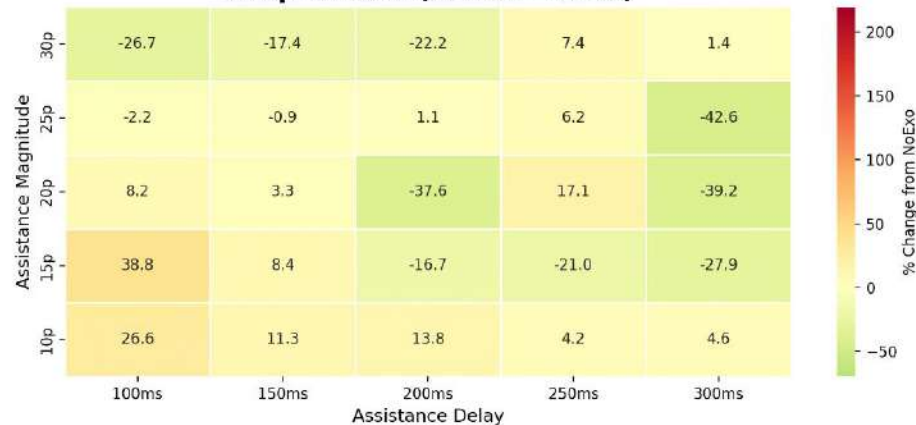


Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

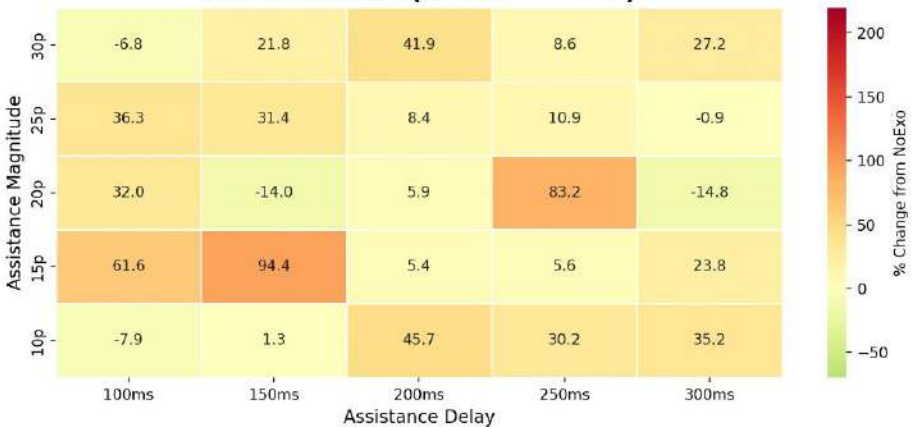
Tibialis Anterior (NoAssi = 2.8%)



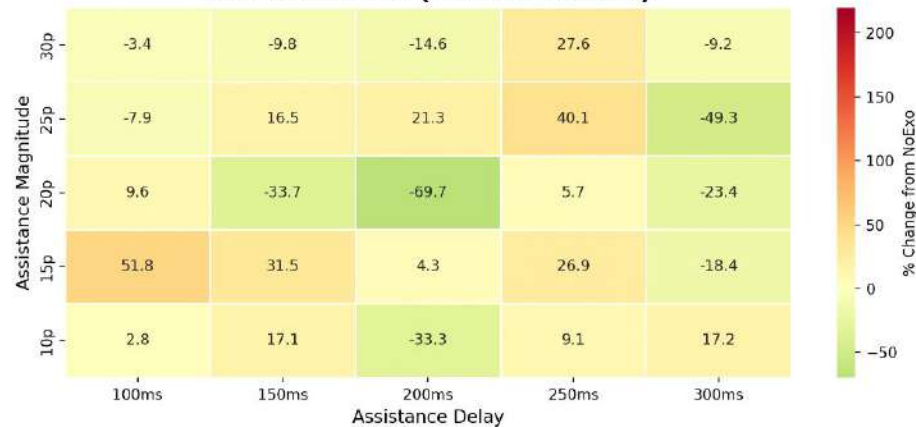
Bicep Femoris (NoAssi = 6.1%)



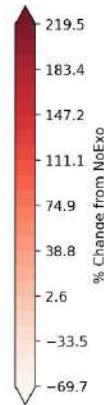
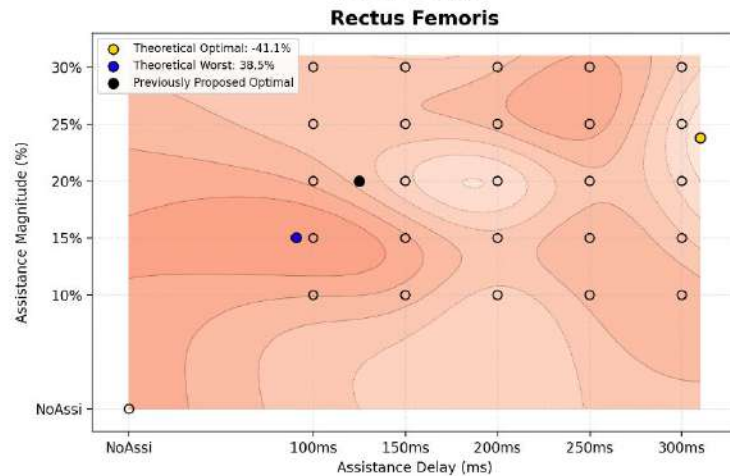
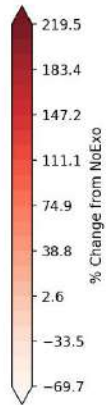
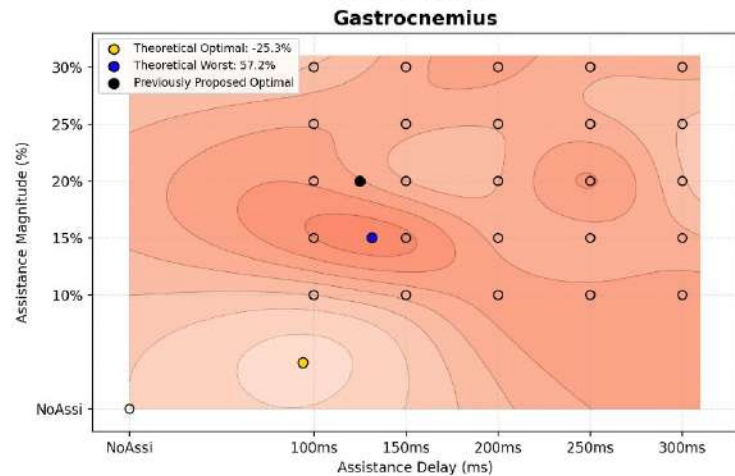
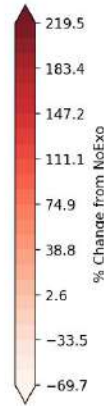
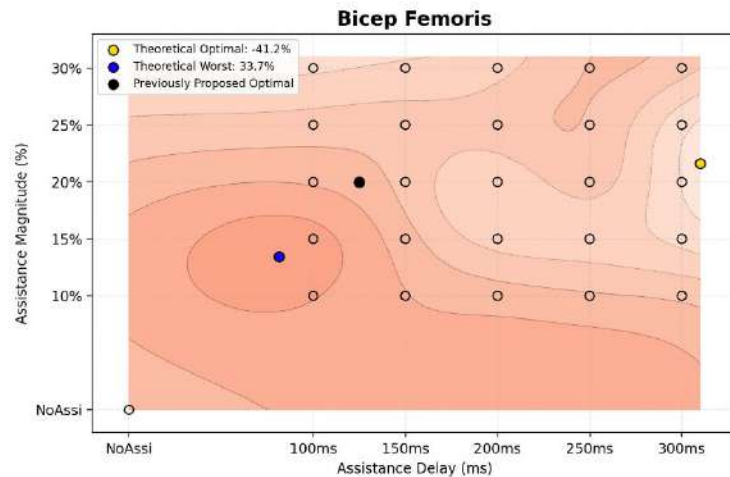
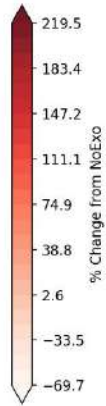
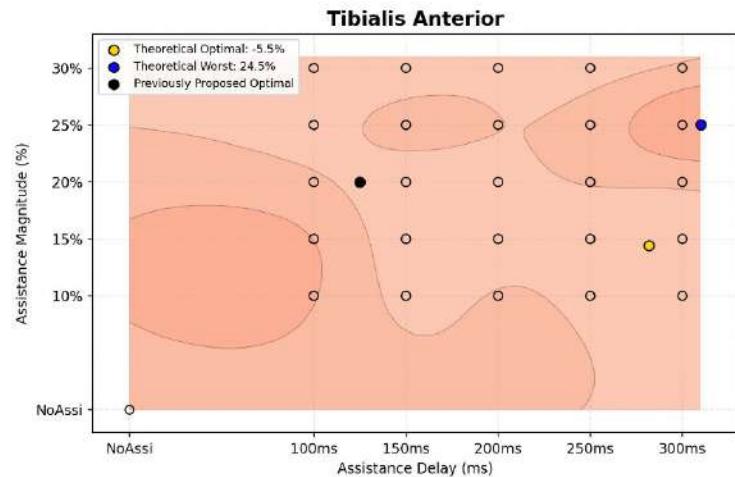
Gastrocnemius (NoAssi = -8.6%)



Rectus Femoris (NoAssi = 23.9%)



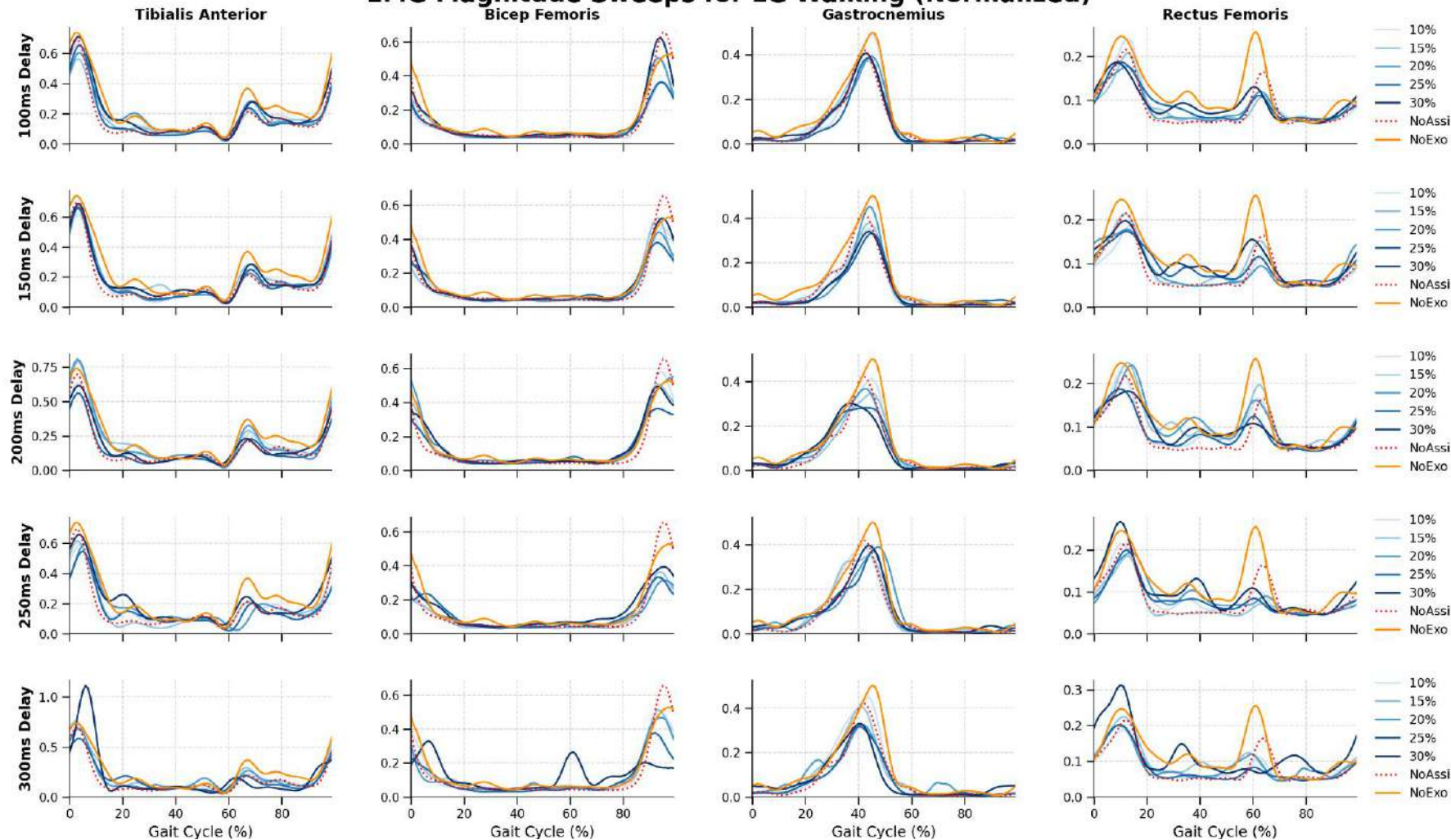
Peak EMG Activation Contour Map - RD Walking



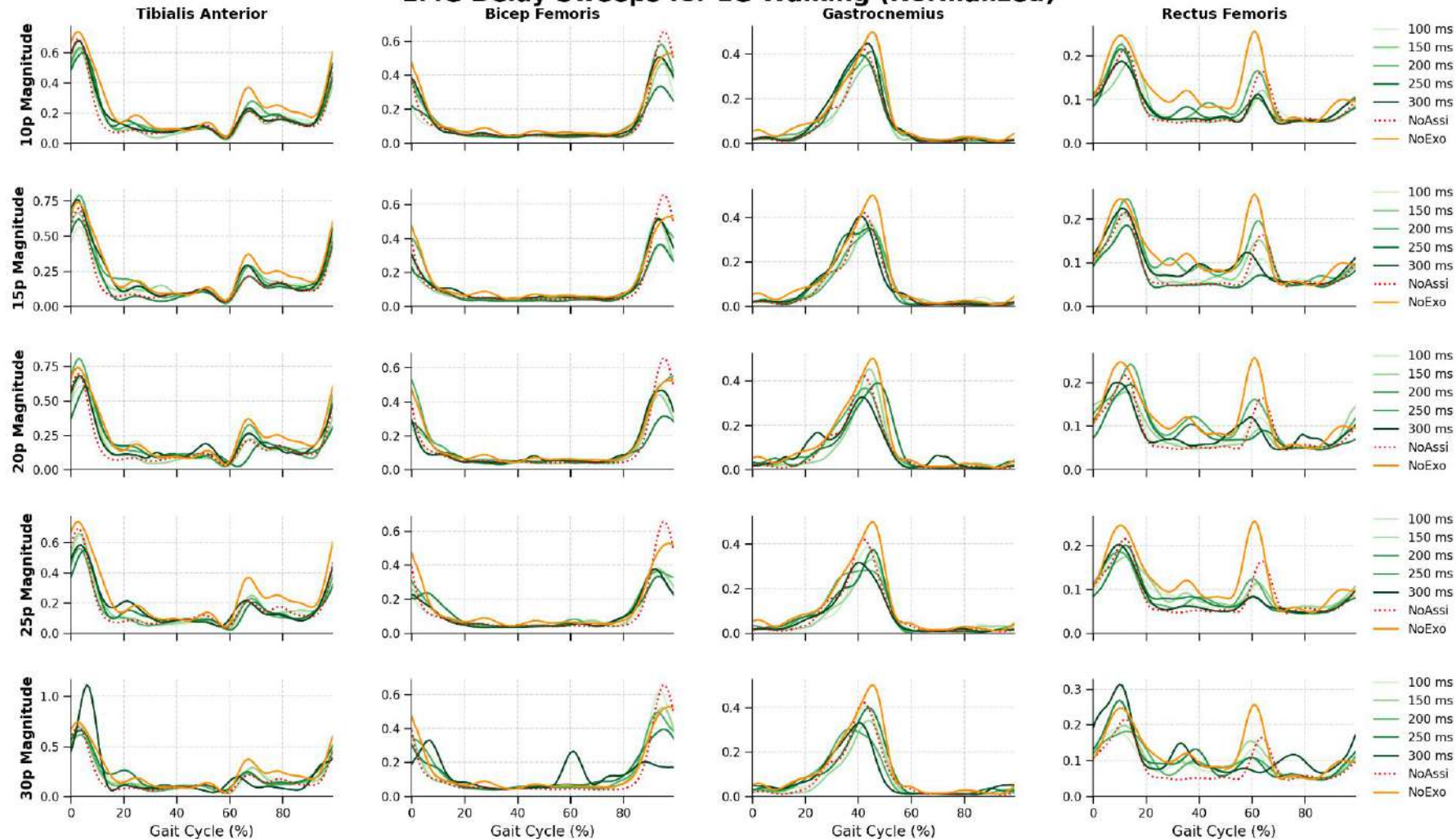
Subject: AB03_Amy

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

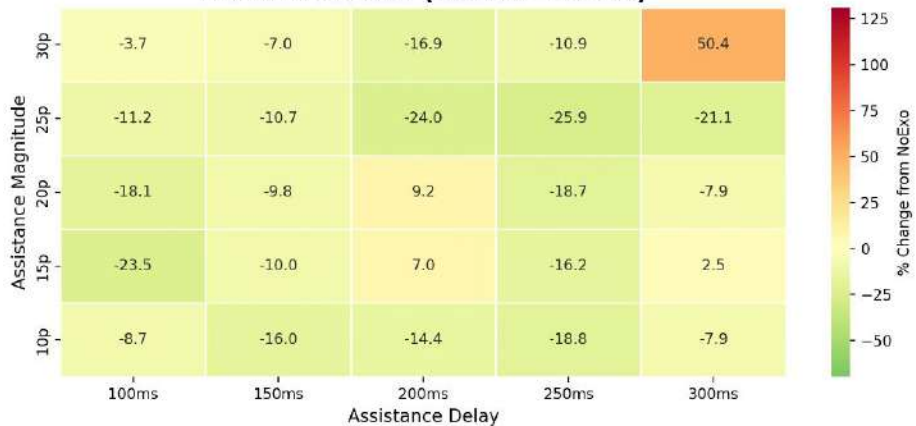


EMG Delay Sweeps for LG Walking (Normalized)

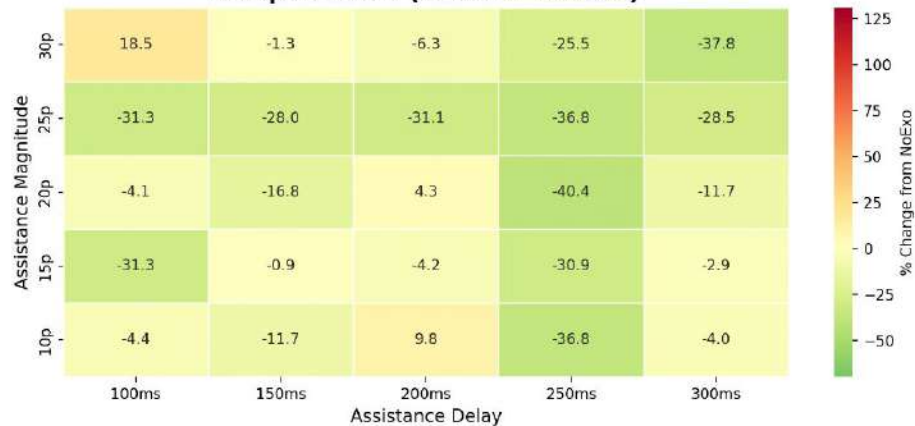


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

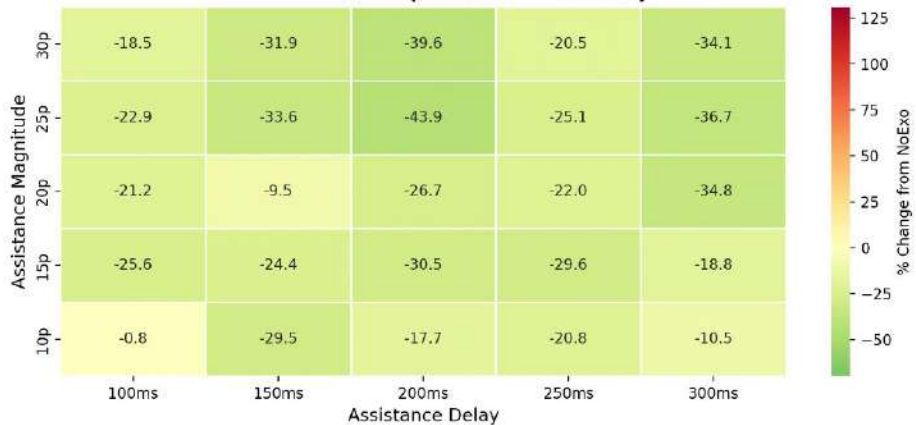
Tibialis Anterior (NoAssi = -5.6%)



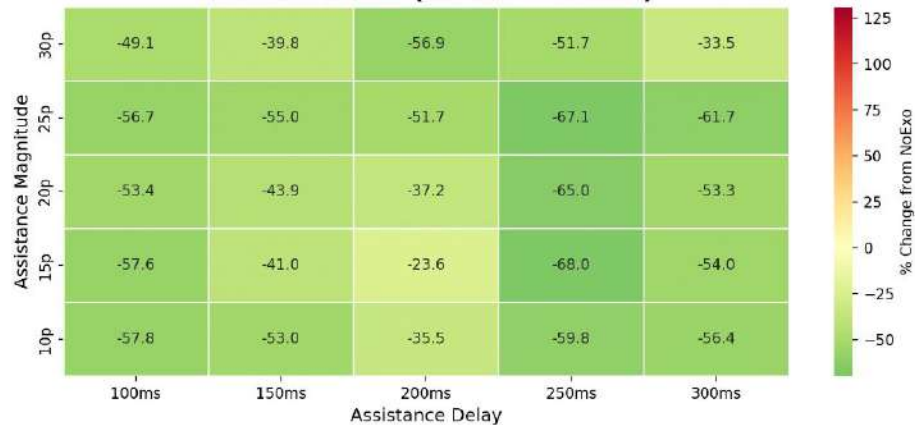
Bicep Femoris (NoAssi = 24.0%)



Gastrocnemius (NoAssi = -15.4%)

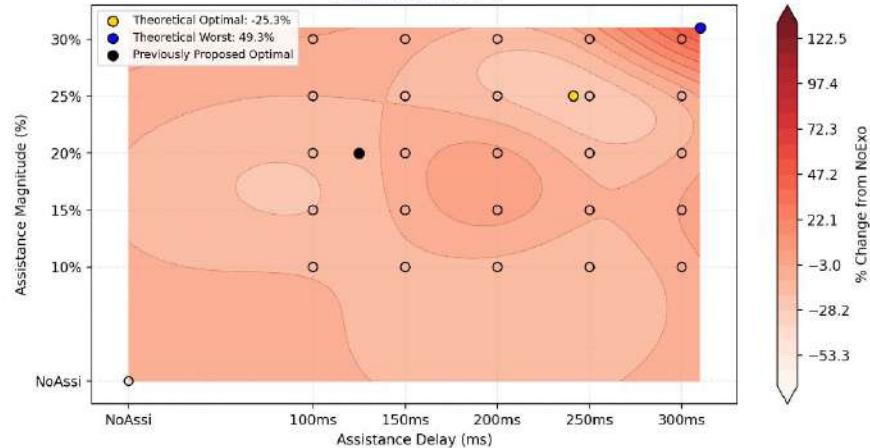


Rectus Femoris (NoAssi = -36.1%)

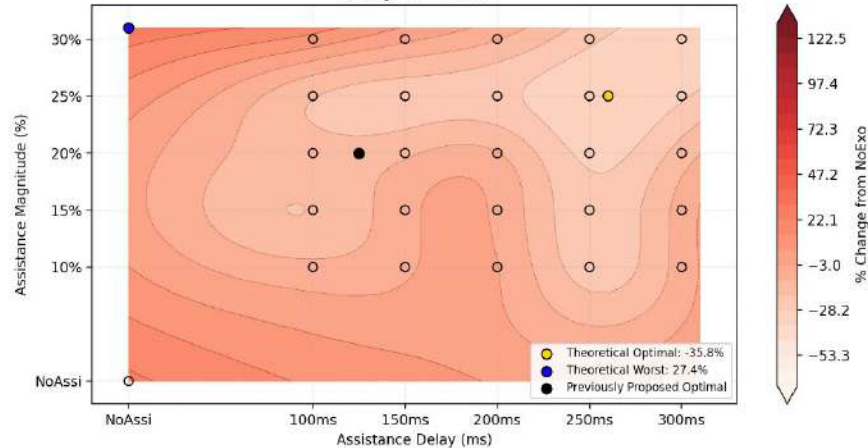


Peak EMG Activation Contour Map - LG Walking

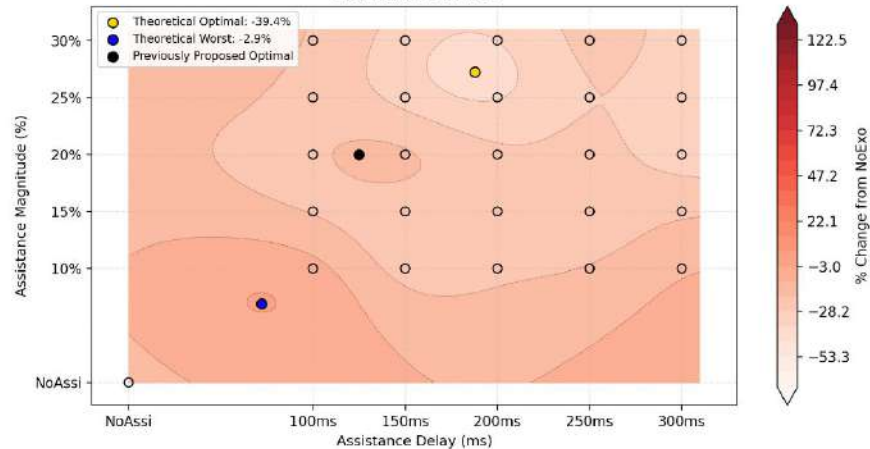
Tibialis Anterior



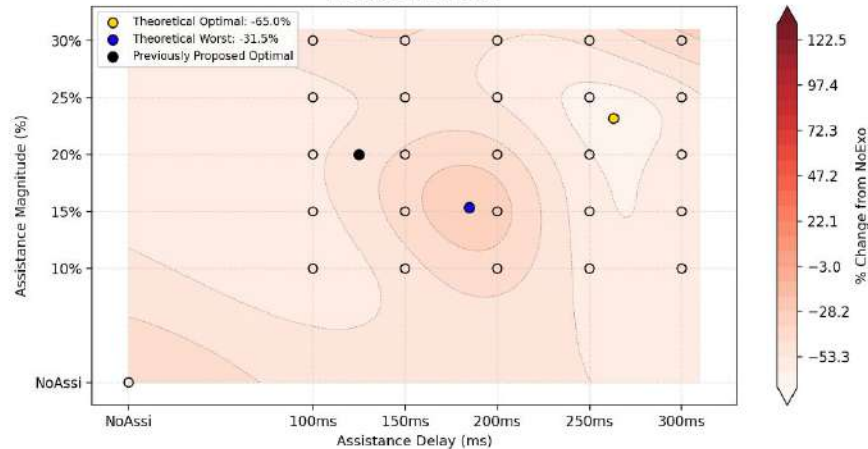
Bicep Femoris



Gastrocnemius

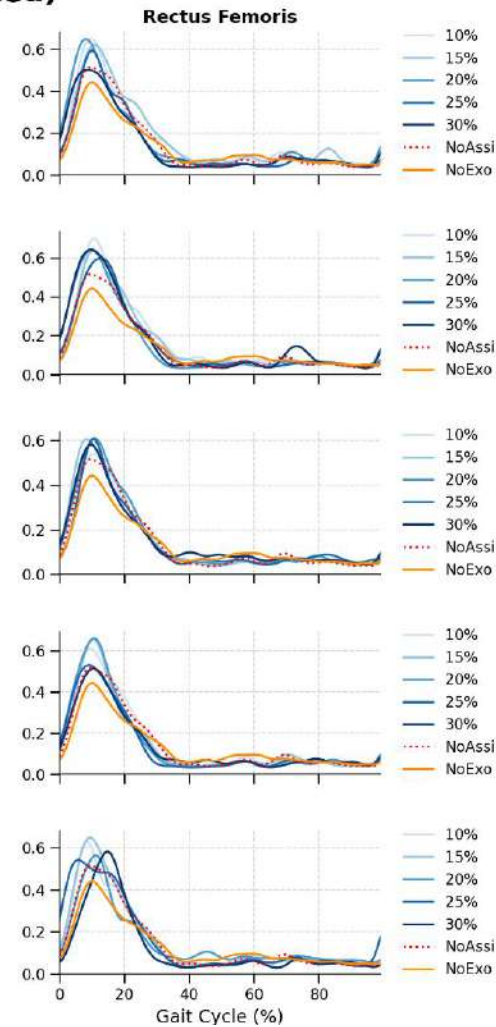
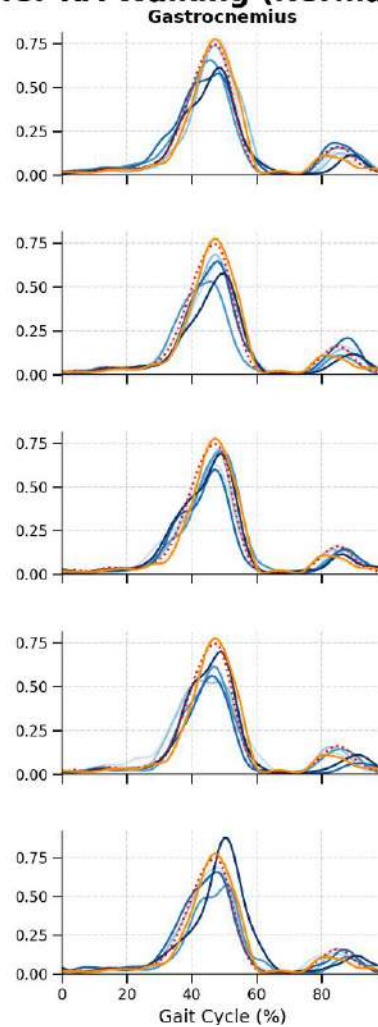
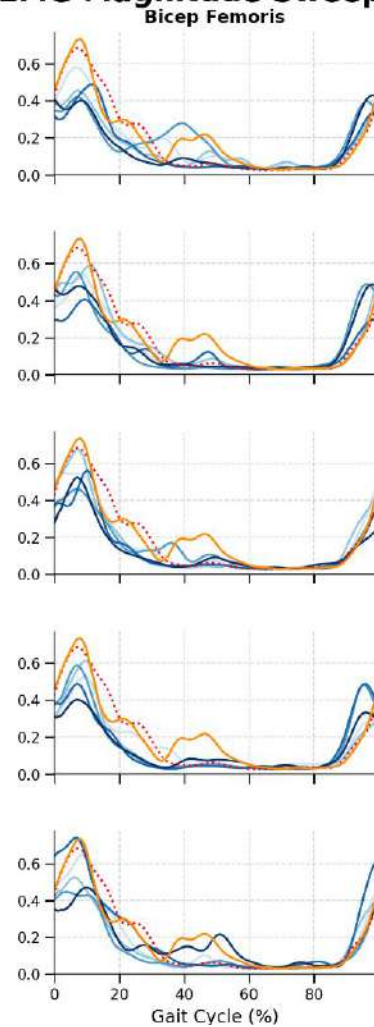
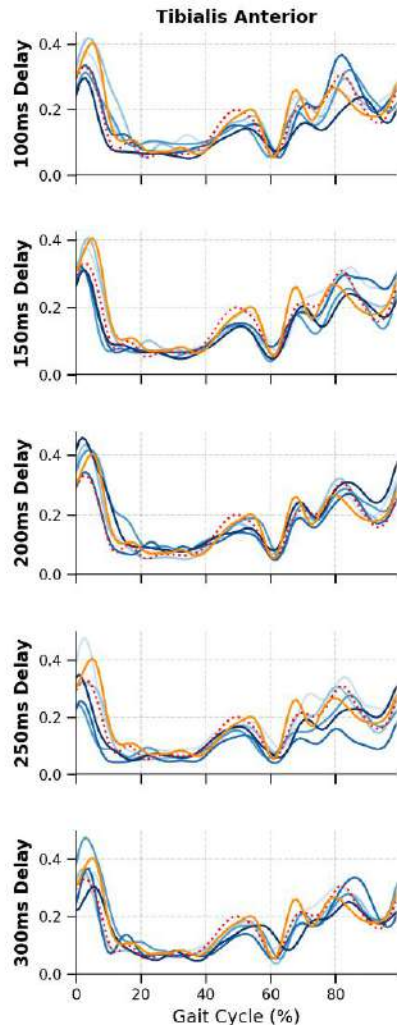


Rectus Femoris

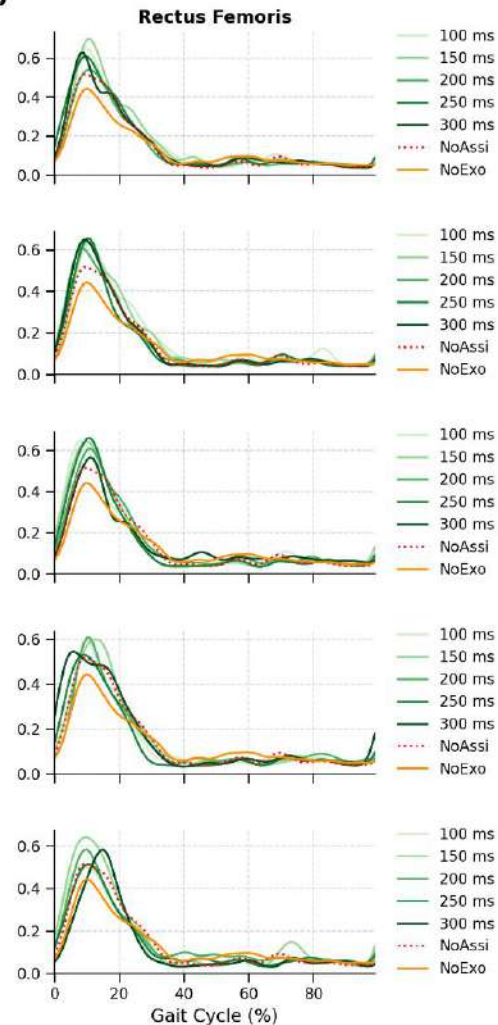
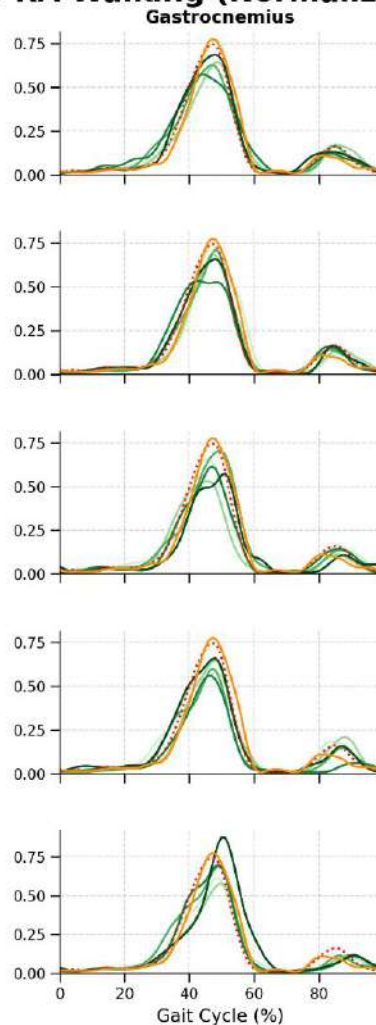
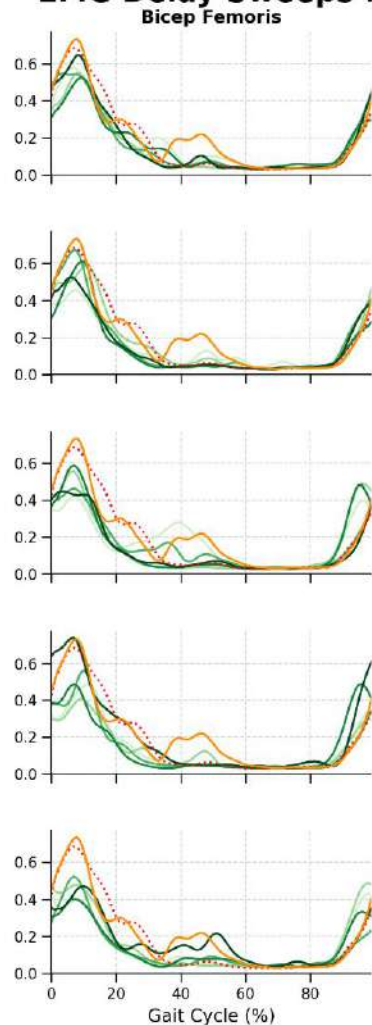
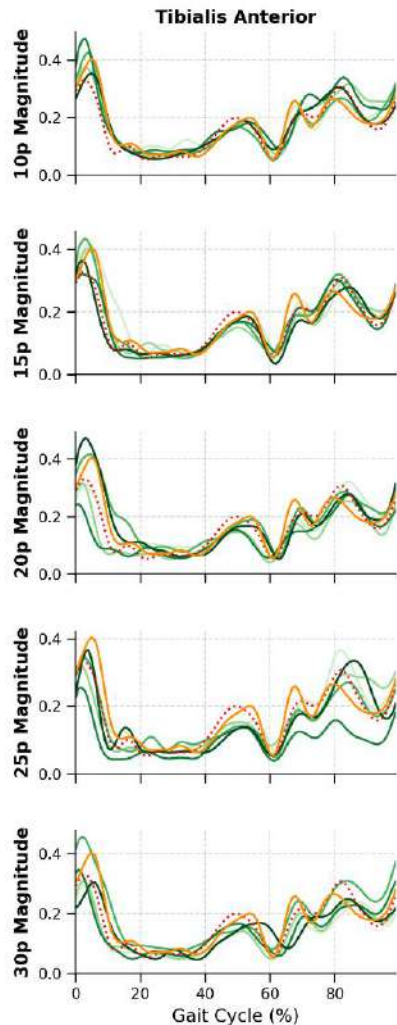


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

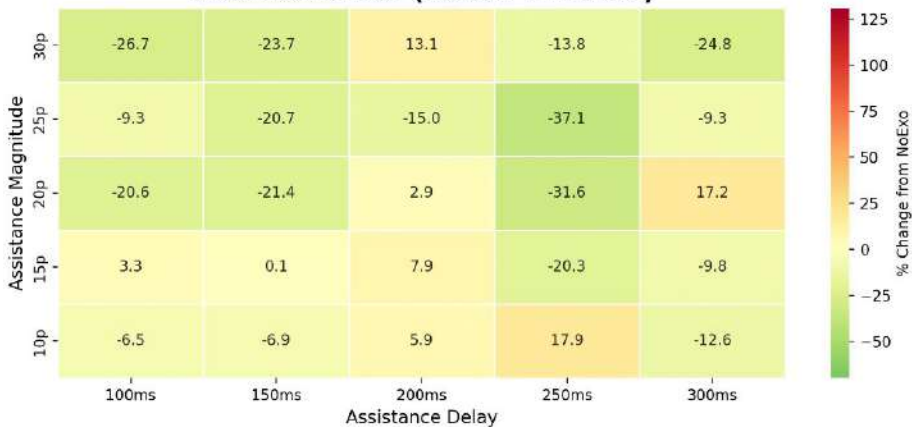


EMG Delay Sweeps for RA Walking (Normalized)

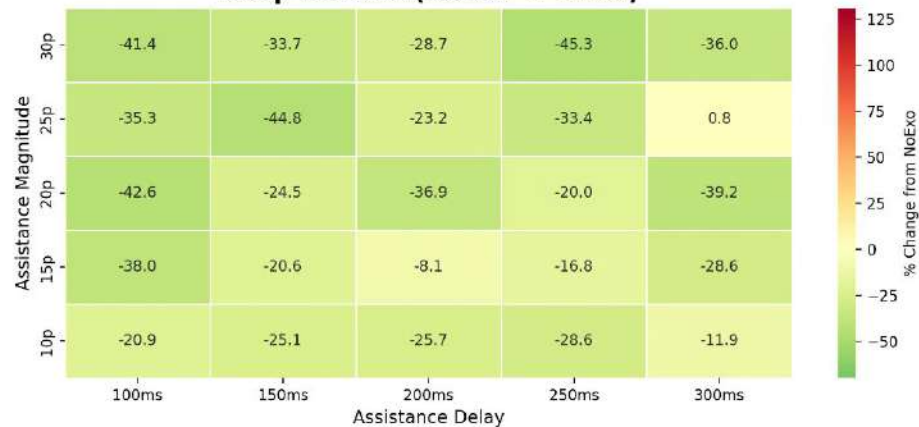


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

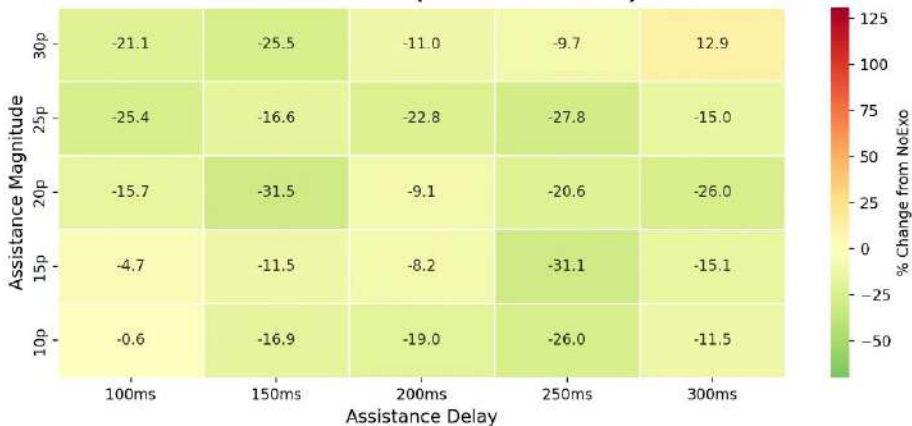
Tibialis Anterior (NoAssi = -18.2%)



Bicep Femoris (NoAssi = -6.8%)



Gastrocnemius (NoAssi = -3.7%)

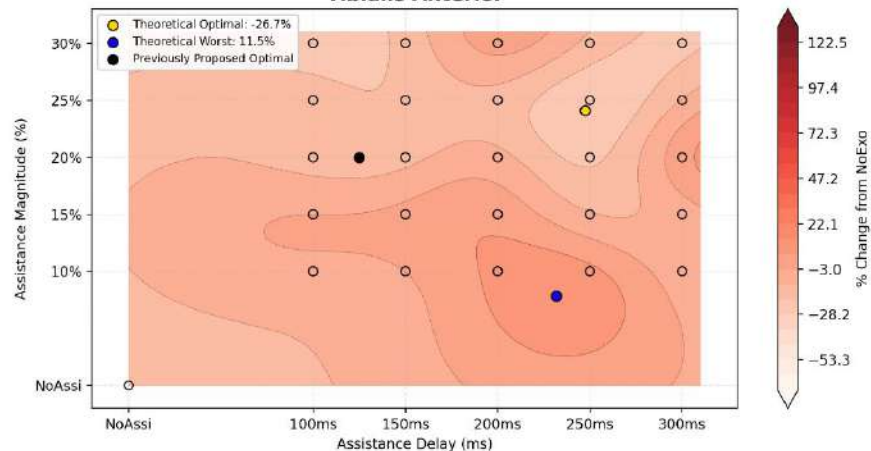


Rectus Femoris (NoAssi = -1.9%)

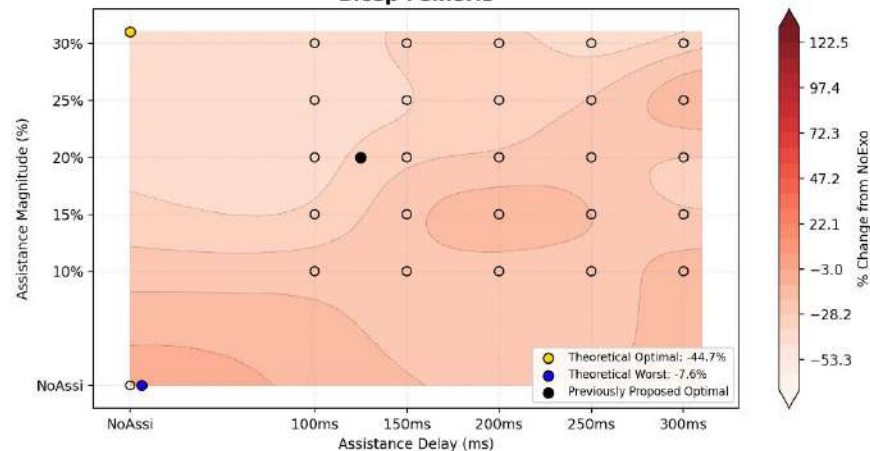


Peak EMG Activation Contour Map - RA Walking

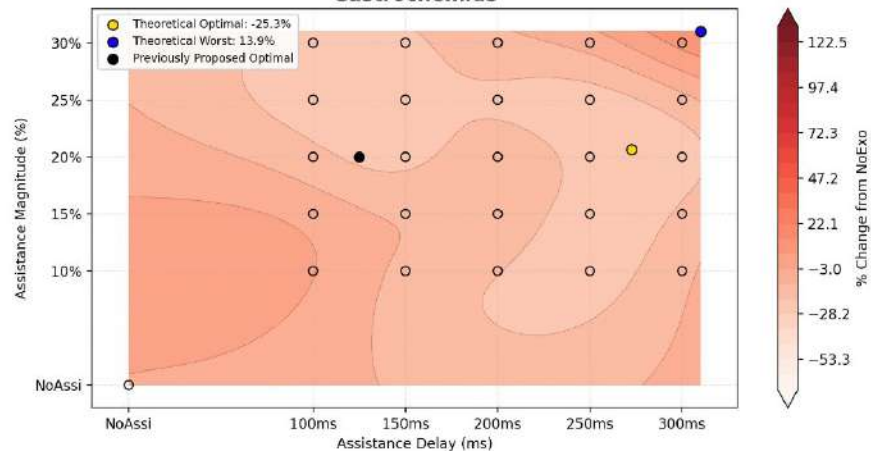
Tibialis Anterior



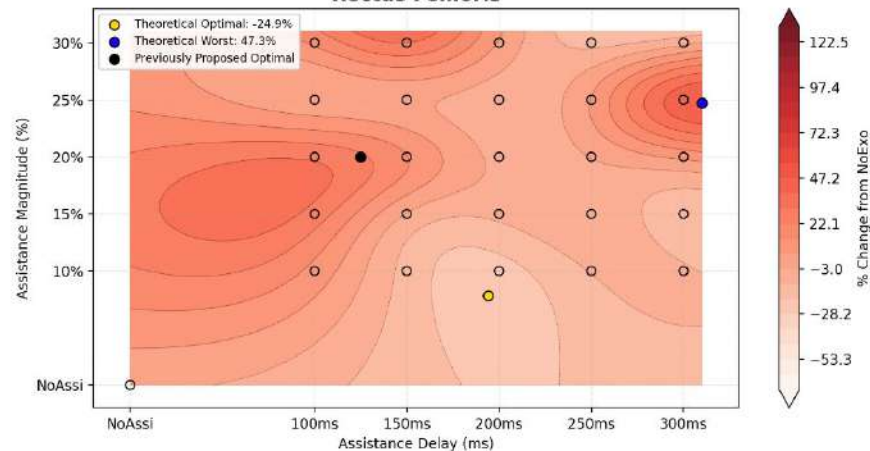
Bicep Femoris



Gastrocnemius

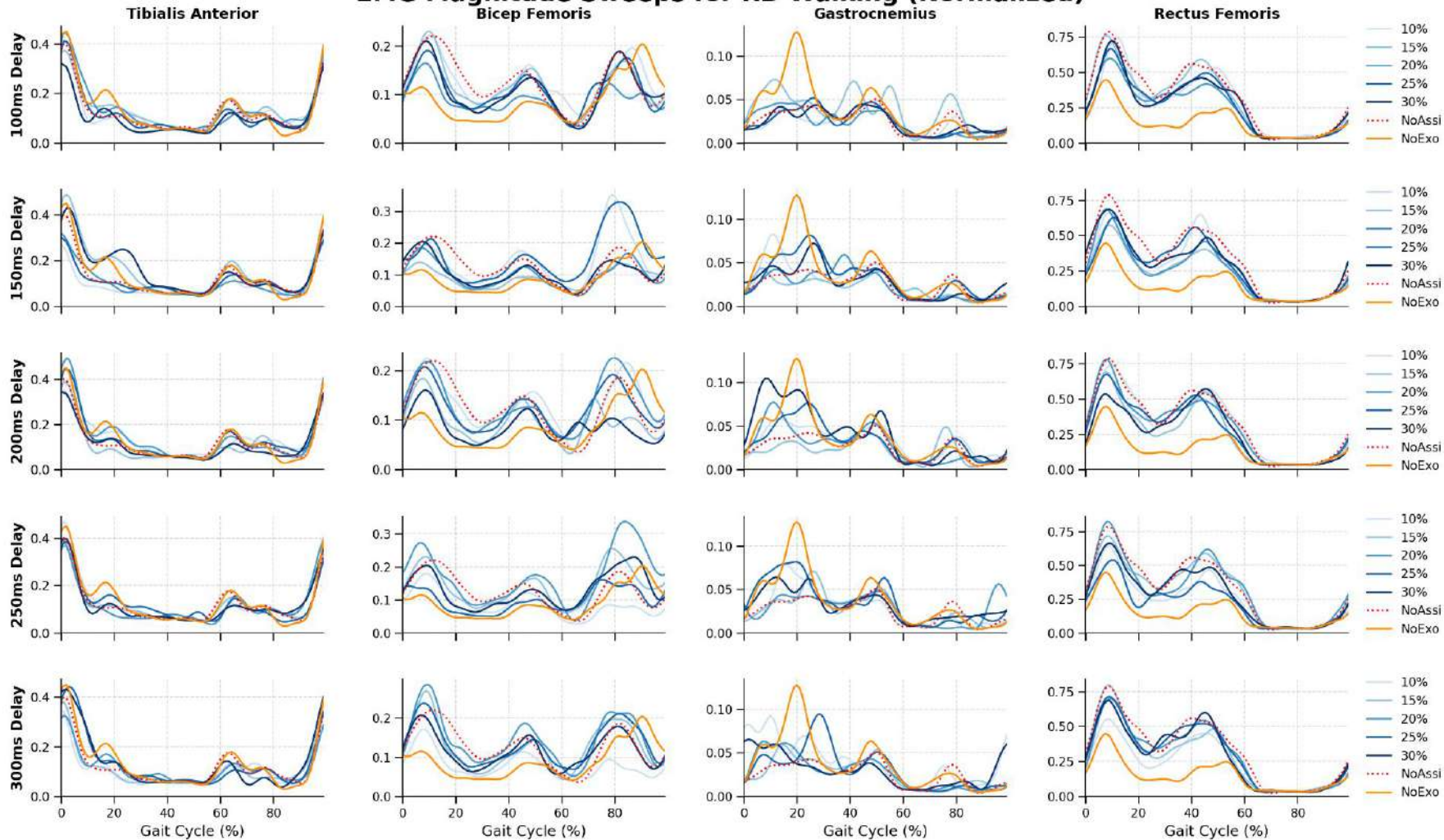


Rectus Femoris



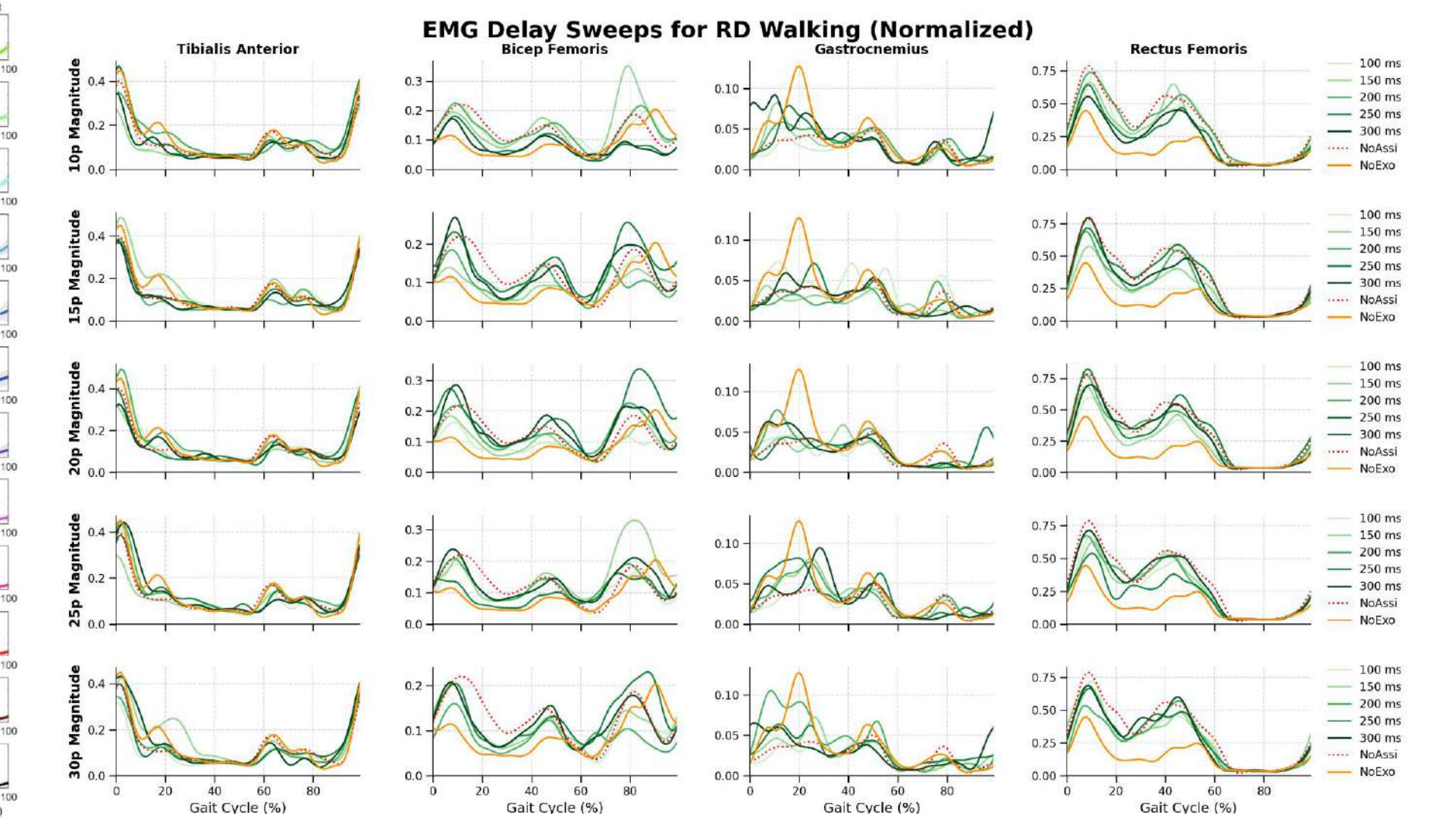
Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)



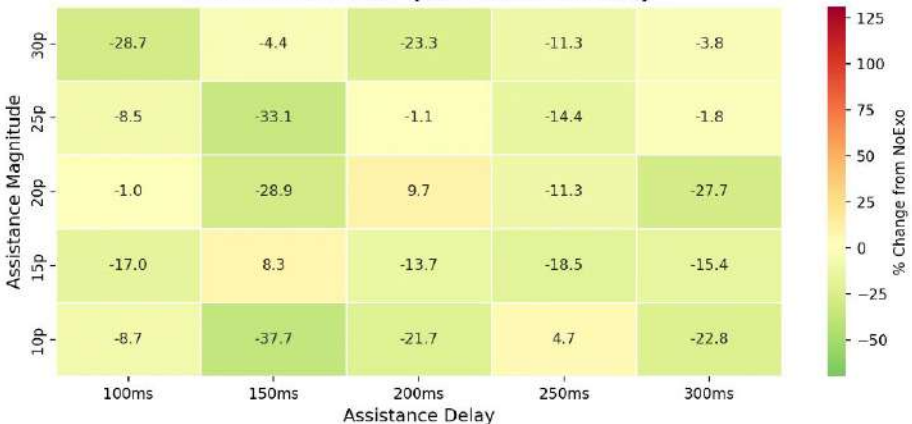
EMG Delay Sweeps for RD Walking (Normalized)

The figure displays a 5x4 grid of line plots showing normalized EMG magnitude for four muscles: Tibialis Anterior, Bicep Femoris, Gastrocnemius, and Rectus Femoris. The rows represent different delay levels: 100 ms, 150 ms, 200 ms, 250 ms, and 300 ms. The columns represent the muscles. Each plot shows the EMG magnitude (y-axis, 0.0 to 0.4 or 0.75) versus the gait cycle percentage (x-axis, 0 to 100%). The legend indicates two conditions: NoAssi (dotted lines) and NoExo (solid lines). The plots show that the EMG magnitude generally decreases as the delay increases, and the NoExo condition often shows higher magnitude than the NoAssi condition.



Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

Tibialis Anterior (NoAssi = -10.8%)



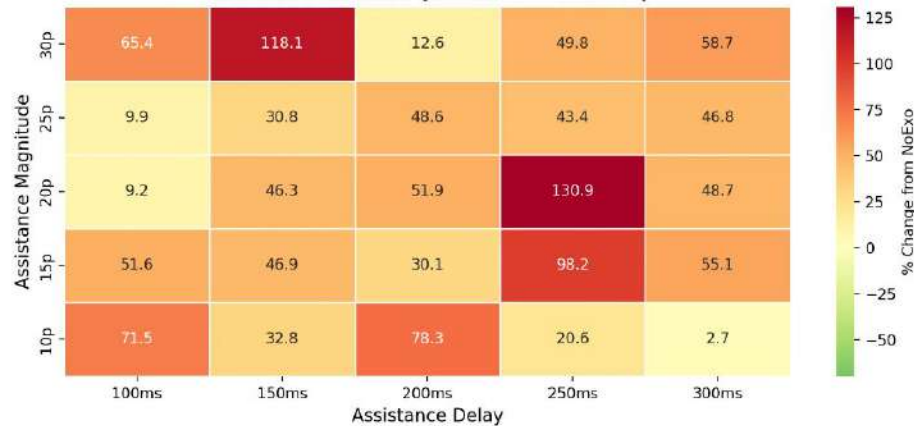
Bicep Femoris (NoAssi = 7.4%)



Gastrocnemius (NoAssi = -60.5%)

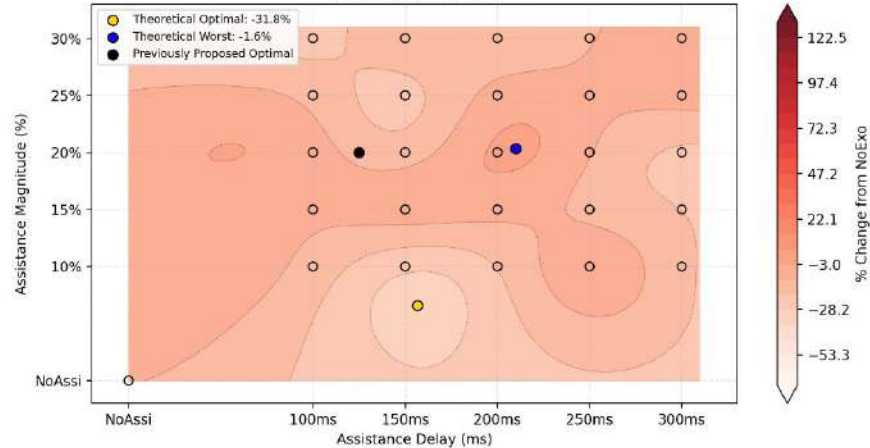


Rectus Femoris (NoAssi = 91.6%)

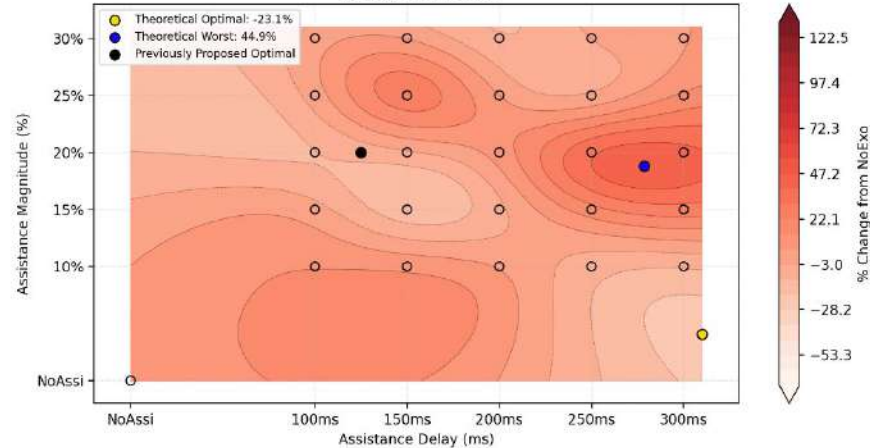


Peak EMG Activation Contour Map - RD Walking

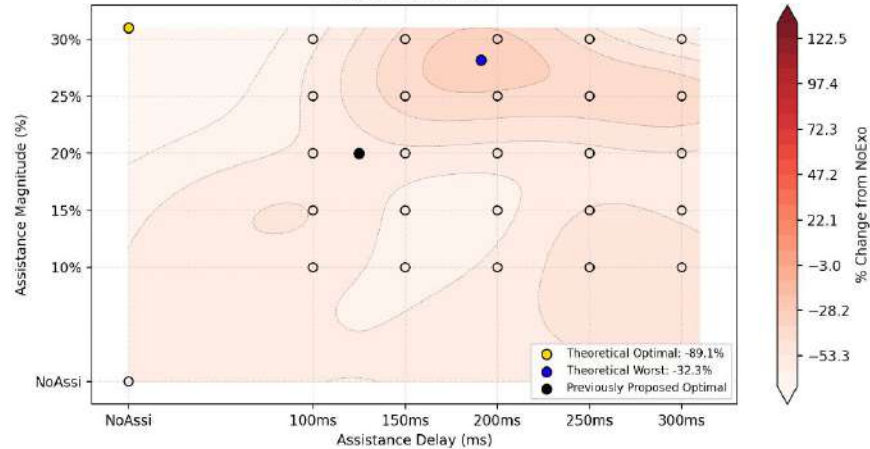
Tibialis Anterior



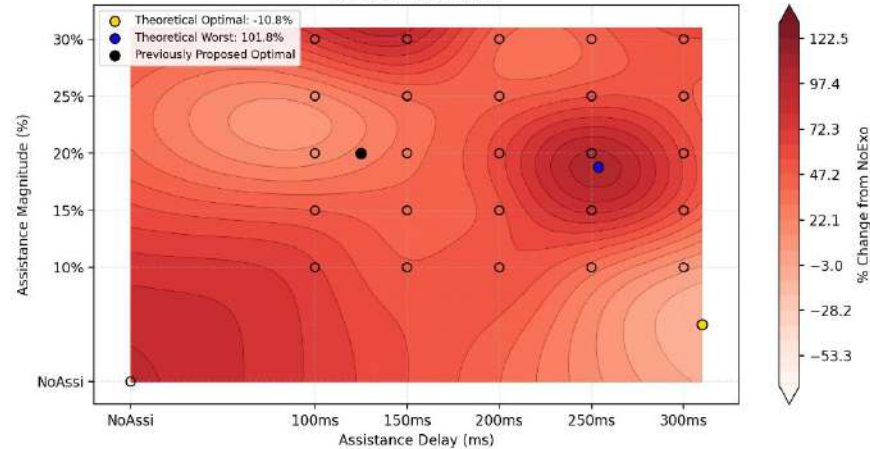
Bicep Femoris



Gastrocnemius



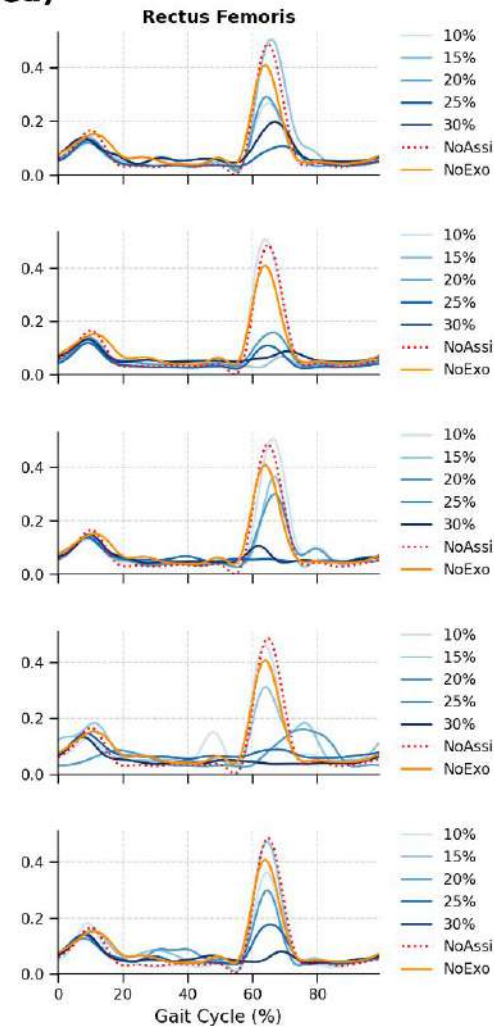
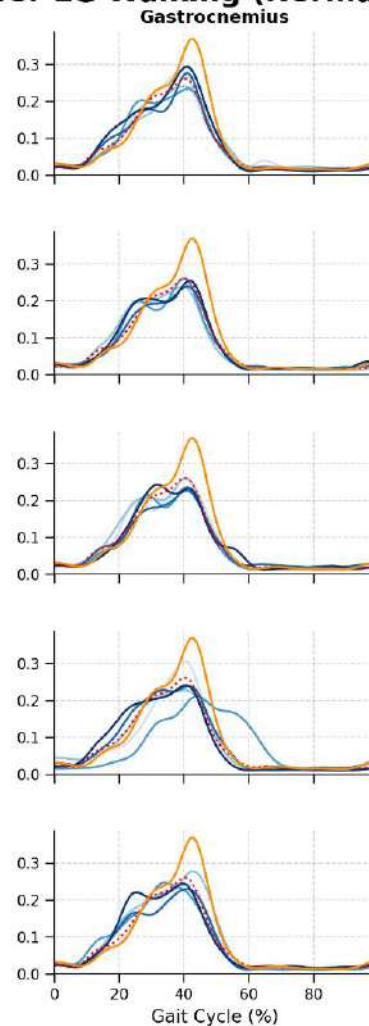
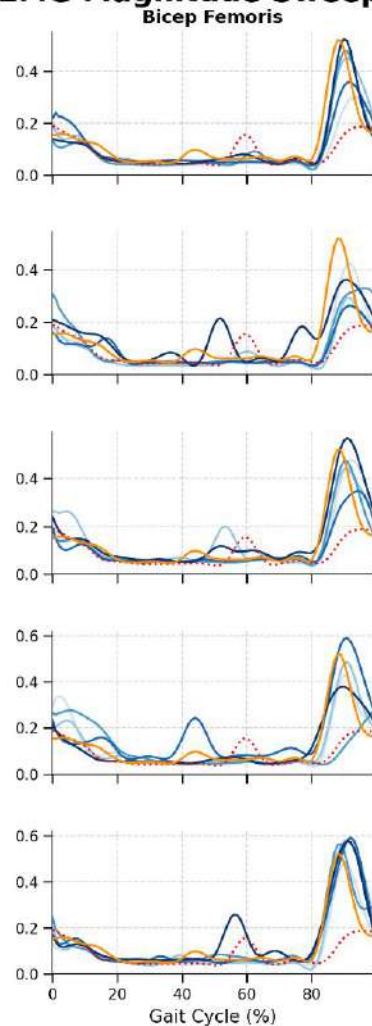
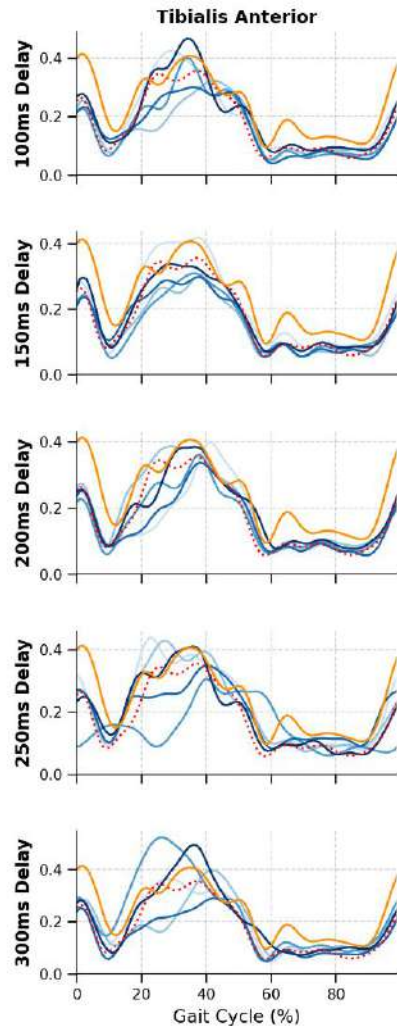
Rectus Femoris



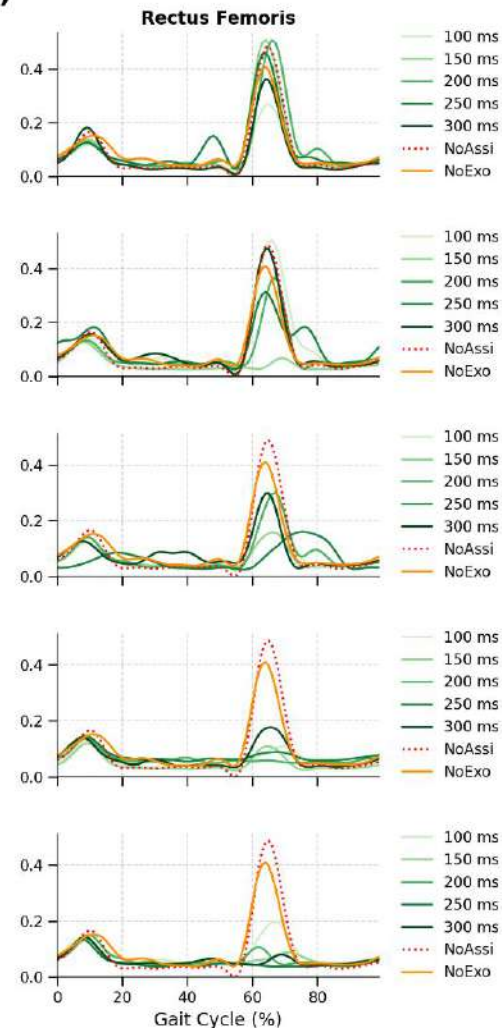
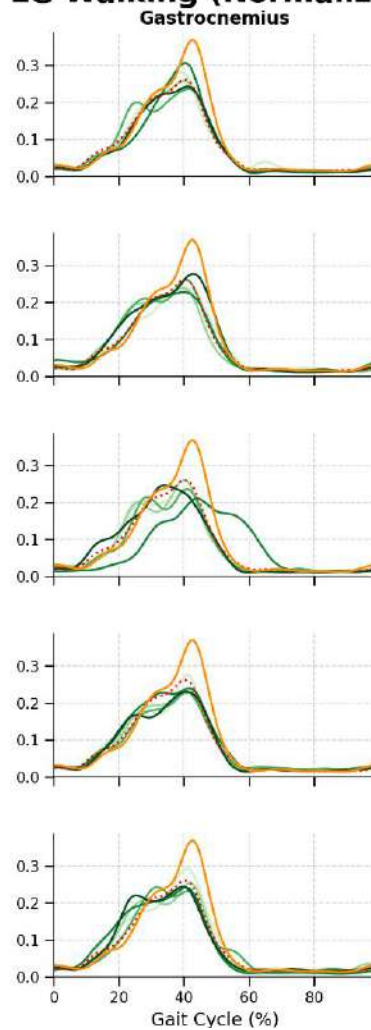
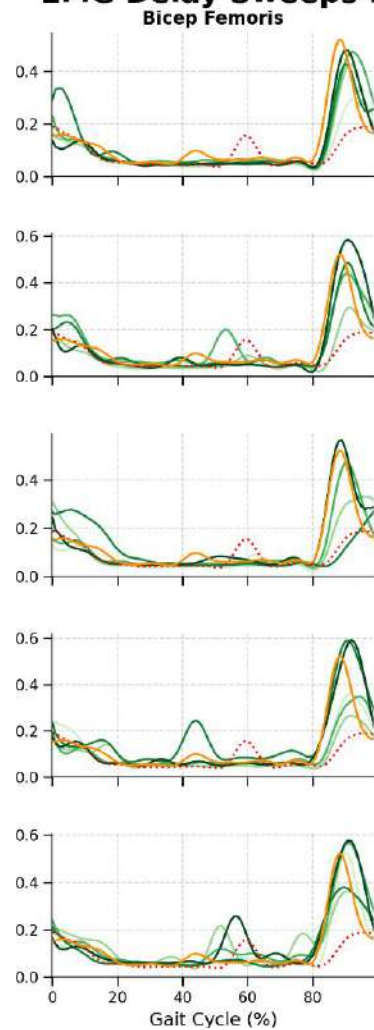
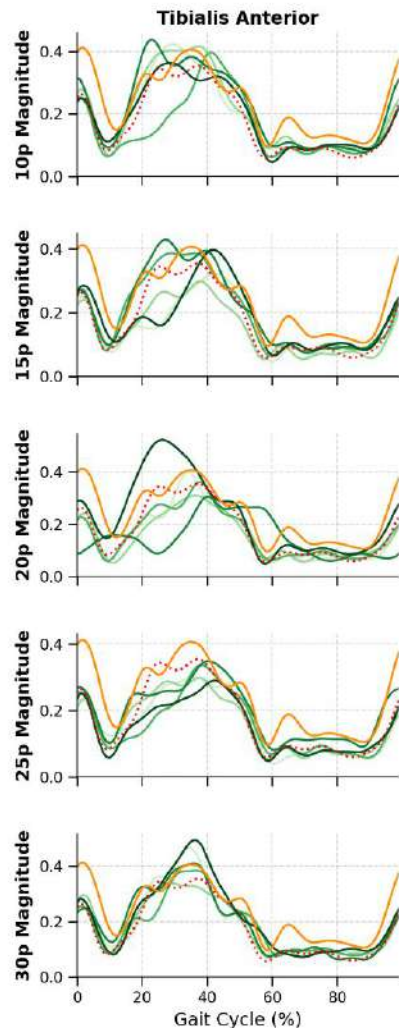
Subject: AB05_Maria

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

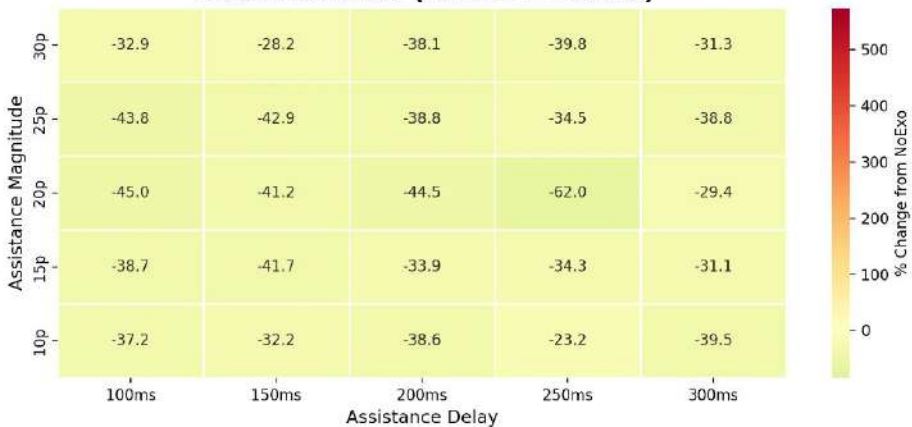


EMG Delay Sweeps for LG Walking (Normalized)

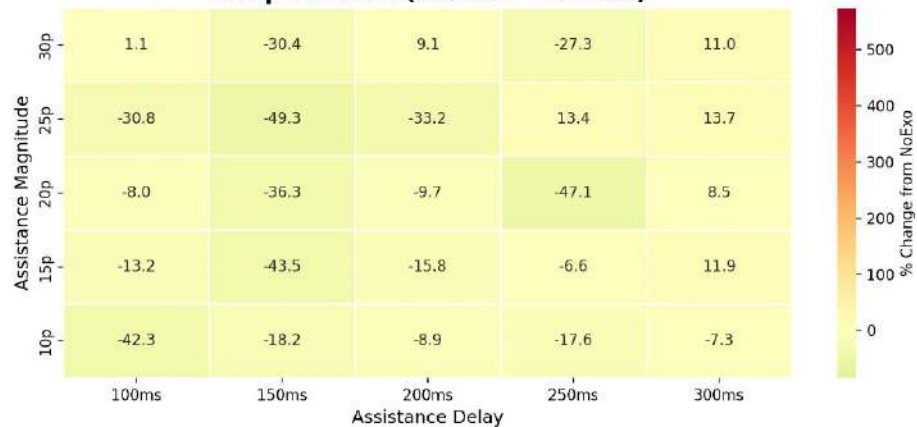


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

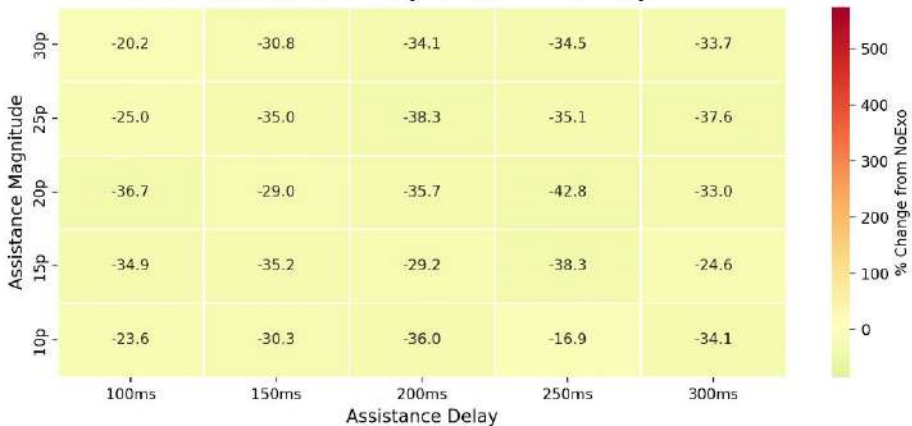
Tibialis Anterior (NoAssi = -35.6%)



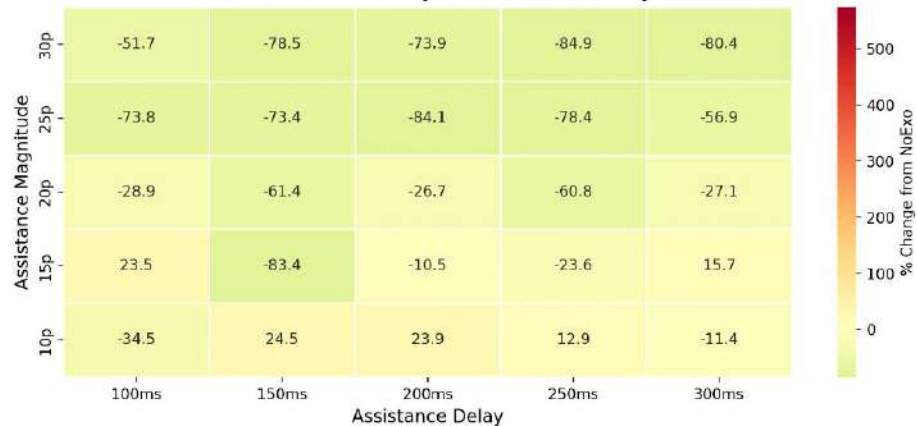
Bicep Femoris (NoAssi = -63.5%)



Gastrocnemius (NoAssi = -29.2%)

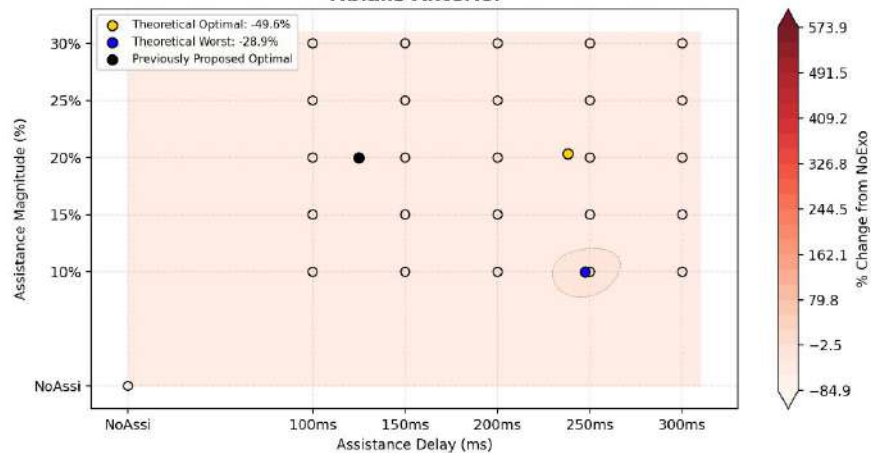


Rectus Femoris (NoAssi = 18.9%)

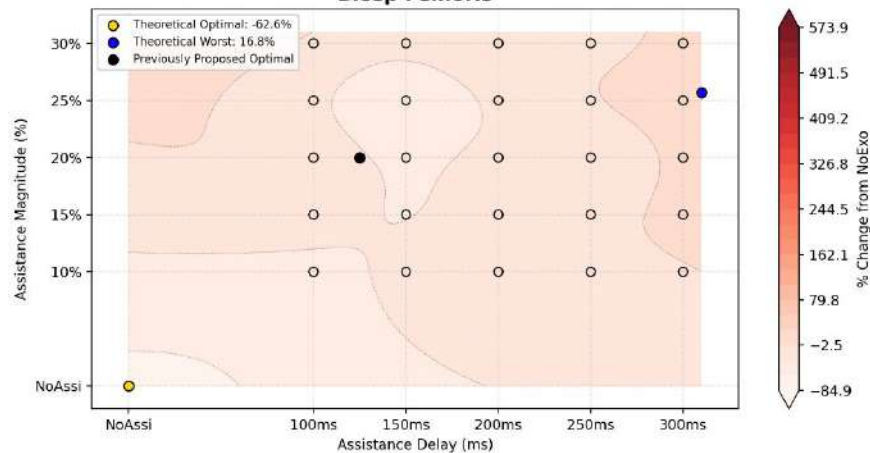


Peak EMG Activation Contour Map - LG Walking

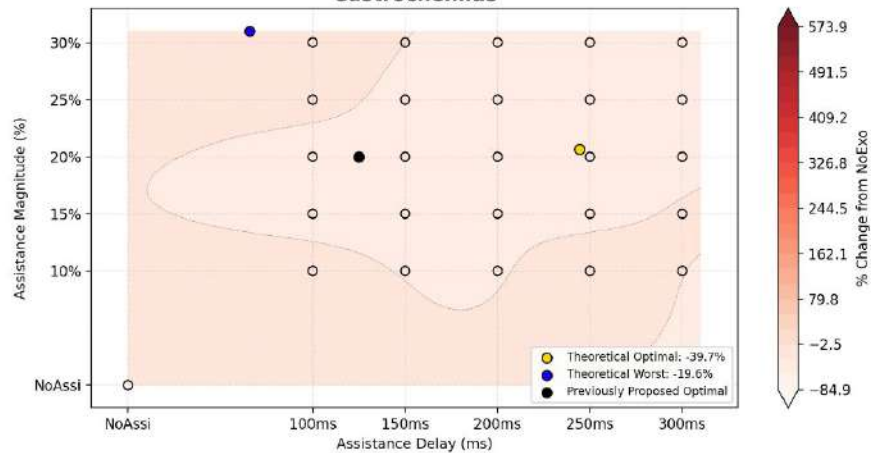
Tibialis Anterior



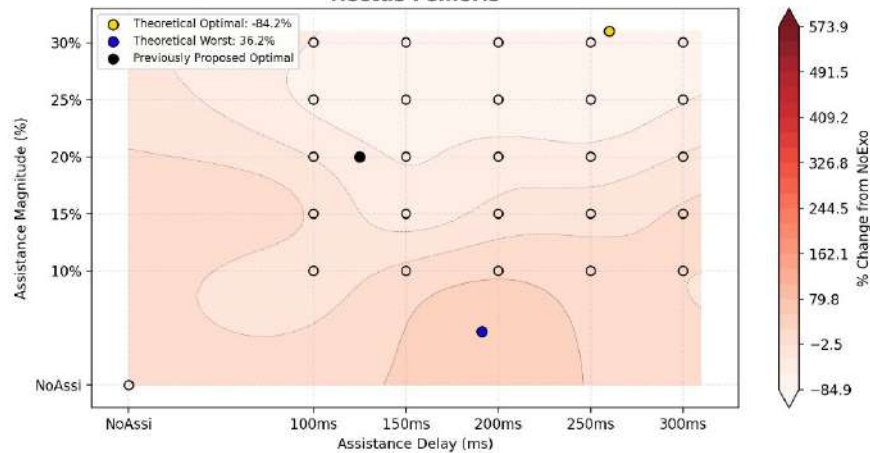
Bicep Femoris



Gastrocnemius

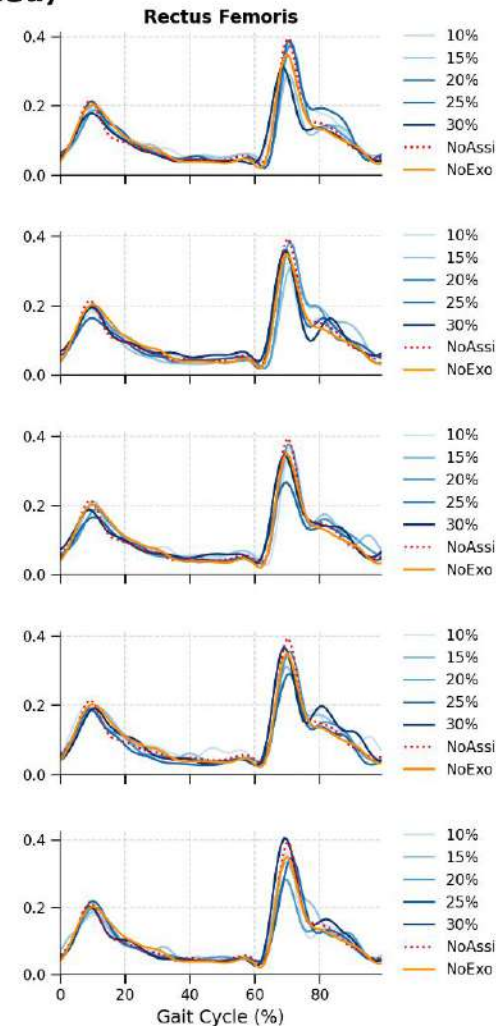
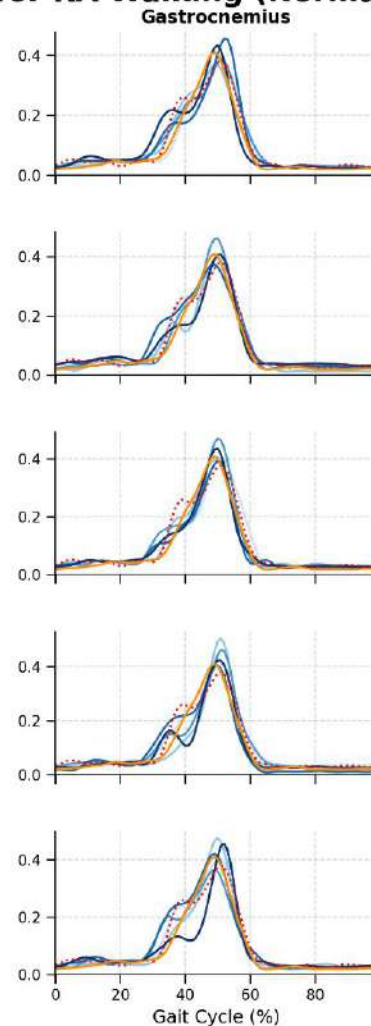
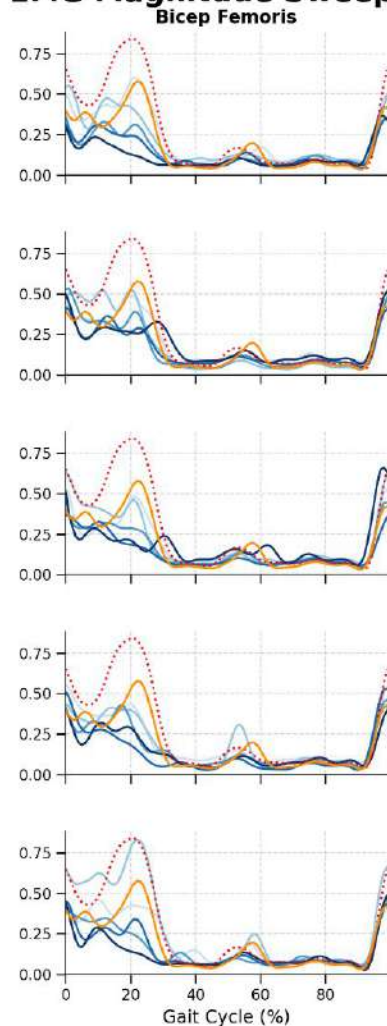
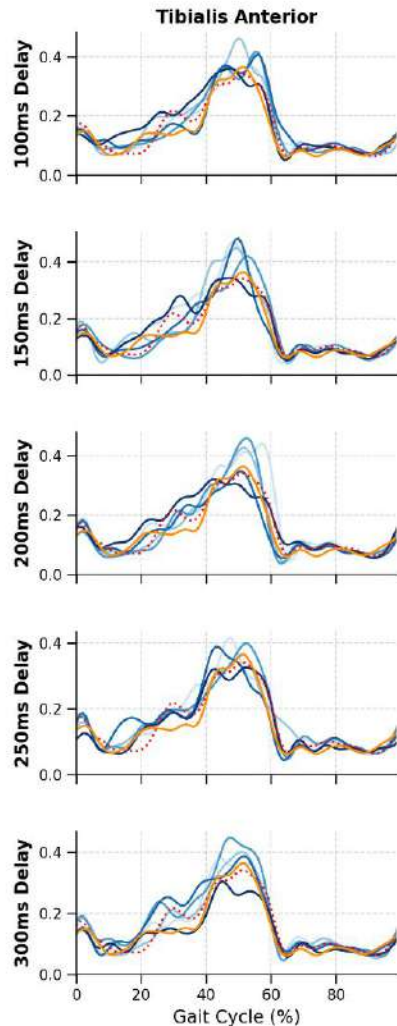


Rectus Femoris

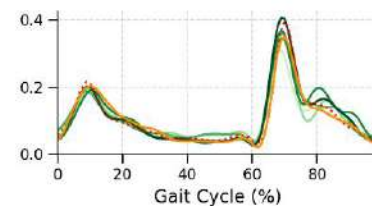
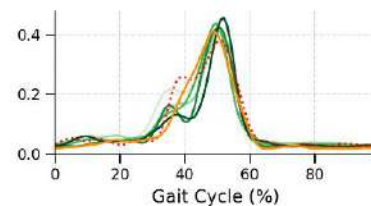
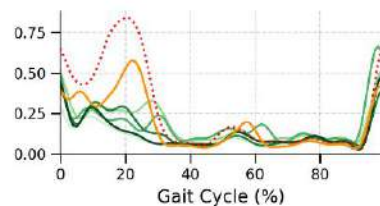
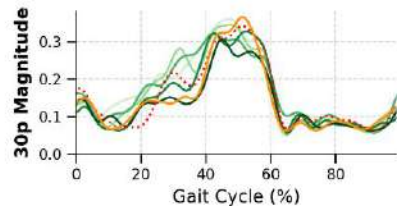
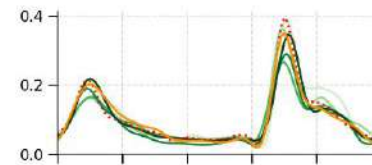
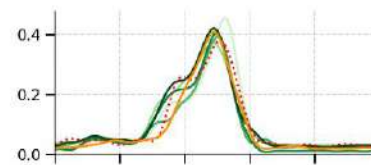
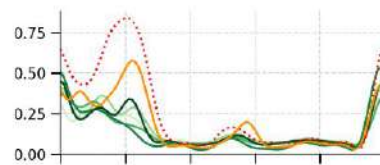
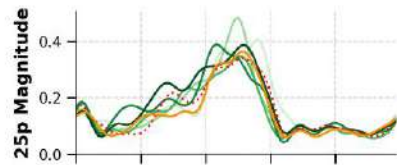
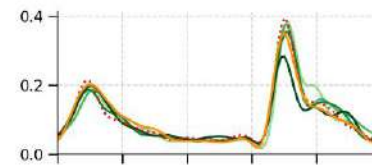
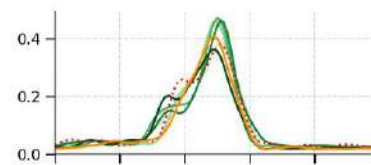
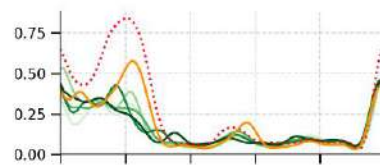
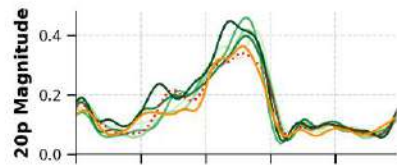
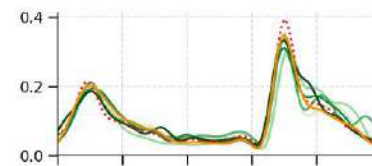
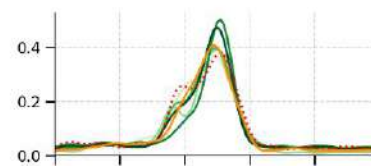
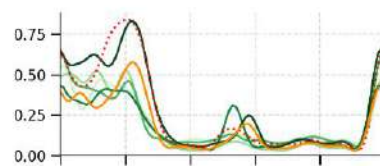
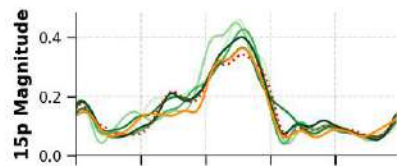
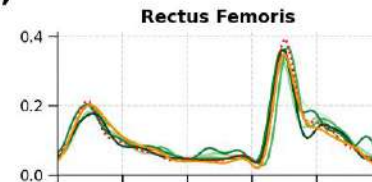
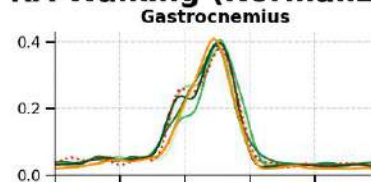
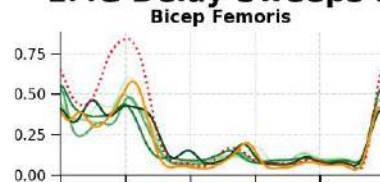
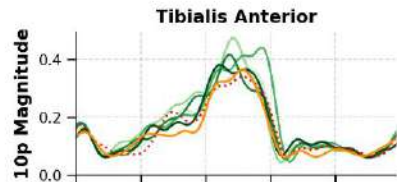


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

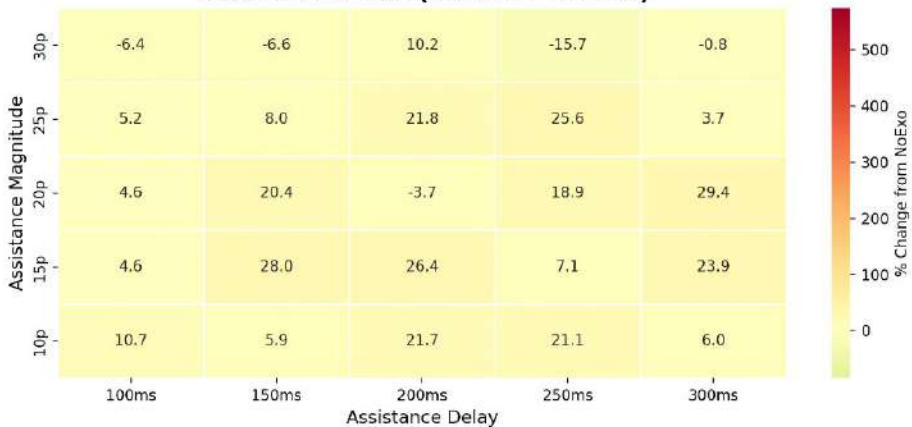


EMG Delay Sweeps for RA Walking (Normalized)

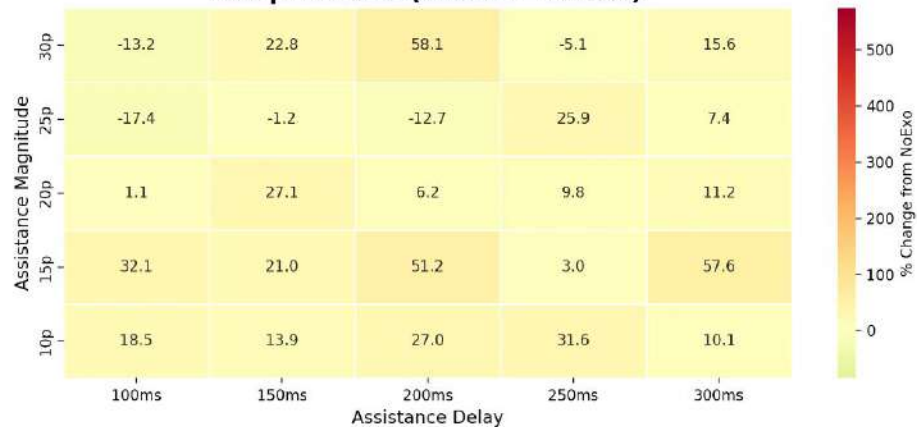


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

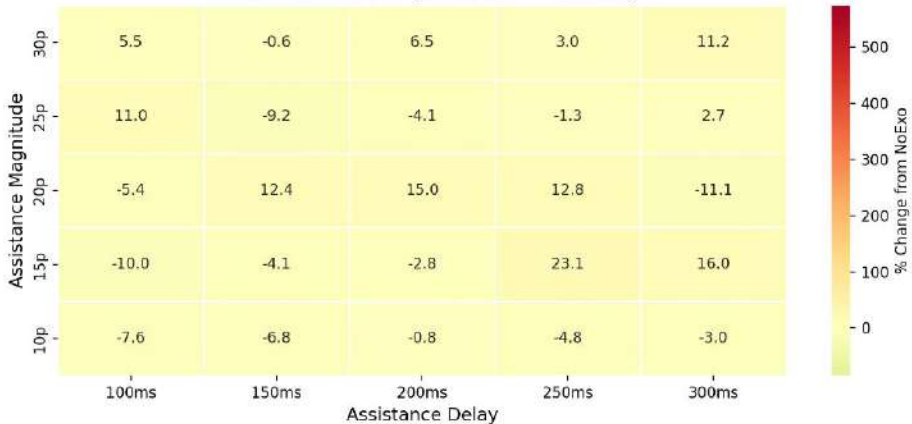
Tibialis Anterior (NoAssi = 17.8%)



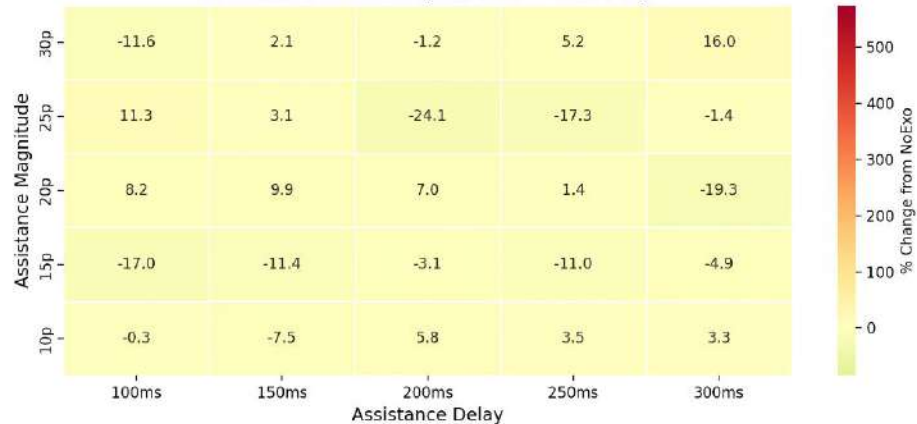
Bicep Femoris (NoAssi = 55.3%)



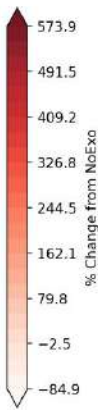
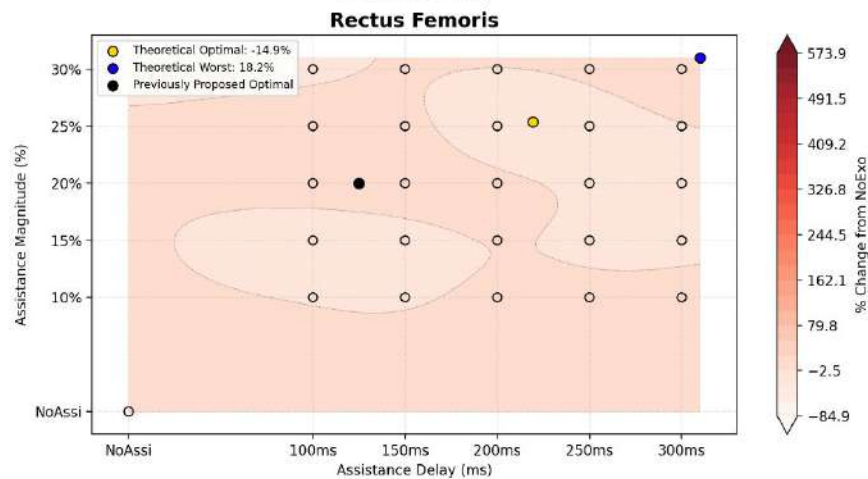
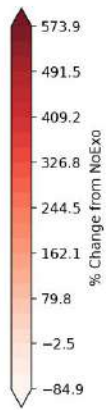
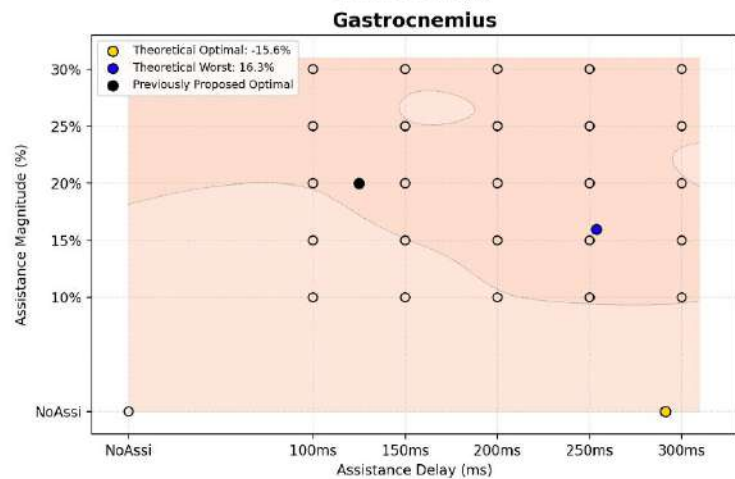
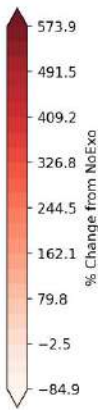
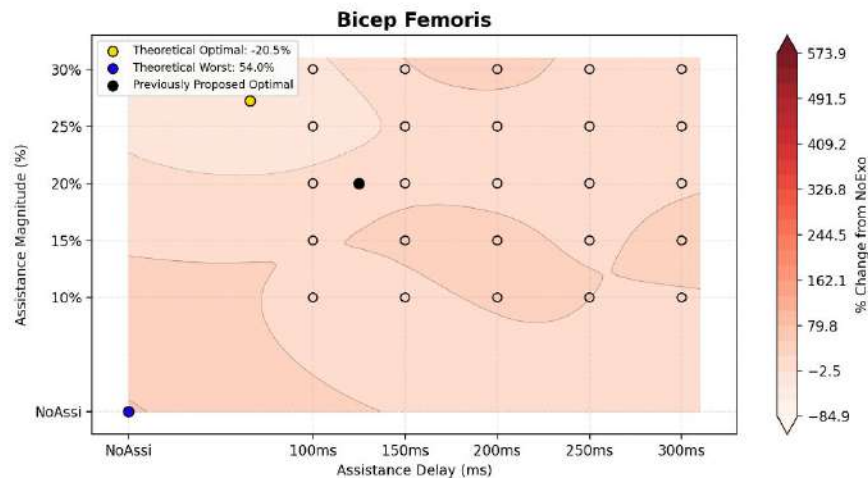
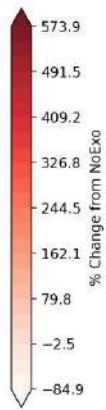
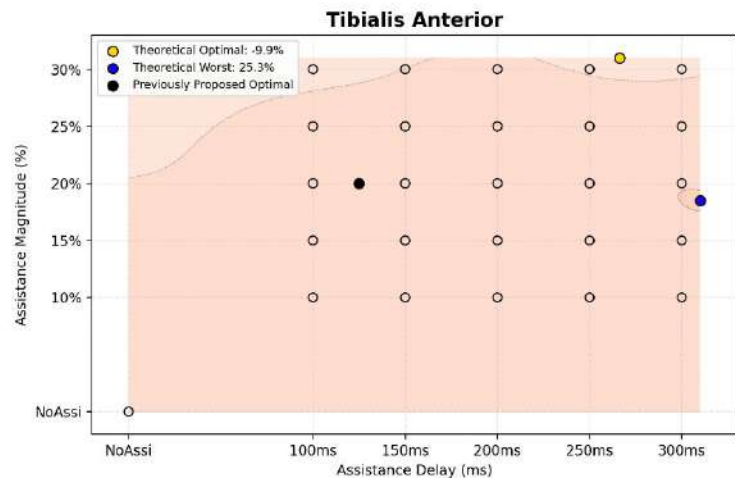
Gastrocnemius (NoAssi = -7.1%)



Rectus Femoris (NoAssi = 12.4%)

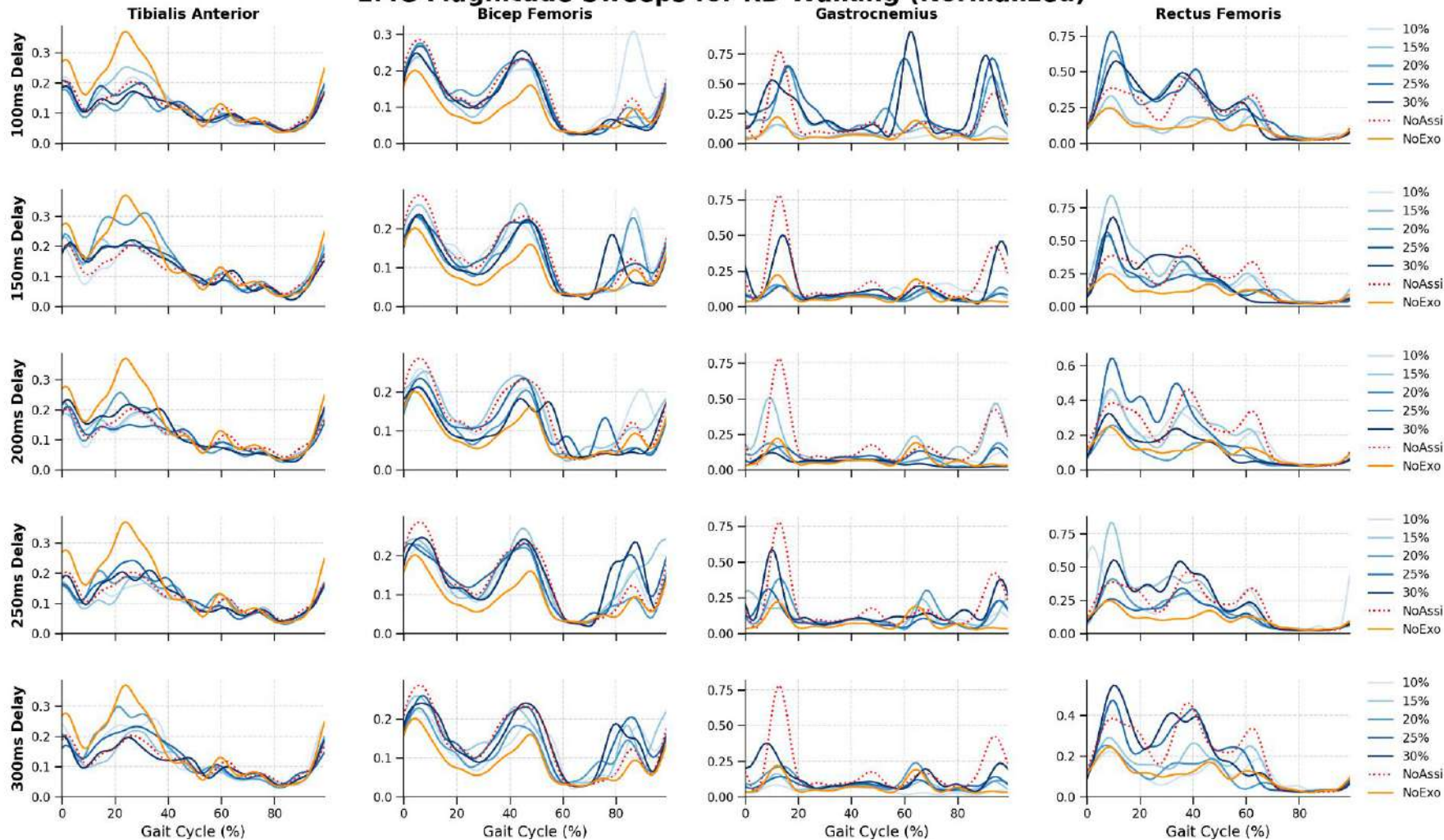


Peak EMG Activation Contour Map - RA Walking

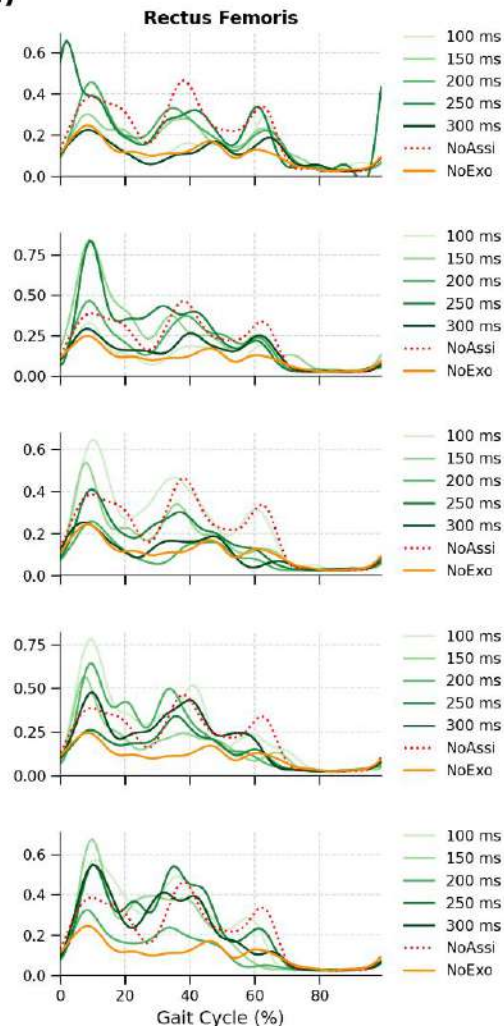
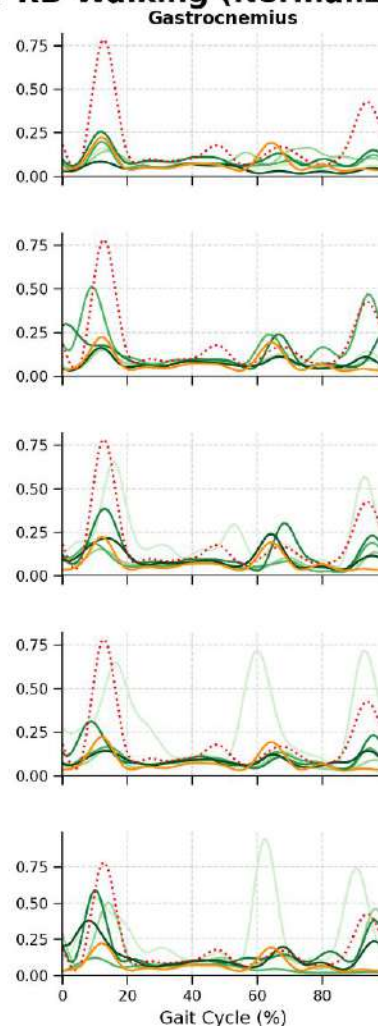
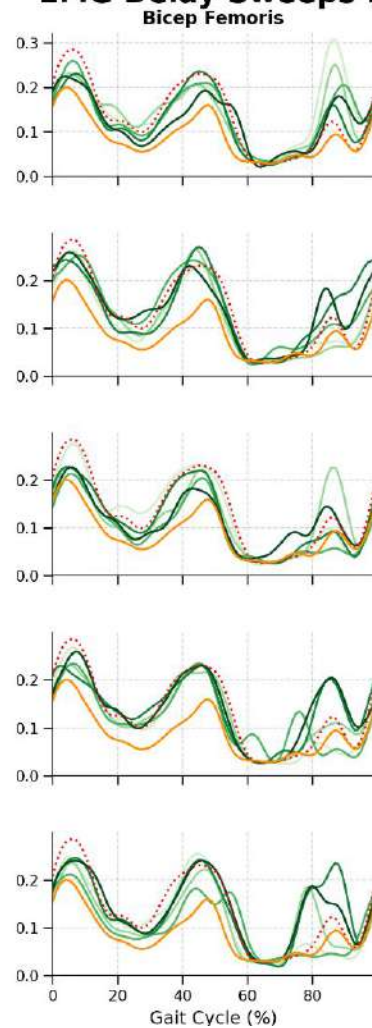
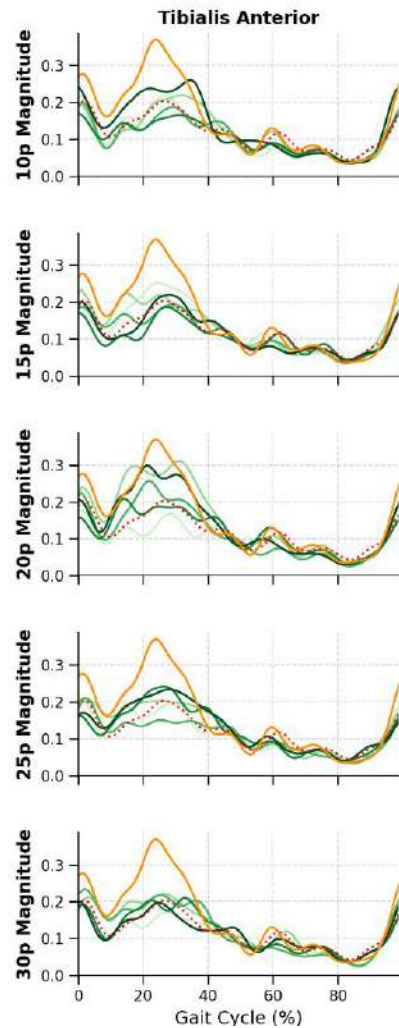


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

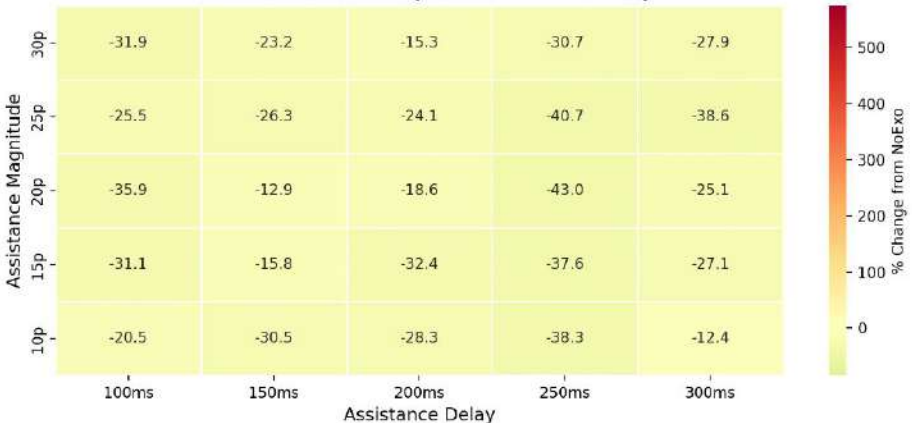


EMG Delay Sweeps for RD Walking (Normalized)

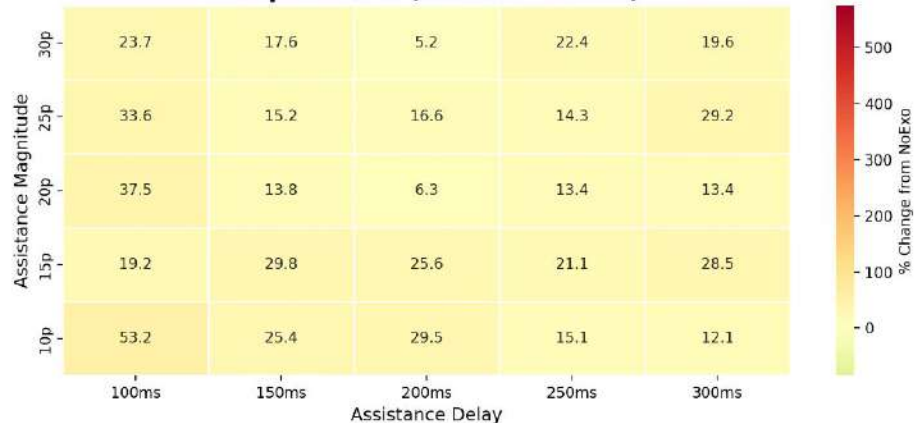


Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

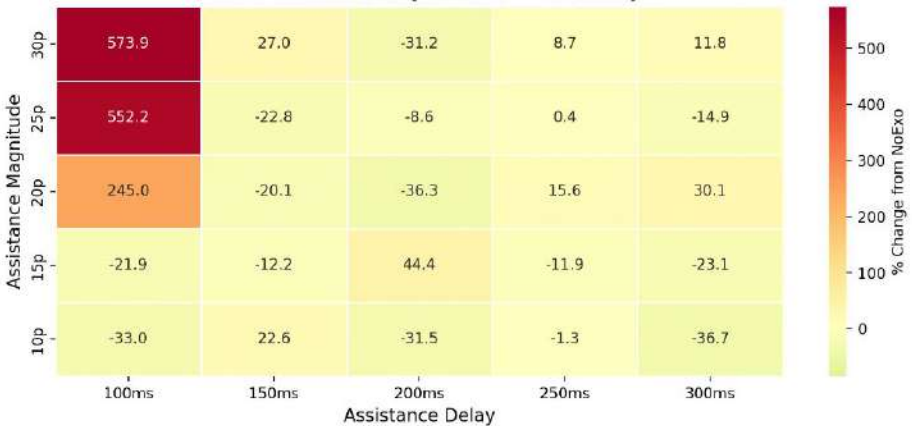
Tibialis Anterior (NoAssi = -26.2%)



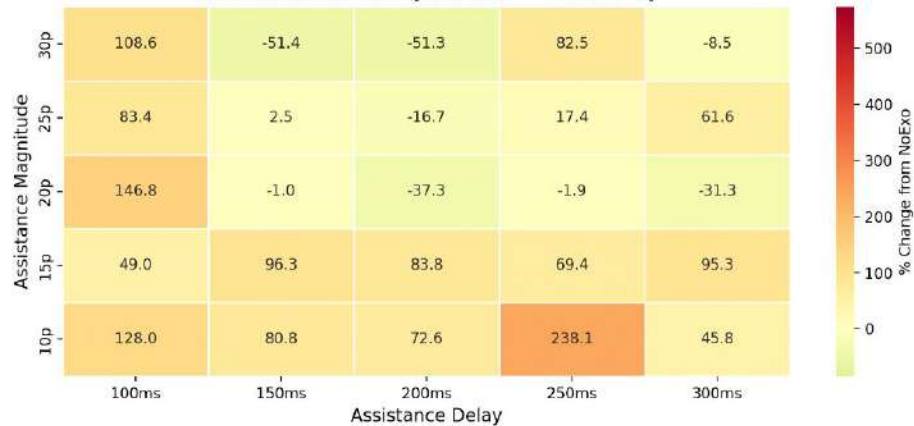
Bicep Femoris (NoAssi = 42.3%)



Gastrocnemius (NoAssi = 60.1%)

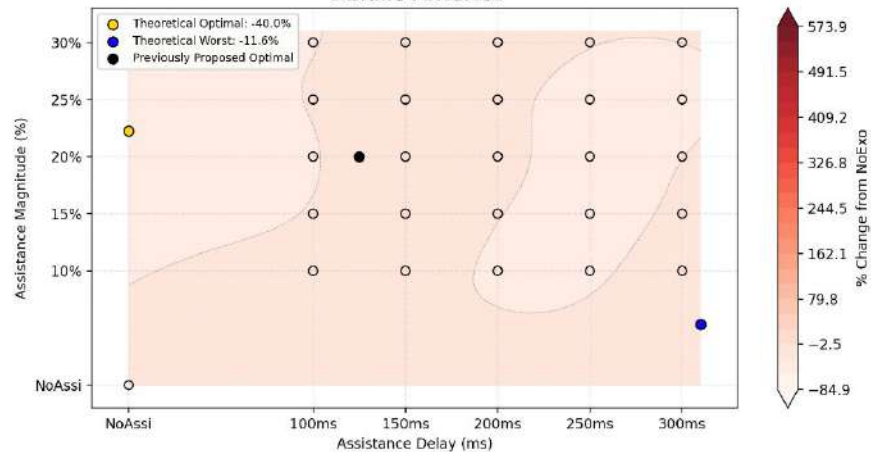


Rectus Femoris (NoAssi = 164.5%)

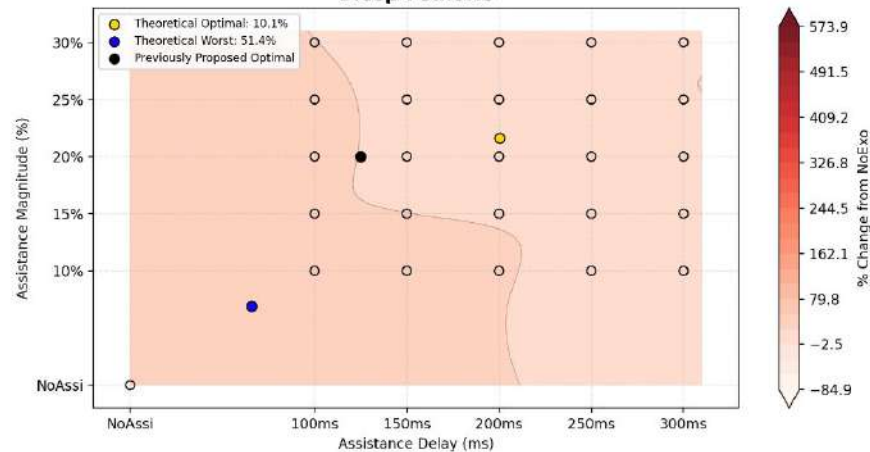


Peak EMG Activation Contour Map - RD Walking

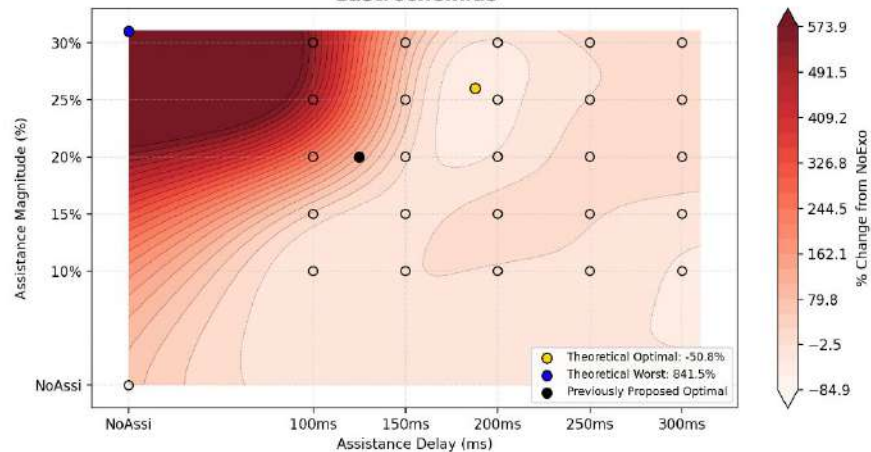
Tibialis Anterior



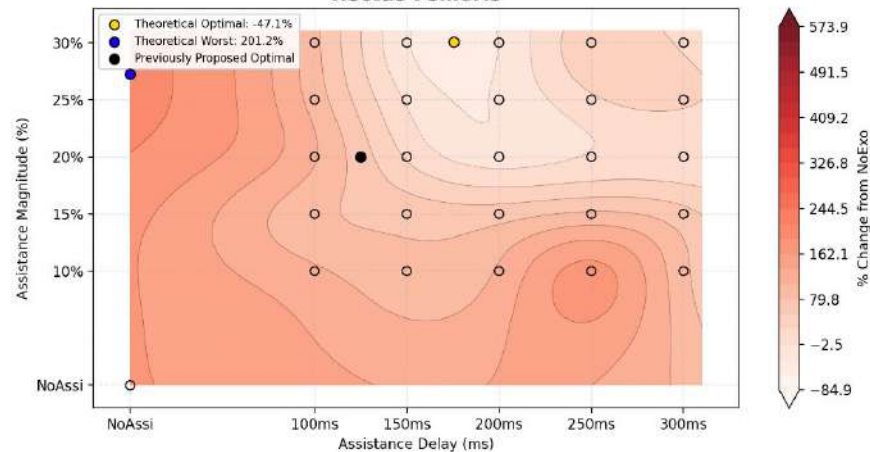
Bicep Femoris



Gastrocnemius



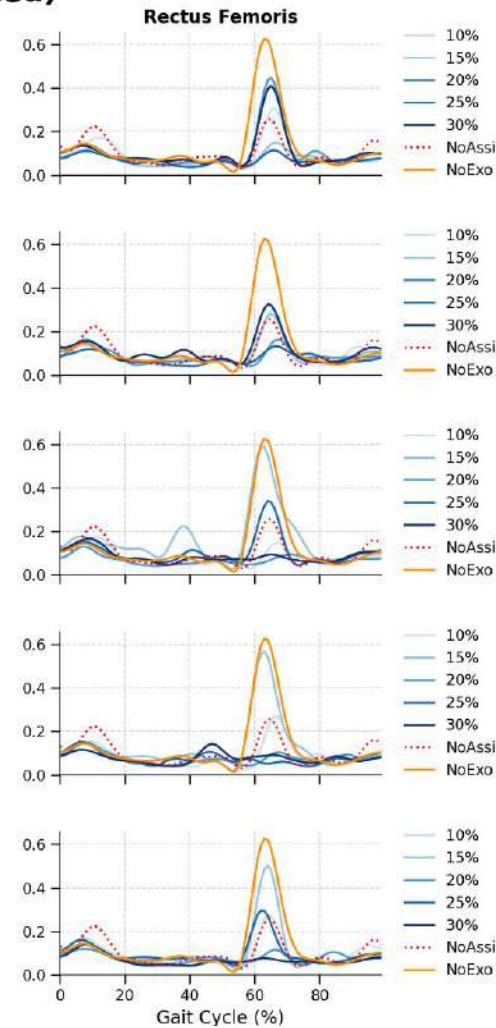
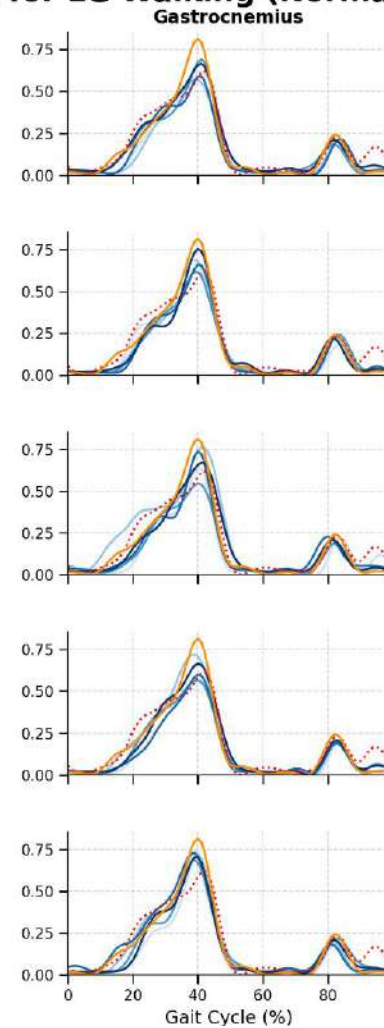
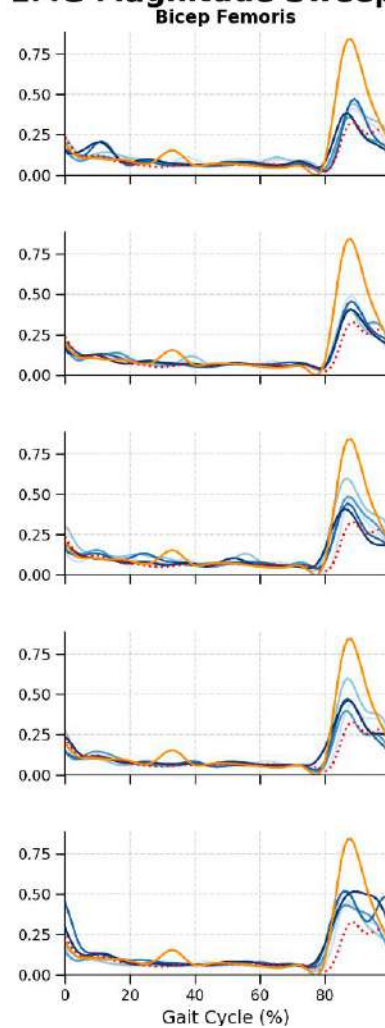
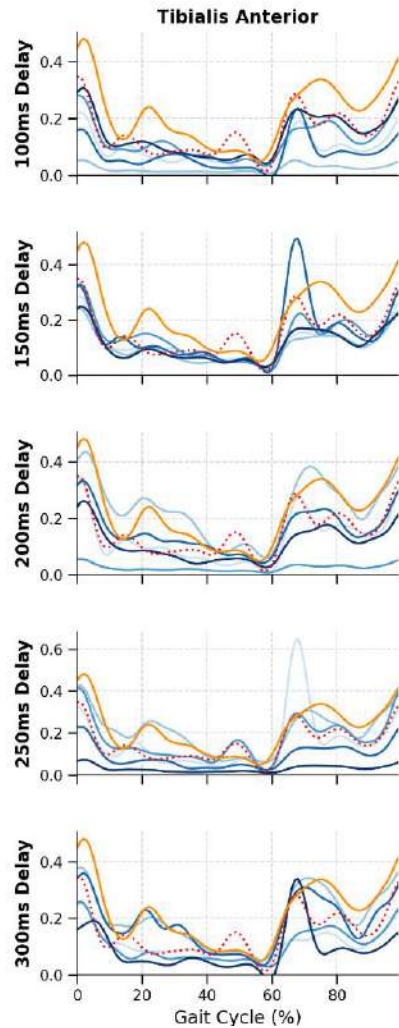
Rectus Femoris



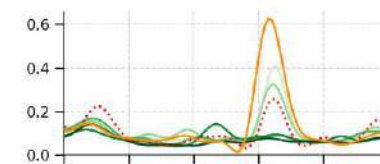
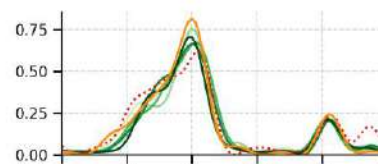
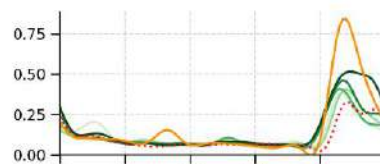
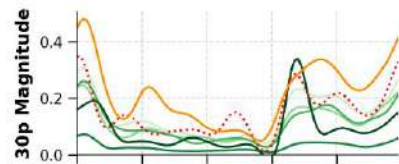
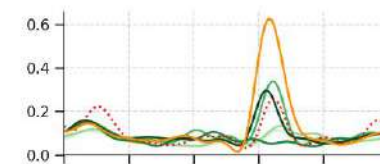
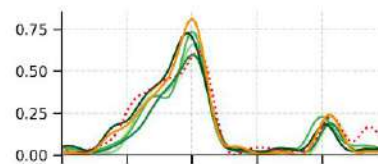
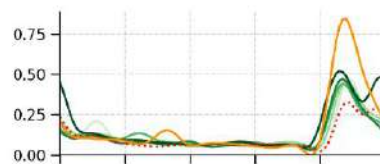
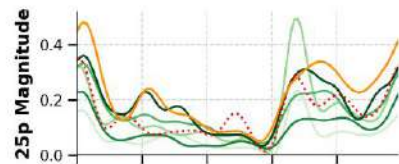
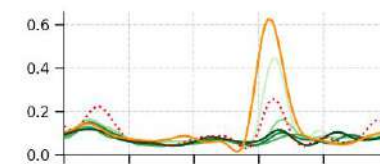
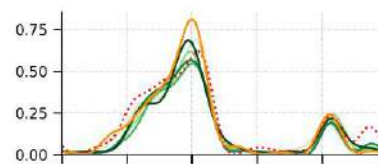
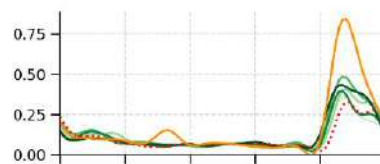
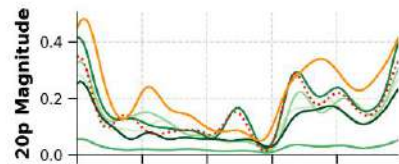
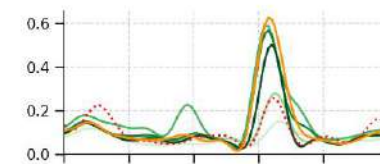
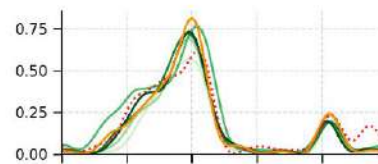
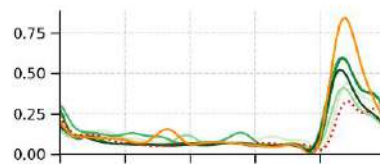
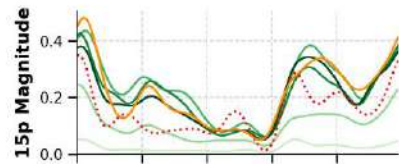
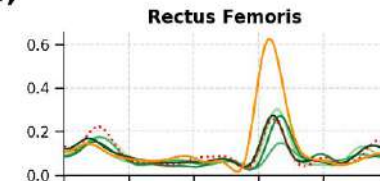
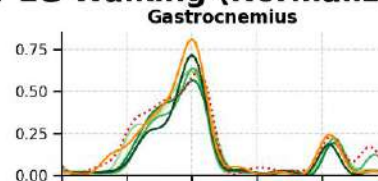
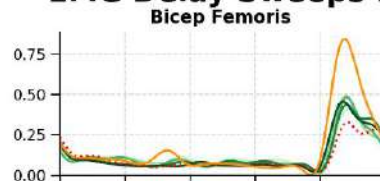
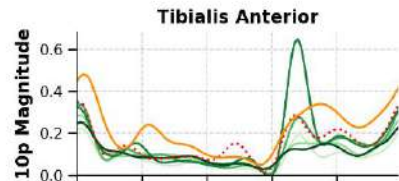
Subject: AB08_Adrian

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)



EMG Delay Sweeps for LG Walking (Normalized)

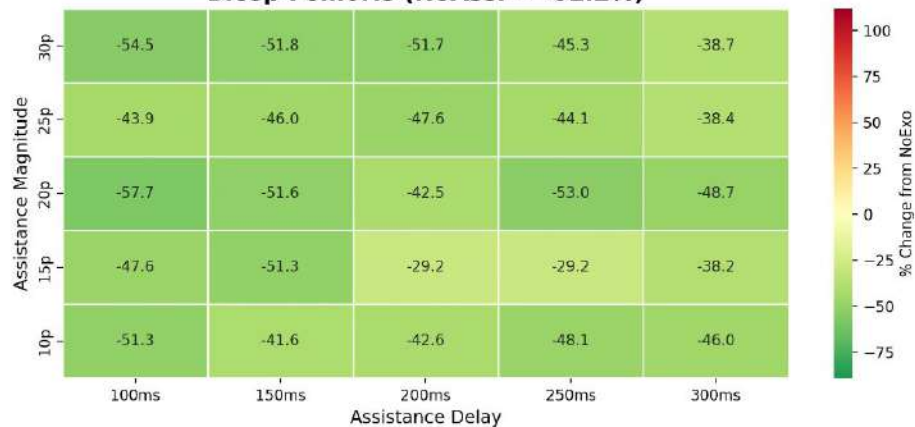


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

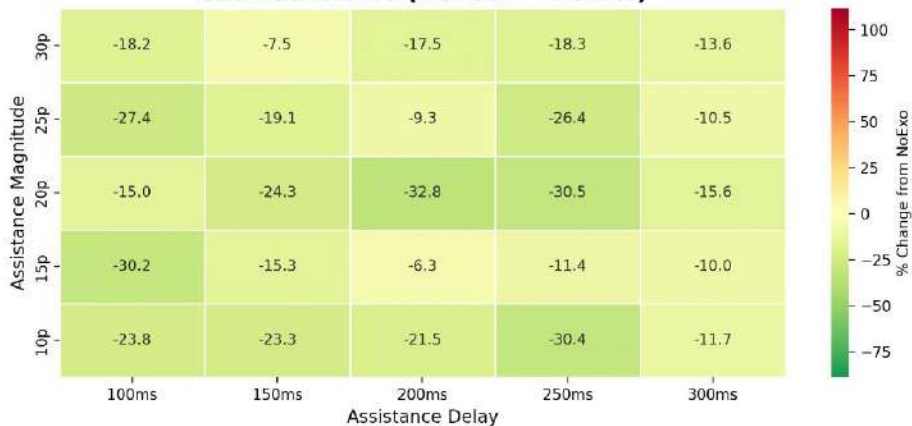
Tibialis Anterior (NoAssi = -27.4%)



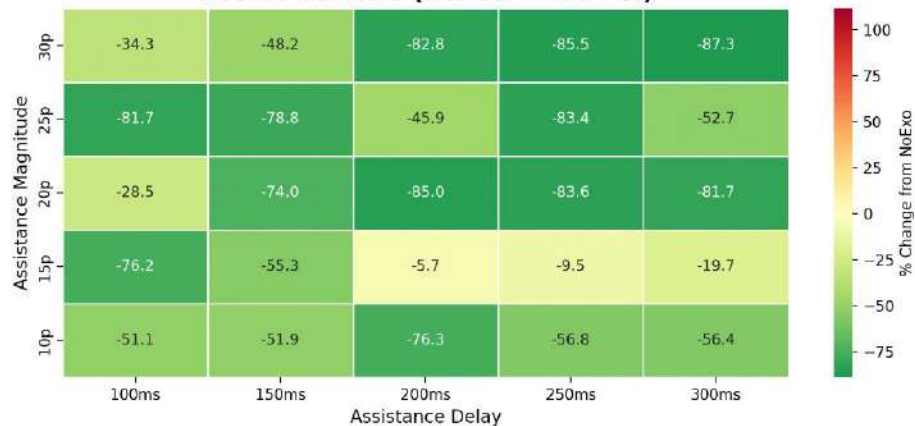
Bicep Femoris (NoAssi = -61.2%)



Gastrocnemius (NoAssi = -23.6%)

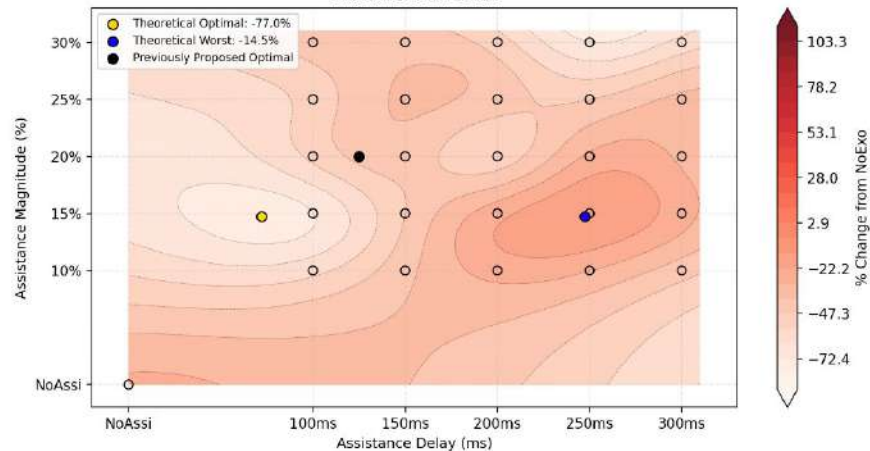


Rectus Femoris (NoAssi = -59.1%)

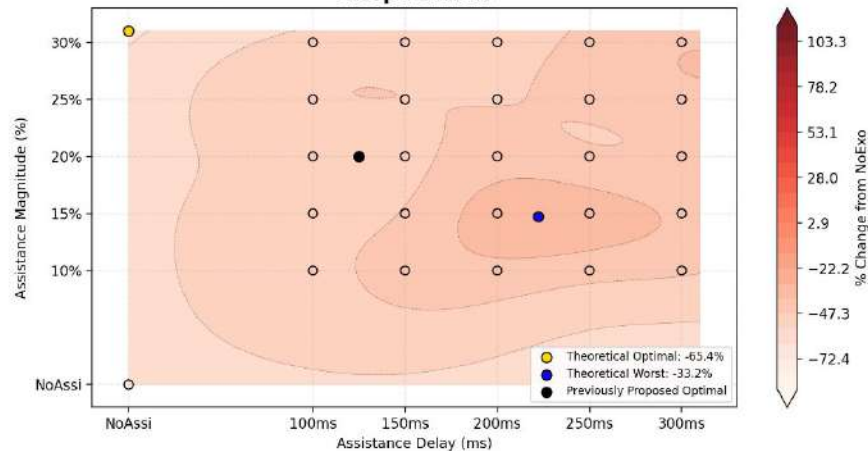


Peak EMG Activation Contour Map - LG Walking

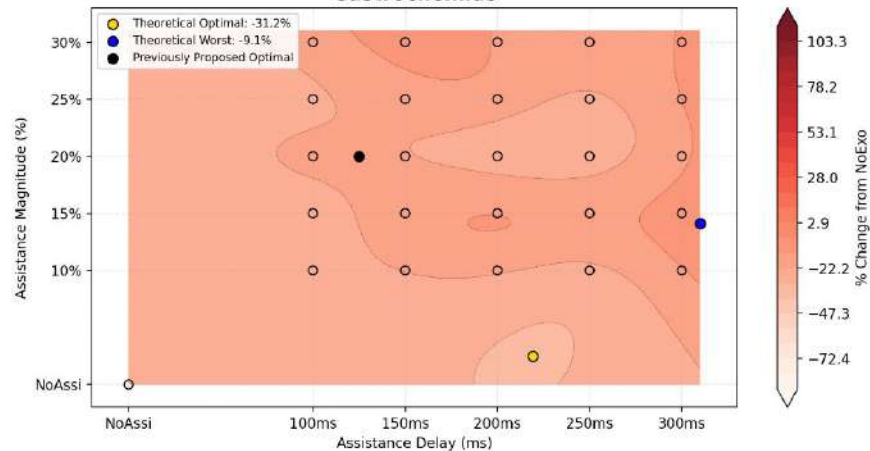
Tibialis Anterior



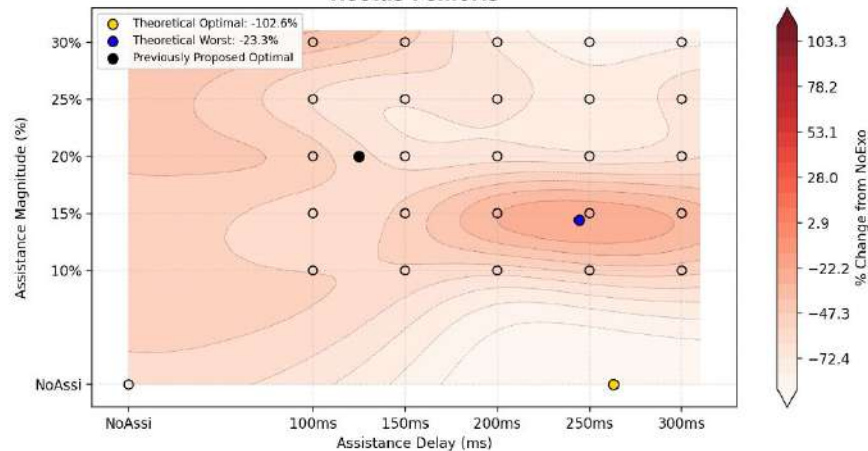
Bicep Femoris



Gastrocnemius

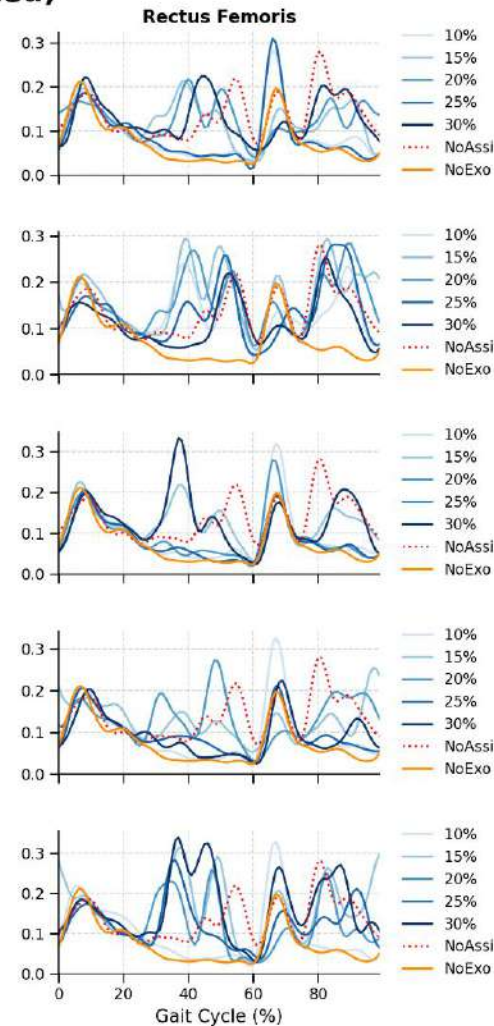
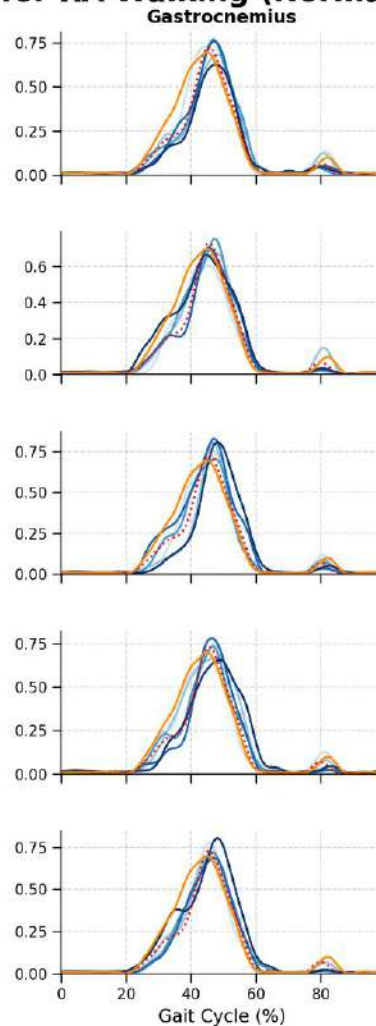
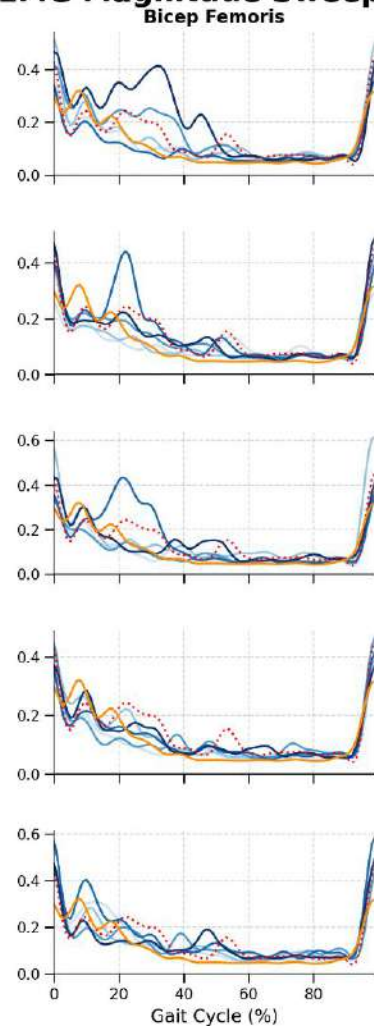
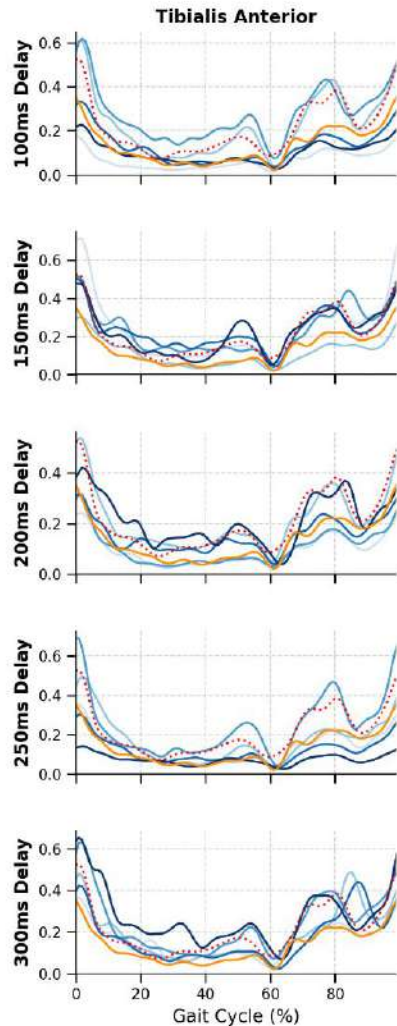


Rectus Femoris

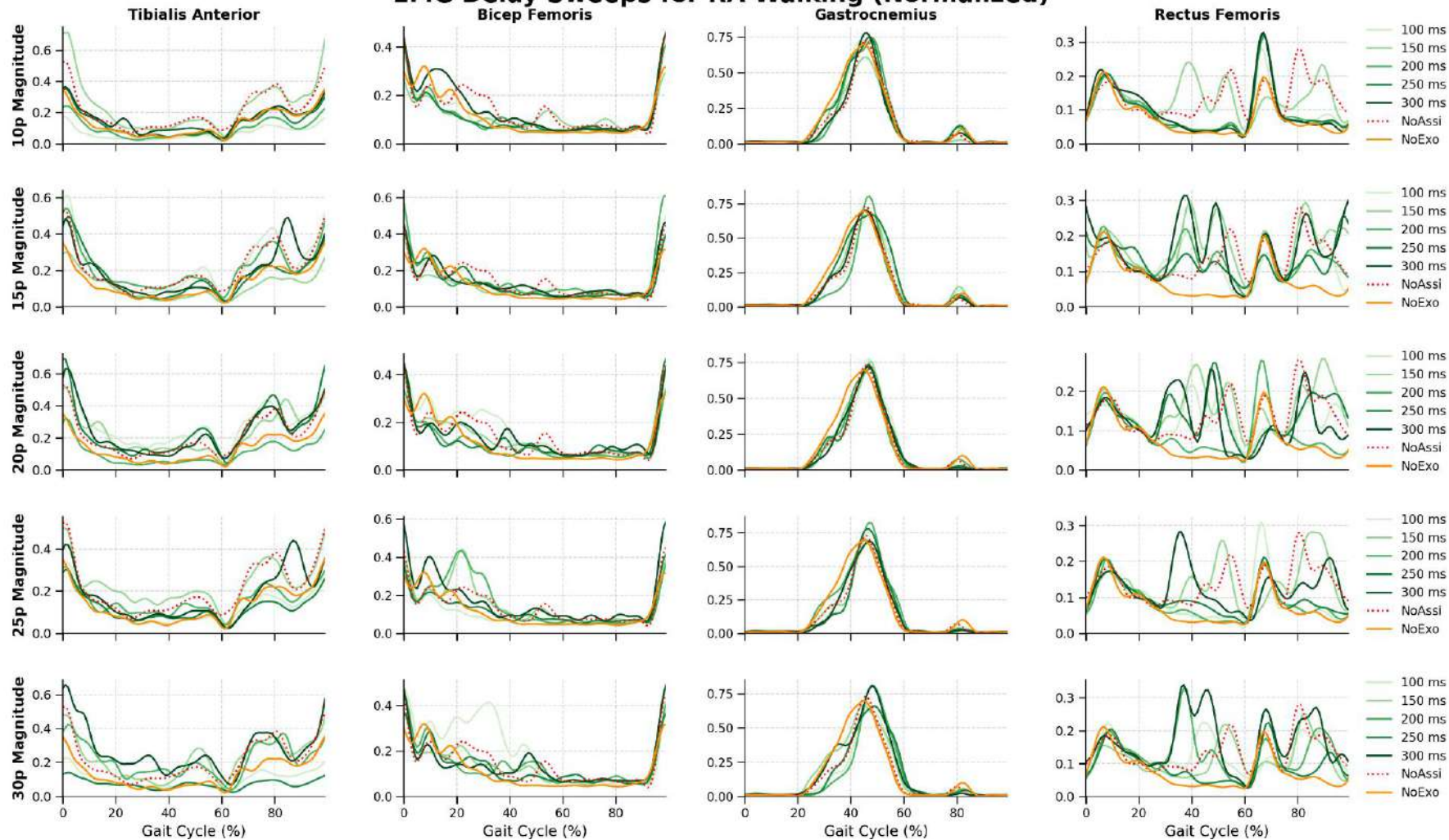


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

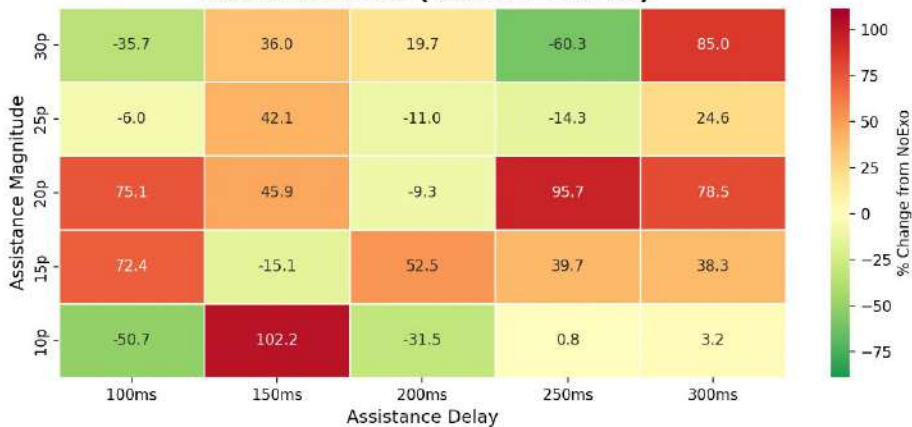


EMG Delay Sweeps for RA Walking (Normalized)

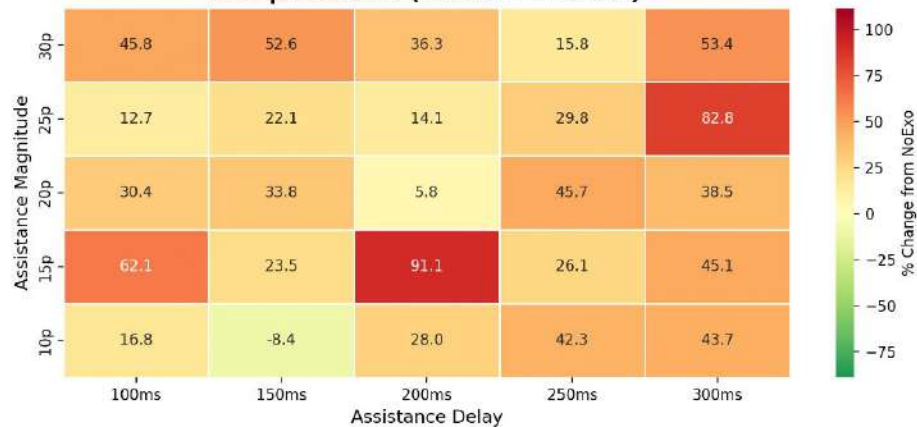


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

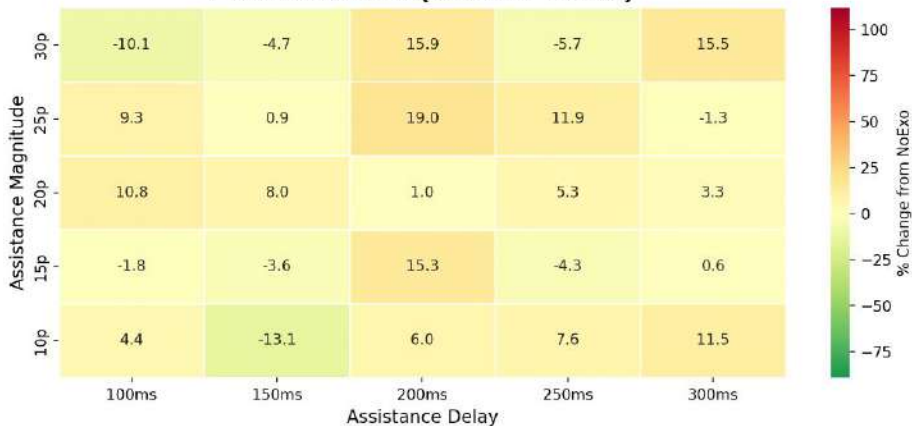
Tibialis Anterior (NoAssi = 49.5%)



Bicep Femoris (NoAssi = 40.9%)



Gastrocnemius (NoAssi = 4.4%)

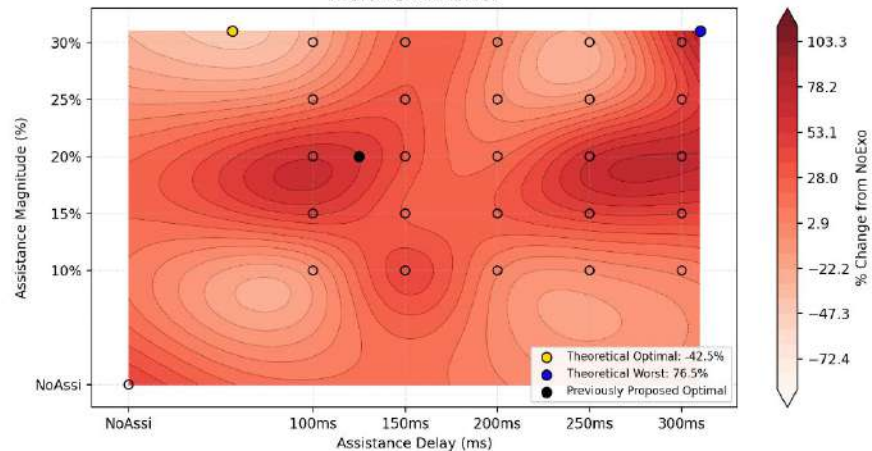


Rectus Femoris (NoAssi = 41.5%)

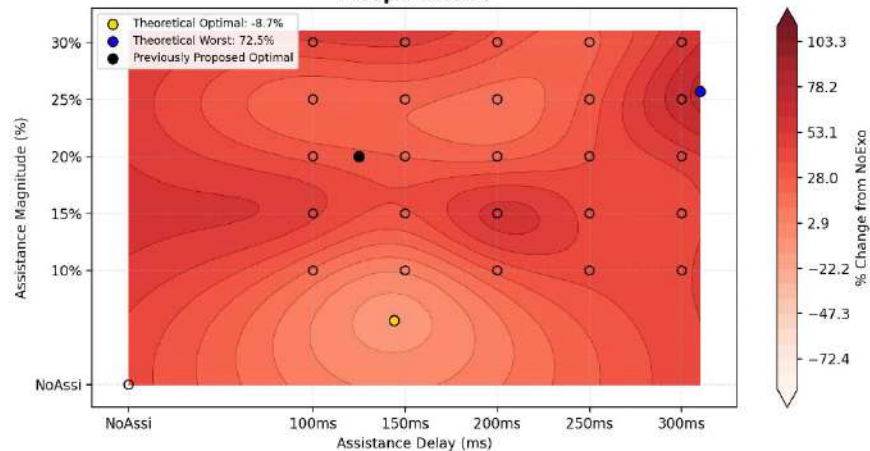


Peak EMG Activation Contour Map - RA Walking

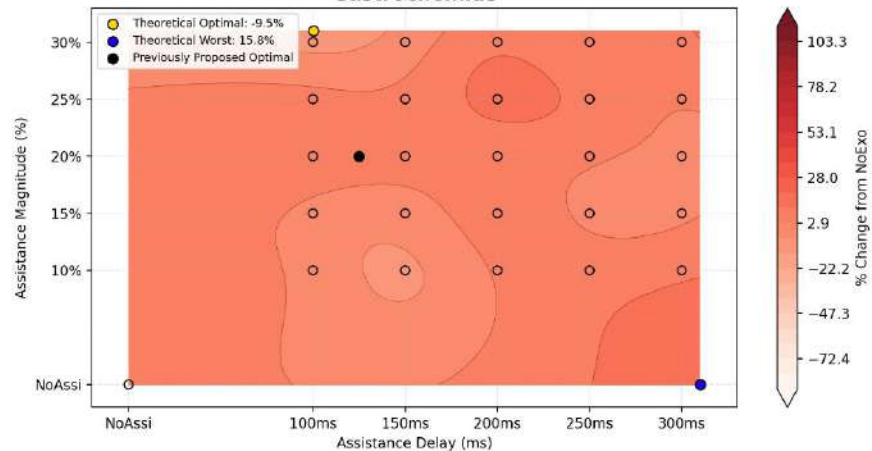
Tibialis Anterior



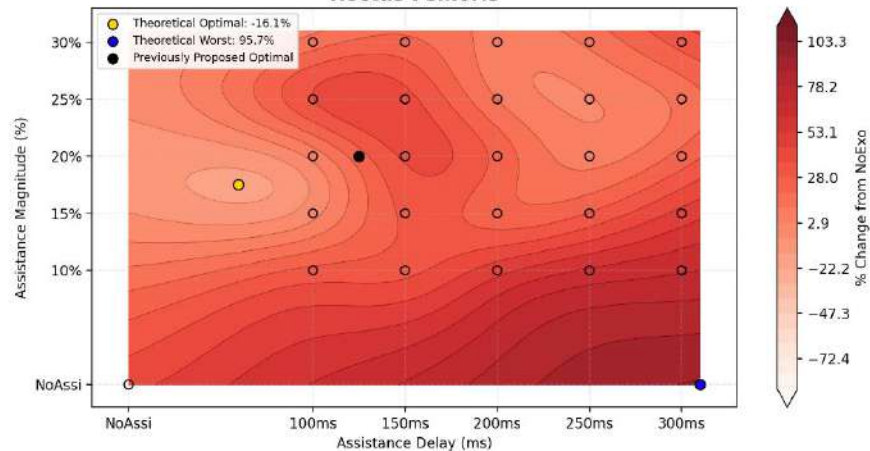
Bicep Femoris



Gastrocnemius

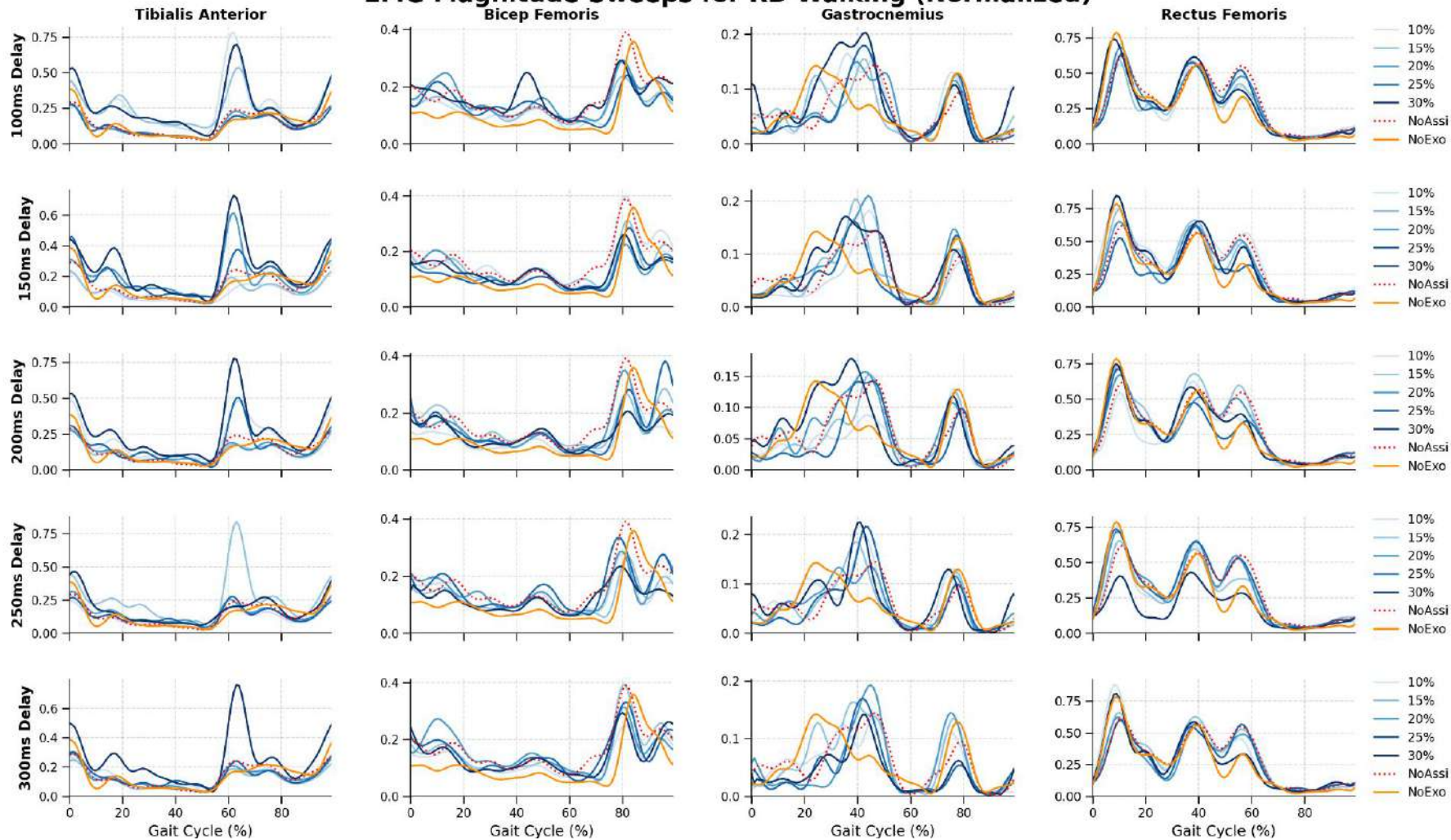


Rectus Femoris

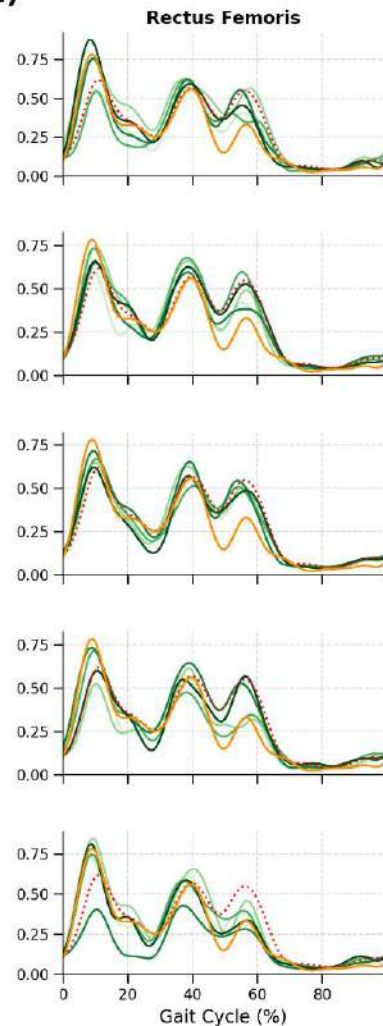
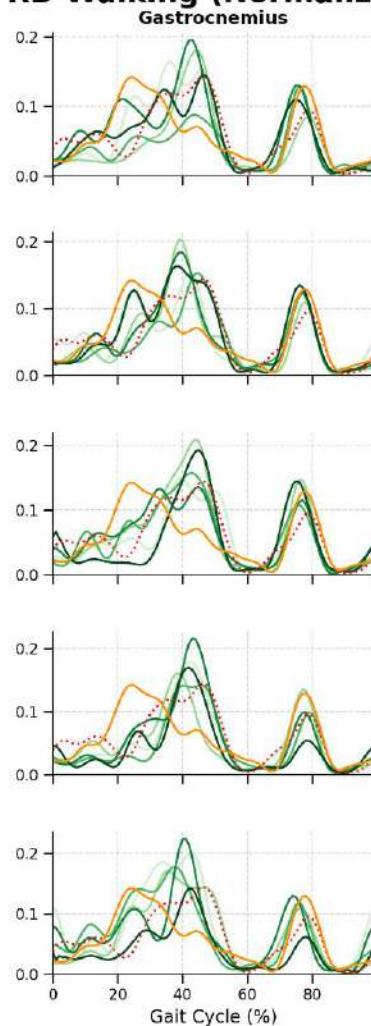
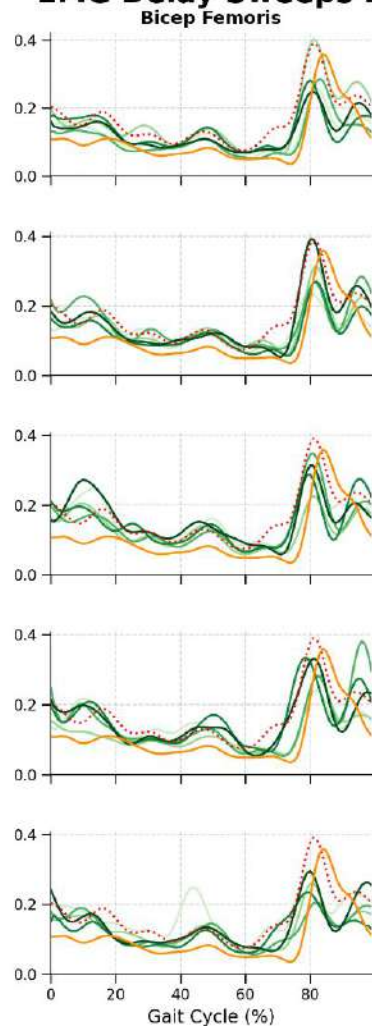
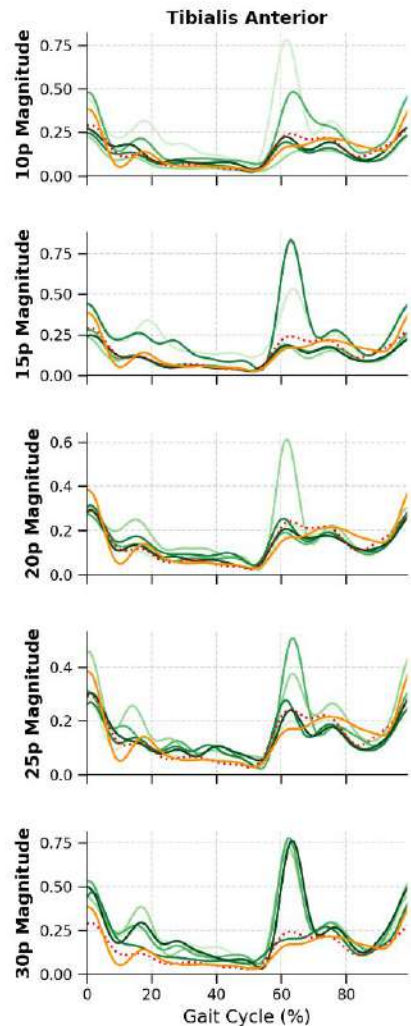


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

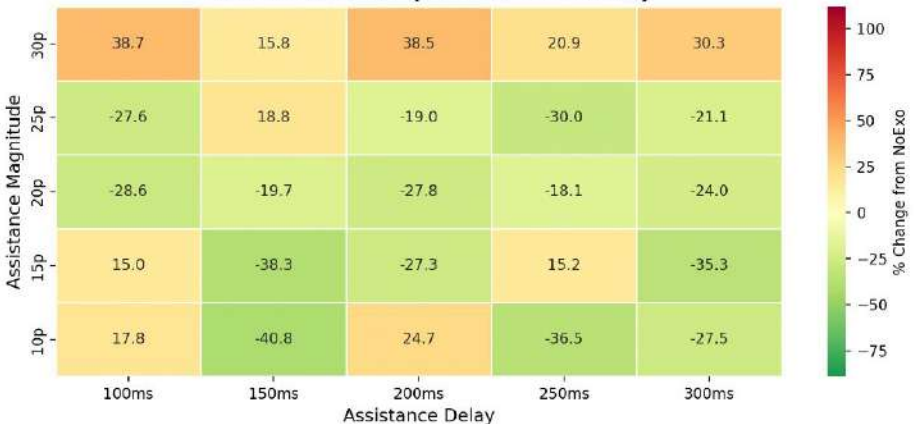


EMG Delay Sweeps for RD Walking (Normalized)



Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

Tibialis Anterior (NoAssi = -24.1%)



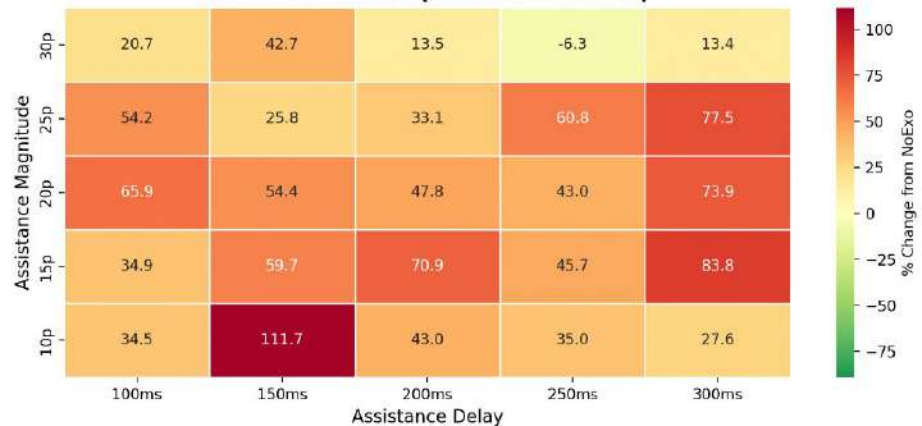
Bicep Femoris (NoAssi = -15.8%)



Gastrocnemius (NoAssi = 1.1%)

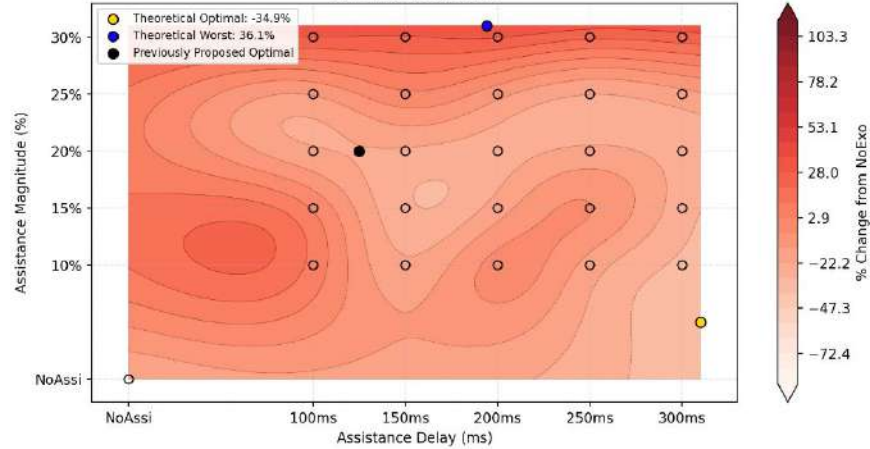


Rectus Femoris (NoAssi = 90.0%)

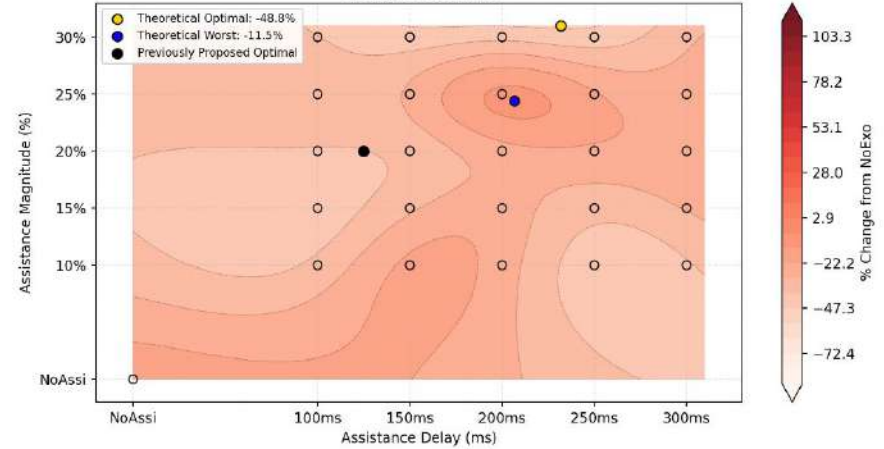


Peak EMG Activation Contour Map - RD Walking

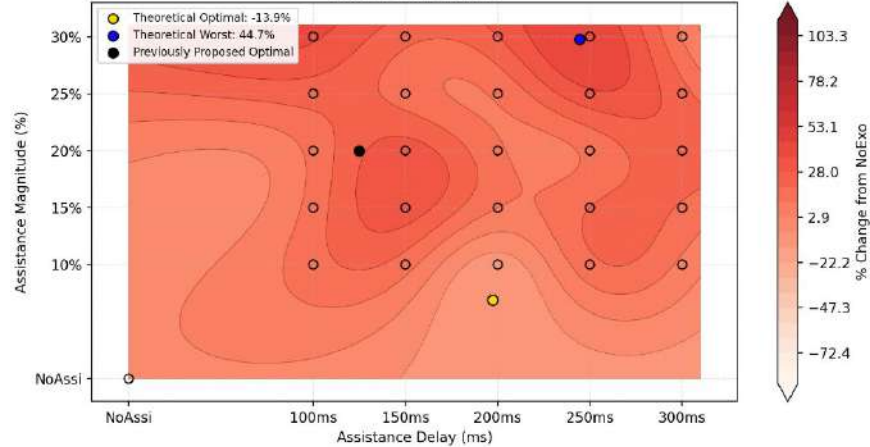
Tibialis Anterior



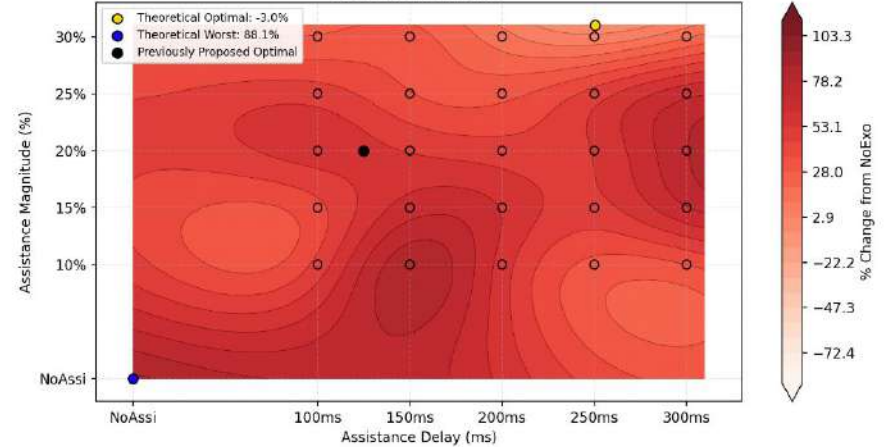
Bicep Femoris



Gastrocnemius



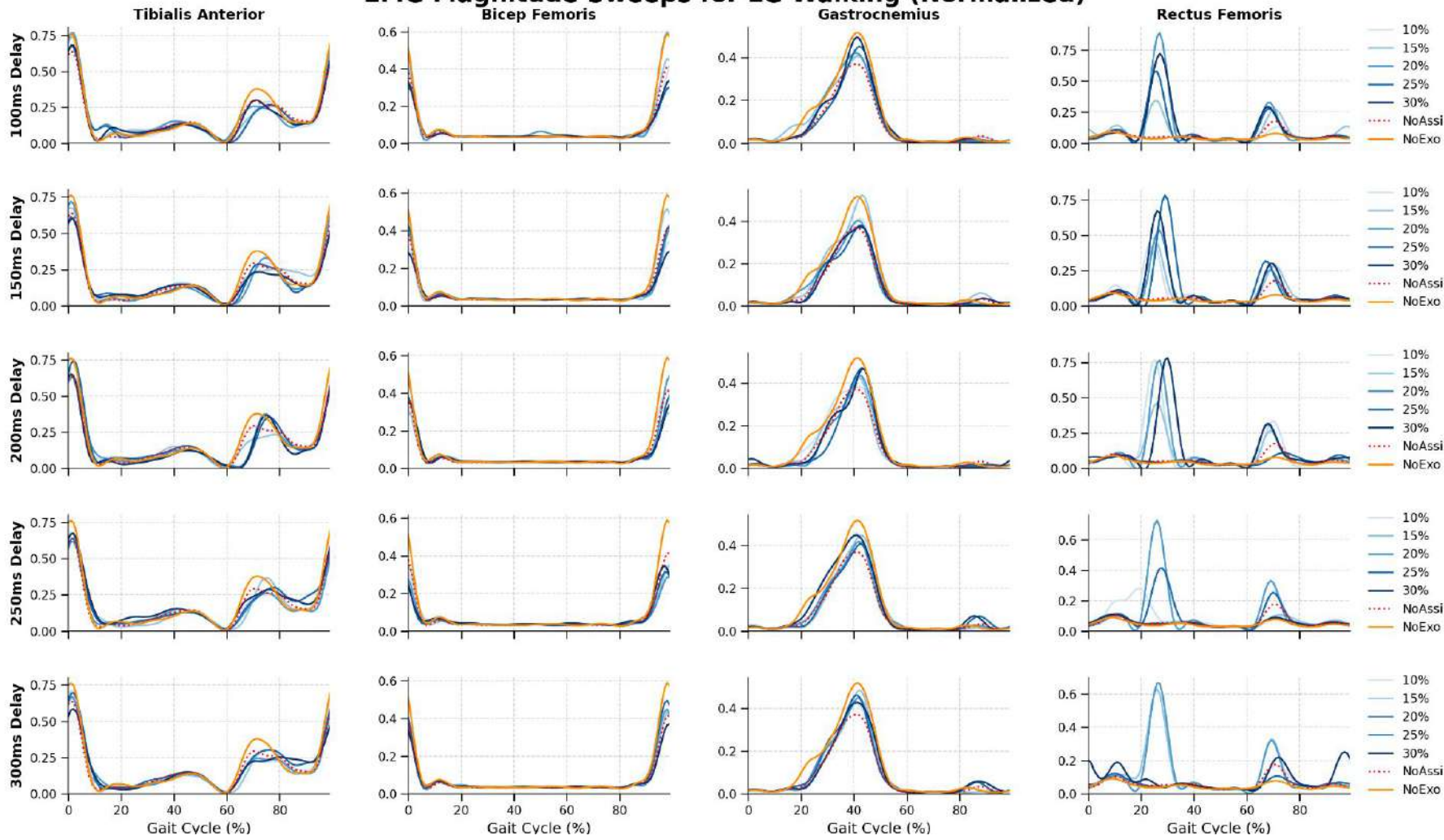
Rectus Femoris



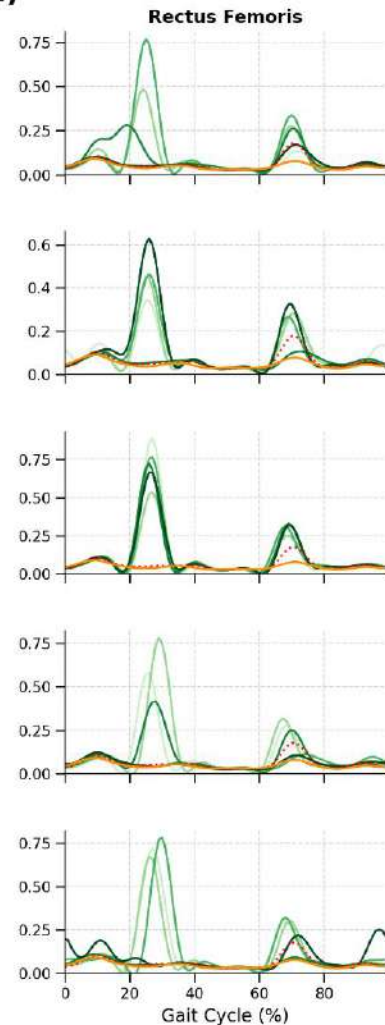
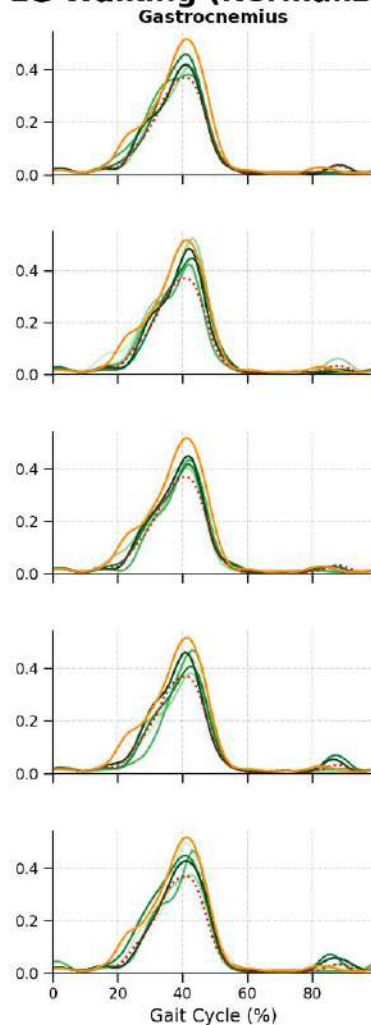
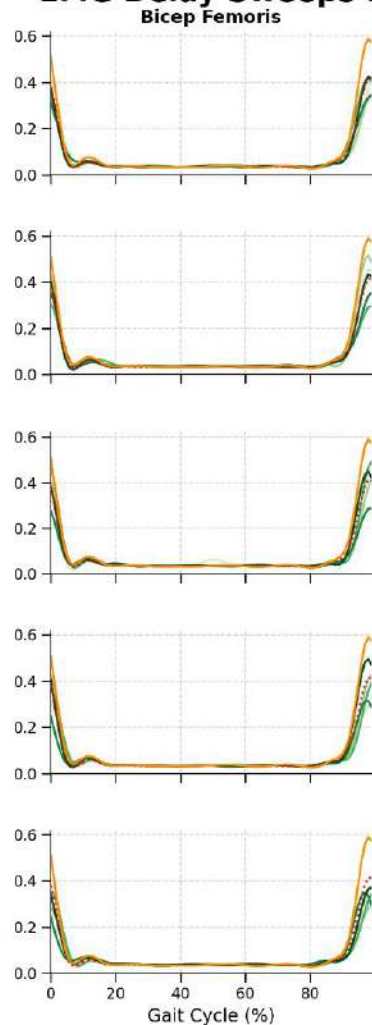
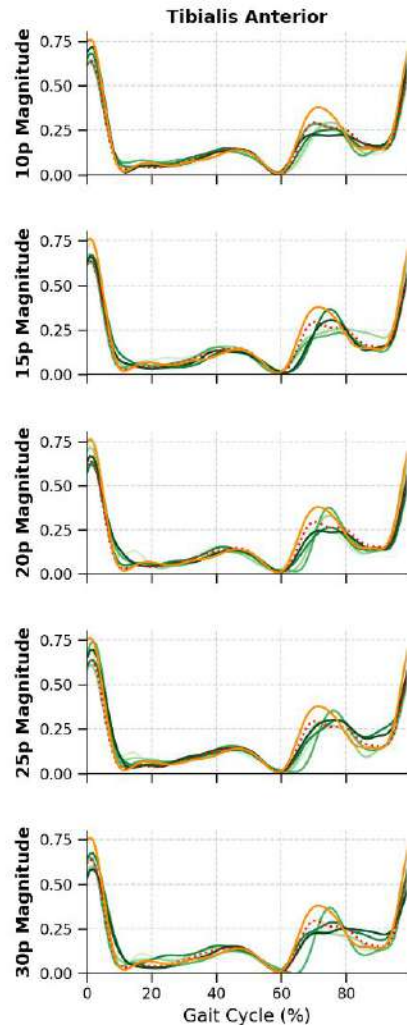
Subject: AB11_Ryan

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

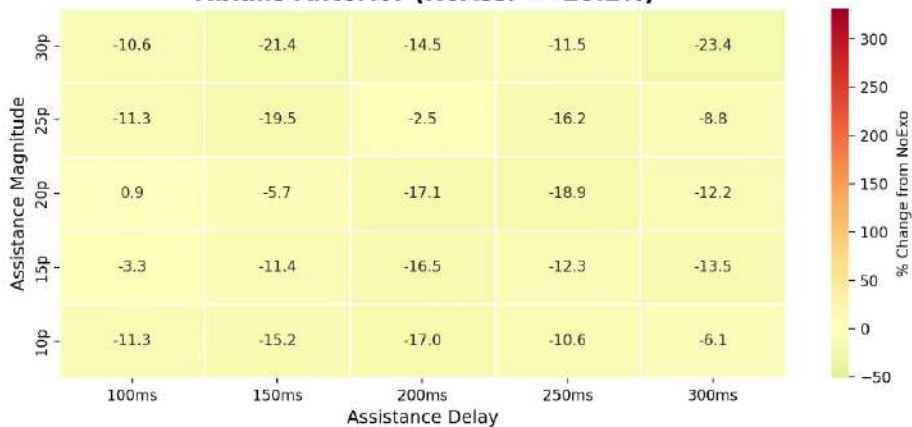


EMG Delay Sweeps for LG Walking (Normalized)

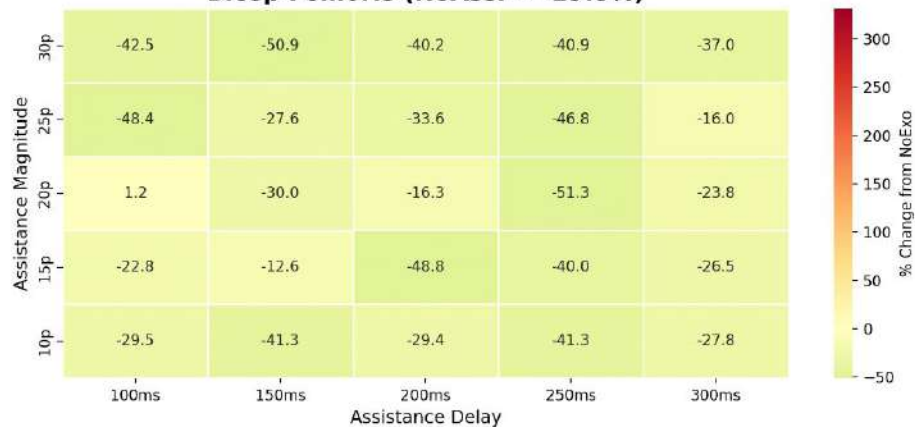


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

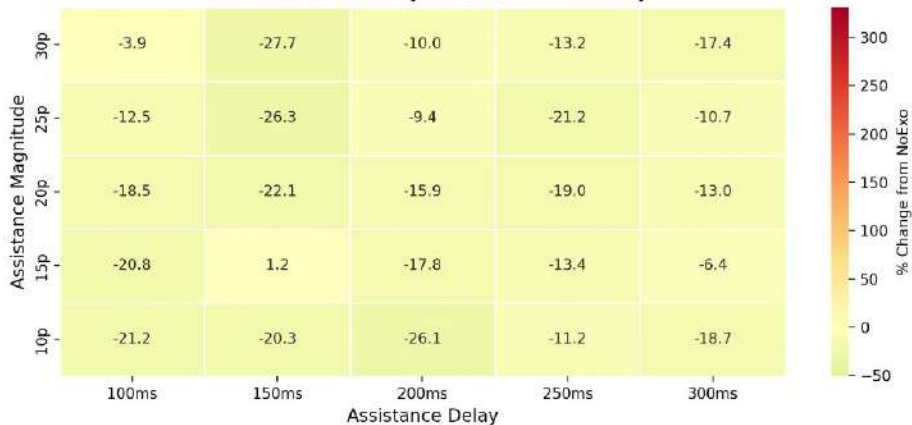
Tibialis Anterior (NoAssi = -16.2%)



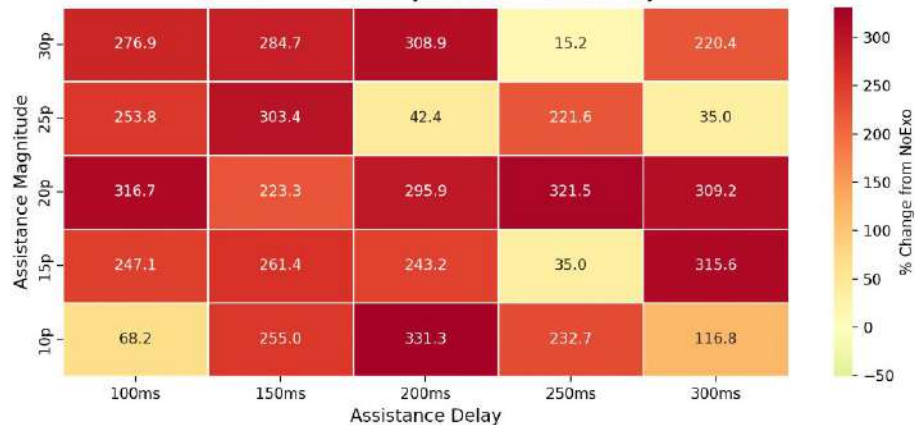
Bicep Femoris (NoAssi = -29.5%)



Gastrocnemius (NoAssi = -28.4%)

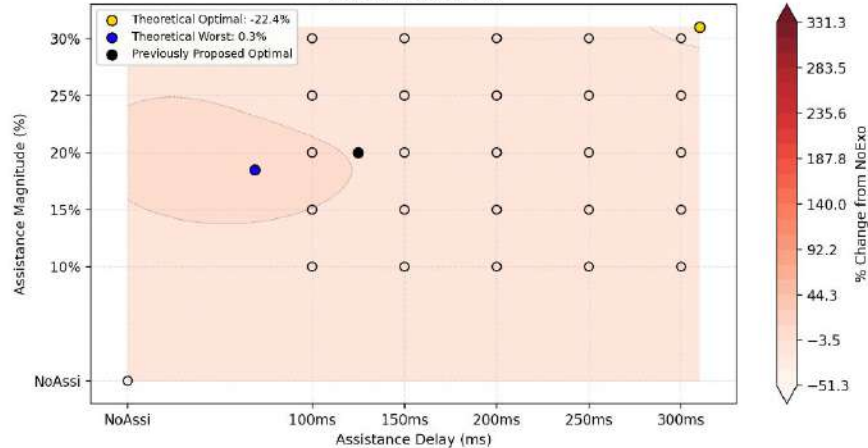


Rectus Femoris (NoAssi = 126.5%)

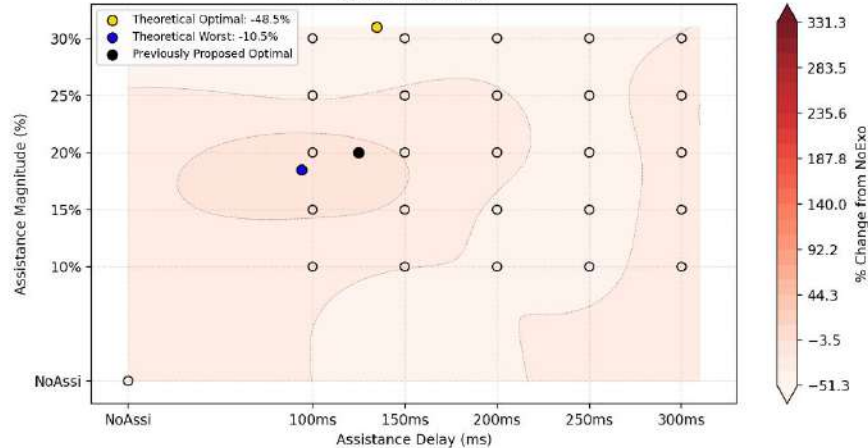


Peak EMG Activation Contour Map - LG Walking

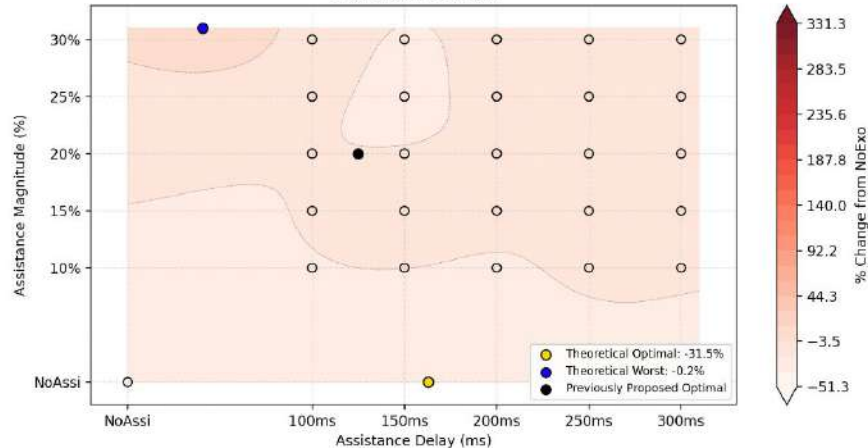
Tibialis Anterior



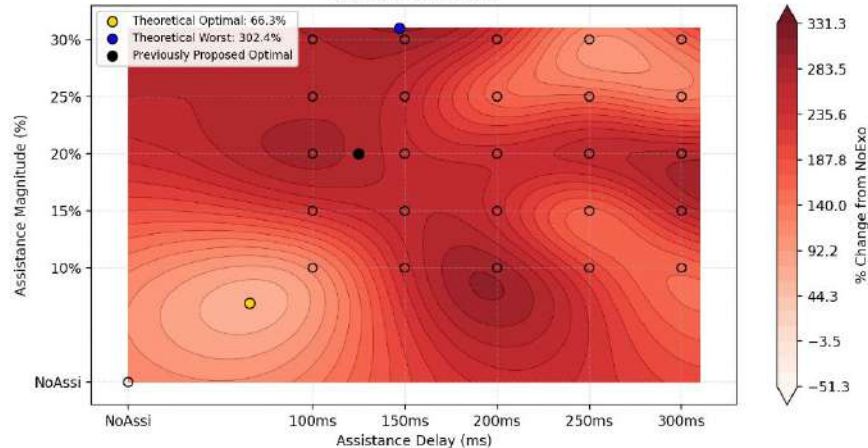
Bicep Femoris



Gastrocnemius

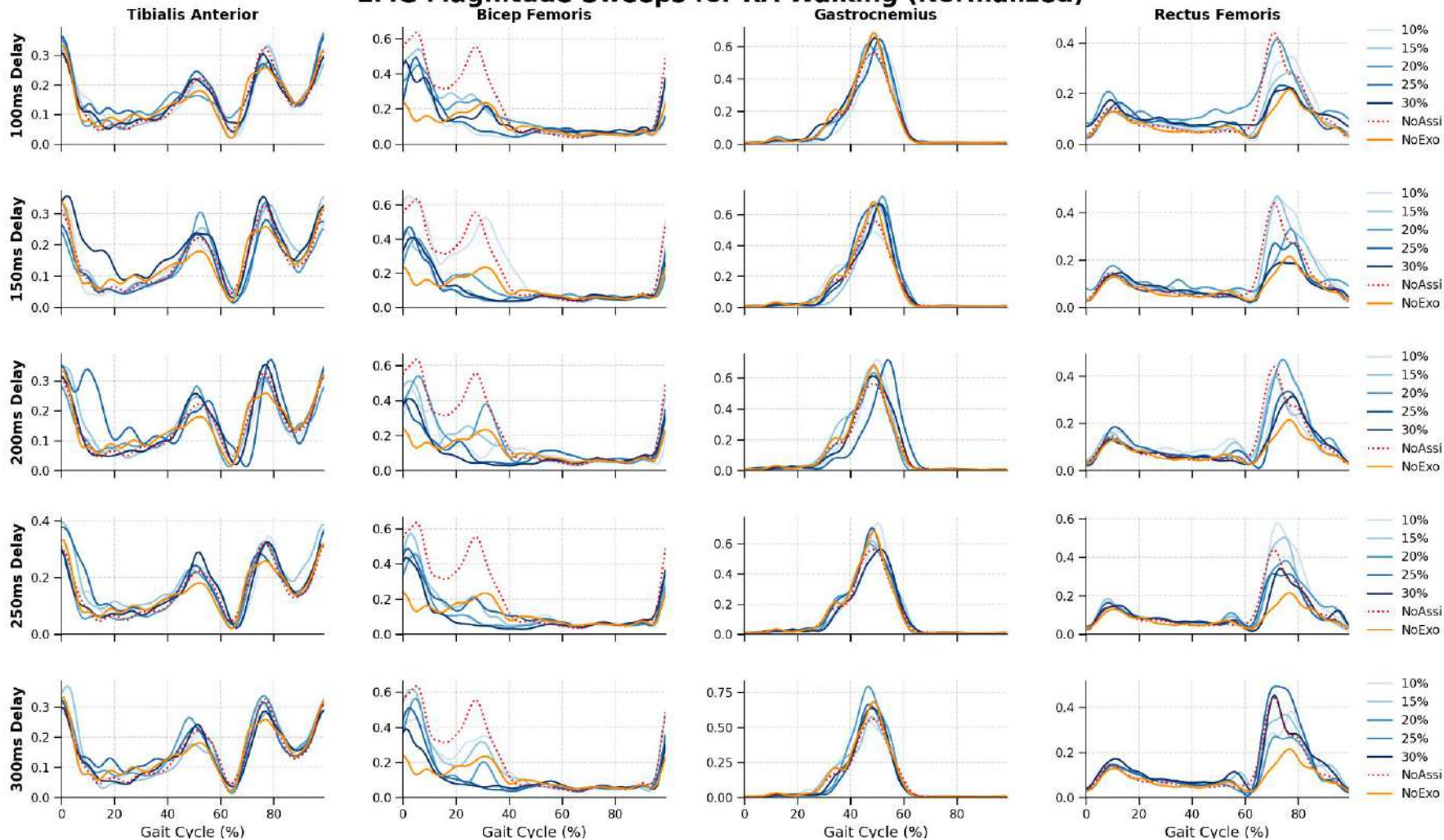


Rectus Femoris

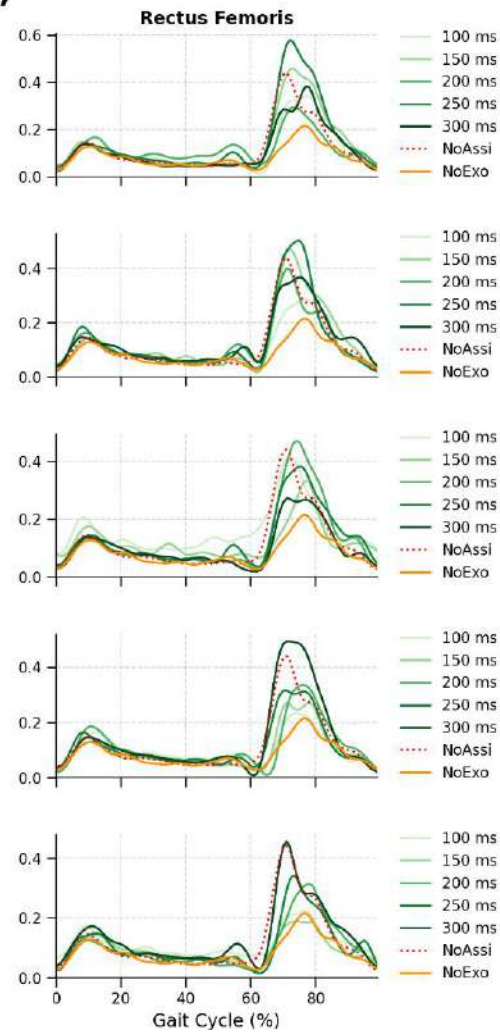
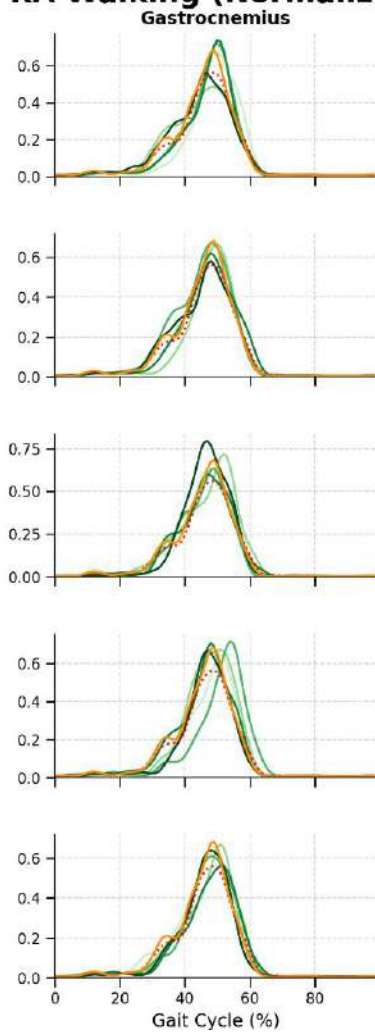
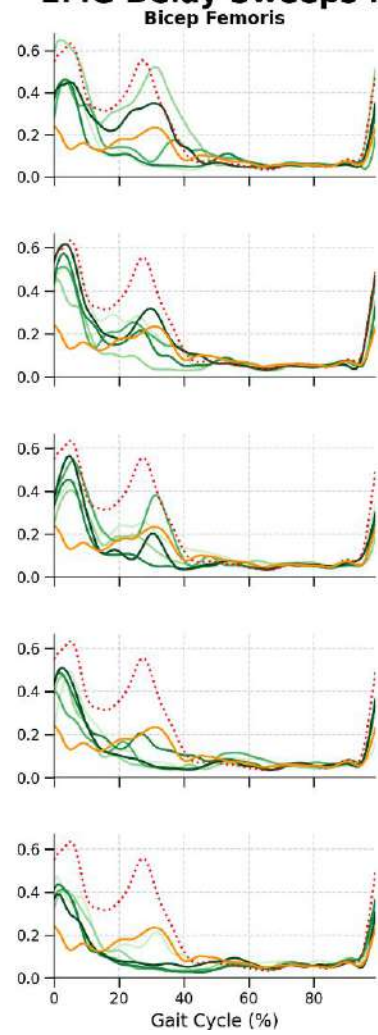
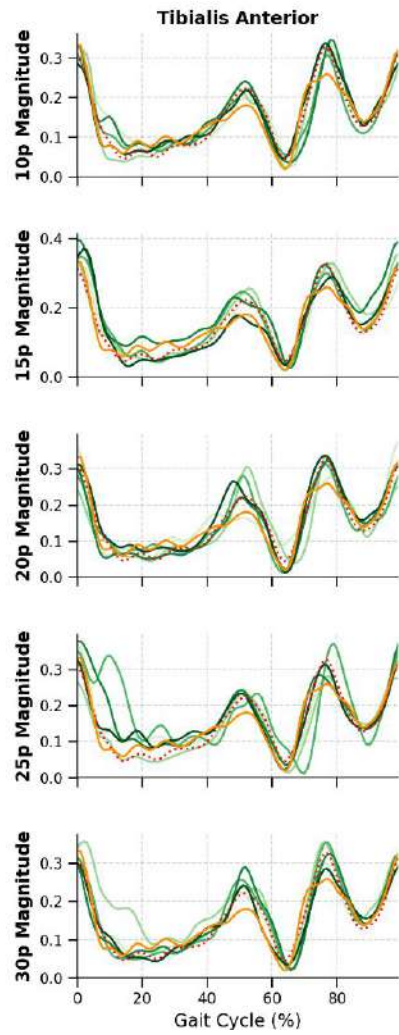


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

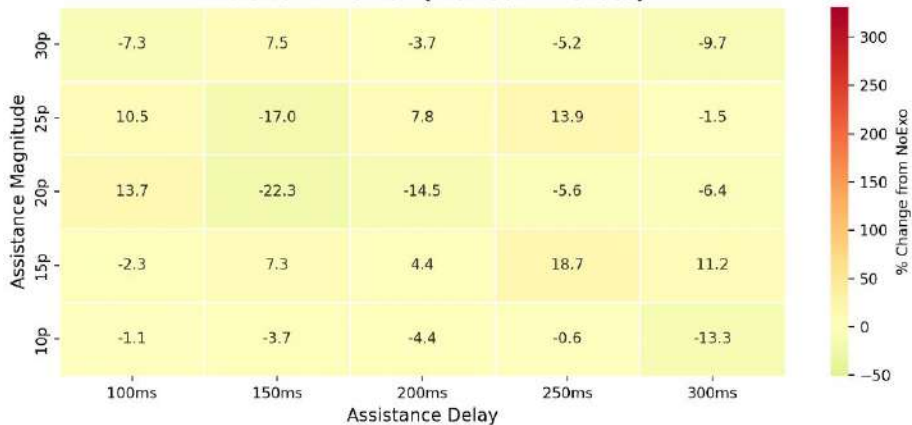


EMG Delay Sweeps for RA Walking (Normalized)

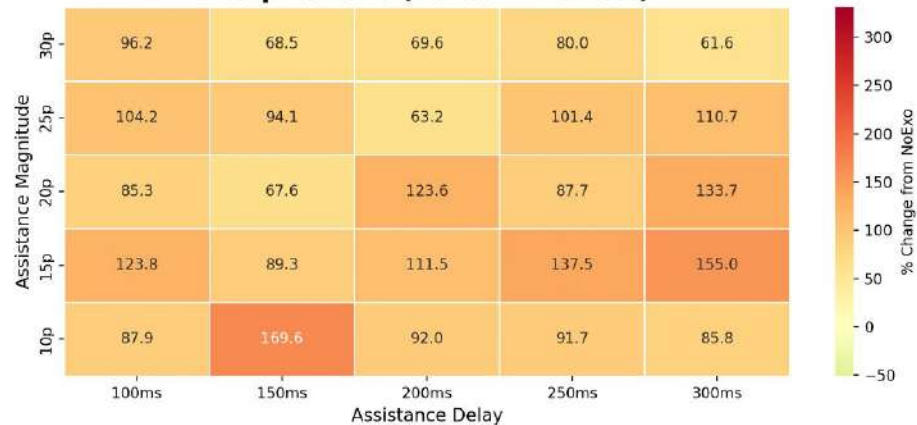


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

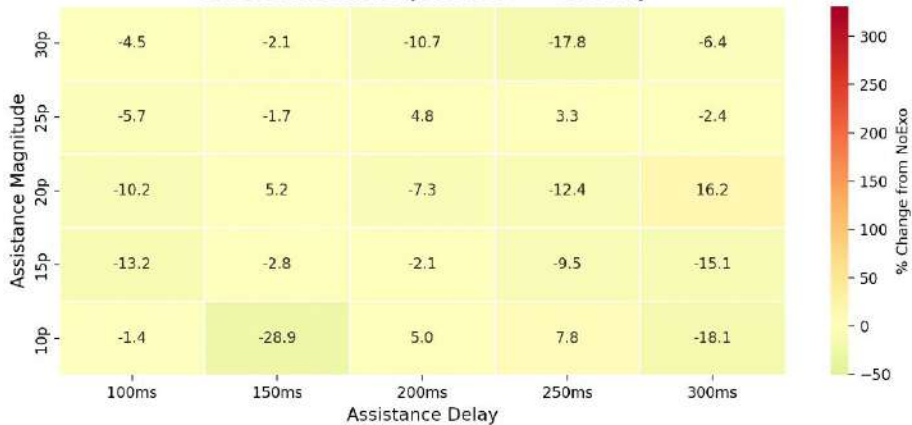
Tibialis Anterior (NoAssi = -5.6%)



Bicep Femoris (NoAssi = 162.6%)



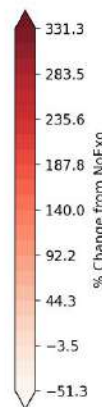
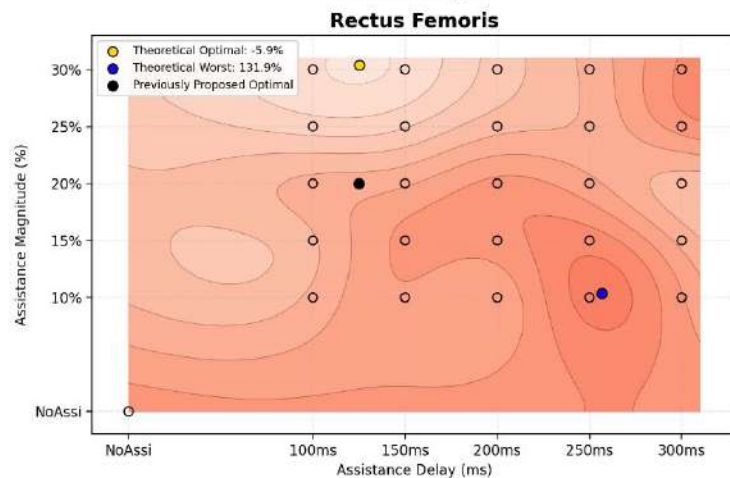
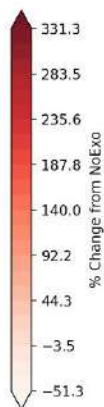
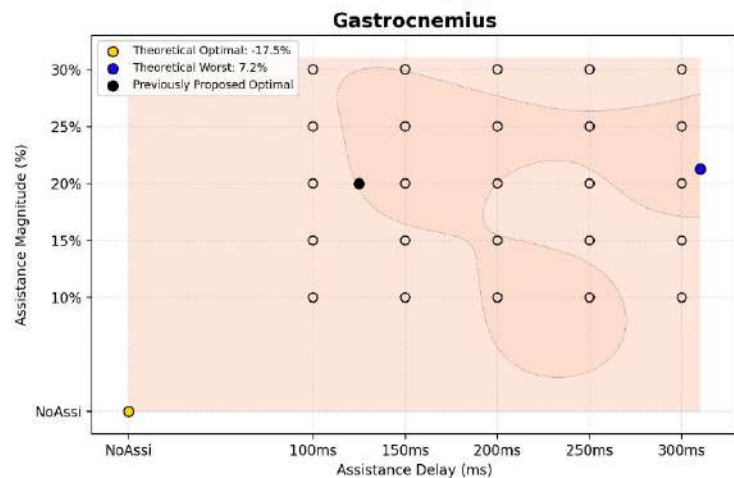
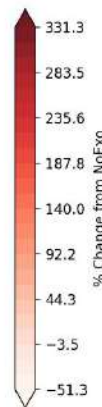
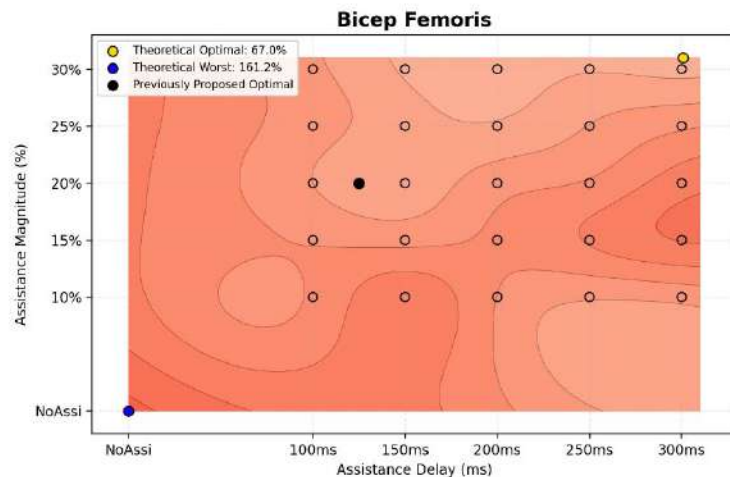
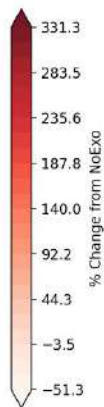
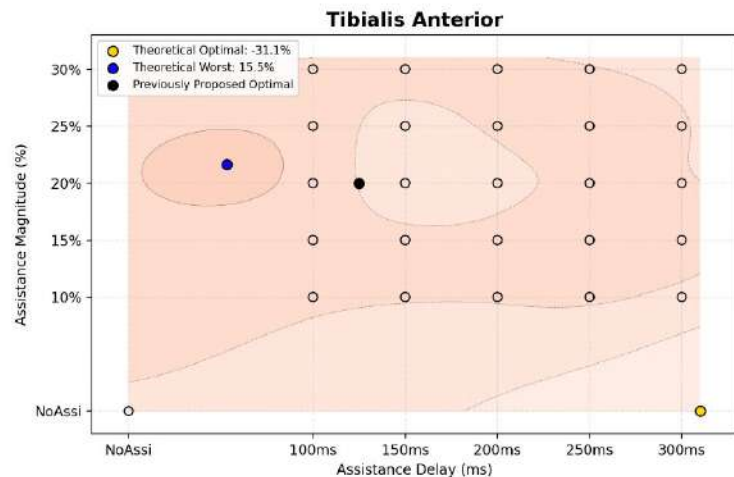
Gastrocnemius (NoAssi = -17.8%)



Rectus Femoris (NoAssi = 104.7%)

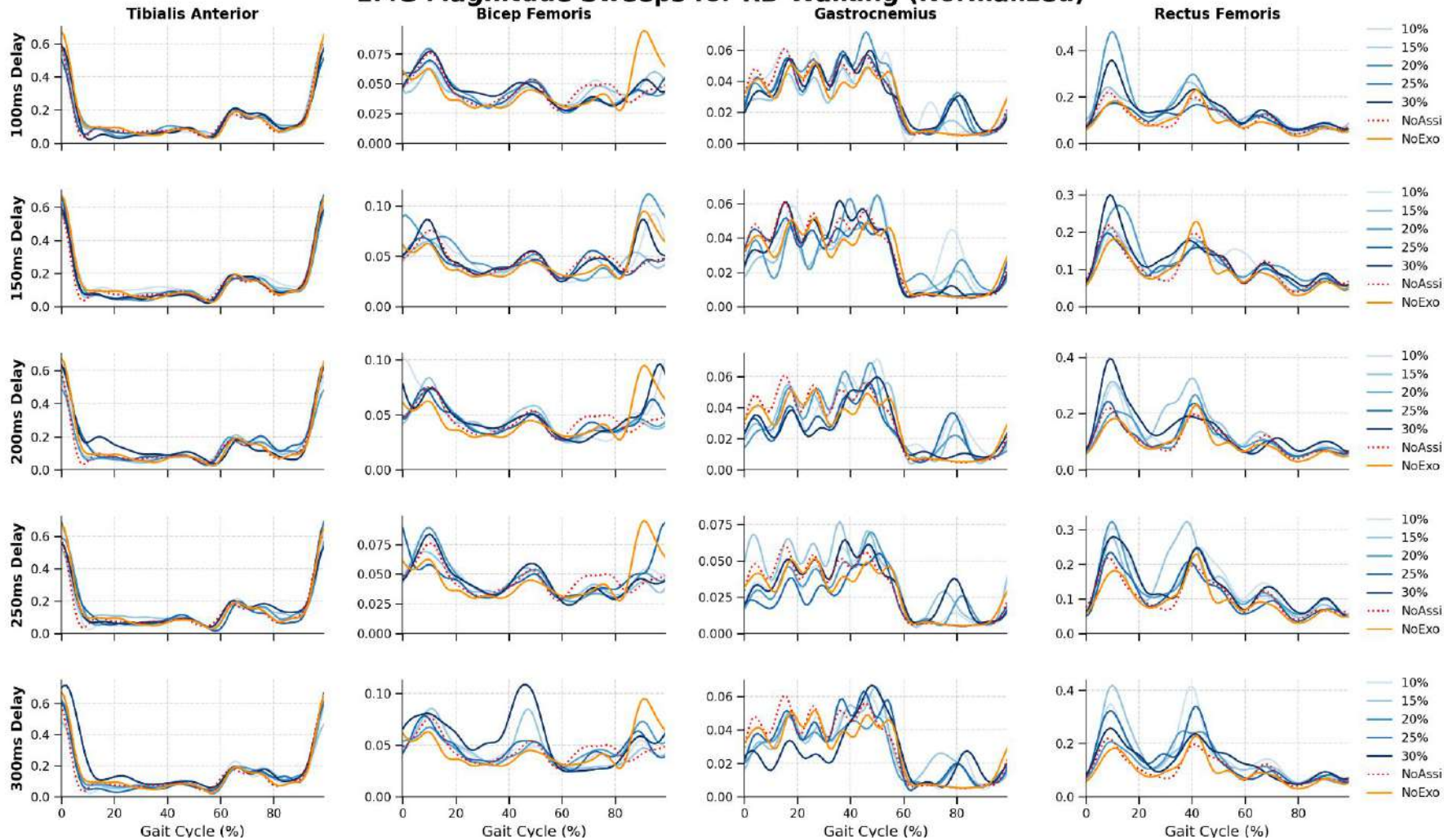


Peak EMG Activation Contour Map - RA Walking

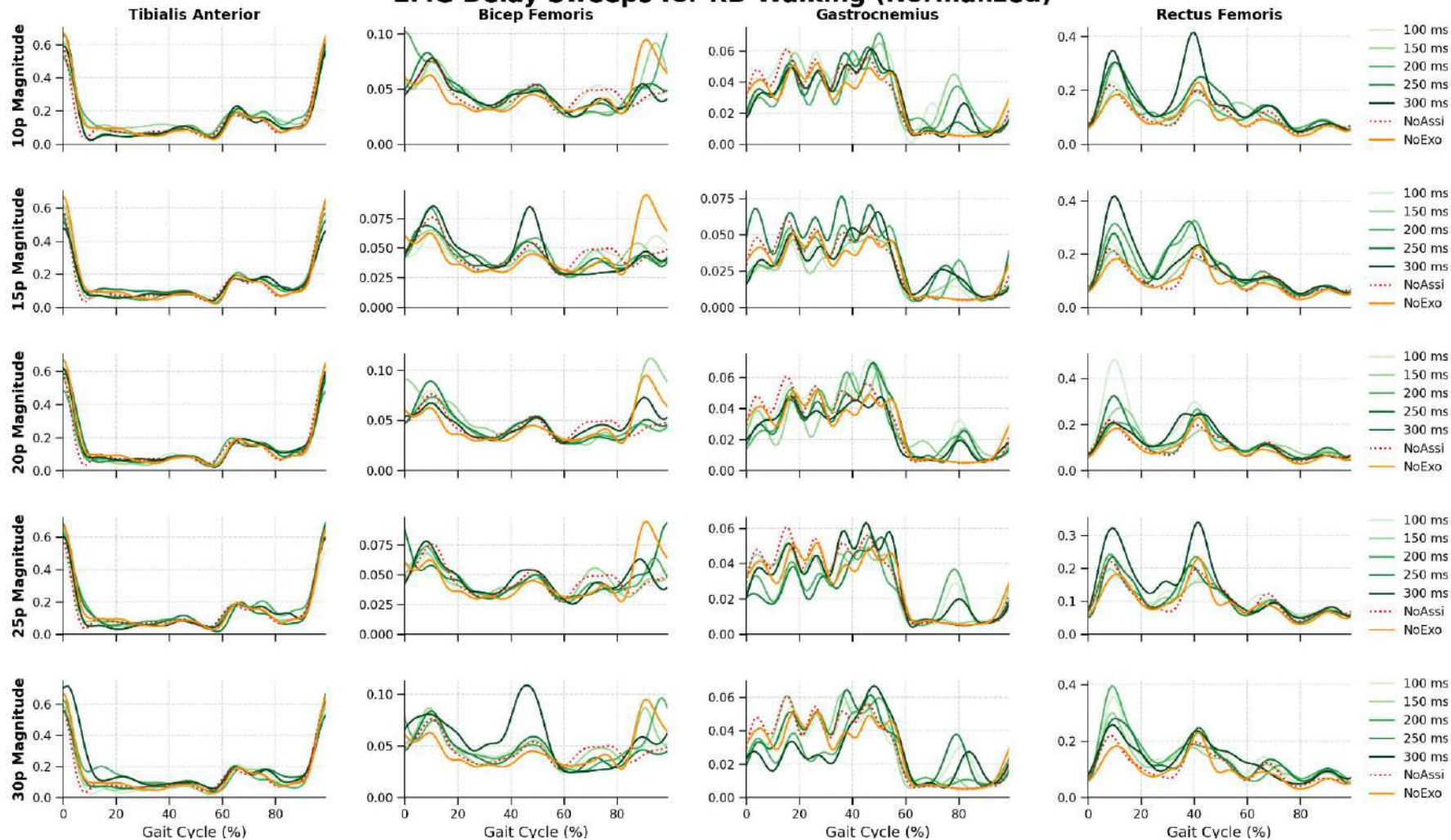


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

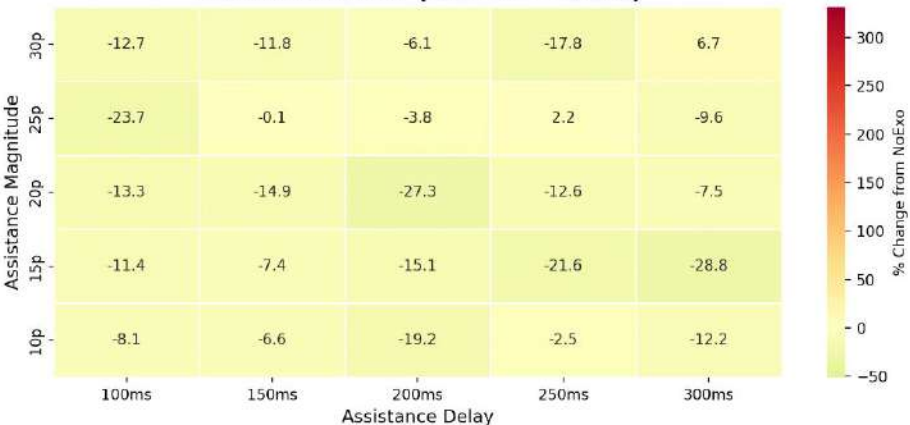


EMG Delay Sweeps for RD Walking (Normalized)

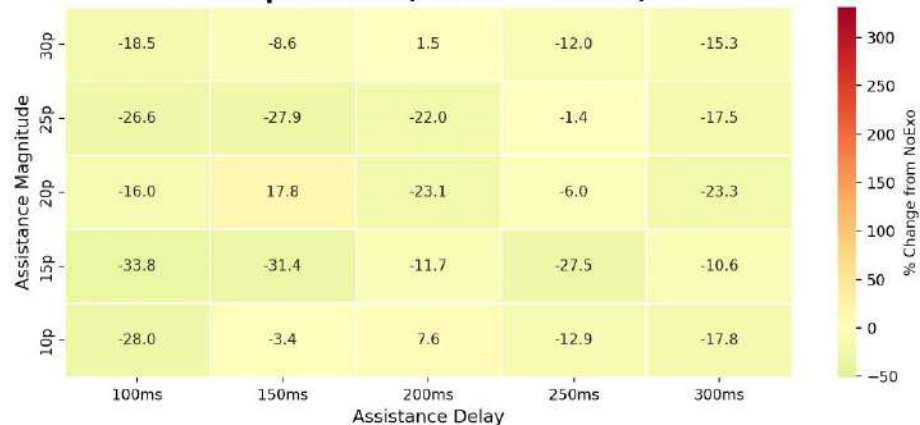


Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

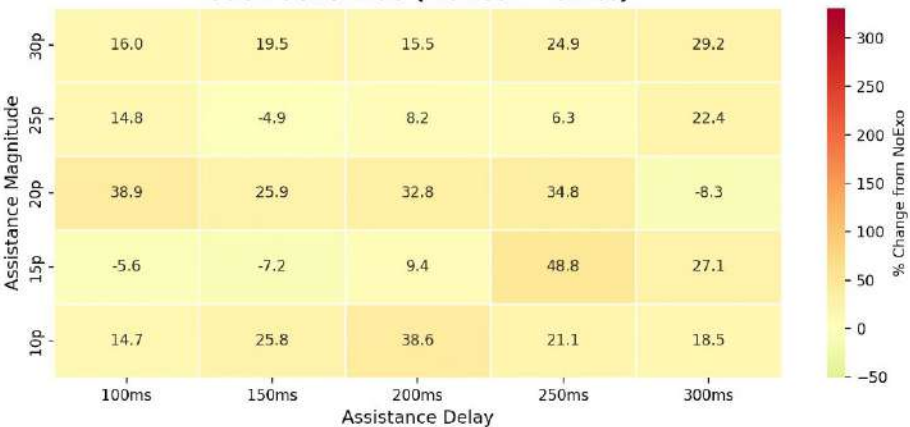
Tibialis Anterior (NoAssi = -8.5%)



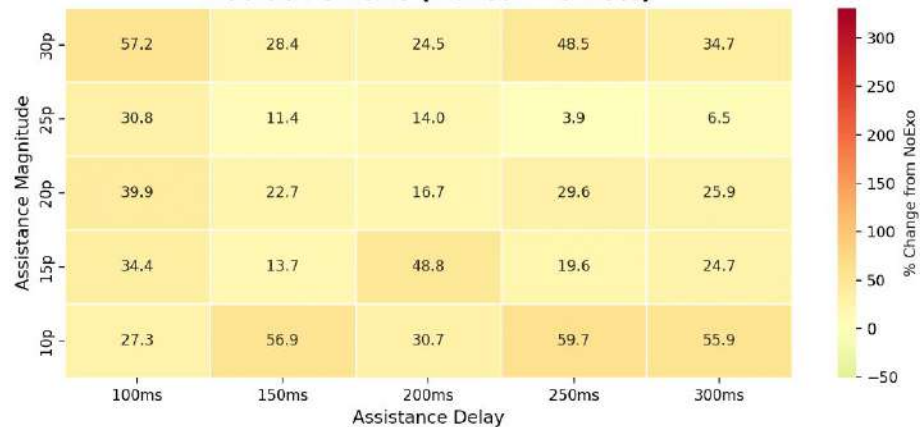
Bicep Femoris (NoAssi = -20.0%)



Gastrocnemius (NoAssi = 8.4%)

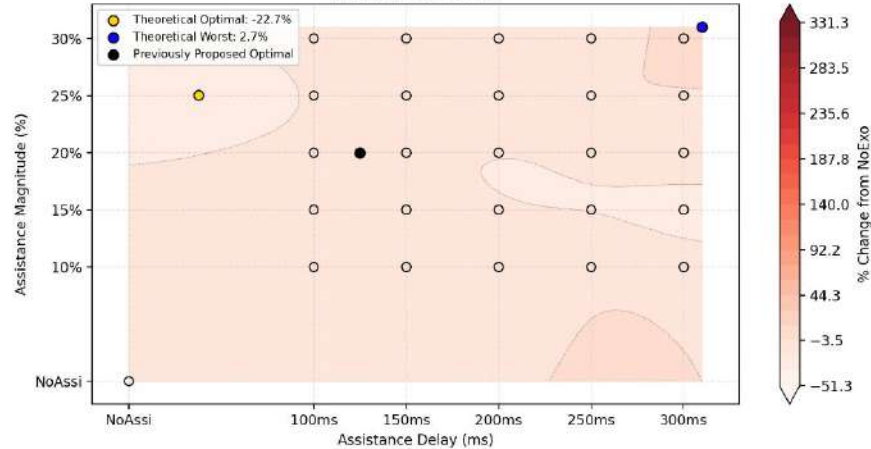


Rectus Femoris (NoAssi = 34.8%)

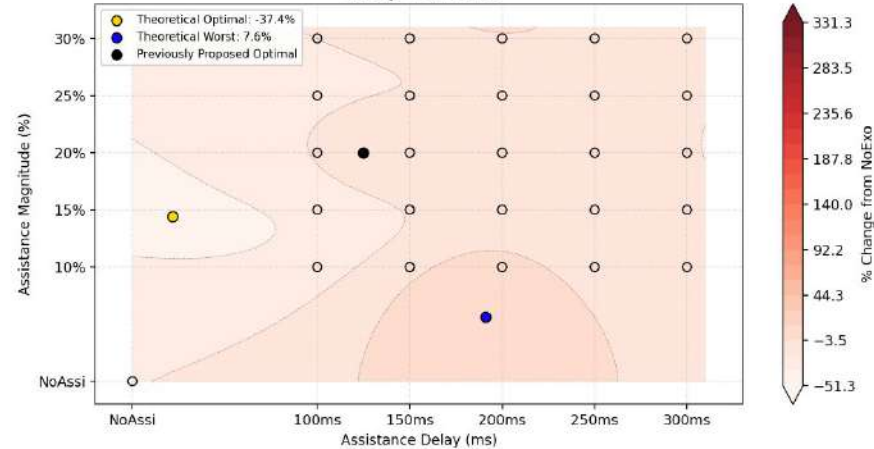


Peak EMG Activation Contour Map - RD Walking

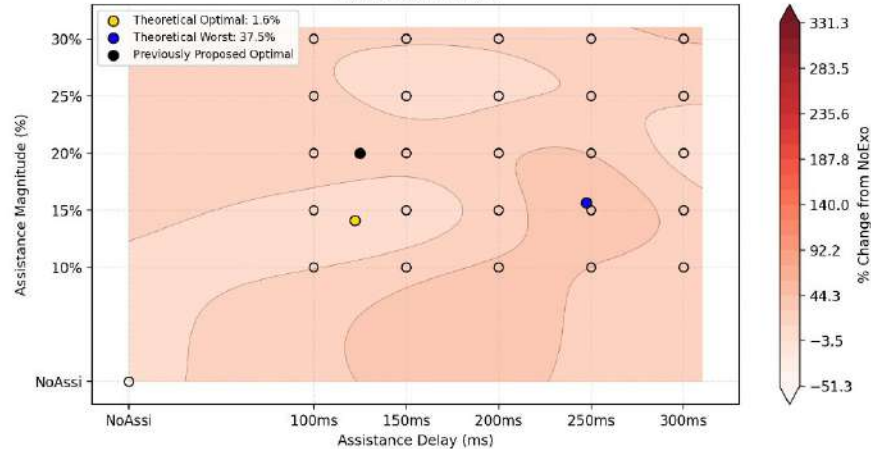
Tibialis Anterior



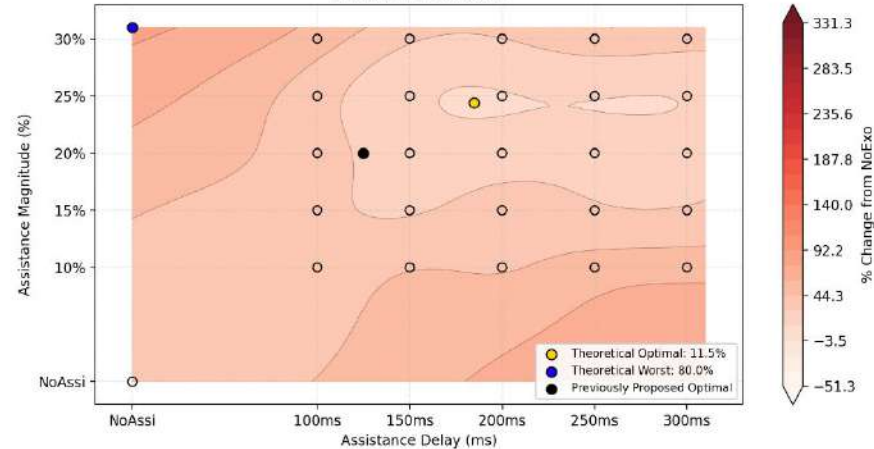
Bicep Femoris



Gastrocnemius



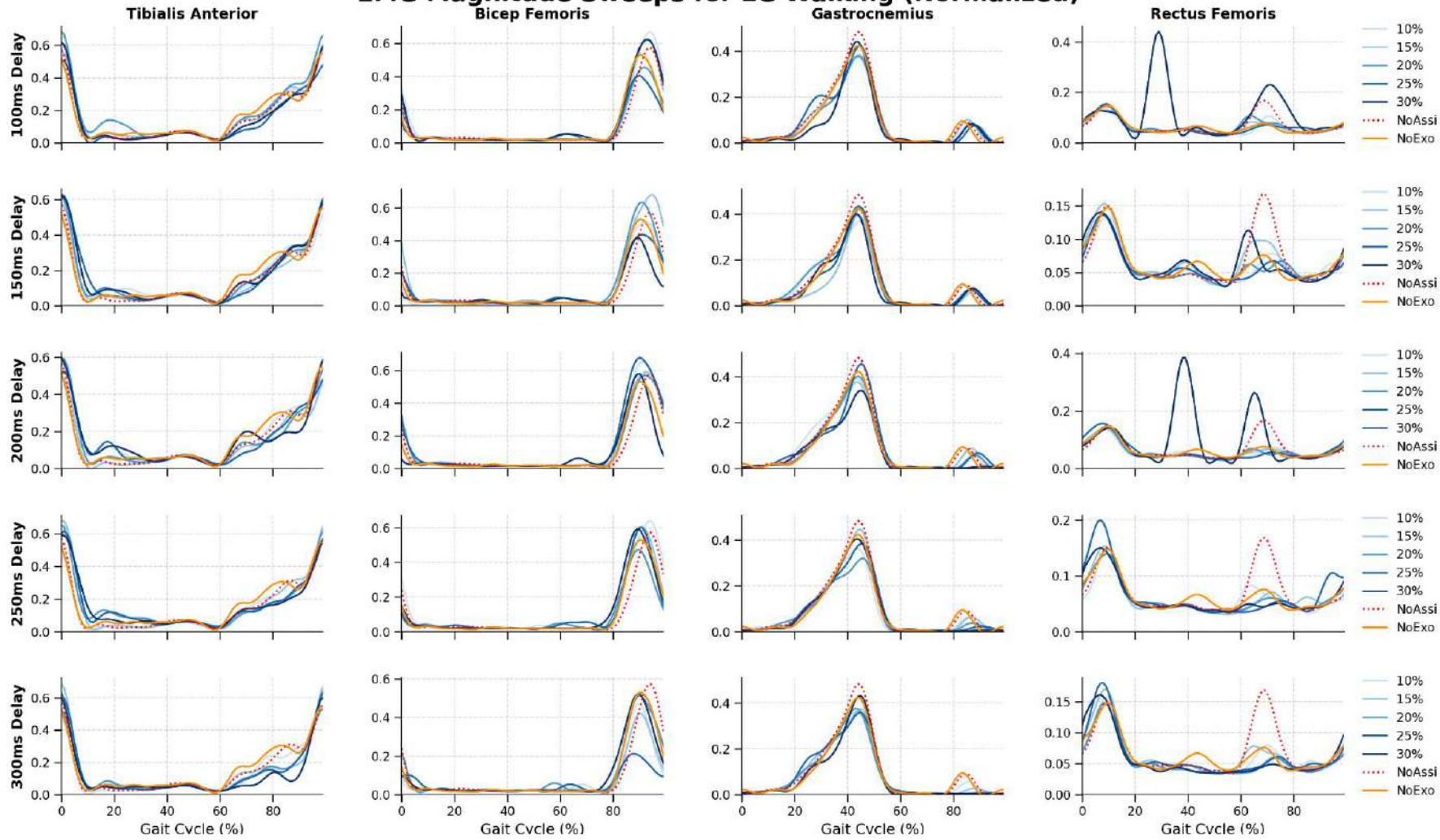
Rectus Femoris



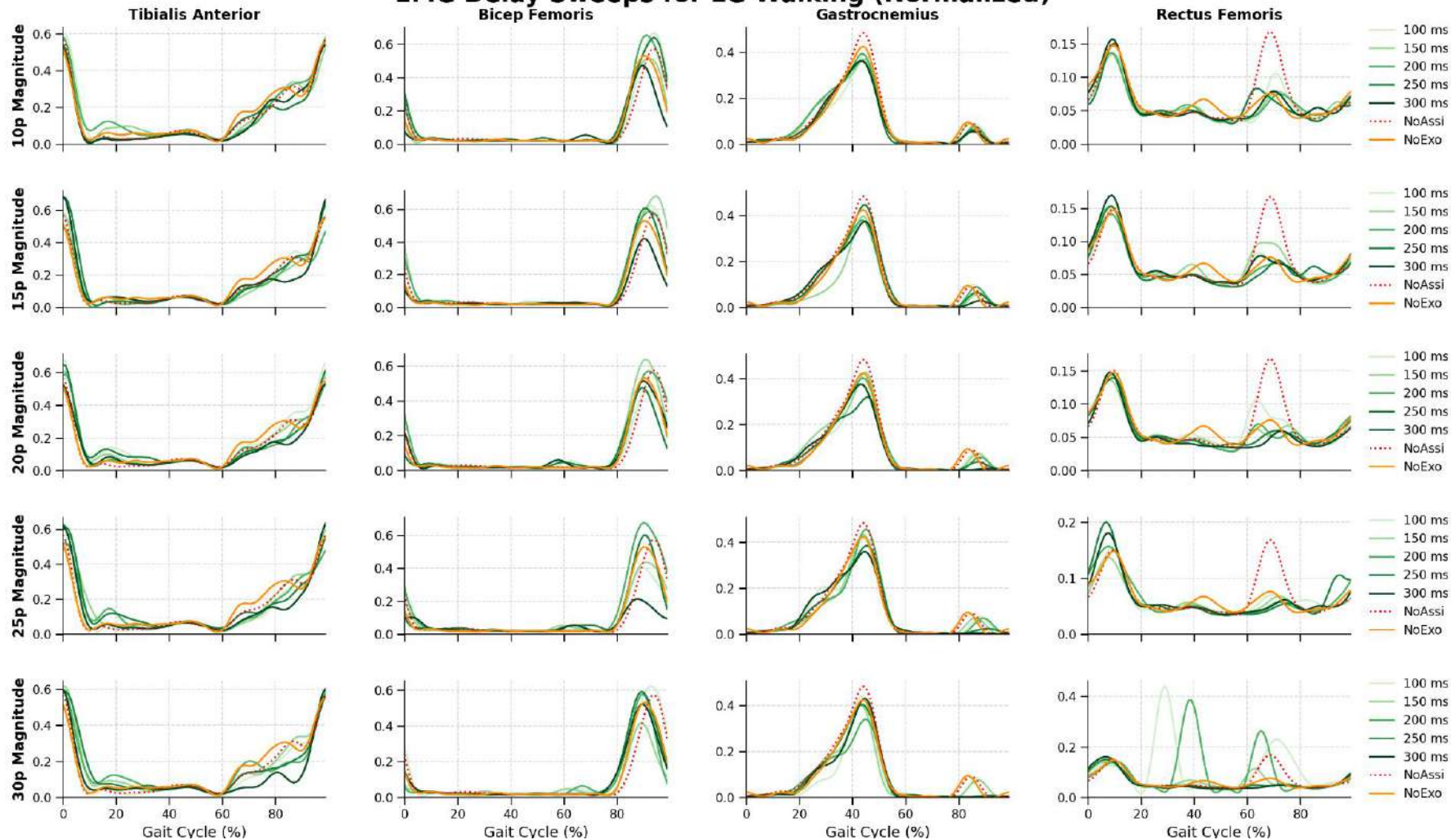
Subject: AB15_Daniel

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

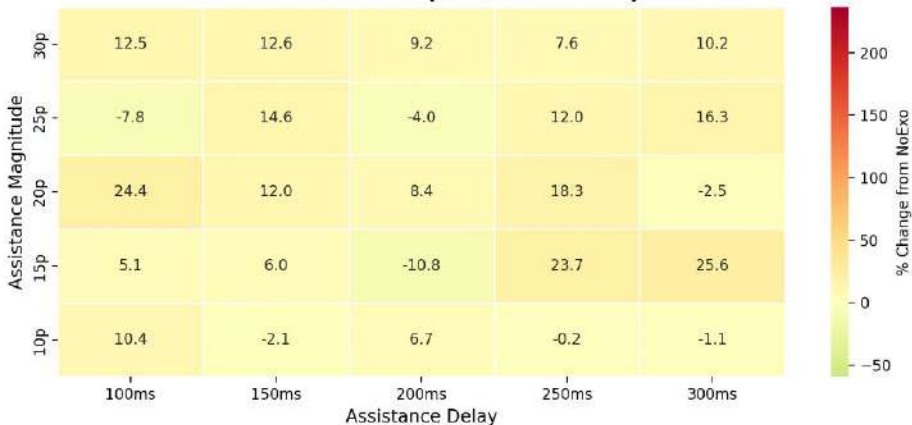


EMG Delay Sweeps for LG Walking (Normalized)

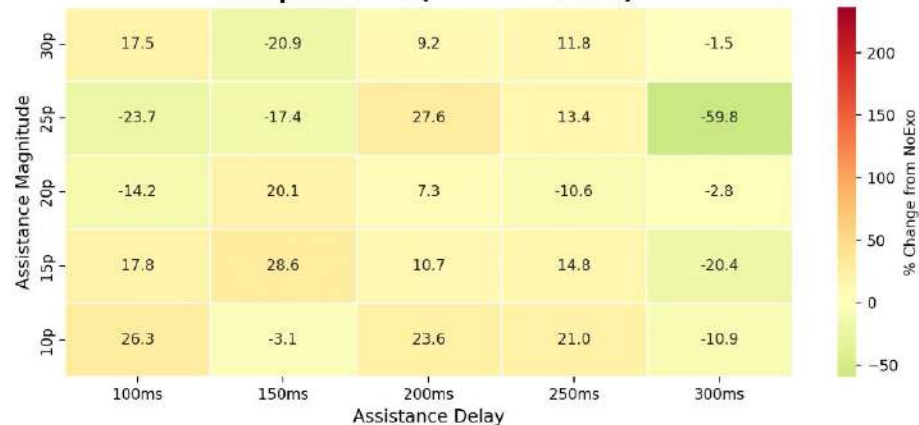


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

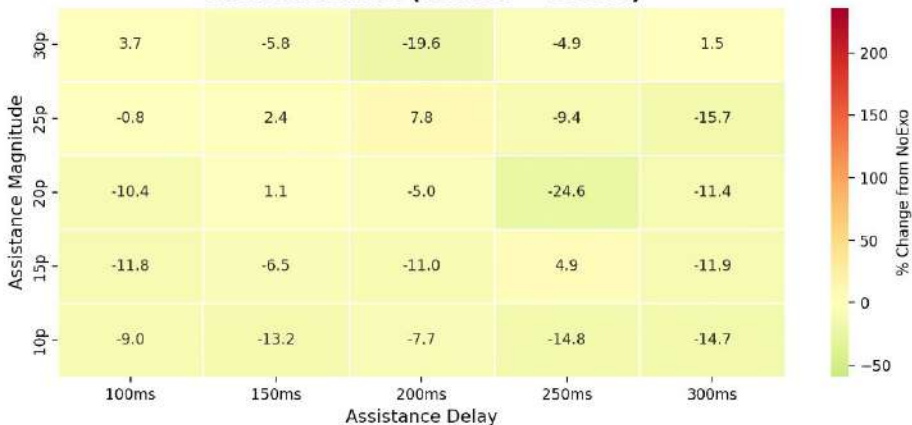
Tibialis Anterior (NoAssi = 3.7%)



Bicep Femoris (NoAssi = 8.5%)



Gastrocnemius (NoAssi = 13.9%)

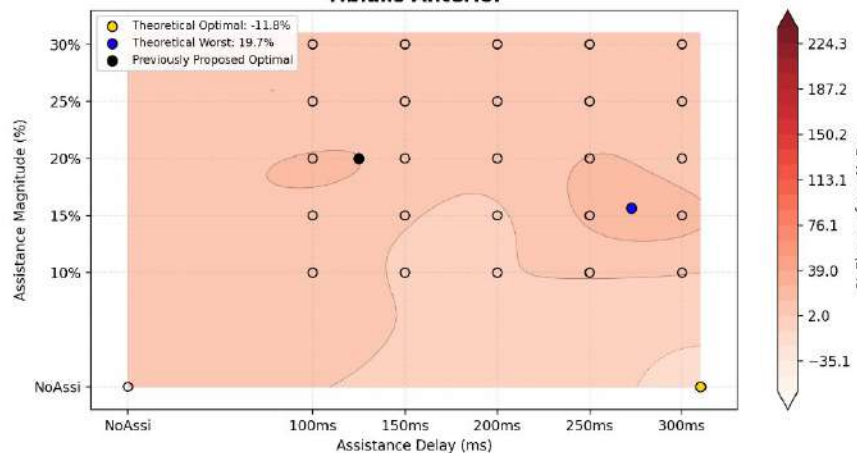


Rectus Femoris (NoAssi = 114.3%)

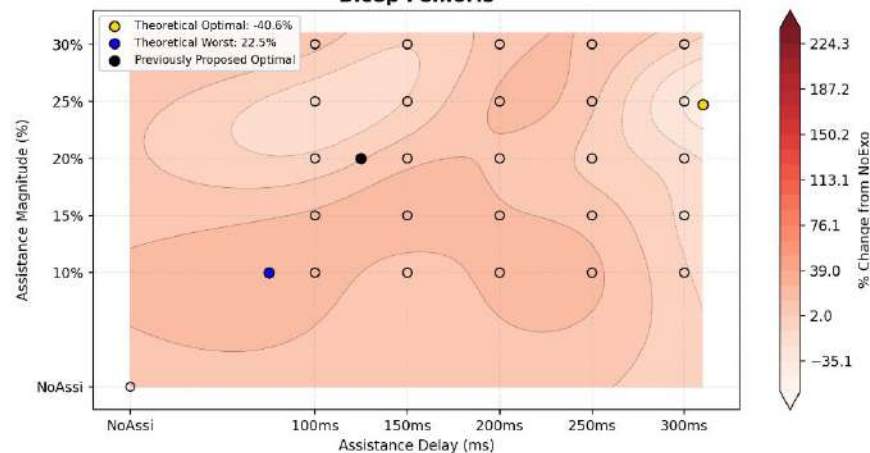


Peak EMG Activation Contour Map - LG Walking

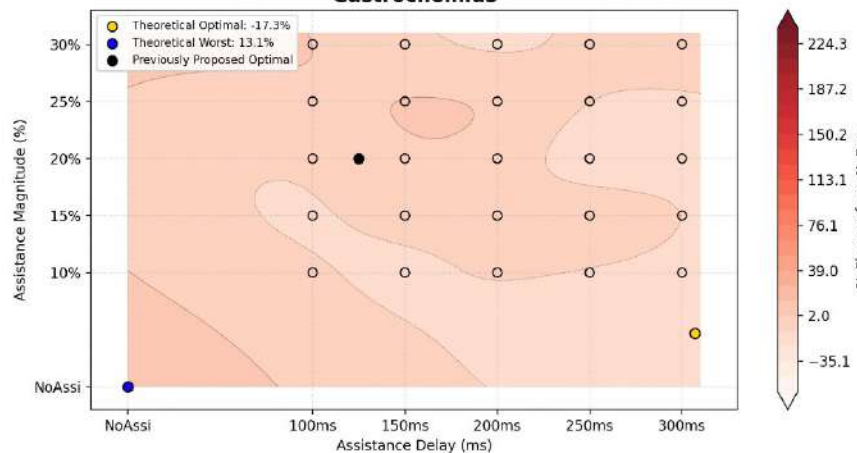
Tibialis Anterior



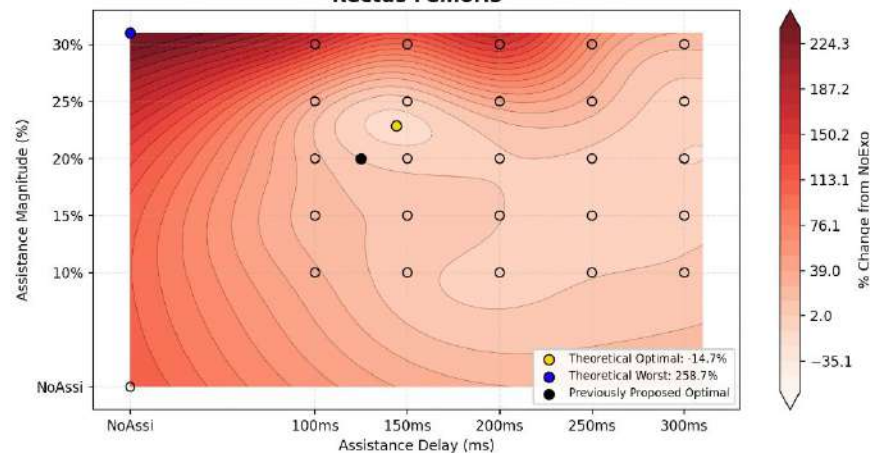
Bicep Femoris



Gastrocnemius

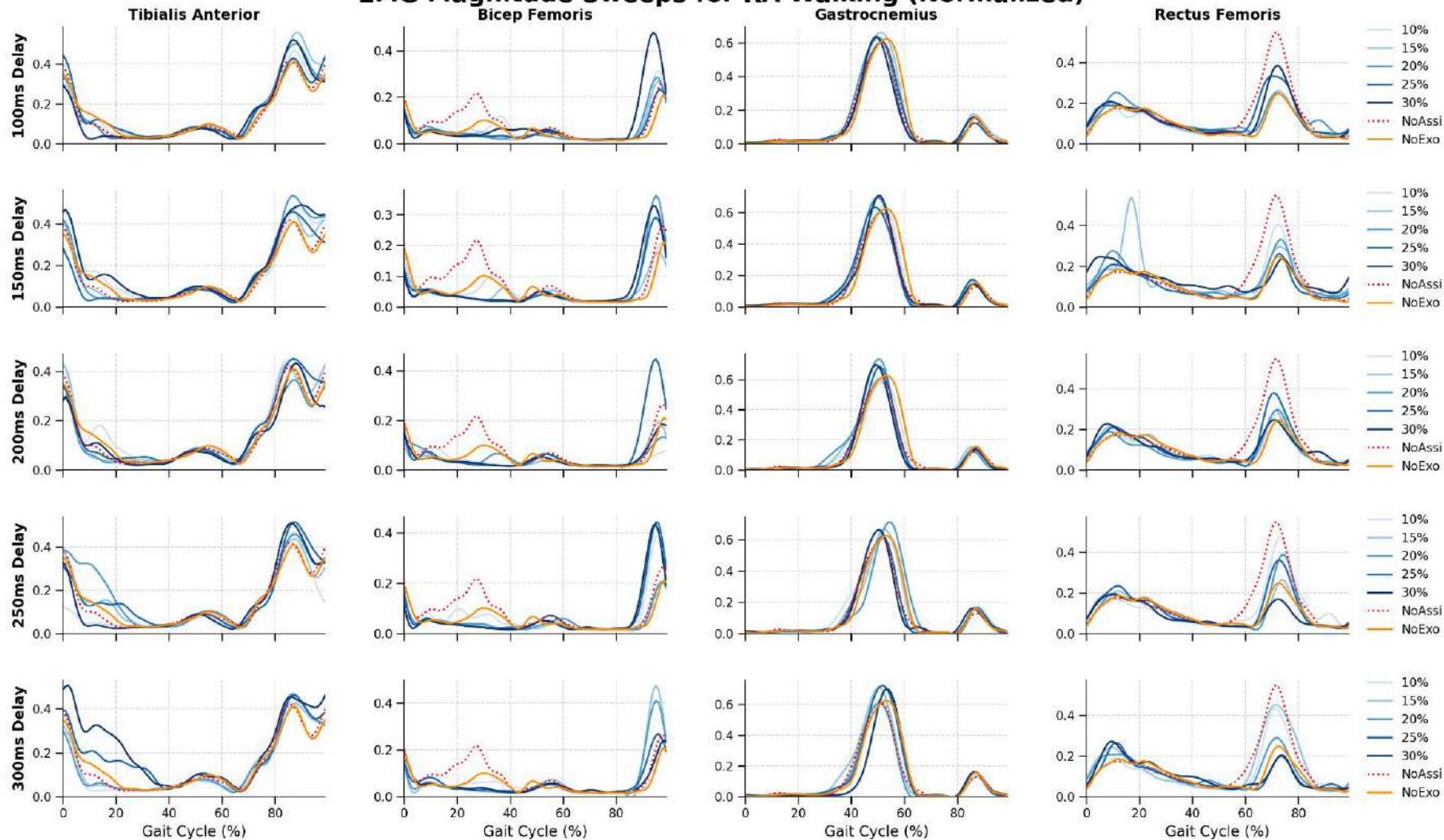


Rectus Femoris

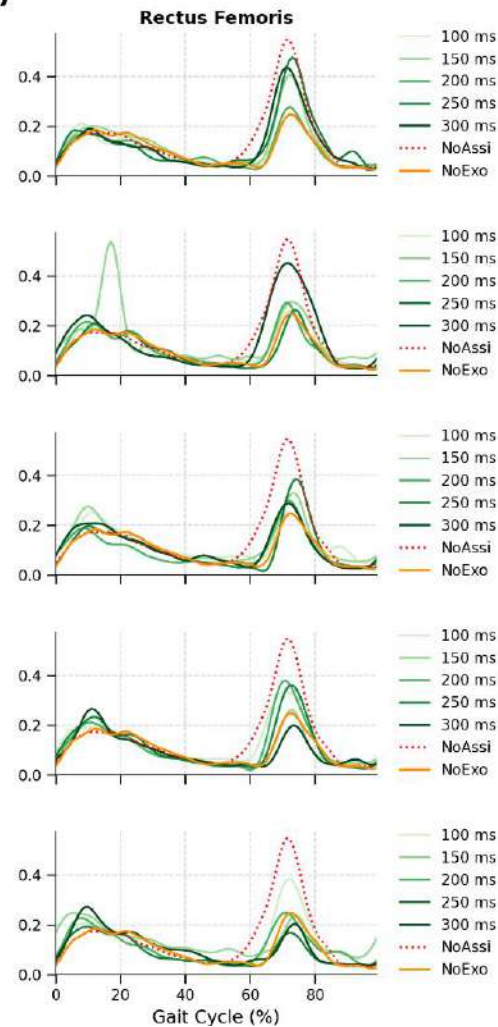
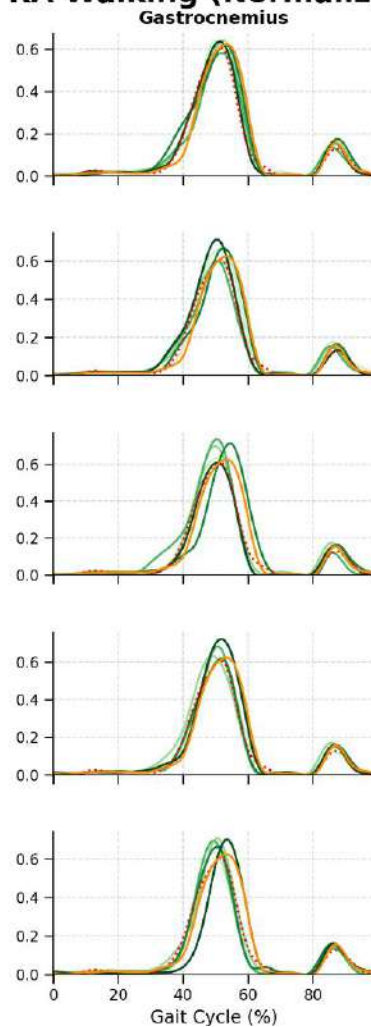
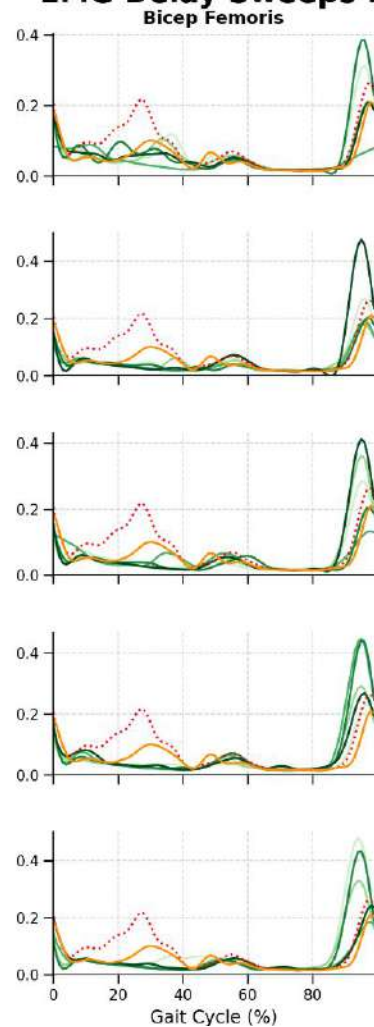
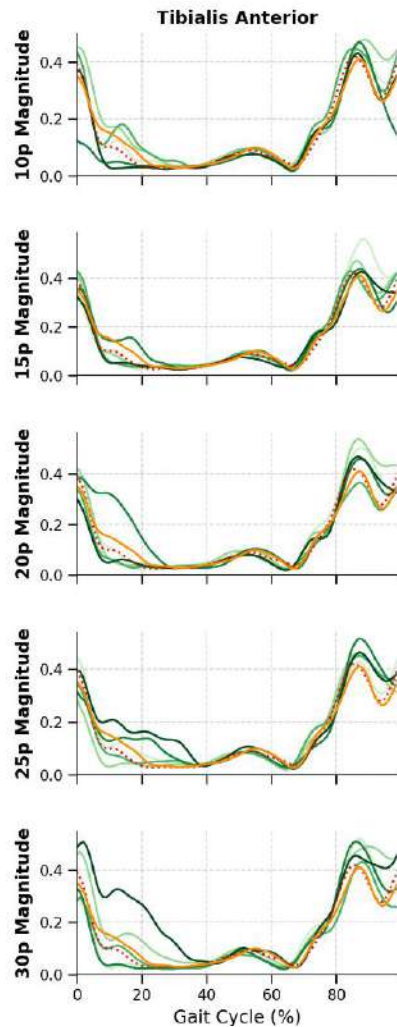


Ramp-Ascent

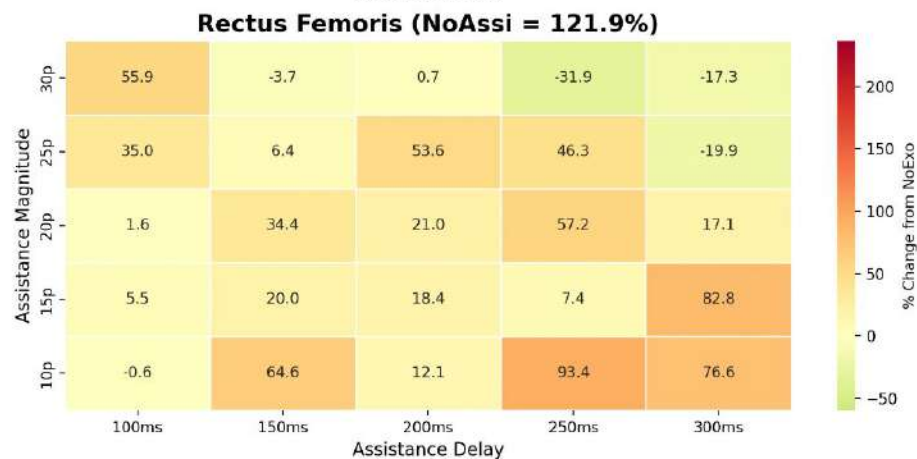
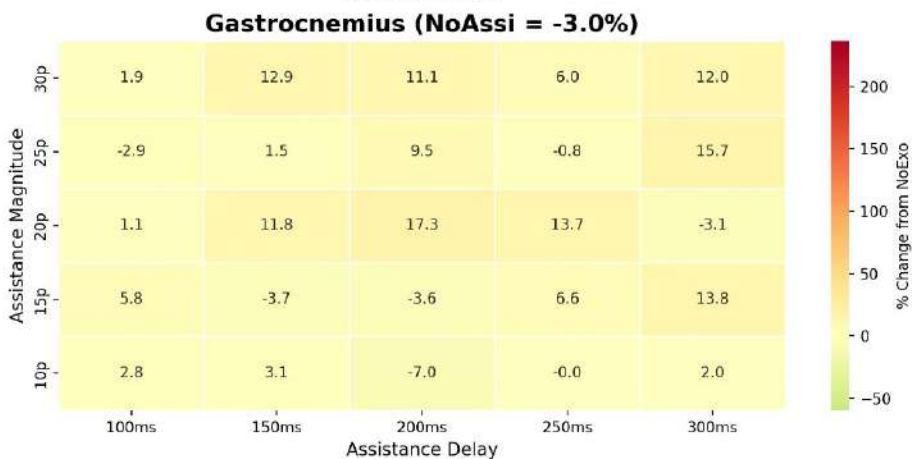
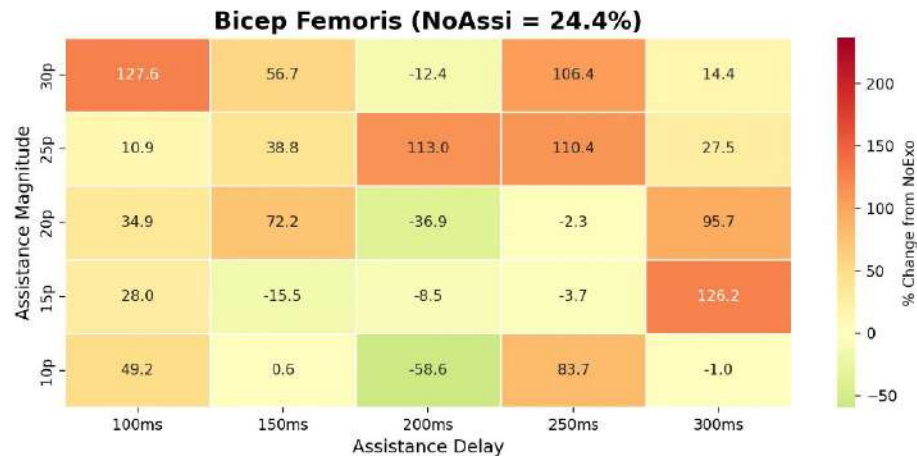
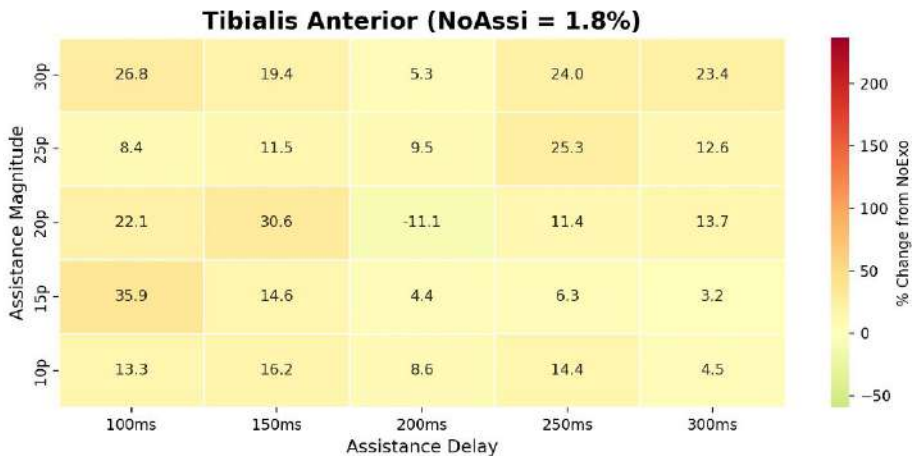
EMG Magnitude Sweeps for RA Walking (Normalized)



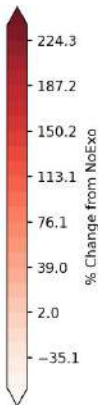
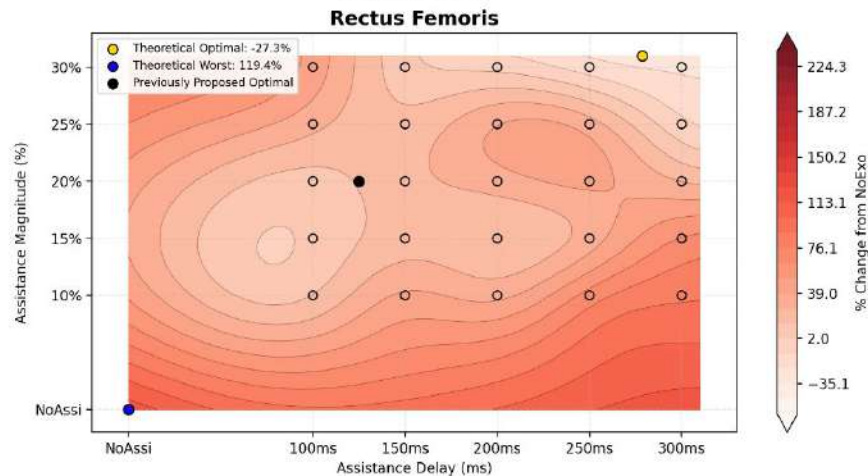
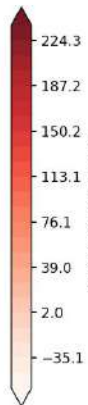
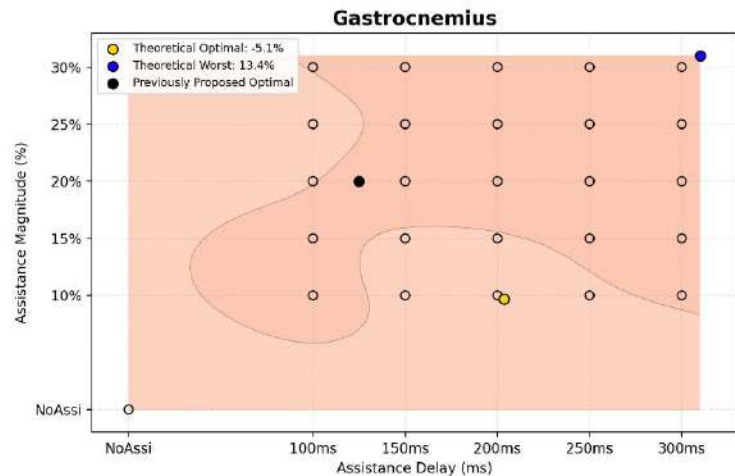
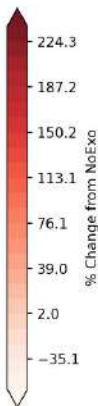
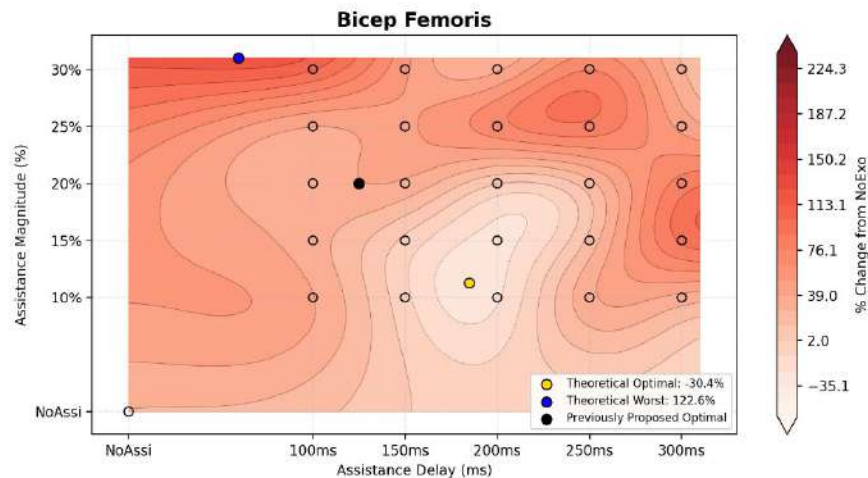
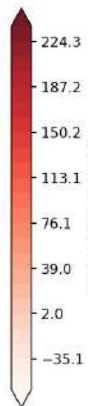
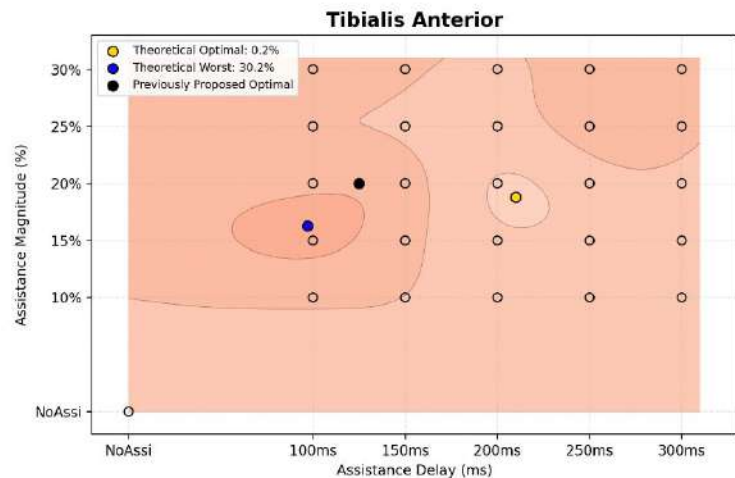
EMG Delay Sweeps for RA Walking (Normalized)



Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

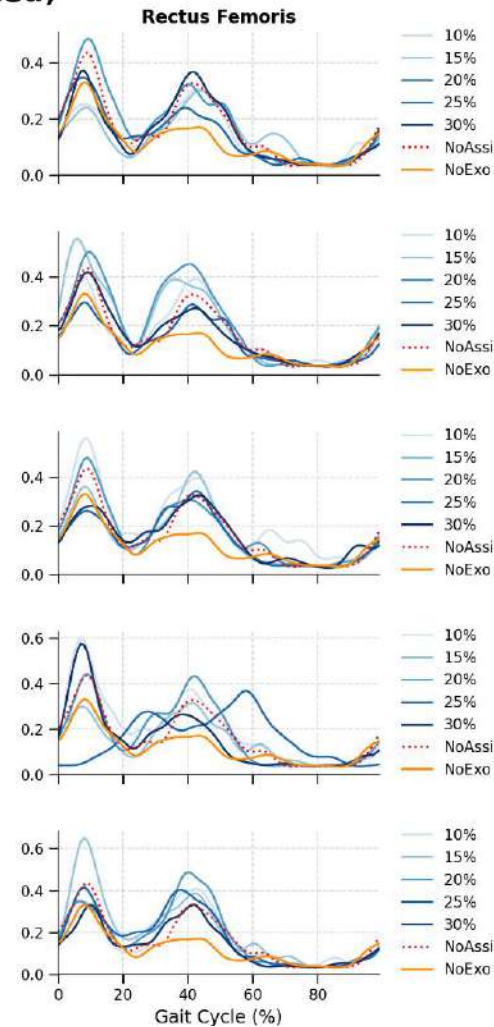
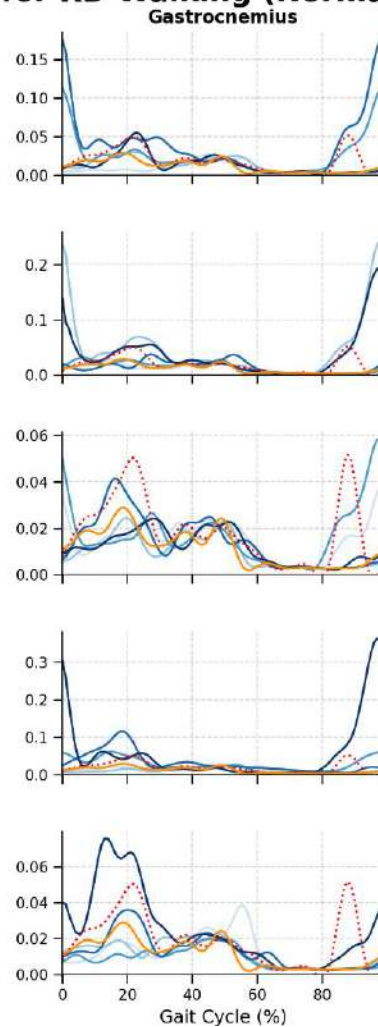
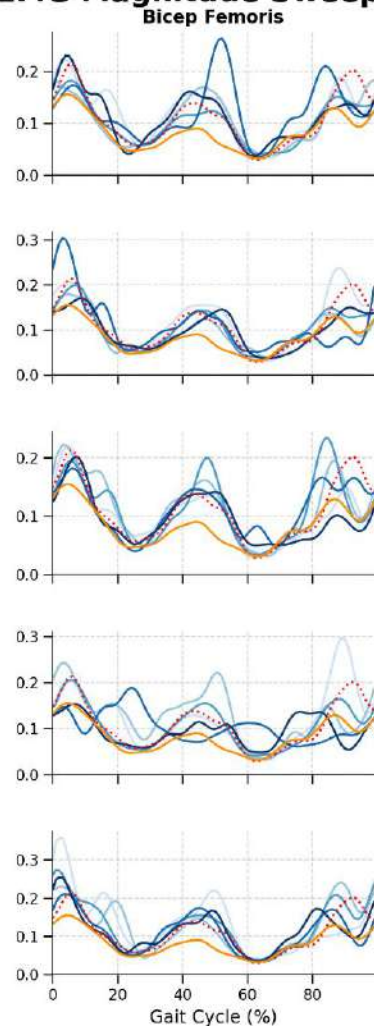
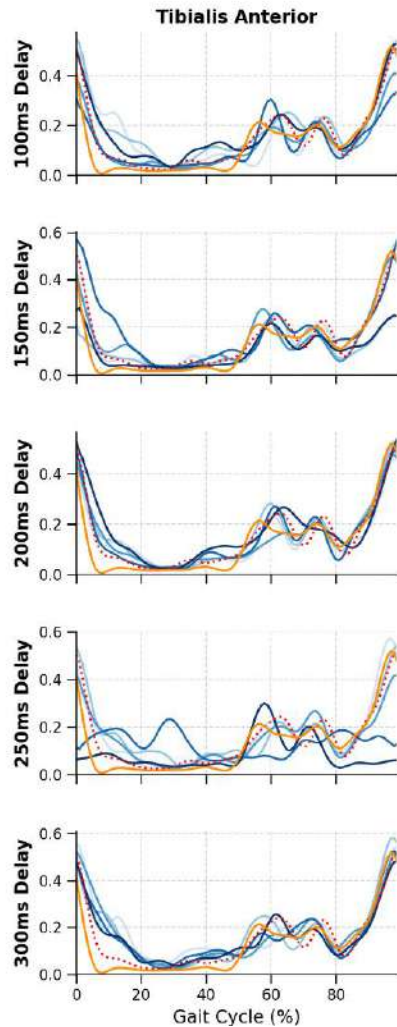


Peak EMG Activation Contour Map - RA Walking

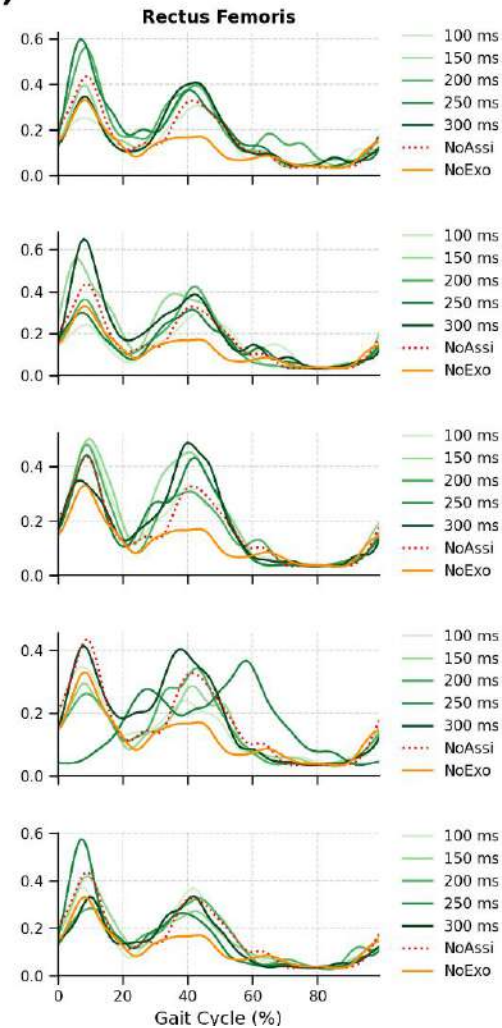
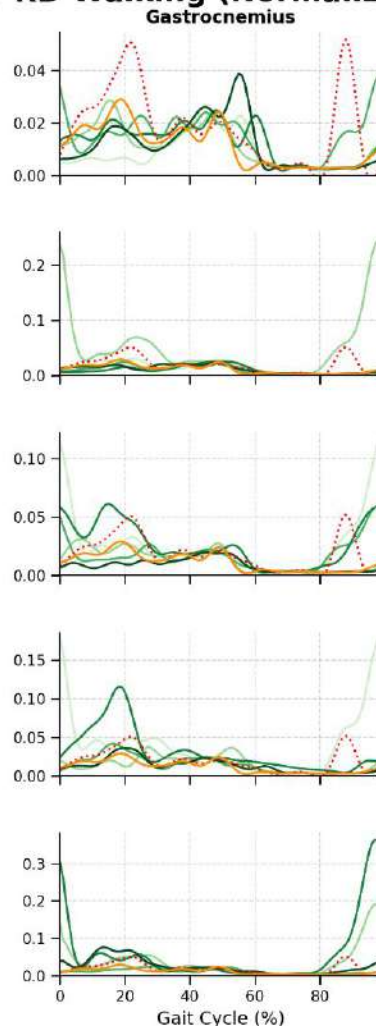
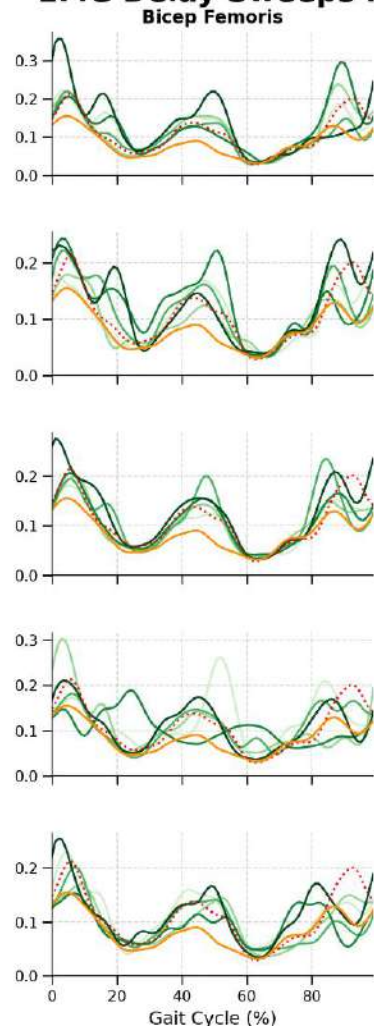
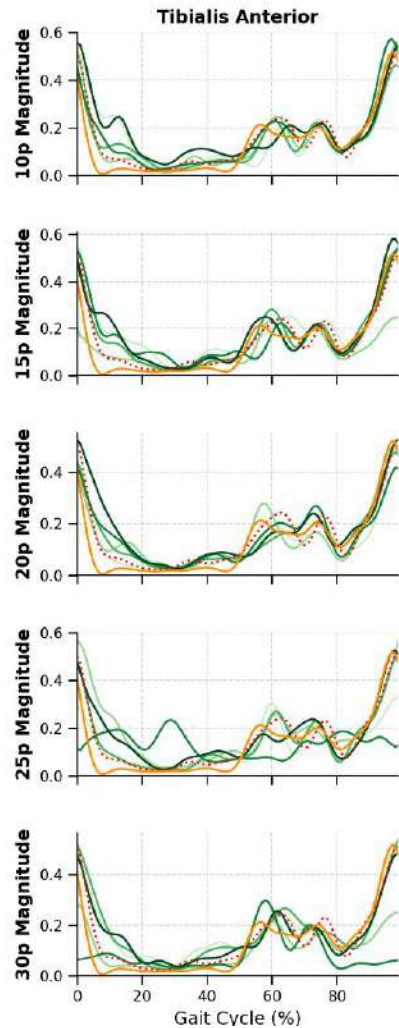


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

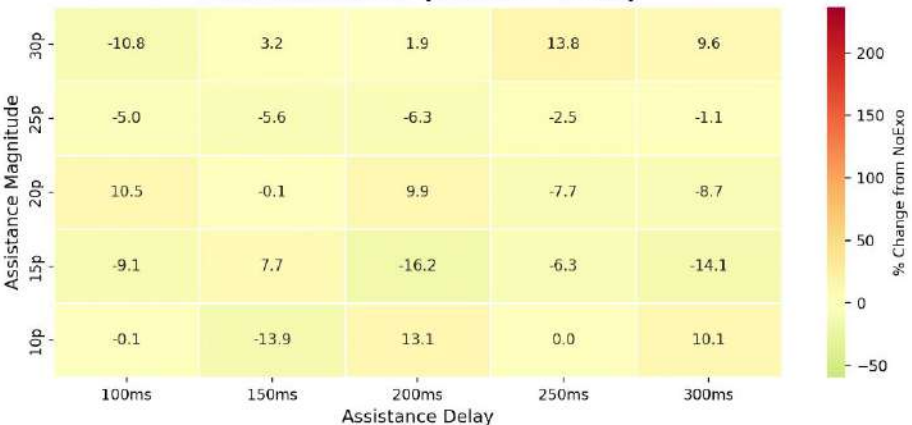


EMG Delay Sweeps for RD Walking (Normalized)

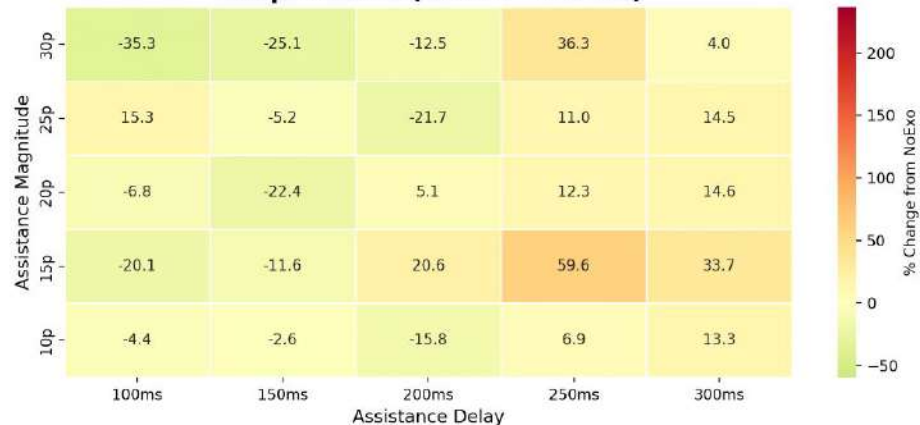


Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

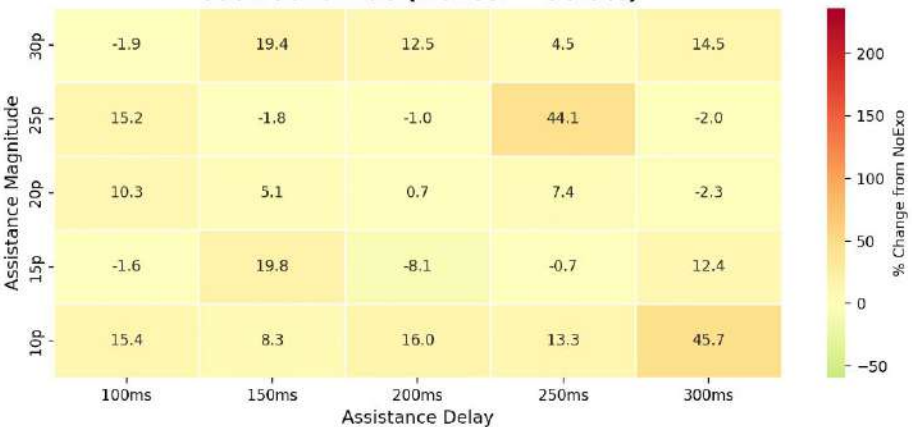
Tibialis Anterior (NoAssi = 7.6%)



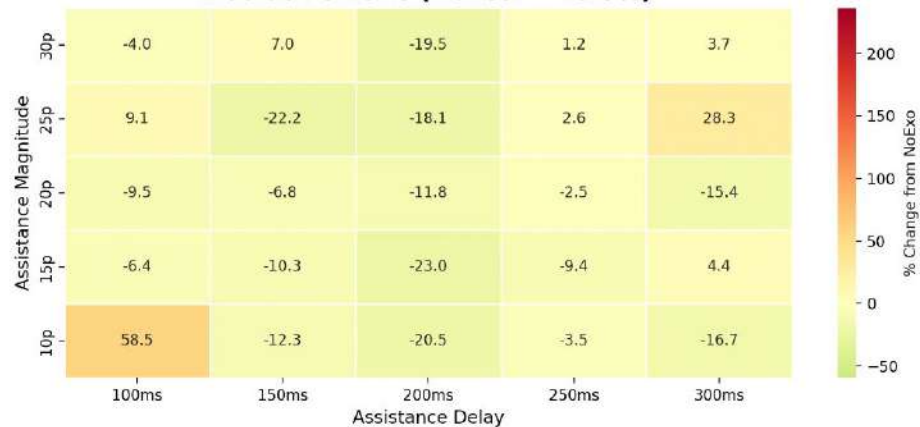
Biceps Femoris (NoAssi = 28.8%)



Gastrocnemius (NoAssi = 36.5%)

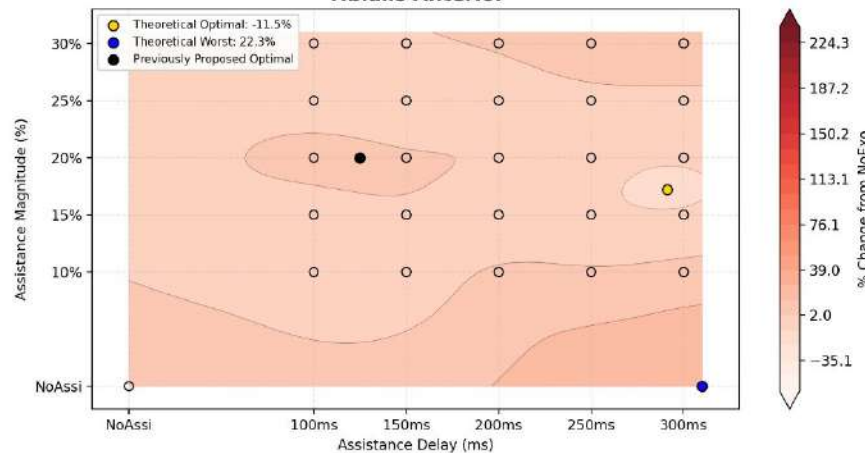


Rectus Femoris (NoAssi = 49.3%)

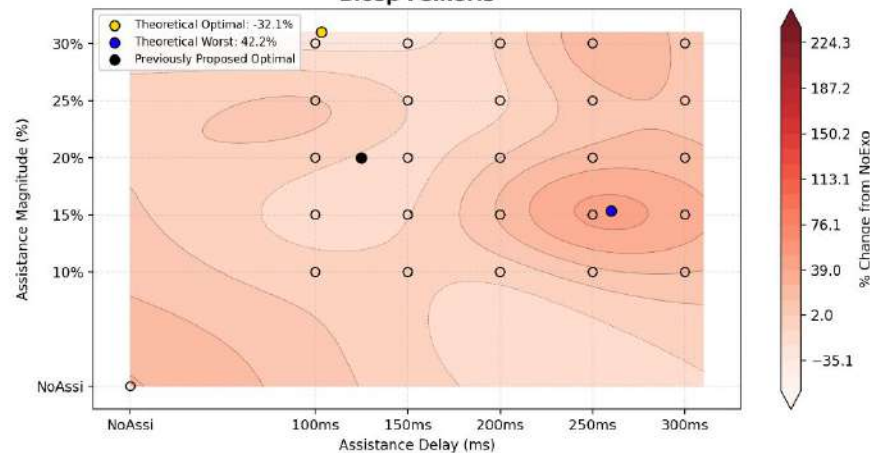


Peak EMG Activation Contour Map - RD Walking

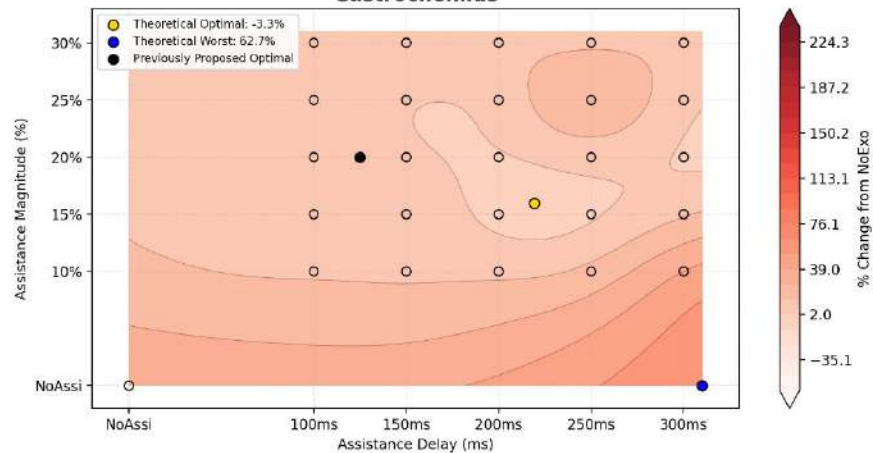
Tibialis Anterior



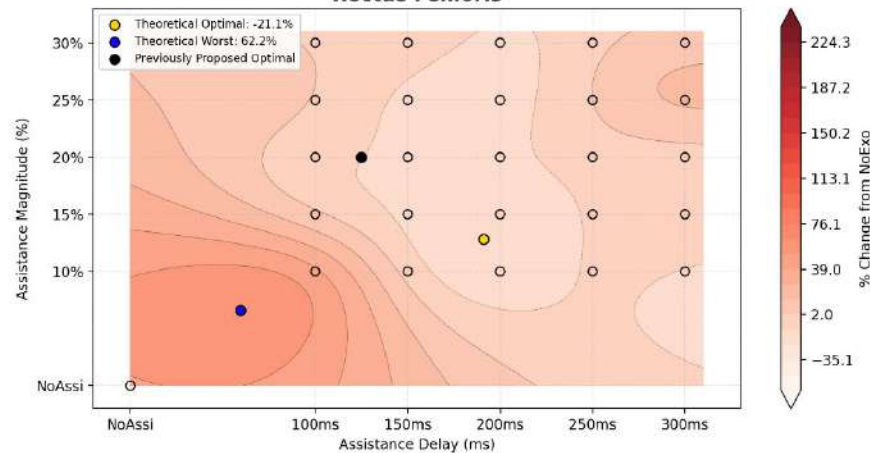
Bicep Femoris



Gastrocnemius



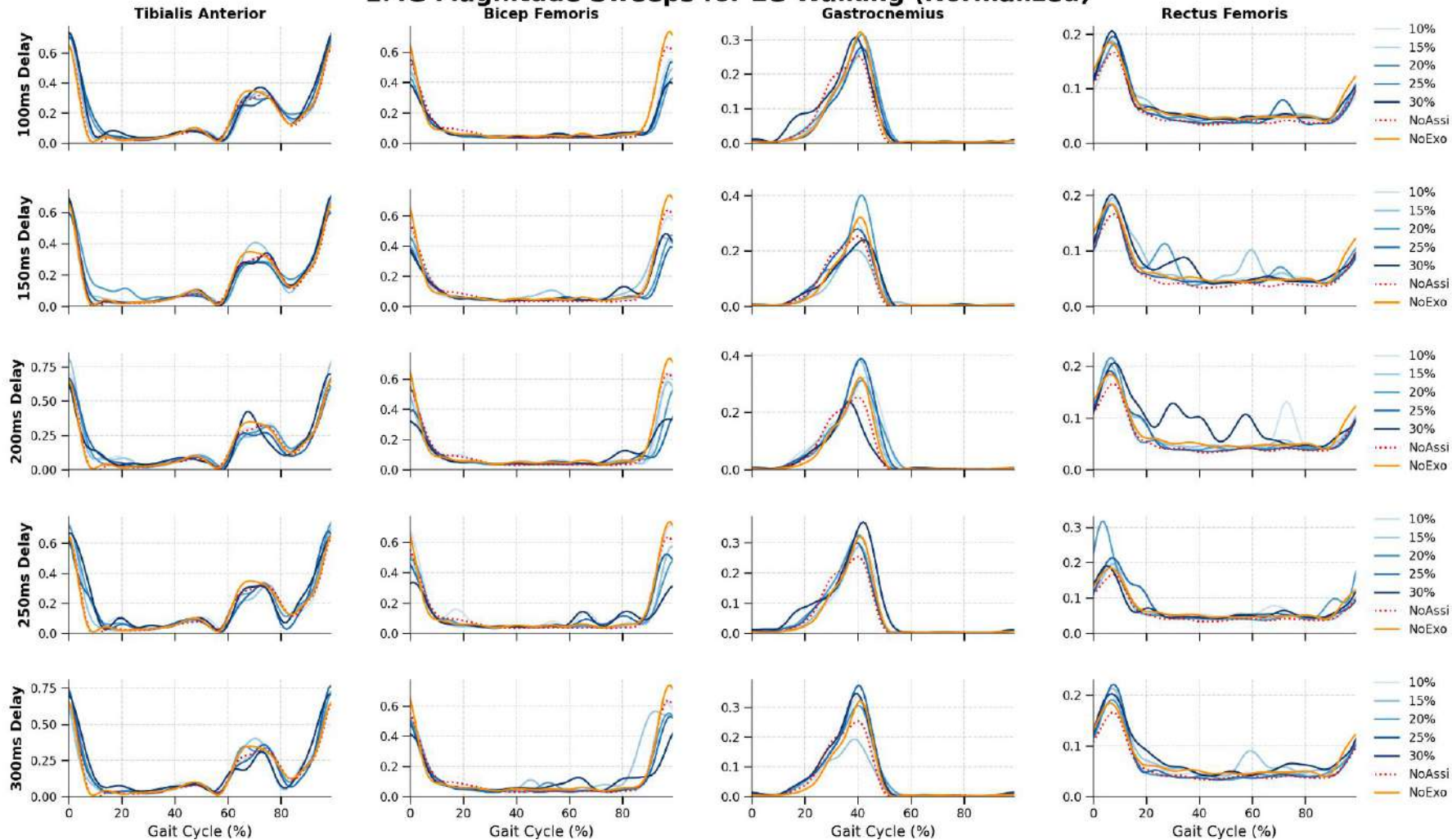
Rectus Femoris



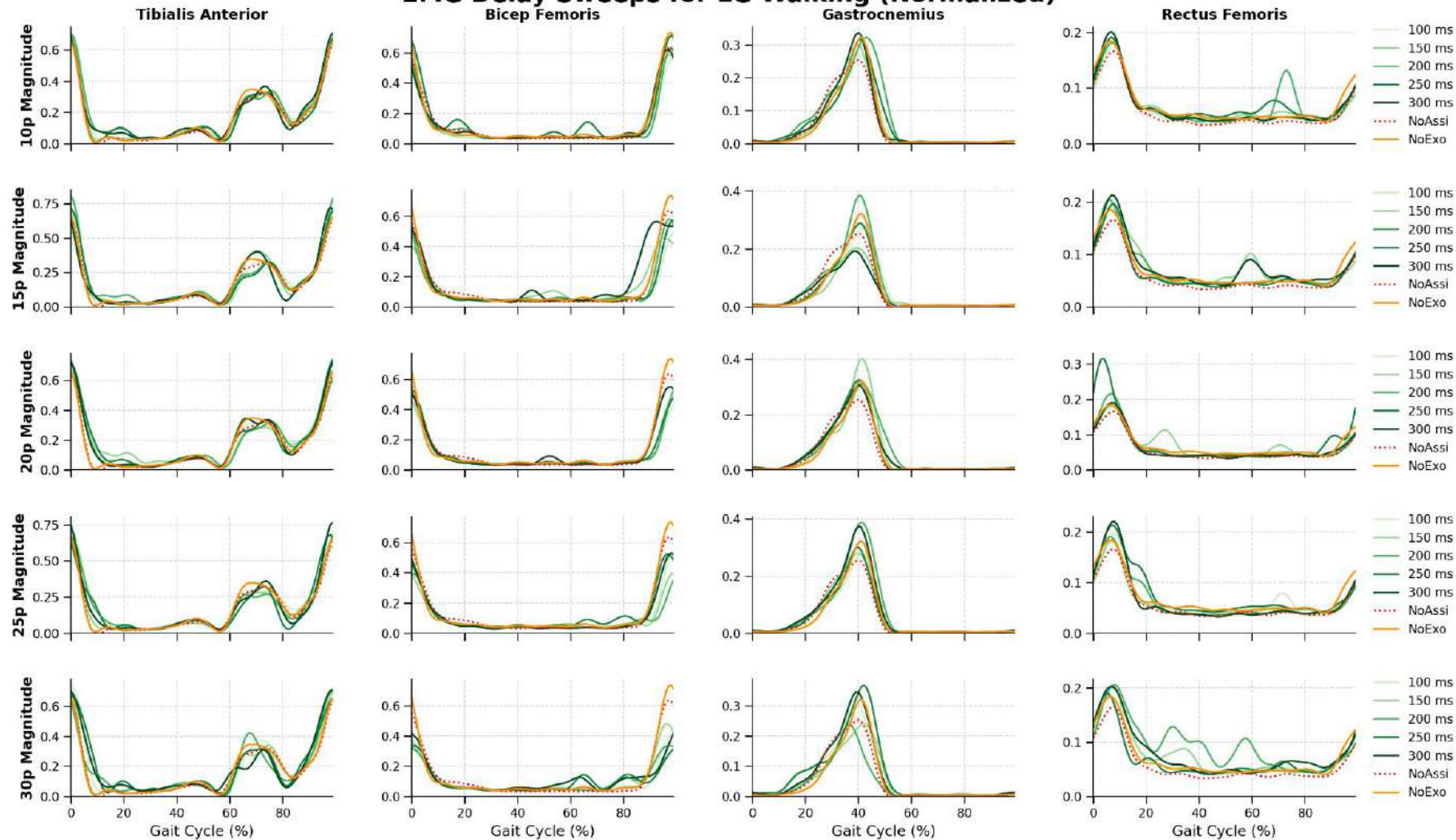
Subject: AB16_Ilseung

Level-Ground

EMG Magnitude Sweeps for LG Walking (Normalized)

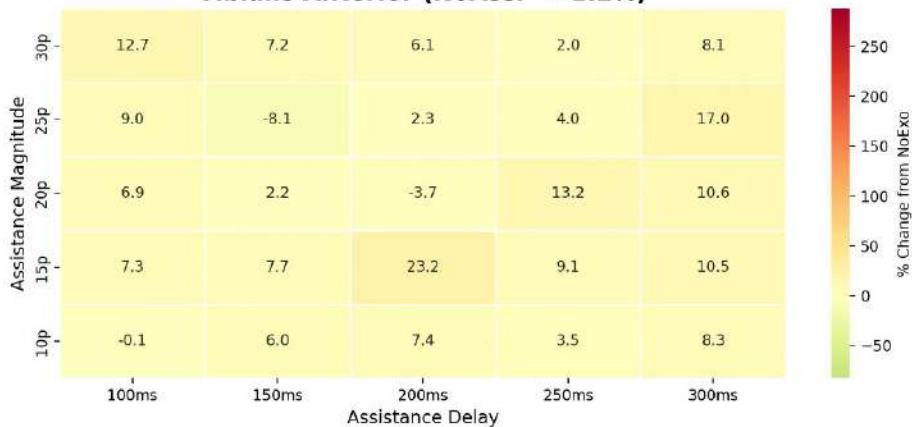


EMG Delay Sweeps for LG Walking (Normalized)

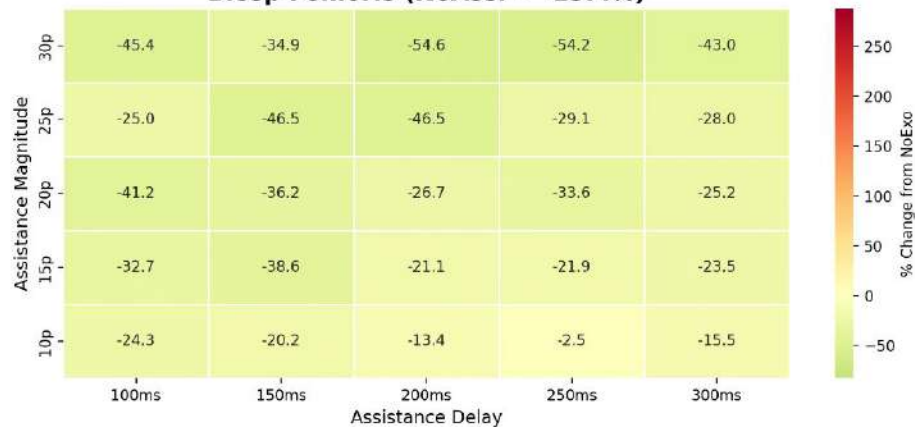


Peak EMG Activation (% Change from NoExo) - LG Walking (Phased)

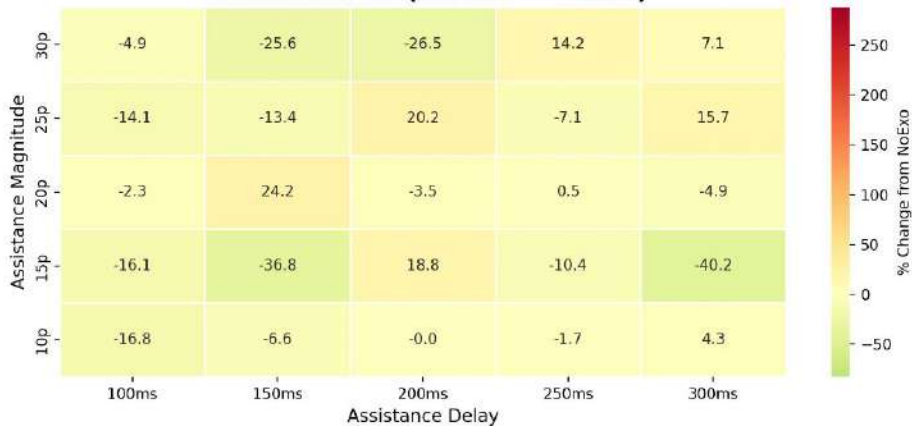
Tibialis Anterior (NoAssi = -1.2%)



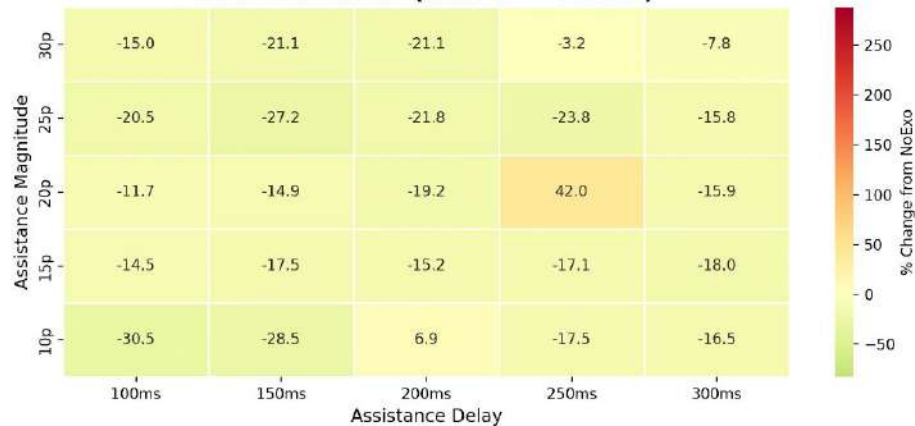
Bicep Femoris (NoAssi = -13.4%)



Gastrocnemius (NoAssi = -21.4%)

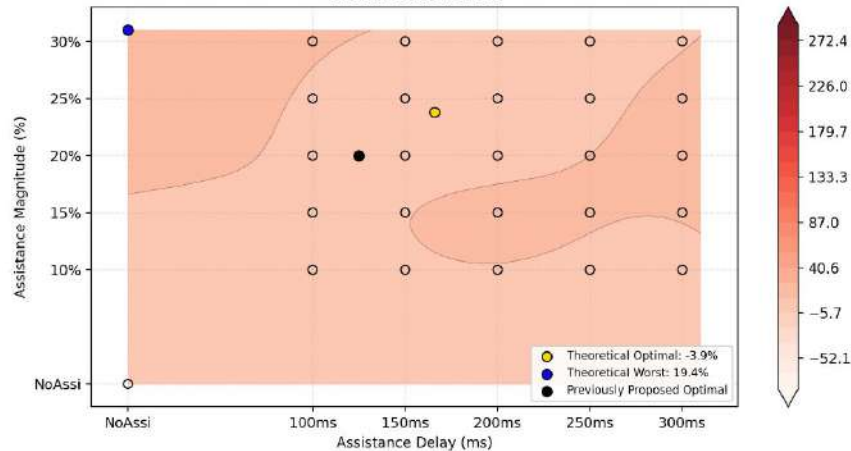


Rectus Femoris (NoAssi = -20.5%)

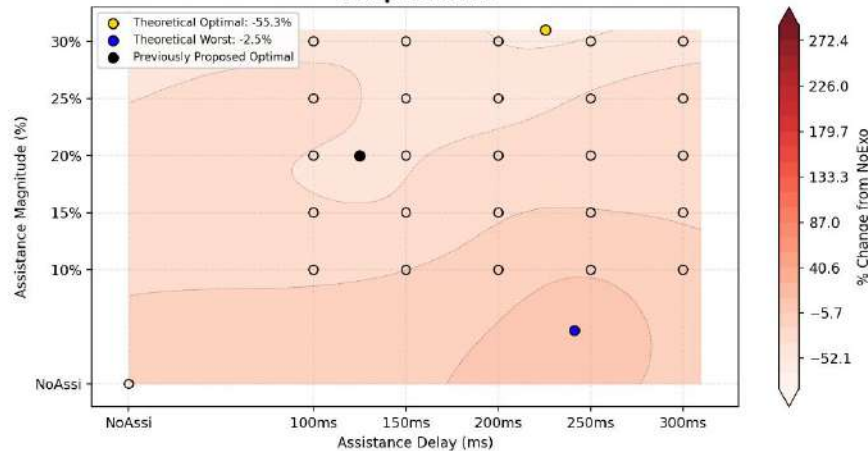


Peak EMG Activation Contour Map - LG Walking

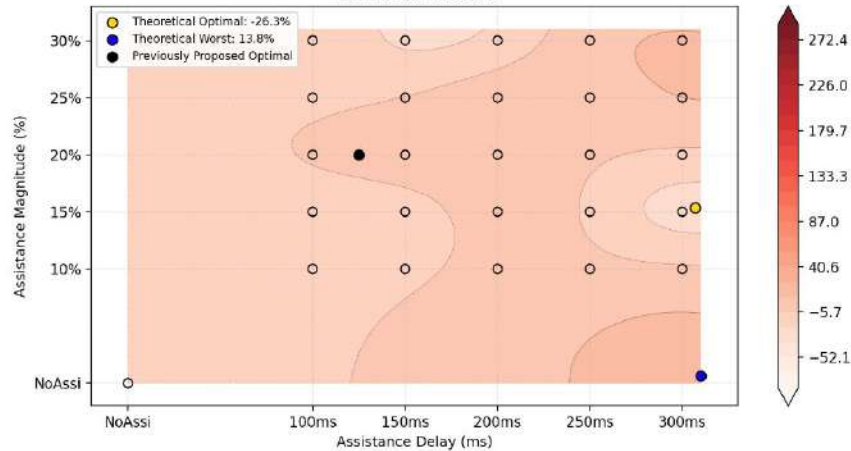
Tibialis Anterior



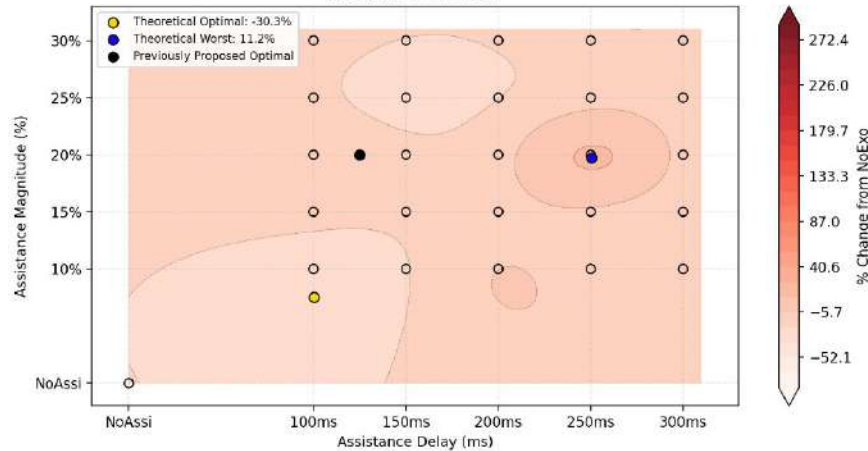
Bicep Femoris



Gastrocnemius

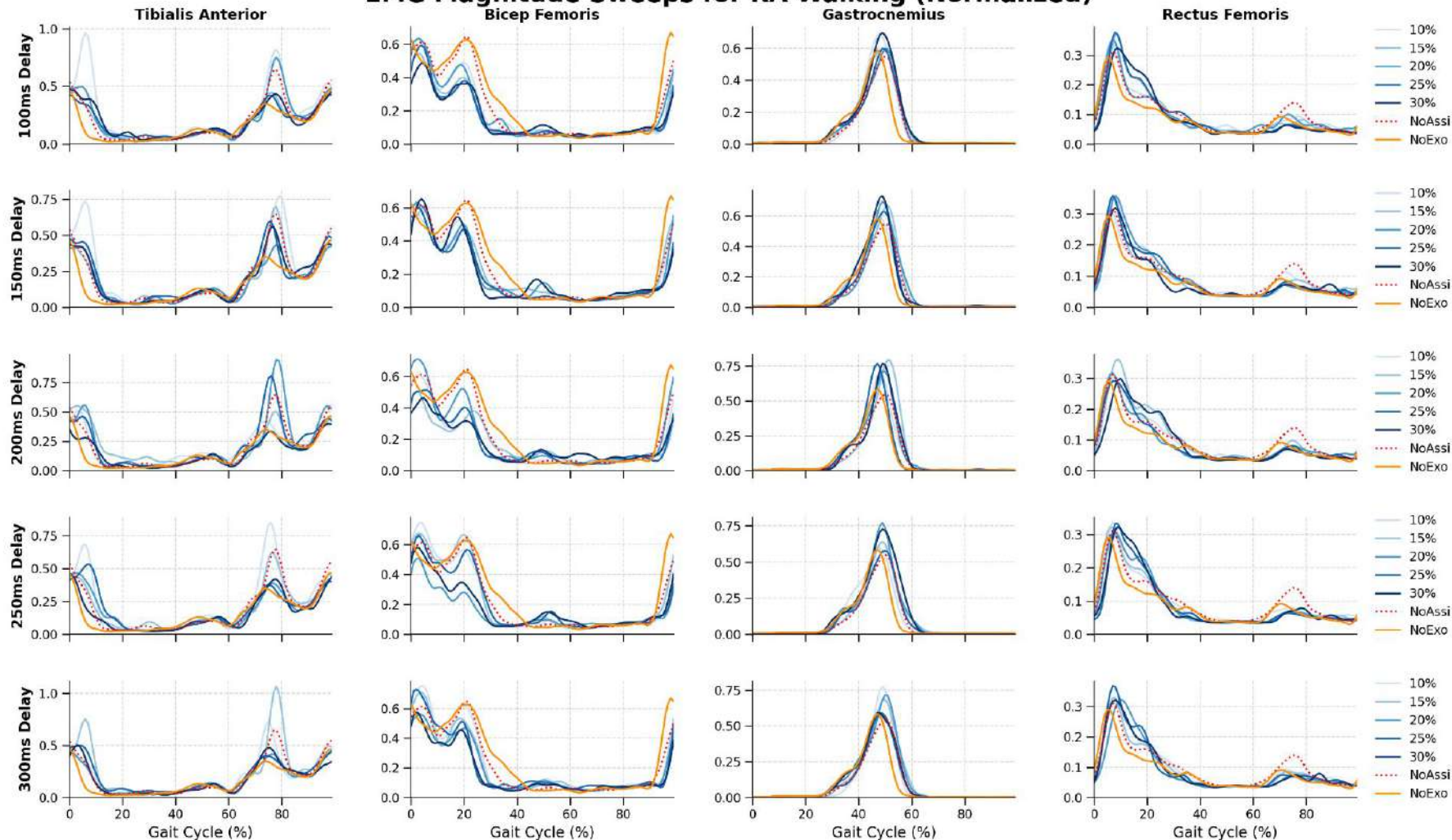


Rectus Femoris

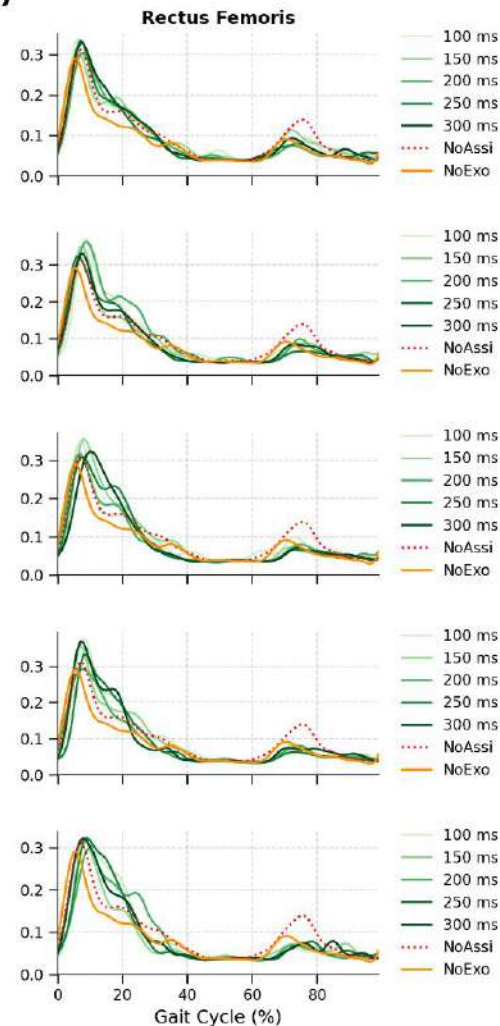
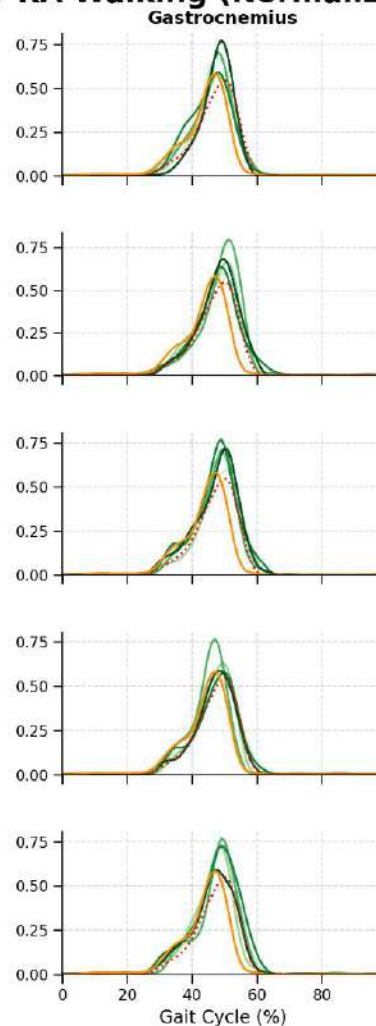
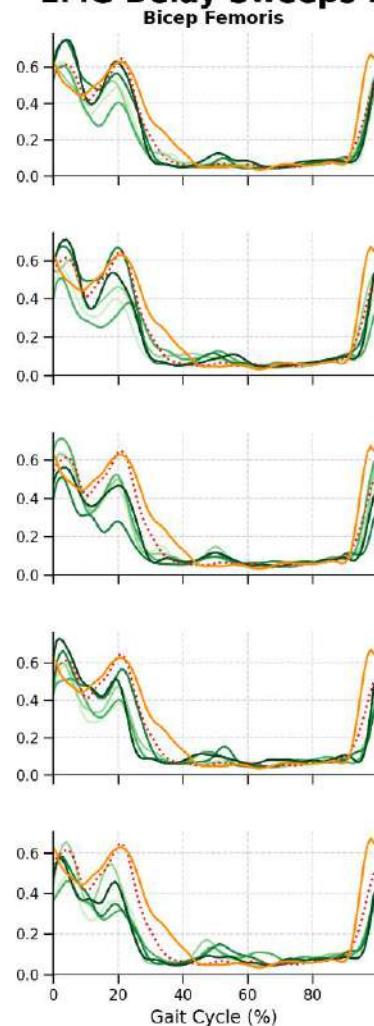
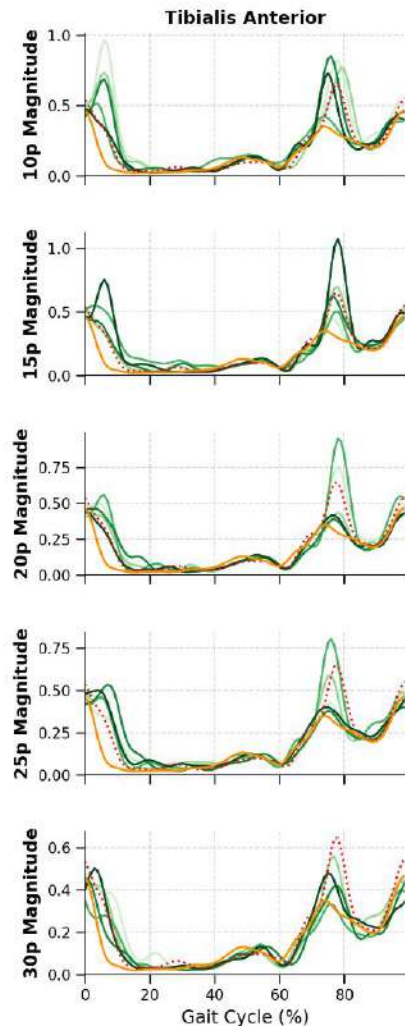


Ramp-Ascent

EMG Magnitude Sweeps for RA Walking (Normalized)

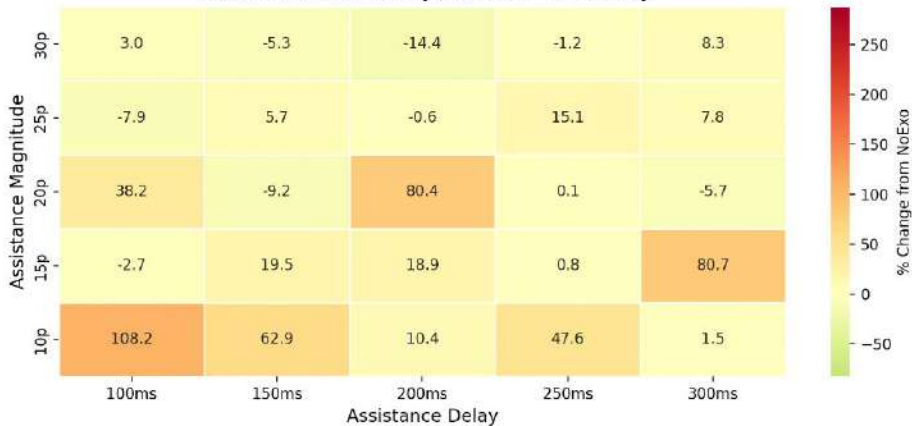


EMG Delay Sweeps for RA Walking (Normalized)

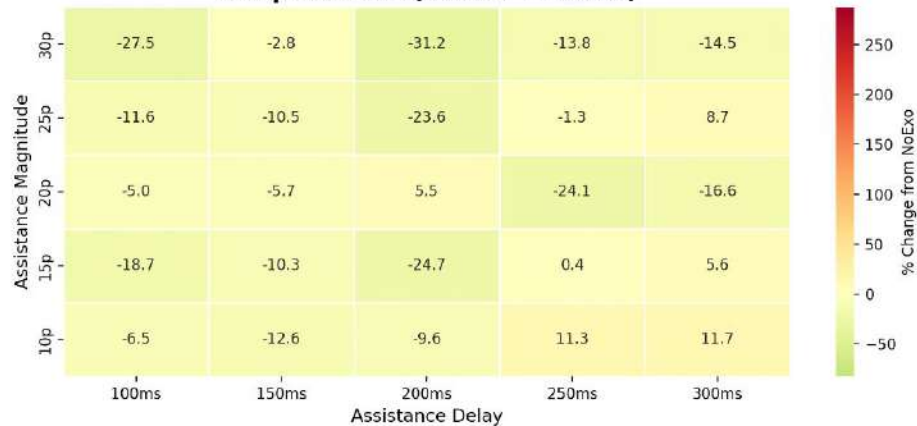


Peak EMG Activation (% Change from NoExo) - RA Walking (Phased)

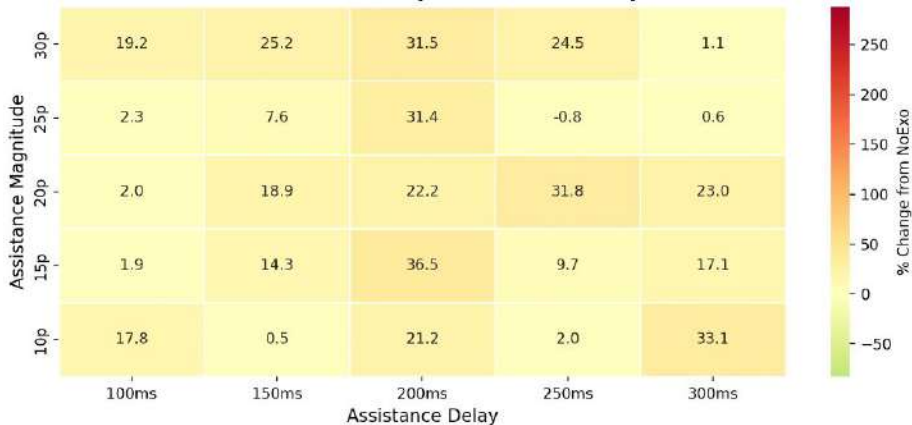
Tibialis Anterior (NoAssi = 19.1%)



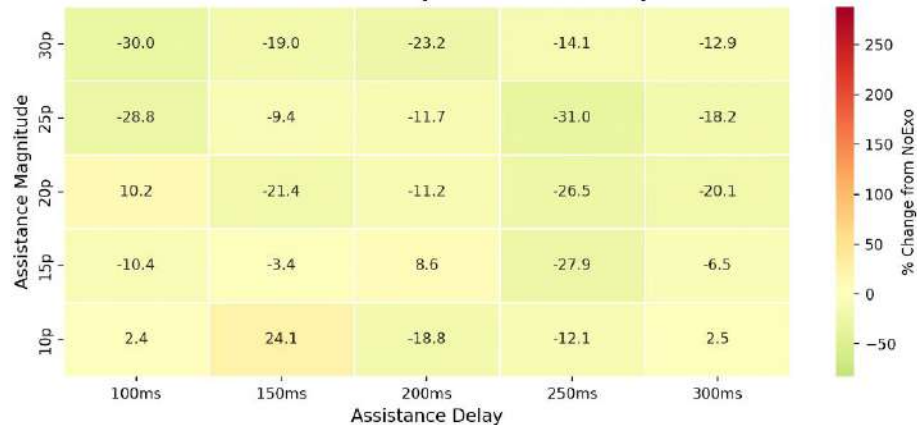
Bicep Femoris (NoAssi = -8.7%)



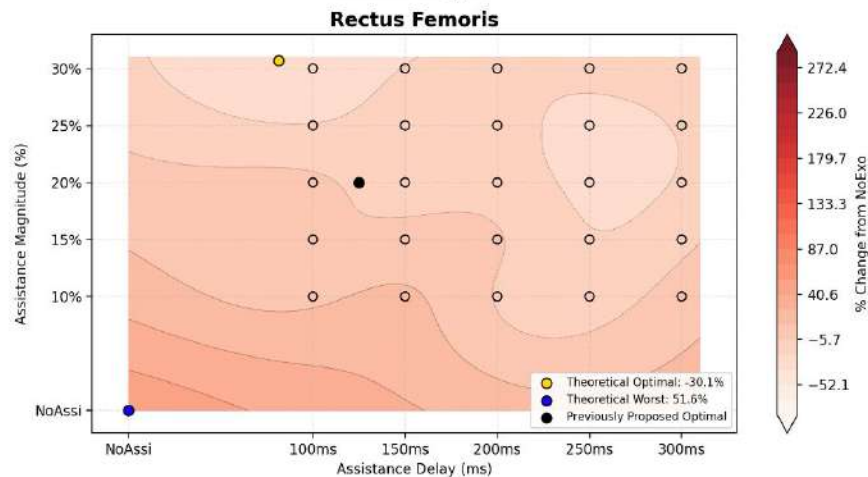
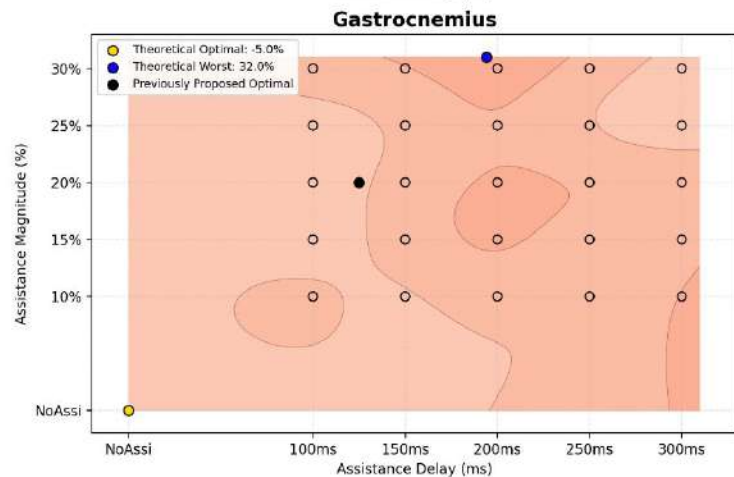
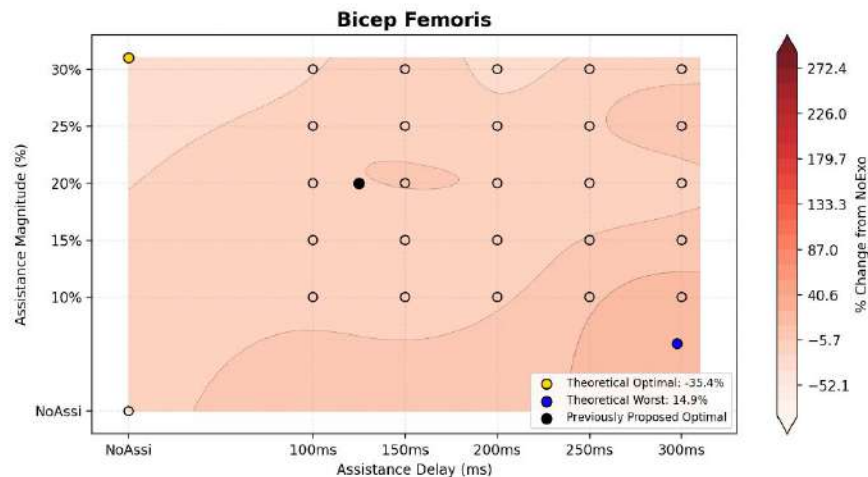
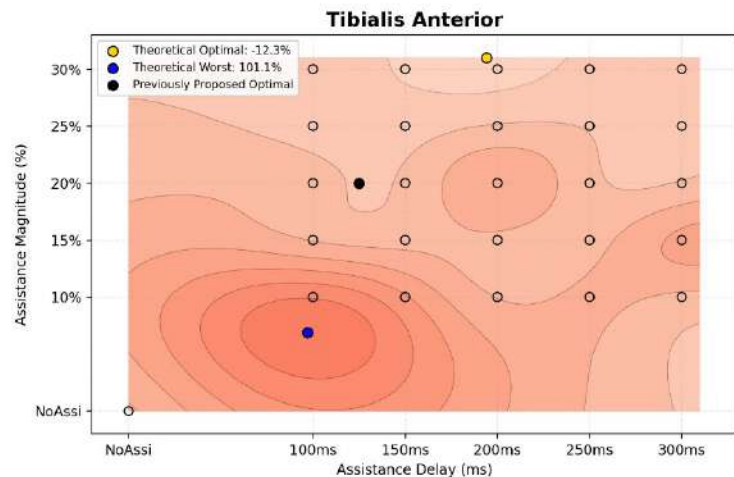
Gastrocnemius (NoAssi = -5.9%)



Rectus Femoris (NoAssi = 52.9%)

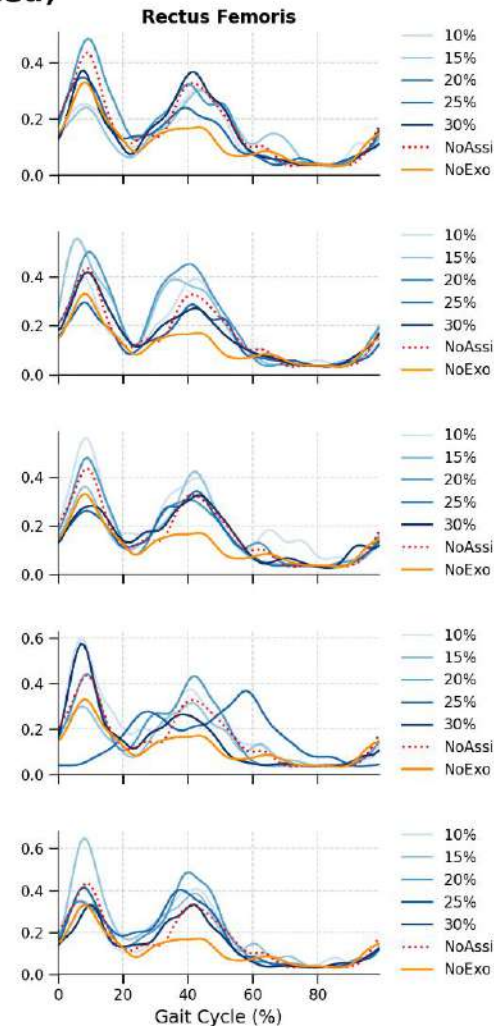
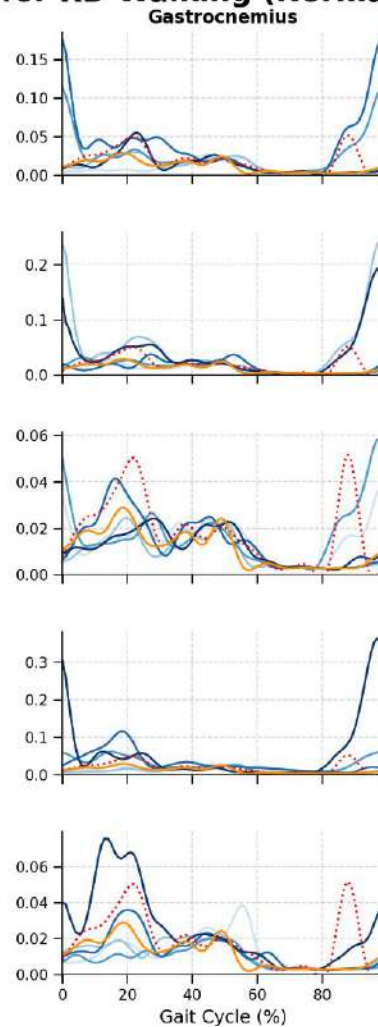
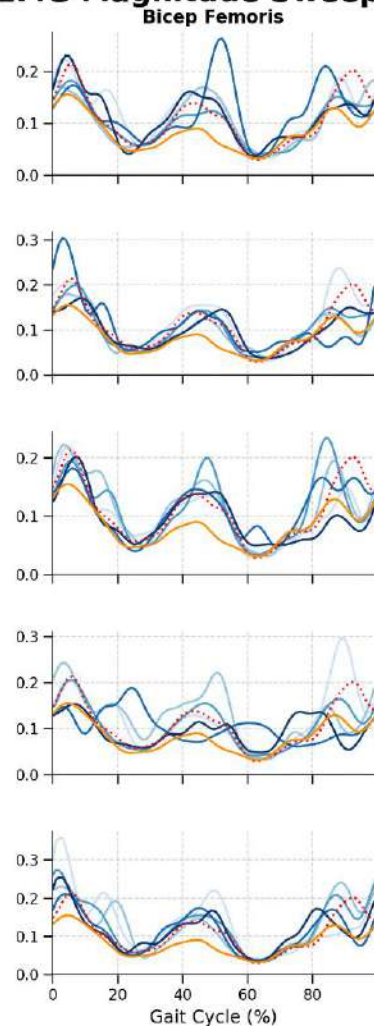
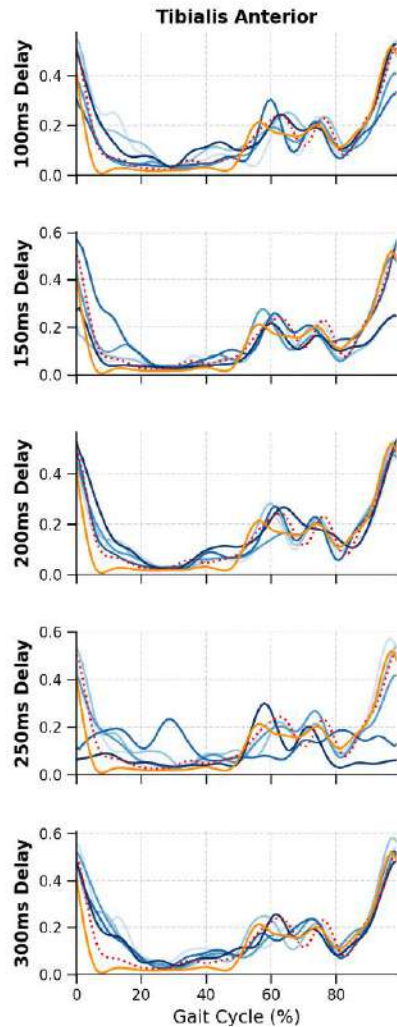


Peak EMG Activation Contour Map - RA Walking

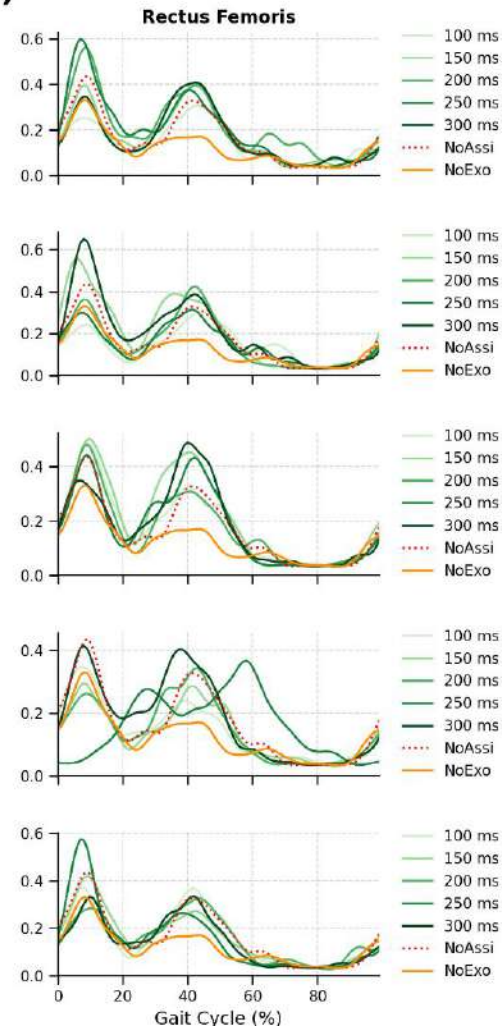
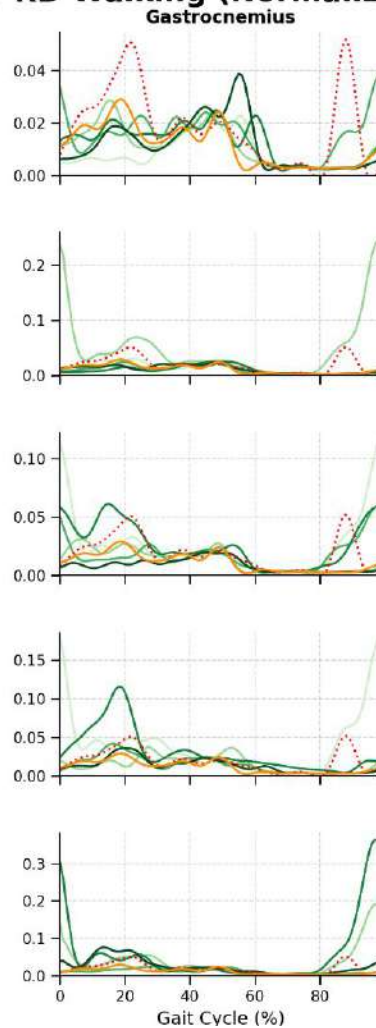
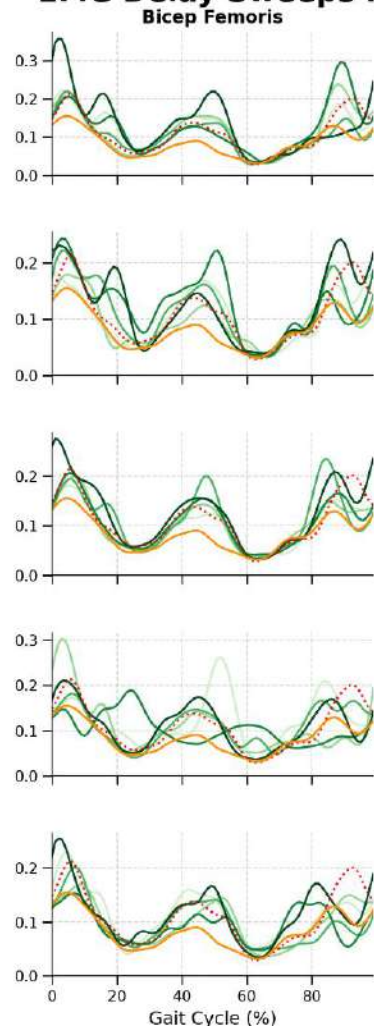
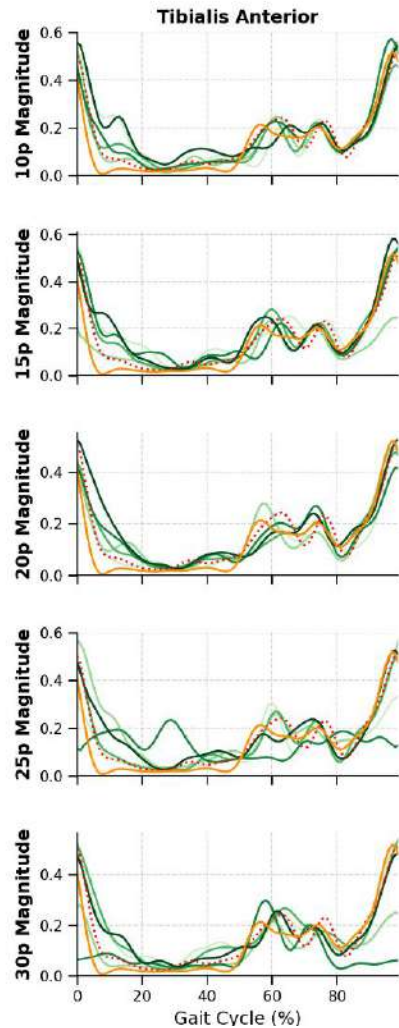


Ramp-Descent

EMG Magnitude Sweeps for RD Walking (Normalized)

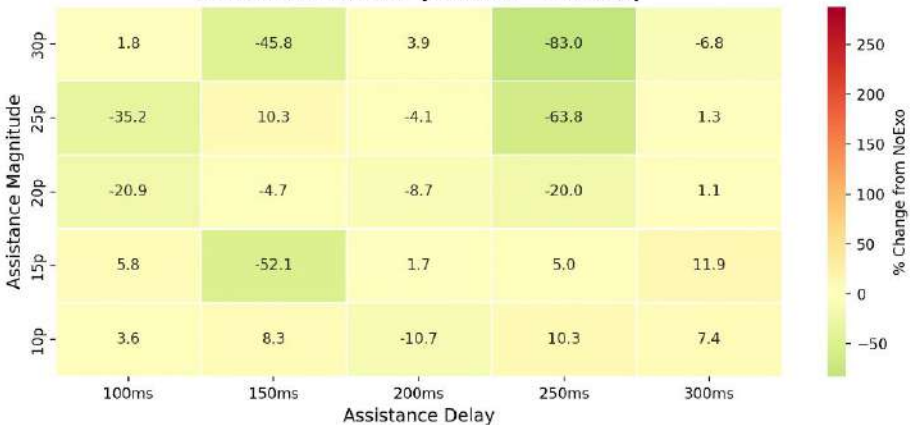


EMG Delay Sweeps for RD Walking (Normalized)



Peak EMG Activation (% Change from NoExo) - RD Walking (Phased)

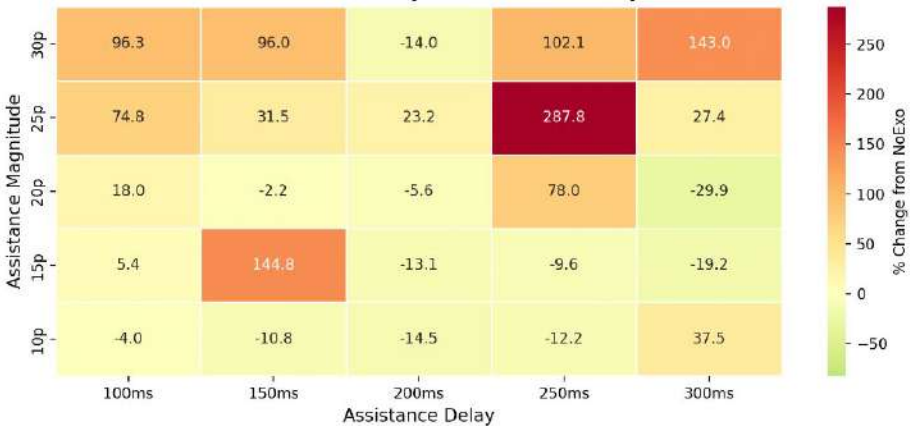
Tibialis Anterior (NoAssi = -2.0%)



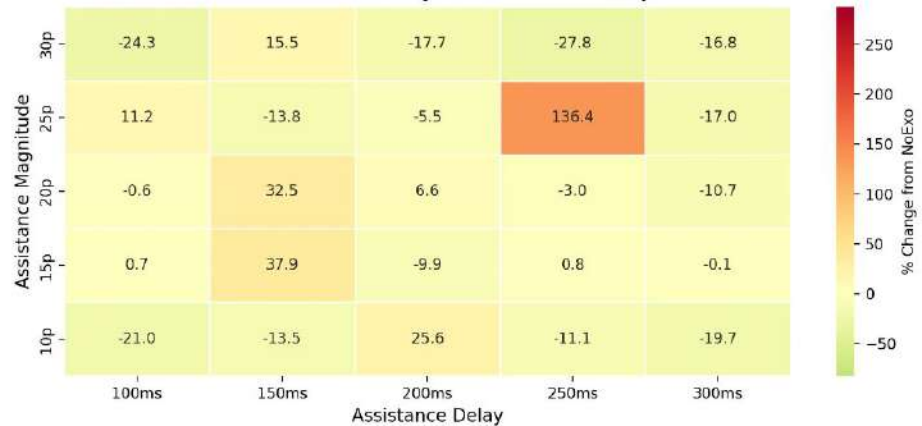
Bicep Femoris (NoAssi = 37.8%)



Gastrocnemius (NoAssi = 80.2%)

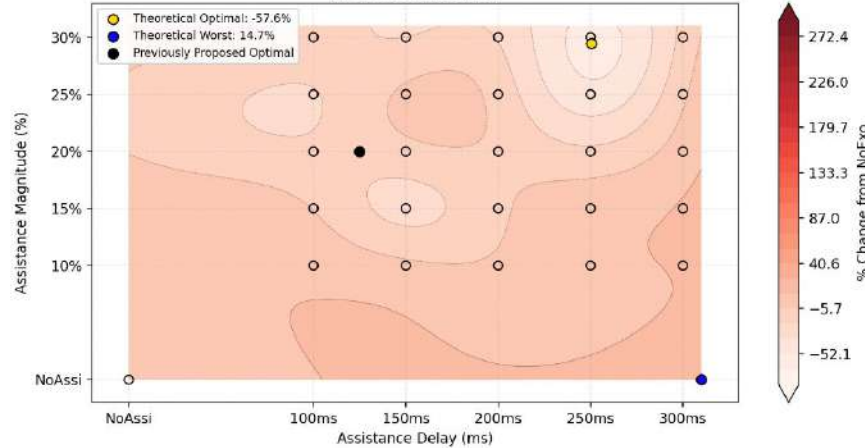


Rectus Femoris (NoAssi = 21.6%)

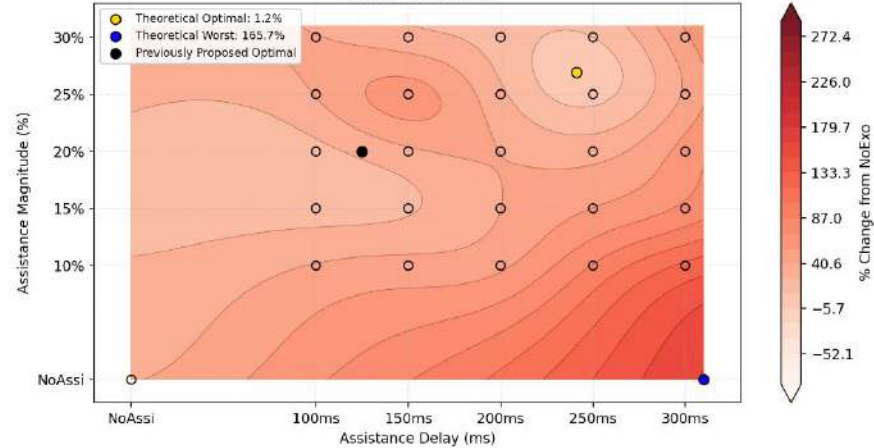


Peak EMG Activation Contour Map - RD Walking

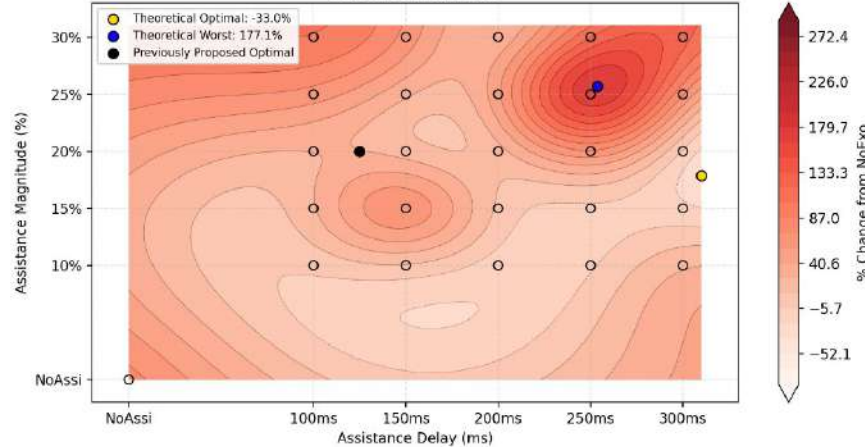
Tibialis Anterior



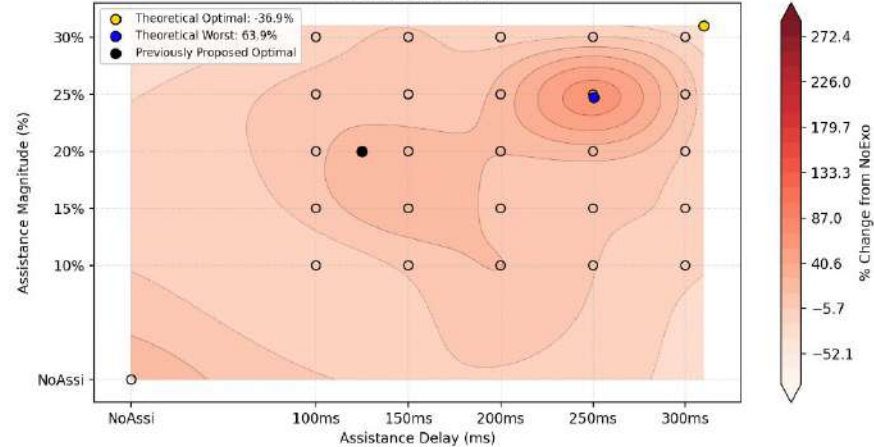
Bicep Femoris



Gastrocnemius



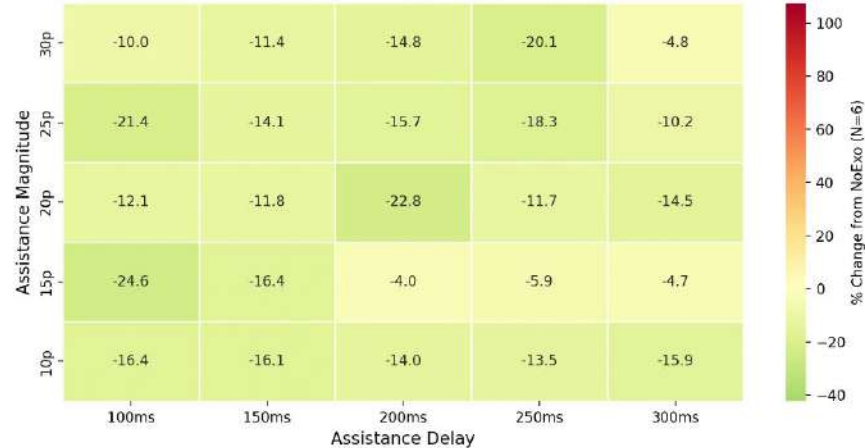
Rectus Femoris



N=6 Average Heat Map

Average Peak EMG (% Change from NoExo) - LG (N=6, Phased)

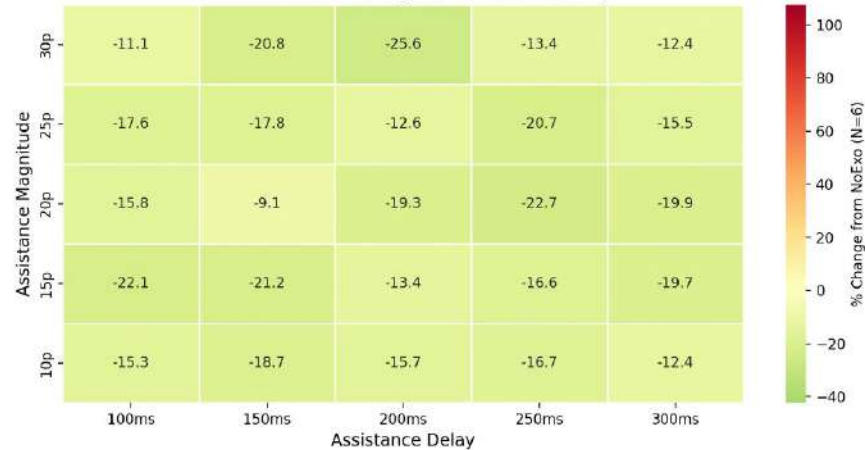
Tibialis Anterior (NoAssi: -12.3%)



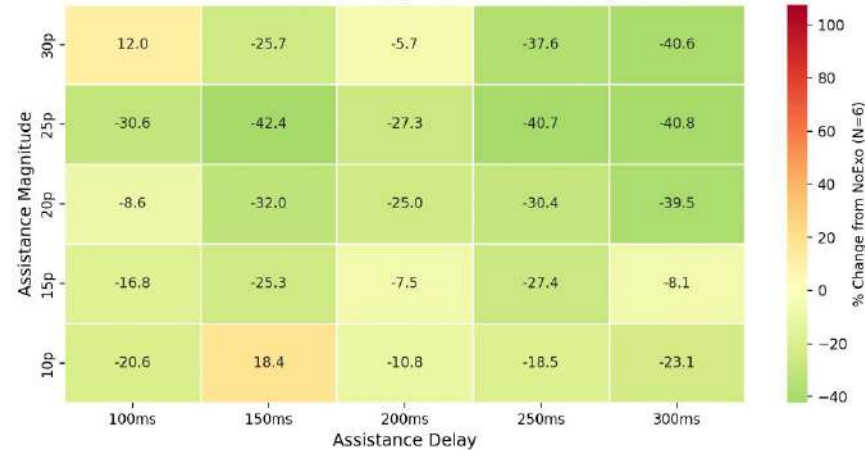
Bicep Femoris (NoAssi: -18.4%)



Gastrocnemius (NoAssi: -13.4%)



Rectus Femoris (NoAssi: +13.0%)

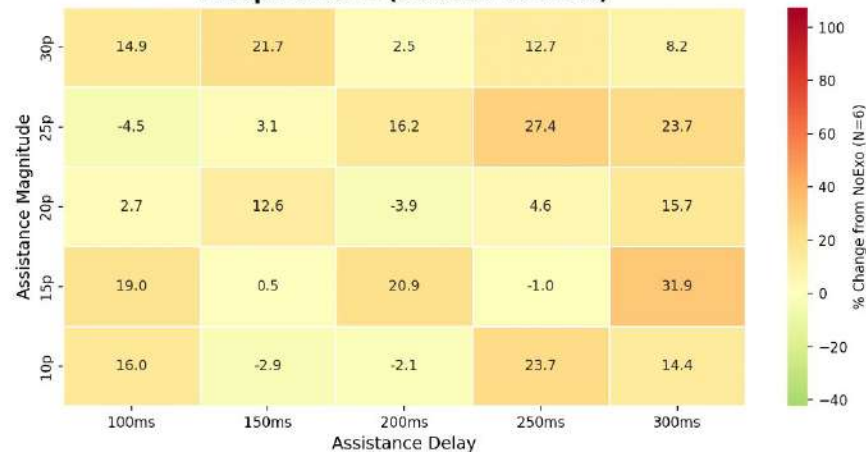


Average Peak EMG (% Change from NoExo) - RA (N=6, Phased)

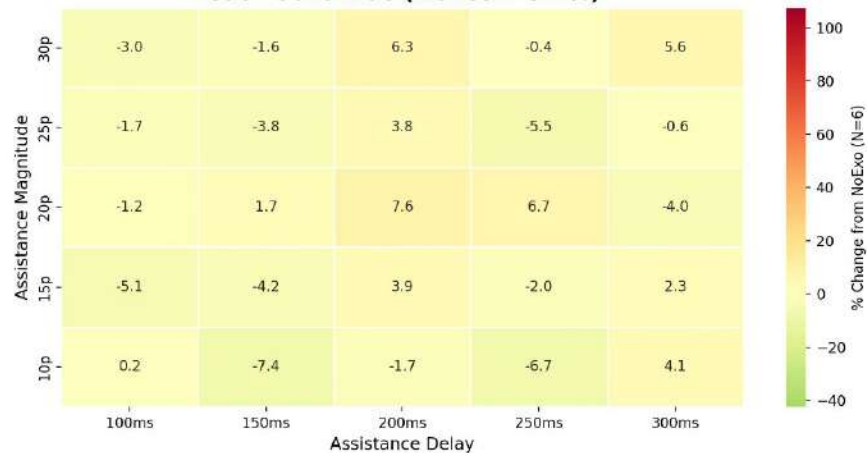
Tibialis Anterior (NoAssi: +11.2%)



Bicep Femoris (NoAssi: +21.2%)



Gastrocnemius (NoAssi: -3.7%)

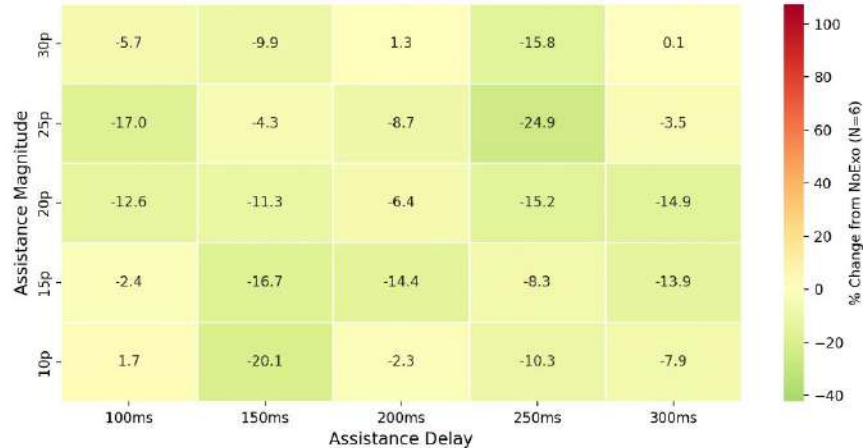


Rectus Femoris (NoAssi: +36.5%)

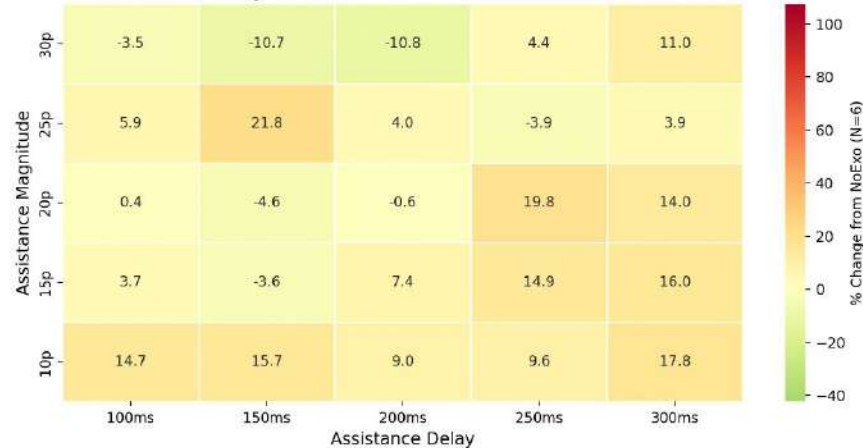


Average Peak EMG (% Change from NoExo) - RD (N=6, Phased)

Tibialis Anterior (NoAssi: -8.8%)



Bicep Femoris (NoAssi: +17.8%)



Gastrocnemius (NoAssi: +18.1%)

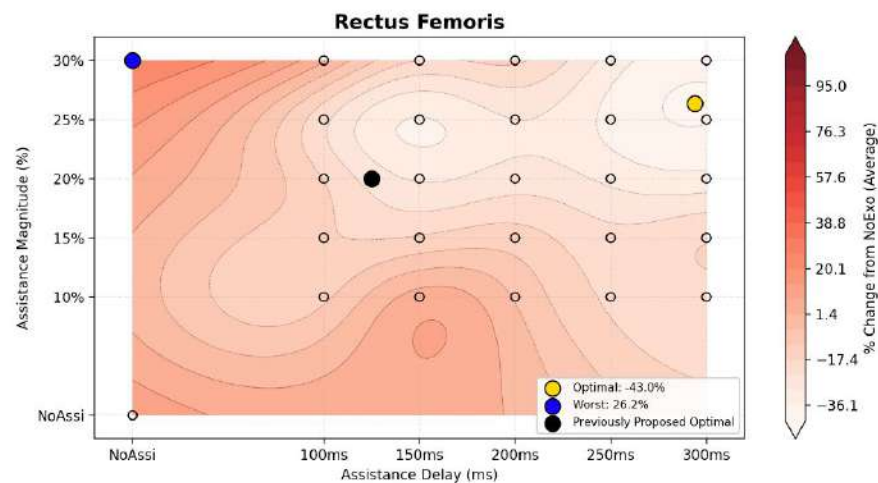
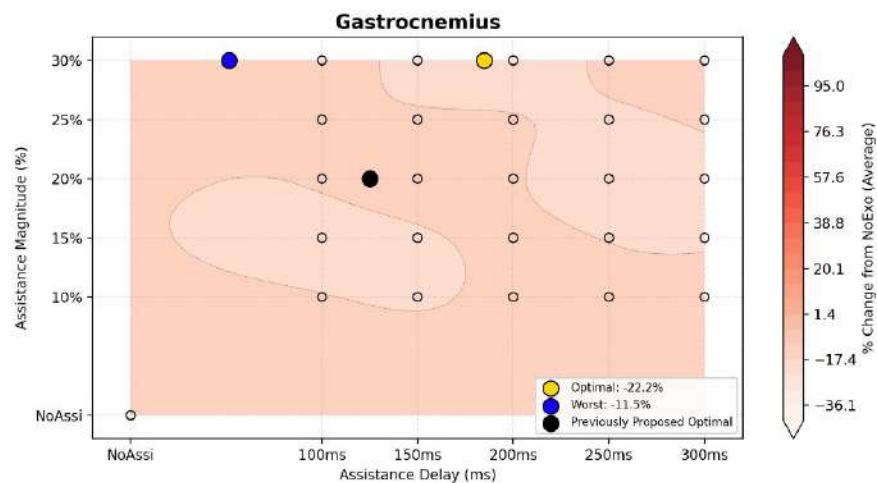
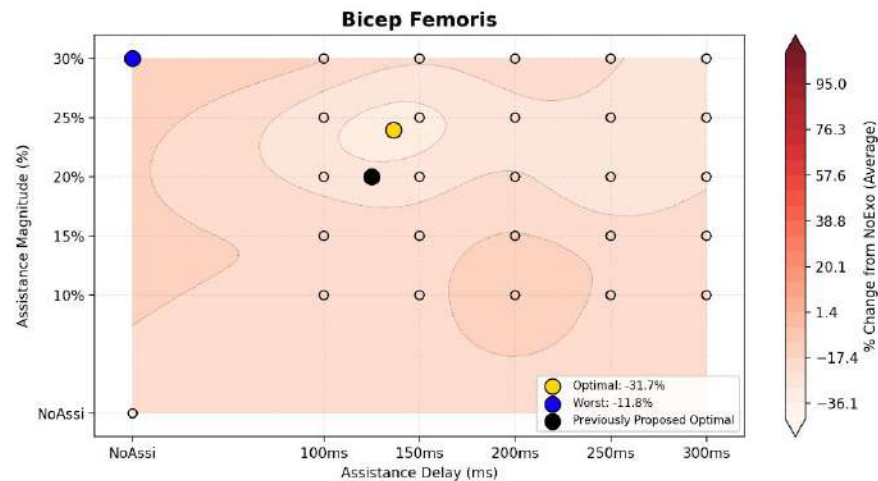
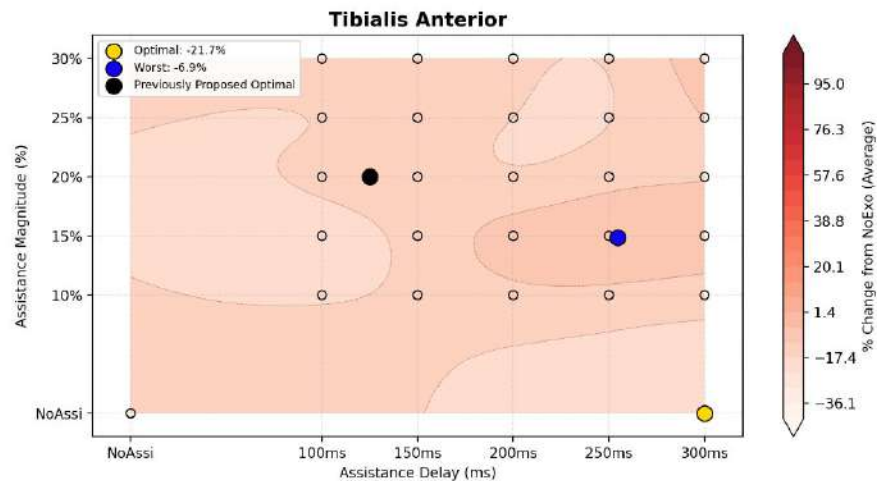


Rectus Femoris (NoAssi: +73.5%)

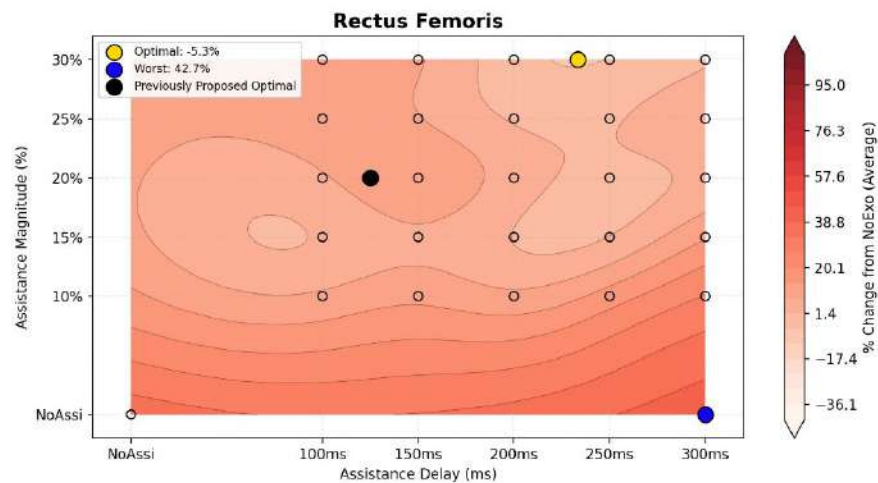
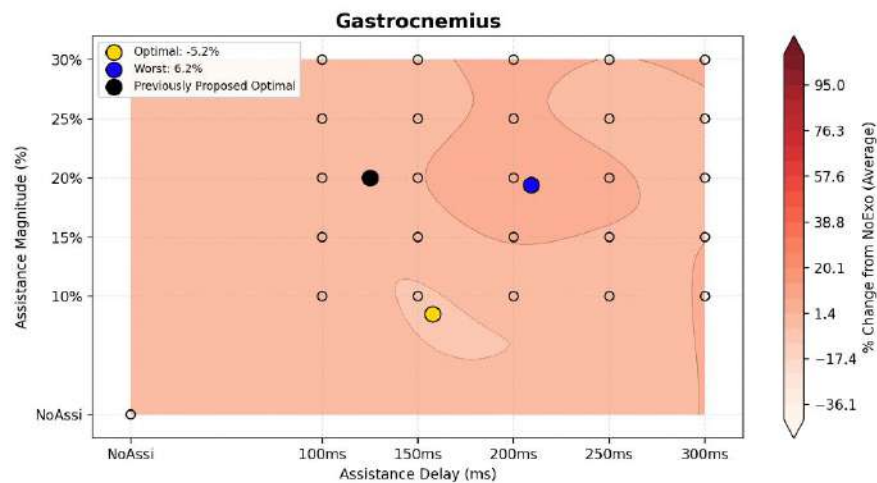
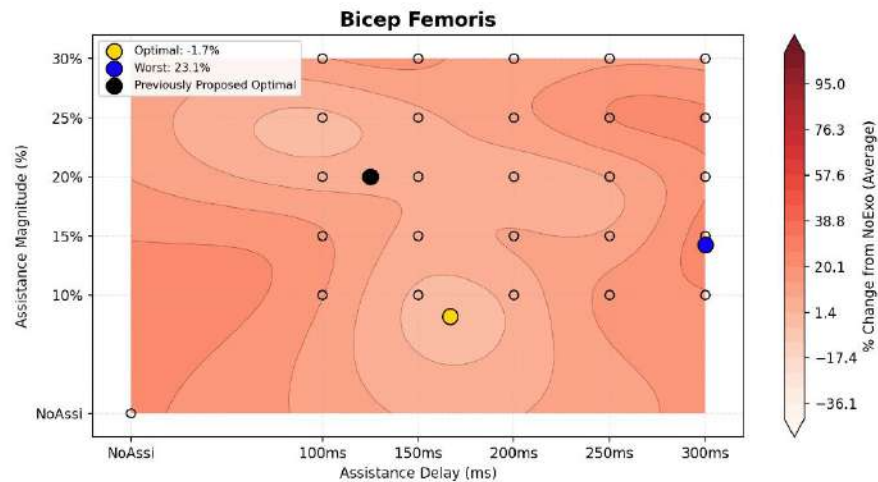
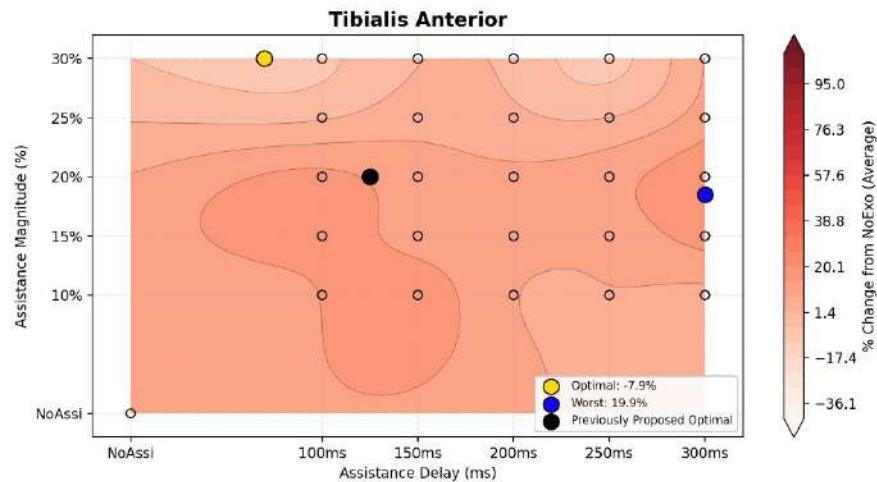


N=6 Average Contour Map

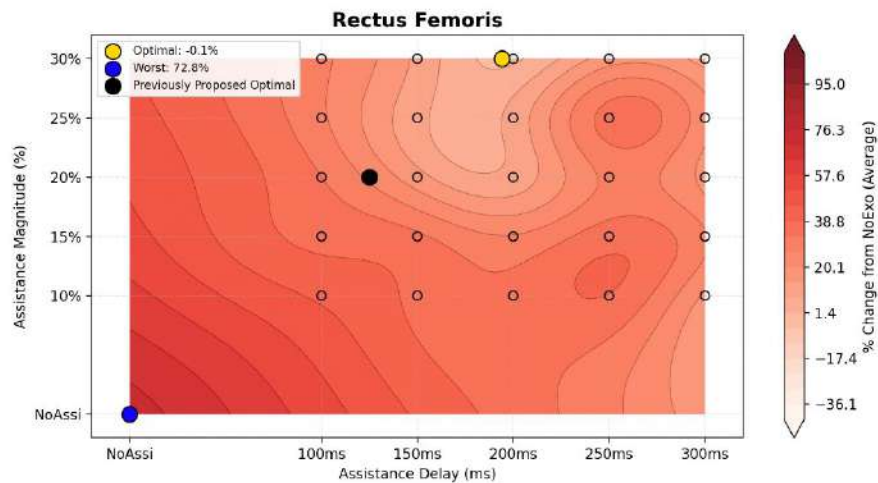
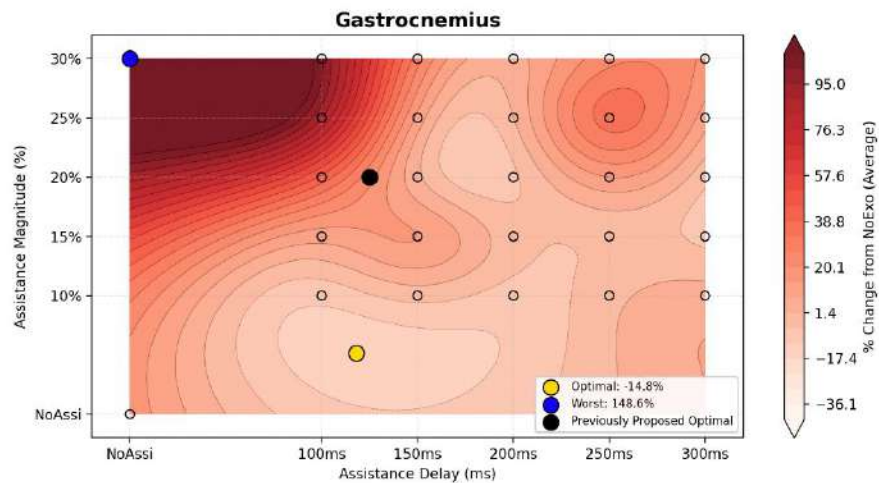
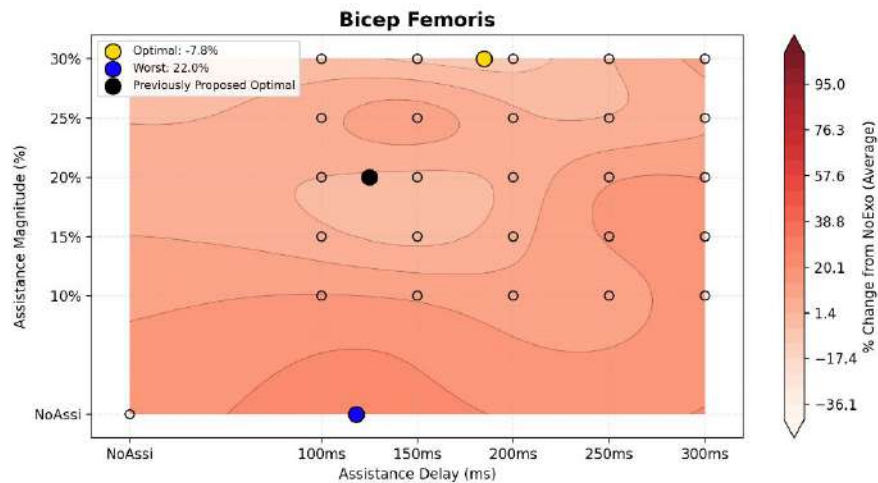
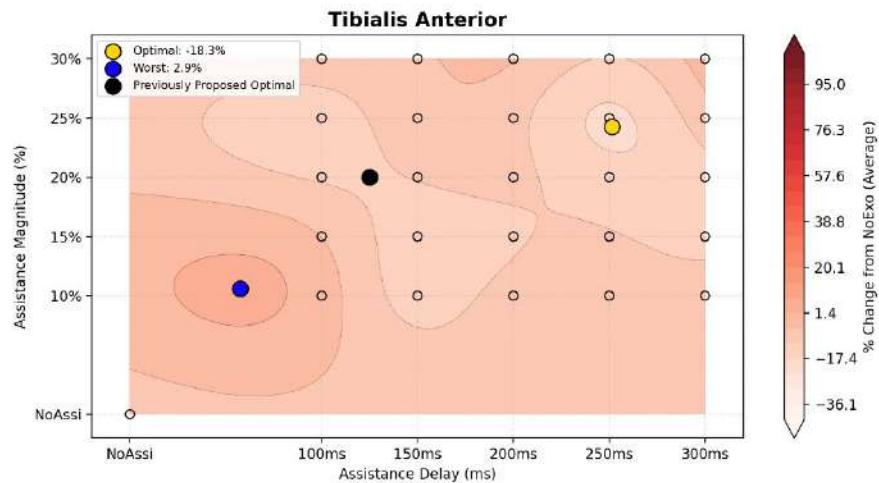
Average Peak EMG Contour Map - LG Walking (N=6)



Average Peak EMG Contour Map - RA Walking (N=6)

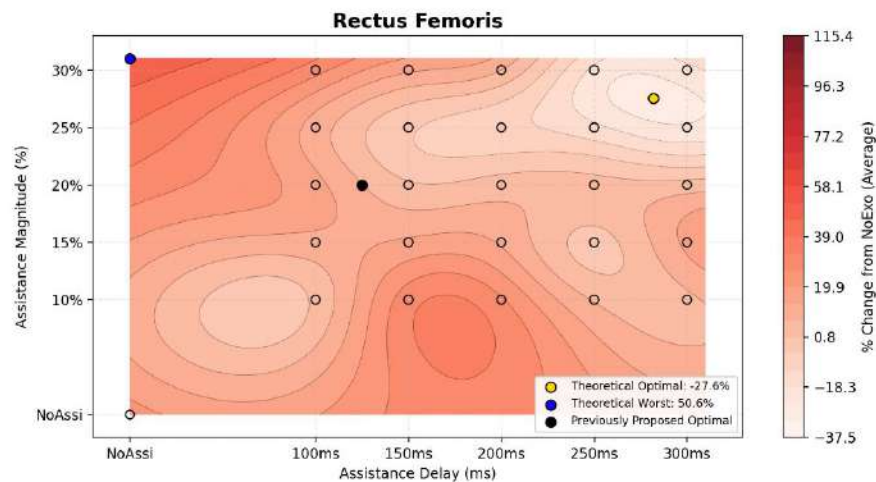
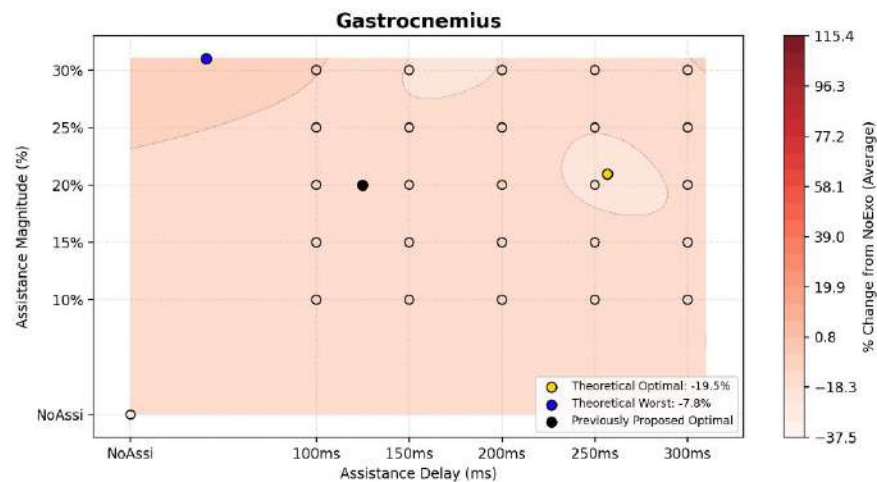
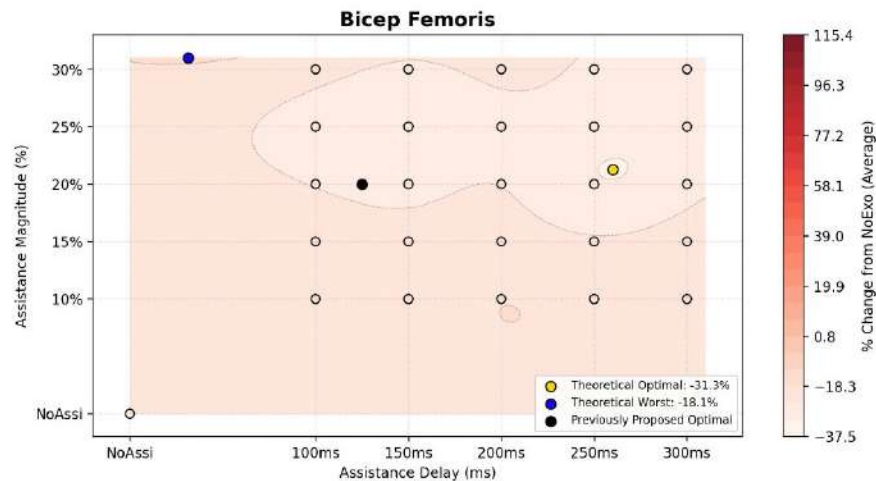
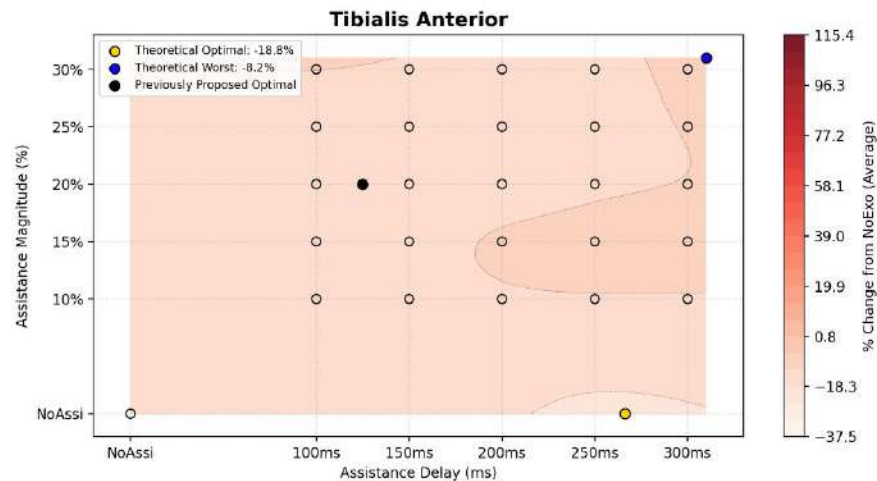


Average Peak EMG Contour Map - RD Walking (N=6)

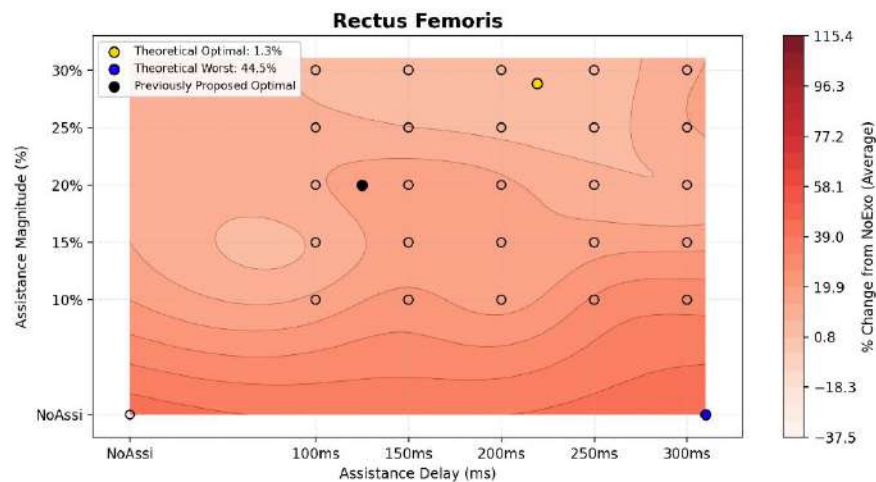
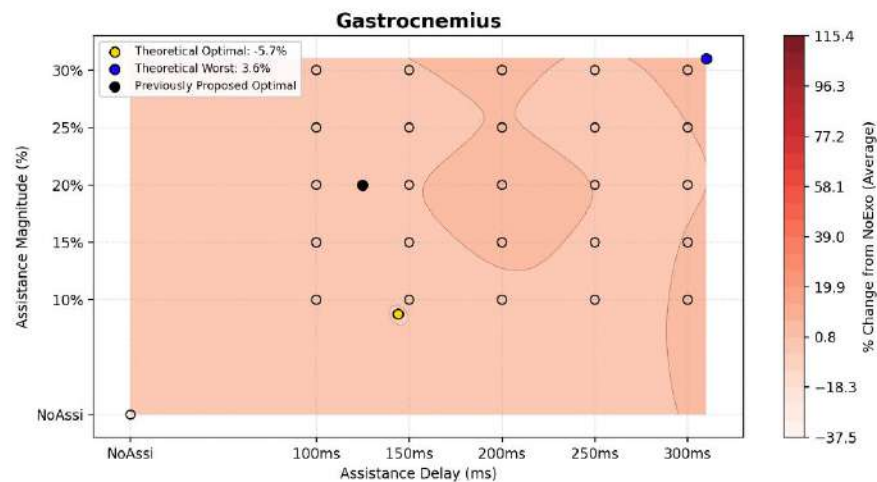
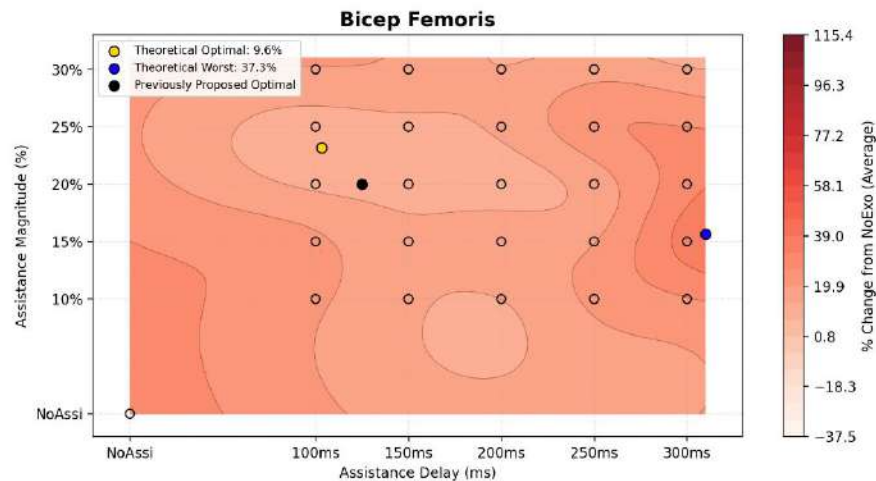
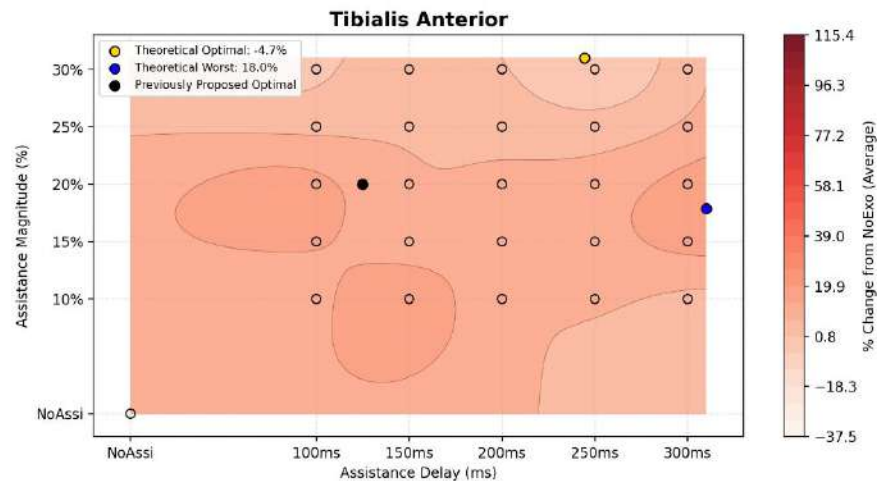


N=8 Average Contour Map

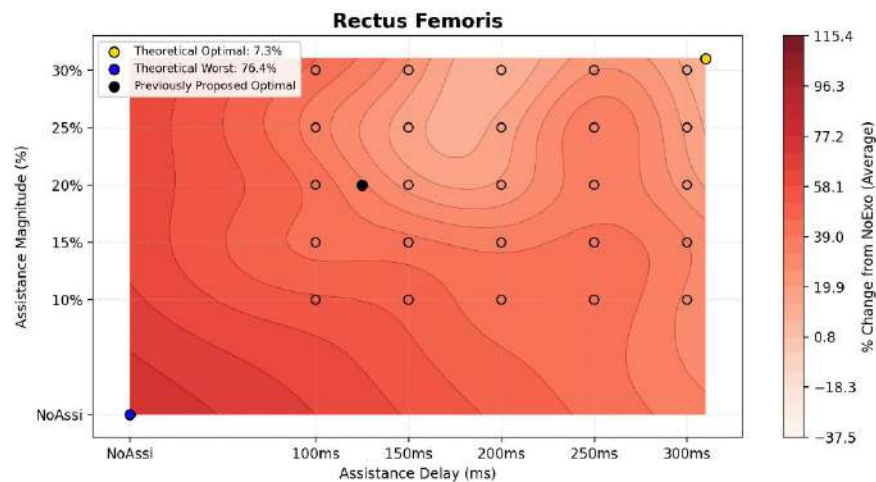
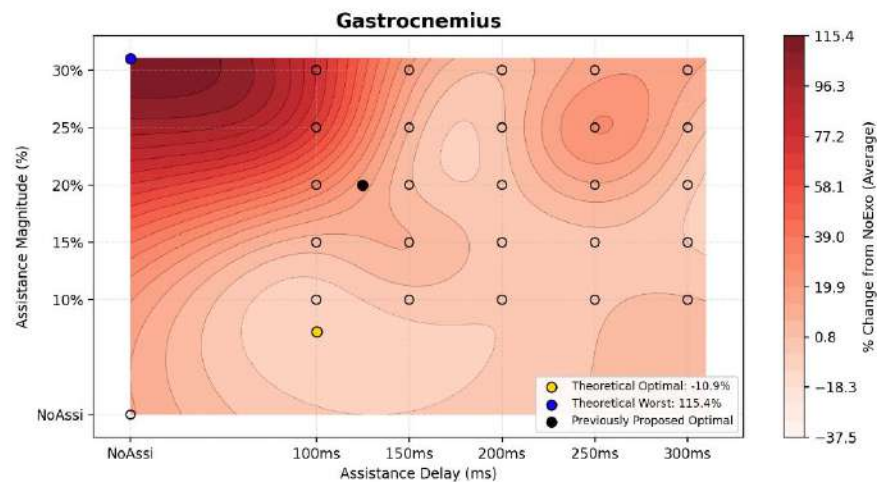
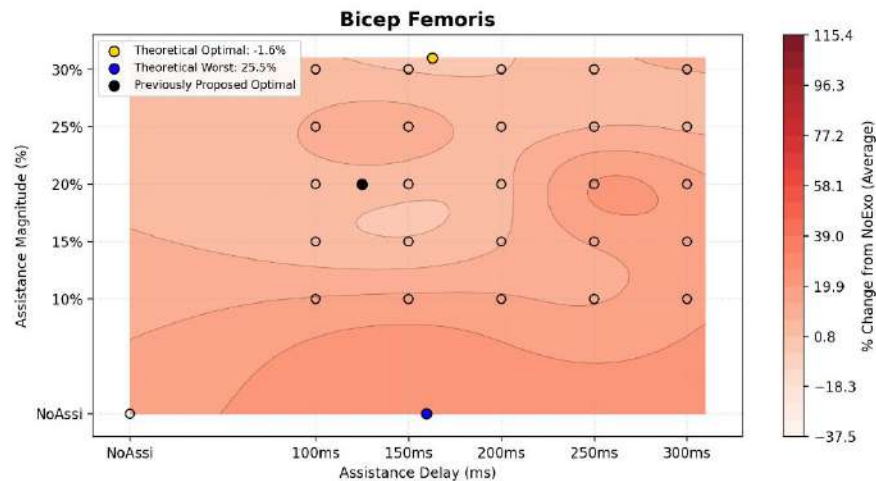
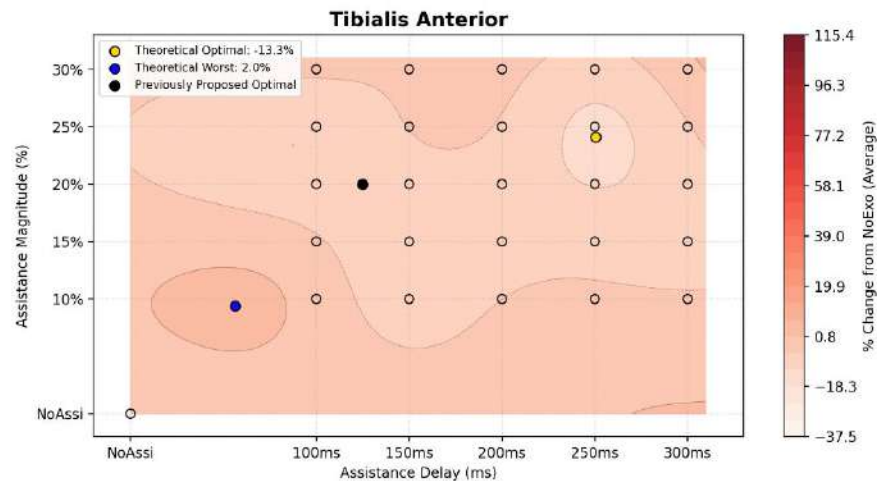
Average Peak EMG Contour Map - LG Walking (N=8)



Average Peak EMG Contour Map - RA Walking (N=8)



Average Peak EMG Contour Map - RD Walking (N=8)



LG Range

RF: Hip Flexor / Knee Extensor (60-100%)

BF: Hip Extensor / Knee Flexor, (85 - 110%)

TA: Ankle Dorsiflexor (80 - 110% → 0-10%, 80-100%)

GM: Ankle plantarflexor, knee flexor (20~60%)

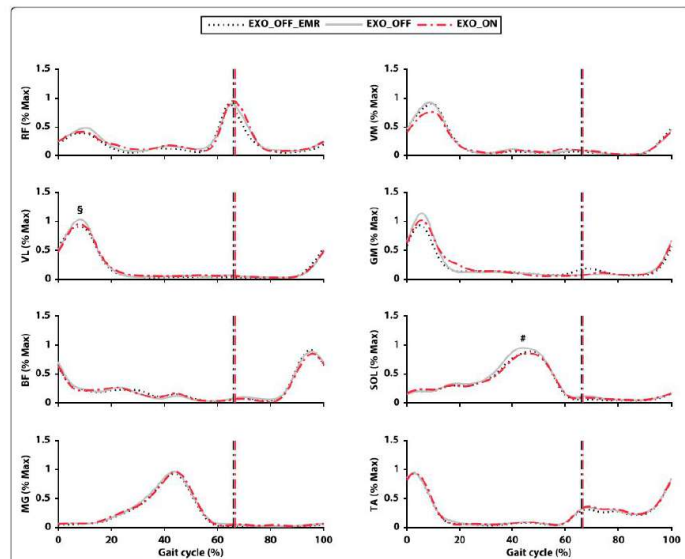
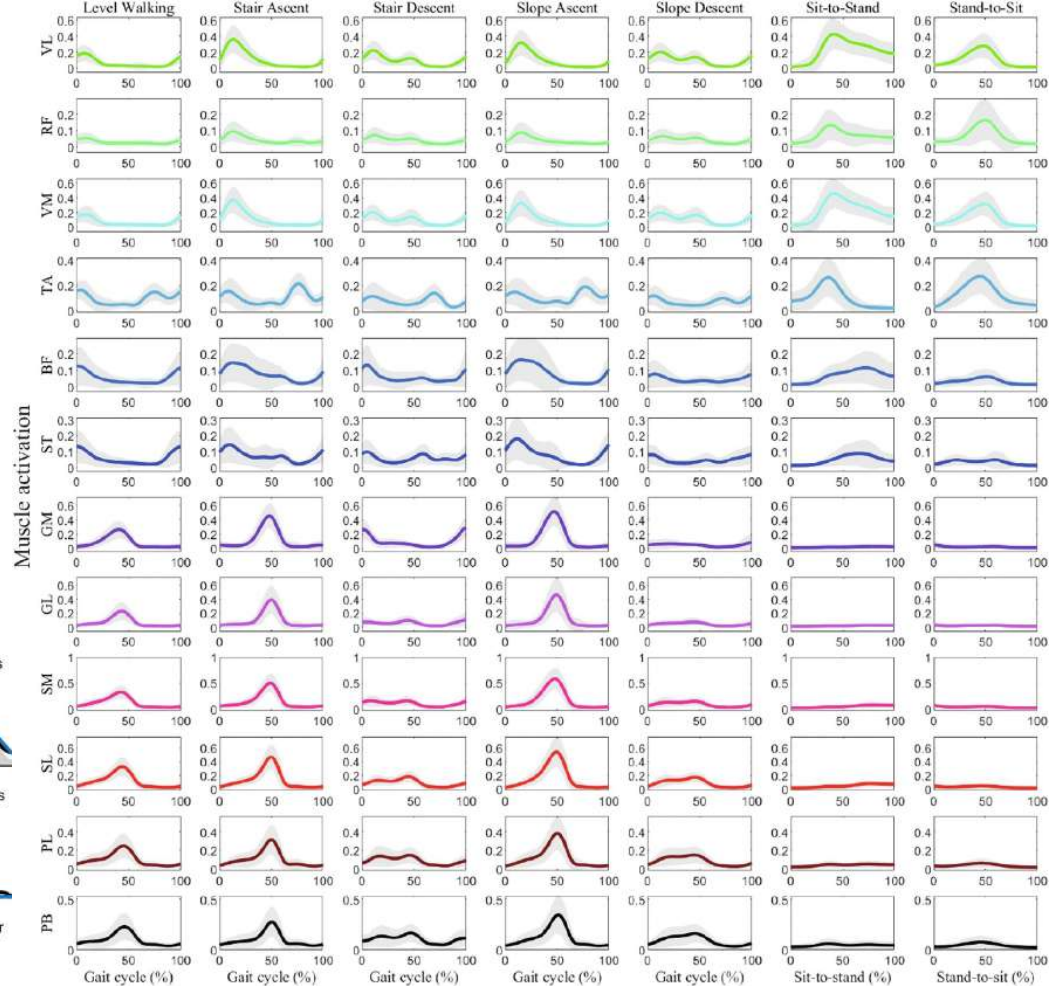
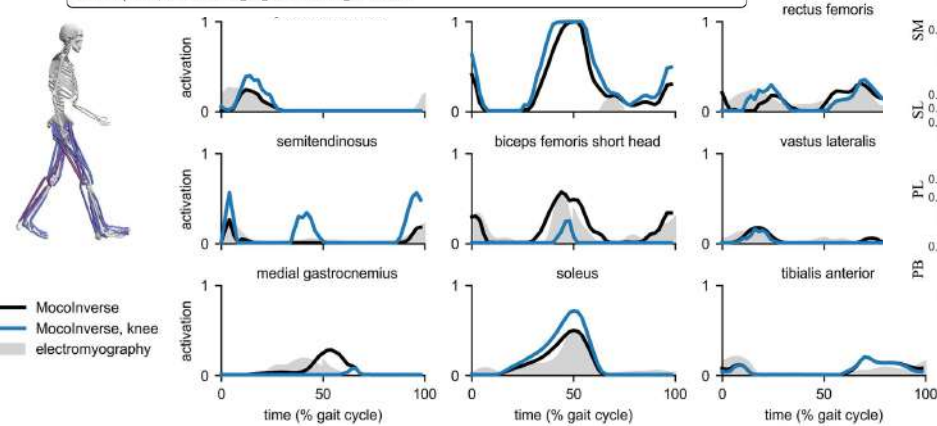


Fig. 6 Muscle activation. Normalized EMG linear envelope as a percent of gait cycle (heel-strike to heel-strike) for the eight muscles examined. The curves represent the three different conditions: EXO_OFF_EMR (dotted black), EXO_OFF (solid grey) and EXO_ON (dashed red). The dotted vertical lines represent toe off of each testing condition. # indicates a significant difference ($p < 0.05$) with respect to the EXO_OFF condition, § indicates a significant difference ($p < 0.05$) between the EXO_OFF_EMR and the EXO_OFF condition.



Muscle activation



— Mocolnverse
— Mocolnverse, knee
— electromyography

